Enhanced Fired-Clay Bricks Incorporating Cigarette Butts

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Doctor of Philosophy

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Declaration

I certify that except where due acknowledgement has been made, the work is that of the author alone; the work has not been submitted previously, in whole or in part, to qualify for any other academic award; the content of the thesis is the result of work which has been carried out since the official commencement date of the approved research program; any editorial work, paid or unpaid, carried out by a third party is acknowledged; and, ethics procedures and guidelines have been followed. I acknowledge the support I have received for my research through the provision of an Australian Government Research Training Program Scholarship.

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Abbreviations

CBs	Cigarette Butts
FCB	Fired Clay Brick
IRA	Initial Rate of Absorption
LCA	Life Cycle Assessment
HHV	Higher Heating Value
MC	Sandy Silty Clay
PSD	Particle Size Distribution
XRF	X-ray Fluorescence
XRD	X-ray Diffraction
TGA	Thermo-Gravimetric Analysis
SEM	Scanning Electron Microscopy
Micro-CT	X-ray micro-Computed Tomography
EPA	Environmental Protection Agency
TLS	Transient Line Source
ASEAN	Association of Southeast Asian Nations
ABLP	Australian Bottle Leaching Procedure
USEPA	United States Environmental Protection Agency
PAHs	Polycyclic Aromatic Hydrocarbons
TCLP	Toxicity Characteristics Leaching Procedure
LEAF	Leaching Environmental Assessment Framework
MW	Moderate Weathering
pH	Power of the Hydrogen Ion
AC	Asphalt Concrete
SMA	Stone Mastic Asphalt
ASTM	American Society for Testing and Materials
ICP-MS	Inductively Couples Plasma-Mass Spectrometry
GC-MS	Gas Chromatography-Mass Spectrometry
VOCs	Volatile Organic Compounds
CHNS	Carbon Hydrogen Nitrogen Sulphur
IS	Internal Standard

Executive Summary

Cigarette butts (CBs) are the world's most prevalent source of waste litter. In the year 2016, 5.7 trillion cigarettes were consumed worldwide, and about 97% of the cigarette filters were composed of cellulose acetate, a modified natural polymer. In Australia, it is estimated that over 20 billion filtered cigarettes are consumed each year, of which an estimated 7 billion are littered into the environment. The littering of CBs is a growing waste management issue and requires specific planning. When CBs are littered, harmful chemicals are leached into the environment, posing major concerns for the global environment and the quality of urban and aquatic life. In cigarette smoke, it is possible to classify more than 4,000 chemical components that are generated during burning or that are distilled from tobacco, of which 69 are carcinogenic. Cigarettes are made of cellulose acetate filters, which are designed to fully or partially retain particulate smoke components. Cellulose acetate filters have poor biodegradability and can take up to 18 months to decompose under normal environmental conditions, hence, posing a serious dilemma in terms of toxic waste and litter disposal. Based on several years of research, Mohajerani et al. (2016) proposed that the CB conundrum can potentially be eliminated by incorporating 1% CBs into 2.5% of the world's total fired clay brick (FCB) production. This research is a continuation and extensive expansion of the parent research on recycling CBs in FCBs. The initial stage of the research involved a comprehensive laboratory investigation to develop techniques to recycle CB waste in FCBs without compromising the physical and mechanical properties. FCB samples were prepared incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs and were tested for shrinkage, water absorption, initial rate of absorption, density, compressive strength, and efflorescence. The results from this investigation met the Australian Standards for masonry building blocks. The second part of the research investigated the thermal conductivity and energy savings of FCBs incorporating CBs, whereby the results from the calculation and measurement method were compared. For the energy savings study, a novel method was developed to monitor the energy consumption of the furnace during the firing process of the FCB samples incorporating CBs. An increase in CB content positively resulted in a decrease in density and thermal conductivity and an increase in energy savings. X-ray micro-Computed Tomography (micro-CT) imaging was utilized to characterize the quantity, distribution, size, and shape of the micro- and macro-scale pores generated from the combustion of the CBs during the firing process. The 3D models demonstrated a homogenous distribution of pores within the brick structure. The third part of the investigation involved a comprehensive leachate analysis of heavy metals for used, unused, and shredded used CBs, paraffin wax and bitumen encapsulated CBs, and unfired and fired clay bricks incorporating CBs. The leachate analysis was conducted according to the Australian Bottle Leaching

Procedure (ABLP) for pH values 2.9, 5.0, and 9.2 using the Inductively Coupled Plasma Mass Spectrometry (ICP-MS). The leaching behavior of fired bricks incorporating CBs was considerably lower than that for the unfired bricks due to the immobilization of heavy metals during the firing process. Moreover, it was found the bitumen and wax encapsulation process successfully trapped the heavy metals present in the CBs. The leachate results for all samples were found to be below the regulatory limits set by the EPA Victoria solid industrial waste guidelines. To investigate the potential residual remaining in FCBs incorporating CBs a comprehensive polycyclic aromatic hydrocarbon (PAH) extraction analysis was conducted in the fourth part of the investigation using the Soxhlet extractor and rotary evaporator. The samples were analysed using gas chromatography-mass spectrometry (GC–MS). The ΣPAH concentrations for all five FCB–CB mixes were well below the EPA Victoria solid waste hazard categorization threshold for industrial waste. Moreover, the samples were studied for their carbon content using the carbon, hydrogen, nitrogen, and sulfur (CHNS) analyzer and thermogravimetric analysis (TGA). The results confirm an almost 100% combustion process of CBs during the firing process. The following part of the research involved the implementation of recycling CBs in bricks on an industrial scale. A bacteriological study was conducted, which involved the investigation of the presence of ten common bacteria in two pilot studies, various sterilisation methods for CBs were examined, and three processing methods were suggested for the implementation of recycling CBs in FCBs on an industrial scale. Contact of naphthalene balls with CBs had a significant disinfecting effect on *Bacillus* spp. The final part of the research involved investigating the issues relative to the strong odor of CBs and the potential release of volatile organic compounds (VOCs) during the storage and manufacturing process of CBs in FCBs. The aim was to determine techniques to detect and analyze the emission of VOCs generated from CBs, along with investigating methods to minimize or eradicate the odors generated from CBs. Charcoal sorbent tubes were used to collect the gas components, and gas chromatography-mass spectrometry (GC-MS) was used to identify and quantify the VOCs generated. Ultraviolet-C (UVC) light was implemented to determine its effect on odor reduction and mitigation. The results confirm the pollutant gas production in UVC light treated CB samples is significantly less than that of untreated CB samples. This research has discussed the world CB pollution issue, potential solutions, and the scope of expansion and further development.

Chapter 1: Introduction

1.1 Background

Cigarette butts (CBs) are the world's most prevalent source of waste litter. The Tobacco Atlas [4] reported that during the year 2016, 5.7 trillion cigarettes were consumed worldwide and that about 97% of the cigarette filters were composed of cellulose acetate, a modified natural polymer [5]. In Australia, it is estimated that over 20 billion filtered cigarettes are consumed each year, of which an estimated 7 billion are littered into the environment [6]. That is 1 out of every 3 smoker litter their CBs into the environment.

The Cigarette Butt Pollution Project in the USA estimated that one-third of the total production of cigarettes was littered to the environment [7]. While Clean Up Australia [8] reported CBs as the most common source of rubbish collected in Australia under the miscellaneous category, representing 91.5%. The littering of CBs is a growing waste management issue and requires specific planning. As climate change is becoming the world's biggest challenge, recycling wastes for sustainable development is becoming more crucial [9].

There has been a great attempt by many Australian companies to develop CB disposal systems. The system involves placing receptacles that come in the form of wall-mounted, post-mounted, and freestanding in areas easily accessible by the public to dispose of their CB waste. Servicing regimes are also provided by companies. The CBs are then collected from the receptacles and disposed of. Landfilling and incineration are the two disposal methods for the collected CBs, however, both methods are costly and pose high exposure risks from hazardous fumes, which can be severely detrimental to the environment and human health [10].

When CBs are littered, harmful chemicals are leached into the environment, posing major concerns for the global environment and the quality of urban and aquatic life. In cigarette smoke, it is possible to classify more than 4,000 chemical components that are generated during burning or that are distilled from tobacco, of which 69 are carcinogenic [11, 12]. The main toxic agents include polycyclic aromatic hydrocarbons, formaldehyde, argon, carbon monoxide, nitrogen oxides, benzene, hydrogen cyanide, ammonia, phenol, acetaldehyde, argon, acetone, and pyridines [11].

Cigarettes are made of cellulose acetate filters, which are designed to fully or partially retain particulate smoke components, including tar and toxic chemicals [13]. Cellulose acetate filters have poor biodegradability and can take up to 18 months to decompose under normal environmental conditions,

hence, posing a serious dilemma in terms of toxic waste and litter disposal [14, 15]. CBs littered on streets commonly get washed away into storm-water drains and end up on beaches, or in rivers and harbors, leaching toxic chemicals [16]. Marine life regularly mistake CBs for food, and they have been discovered in the stomachs of sea turtles, birds, fish, and other creatures in which they have caused severe digestive problems that have led to death [17, 18]. Being a major environmental issue, this has led researchers to explore feasible and practical recycling techniques. The incorporation of CBs in construction materials, particularly in fired clay bricks [1] and asphalt concrete [19-21], has gained pervasive interest.

A number of studies have been conducted on recycling wastes in FCBs to investigate the relationship between the use of additives in fired clay bricks and the micro-pores and nano-pores formed as a result. Such additives include waste sawdust [15], desulfurization slag and basic oxygen furnace aggregates [16], marble powder [17], biomass incineration [18], recycled paper processing residues [4], bio-briquette ash [19], sugarcane bagasse ash [20], and cigarette butts [21–27]. A recent study revealed the substructures produced by the additive olive pomace (10% weight) that resulted in a 14.4% decrease in thermal conductivity [28]. Comparing this to expanded vermiculite (10% content), a pore-forming material that expands when heated, declined from 0.96 to 0.65 W mK^{-1}, which is a 32% drop in thermal conductivity [3]. While olive mill solid residue (OMSD) demonstrated the highest decrease in thermal conductivity of 70% with the use of 15% OMSD in fired clay bricks [29].

In 2016, Mohajerani et al. [1] developed a novel method involving the recycling of CBs in fired clay bricks (FCBs) and proposed by incorporating 1% CBs into 2.5% of the worlds brick production, the global waste catastrophe can potentially be solved [1, 22-24]. With the addition of 2.5% CBs in fired clay bricks, density decreased by 8.3%, while bricks incorporating 1% CBs, the compressive strength was determined to be 19.53 MPa. All the mechanical and physical property values were within the acceptable range of the Australian Standards [1]. Through the addition of CBs, porous bricks were produced, and a drastic decrease in thermal conductivity and an increase in energy savings was observed with the increase in CB content [1].

1.2 Research Objectives

This study aims to investigate and develop environmentally and economically sustainable and practical techniques to recycle CB waste in FCBs. To major objectives are:

1. To investigate and develop practical techniques to recycle CB waste in FCBs without compromising the physical and mechanical properties and determine the optimum percentage of CBs that can be recycled in FCBs.

2. To study energy savings during the firing process of clay bricks incorporating CBs and investigate the reduction of thermal conductivity of FCBs incorporating CBs.

3. To conduct a macro- and micro-structural analysis of FCBs incorporating CBs to quantify the porosity and characterize the size, shape, and distribution of pores.

4. To conduct a comprehensive leachate analysis of; (1) FCBs incorporating CBs and (2) bitumen and wax encapsulated CBs for heavy metals at three pH values.

5. To conduct an extraction analysis of Polycyclic Aromatic Hydrocarbons (PAHs) from FCBs incorporating CBs and investigate the carbon content in FCBs recycled with CBs.

6. To develop practical procedures and guidelines for the implementation of the incorporation of CBs in FCBs on an industrial scale.

7. To investigate and develop techniques to minimize or eradicate the odors generated during the production processes of FCBs incorporating CBs.

1.3 Significance of the Research

1. The research will develop and establish techniques to recycle CBs in FCBs.

2. The research will potentially reduce the pollution caused by littered CBs.

3. Lightweight, low-cost, energy-efficient, and environmentally friendly FCBs will be produced with the addition of CBs.

4. The addition of CBs will partially replace the main raw material (clay) used in making FCBs.

1.4 Thesis Structure

This study developed an environmentally and economically sustainable and practical technique to recycle CB waste in FCBs. All stages of the proposed application on an industrial scale were addressed and investigated. This thesis is divided into seven chapters.

Chapter 1 provides a comprehensive background of the research study, research objectives, significance of the research, and thesis structure.

Chapter 2 provides a comprehensive review through its discussion of the various studies that have been carried out on the toxicity and valorization of cigarette butt waste and investigates the feasibility and sustainability of the recycling methods adopted.

Chapter 3 investigates the physical and mechanical properties of fired clay bricks incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by mass. The energy-saving percentage during the firing process of bricks and the reduction in thermal conductivity are presented as a function of the CBs present in the mixtures. Additionally, water absorption, compressive strength, efflorescence, initial rate of absorption (IRA), density, and shrinkage were tested.

Chapter 4 investigates the thermal properties, energy savings, porosity, and micro- and macro-structure of fired clay bricks incorporating various percentages of CBs. A novel method was developed to monitor the energy consumption of the furnace during the firing process, and Scanning Electron Microscopy (SEM) and X-ray micro-Computed Tomography (micro-CT) imaging were used to characterize the quantity, distribution, size, and shape of the micro- and macro-scale pores of the brick samples incorporating CBs.

Chapter 5 presents and discusses the leaching behaviour of potentially hazardous metals from used, unused, shredded used CBs, bitumen and paraffin wax encapsulated CBs, and unfired and fired clay bricks incorporating CBs. The leachate analysis was conducted according to the Australian Bottle Leaching Procedure (ABLP) for pH values 2.9, 5.0, and 9.2.

Chapter 6 aims to conduct a comprehensive evaluation of PAHs in FCBs incorporated with CBs to examine the potential environmental impact of utilizing this toxic waste in the production of FCBs. In this study, a direct Soxhlet extraction method was employed, and the extract was analyzed for sixteen US EPA priority PAH compounds using gas chromatography-mass spectrometry (GC–MS). Furthermore, the FCB samples incorporating CBs were studied for their carbon content using the carbon, hydrogen, nitrogen, and sulfur (CHNS) analyzer and thermogravimetric analysis (TGA).

Chapter 7 aims to conduct a bacteriological study involving the investigation of the presence of ten common bacteria in two pilot studies. In addition, to examining various sterilisation methods for CBs. Moreover, three processing methods are studied and suggested for the implementation of recycling CBs in FCBs on an industrial scale.

Chapter 8 aims to determine techniques to detect and analyze the volatile organic compounds (VOCs) generated from CBs, along with investigating methods to minimize or eradicate the odors generated during the production processes of FCBs incorporating CBs. Several industrial methods and techniques to eliminate the emissions on an industrial scale were reviewed and analysed, including ultraviolet-C (UVC) light, ozone, and vaporized hydrogen peroxide.

Chapter 9 summarises the key findings, provides conclusions based on this study and offers future recommendations for possible studies in this area of research.

Chapter 2: The Toxicity and Valorization Options of Cigarette Butts

Abstract

Cigarette butts, one of the most abundant forms of waste in the world, contain more than 4,000 toxic chemicals and pose serious risks to the health of wildlife, humans, and marine and freshwater organisms. Although trivial in size, trillions of cigarettes are produced every year worldwide, resulting in the accumulation of tonnes of toxic waste litter. In 2016, a world production of over 5.7 trillion cigarettes was reported with the majority comprising cellulose acetate filters – a polymer with poor biodegradability. Depending on the environmental conditions, cellulose acetate filters can take up to 10 years to decompose during which time they leach heavy metals and toxic chemicals into the environment. Although possible disposal methods for collected cigarette butt waste include incineration and landfilling, both techniques may result in the release of hazardous fumes and can be costly. However, recycling CBs in different materials could be a possible solution for this concurrent environmental pollution. A number of novel studies have been publicized on recycling cigarette butts with encouraging results, and several methods have been studied, including the recycling of cigarette butts in asphalt concrete and fired clay bricks, as a carbon source, sound absorbing material, corrosion inhibitor, biofilm carrier, and many more. Hence, this paper provides a comprehensive review through its discussion of the various studies that have been carried out on the toxicity and valorization of cigarette butt waste and investigates the feasibility and sustainability of the recycling methods adopted. However, for the widespread application of recycling cigarette butt waste, further research and developments are essential.

The research presented in this chapter has been published in the following:

Kurmus, H. and Mohajerani, A., 2020. The toxicity and valorization options of cigarette butts. Waste Management, 104, 104–118. 104-118. https://doi.org/10.1016/j.wasman.2020.01.011.

2.1 Introduction

Cigarette butts (CBs) are the world's most prevalent source of waste litter. The Tobacco Atlas [4] reported that during the year 2016, 5.7 trillion cigarettes were consumed worldwide and that about 97% of the cigarette filters were composed of cellulose acetate, a modified natural polymer [5]. In Australia, it is estimated that over 20 billion filtered cigarettes are consumed each year, of which an estimated 7 billion are littered into the environment [6]. That is 1 out of every 3 smoker litter their CBs into the environment. China is the leading consumer of cigarettes, smoking 41.9% of all cigarettes globally [4]. It is estimated that during the period 2015 to 2025, tobacco smokers are set to increase by 7 million in Nigeria and 24 million in Indonesia [4].

To address the tobacco menace, certain developed countries have implemented strong tobacco control regulations and have seen significant reductions in smoking [25]. However, the decline in smokers appears insignificant when compared with the accruing cigarette consumption around the world.

There is ongoing concern regarding the recycling of cigarette butt (CB) waste. The Cigarette Butt Pollution Project in the USA estimated that one-third of the total production of cigarettes was littered to the environment [7]. While Clean Up Australia [8] reported CBs as the most common source of rubbish collected in Australia under the miscellaneous category, representing 91.5%. The littering of CBs is a growing waste management issue and requires specific planning. As climate change is becoming the world's biggest challenge, recycling wastes for sustainable development is becoming more crucial [9].

There has been a great attempt by many Australian companies to develop CB disposal systems. The system involves placing receptacles that come in the form of wall-mounted, post-mounted, and freestanding in areas easily accessible by the public to dispose of their CB waste. Servicing regimes are also provided by companies. The CBs are then collected from the receptacles and disposed of. Landfilling and incineration are the two disposal methods for the collected CBs, however, both methods are costly and pose high exposure risks from hazardous fumes, which can be severely detrimental to the environment and human health [10].

When CBs are littered, harmful chemicals are leached into the environment, posing major concerns for the global environment and the quality of urban and aquatic life. In cigarette smoke, it is possible to classify more than 4,000 chemical components that are generated during burning or that are distilled from tobacco, of which 69 are carcinogenic [11, 12]. The main toxic agents include polycyclic aromatic hydrocarbons, formaldehyde, argon, carbon monoxide, nitrogen oxides, benzene, hydrogen cyanide, ammonia, phenol, acetaldehyde, argon, acetone, and pyridines [11].

Cigarettes are made of cellulose acetate filters, which are designed to fully or partially retain particulate smoke components, including tar and toxic chemicals [13]. Cellulose acetate filters have poor biodegradability and can take up to 18 months to decompose under normal environmental conditions, hence, posing a serious dilemma in terms of toxic waste and litter disposal [14, 15].

CBs littered on streets commonly get washed away into storm-water drains and end up on beaches, or in rivers and harbors, leaching toxic chemicals [16]. Marine life regularly mistake CBs for food, and they have been discovered in the stomachs of sea turtles, birds, fish, and other creatures in which they have caused severe digestive problems that have led to death [17, 18]. Marinello et al. [26], and Torkashvand and Farzadkia [27] also reviewed multiple solutions to CB management. Marinello et al. [26] have focused on the advantages and disadvantages of the reviewed CB recycling procedures and suggested proposals for future research. While Torkashvand and Farzadkia [27] presented suggestions and models for the control of CB littering.

The objective of this paper is to collect and review all major research studies on the toxicity and valorization of CB waste and investigate the feasibility and sustainability of the recycling methods adopted. Particular attention has been given to the toxicity and historical background of CBs, the leachability aspect of the newly derived materials, and the interpretation of engineering properties of two major potential applications (fired clay bricks and asphalt concrete) with graphical results and variations. Eleven CB recycling proposals and practices have been studied, including; (1) mixing in fired clay bricks, (2) mixing in asphalt concrete, (3) production of cellulose pulp, (4) utilization as corrosion inhibitor, (5) preparation of porous carbon, (6) preparation of activated and charred carbon, (7) utilization as sound absorbing material, (8) productions of superhydrophobic sorbent, (9) use as biofilm carrier, (10) preparation of cellulose nano-crystals and (11) use as vector control.

Valorization of CBs in sustainable products may be an environmentally friendly way to manage a continuously growing pollution calamity and a sustainable development for the health of wildlife, humans, marine and freshwater organisms [9]. In some cases, it even implies an enchantment to the modified material. However, based on the review, further research and developments are recommended. In particular, to the methods adopted in handling the CBs and life cycle assessments for all CB recycling proposals.

This systematic review was undertaken by searching papers in ScienceDirect, PubMed, Web of Science, and ProQuest with the aim of finding major investigations performed on recycling CBs. The paper search was carried out in January 2019.

The research was undertaken by identifying relevant keywords specific to cigarette butts (CBs), CB waste, and the application of recycling CBs to obtain novel materials applicable in civil engineering,

industrial, agricultural, and chemical or environmental cases. The primary keywords, ‘cigarette butt’ or ‘cigarette’ or ‘cigarette filter’, were searched individually and had to appear in the title or abstract. They were then coupled with the secondary keywords ‘recycling’ or ‘toxicity’ or ‘littering’ or health implications’ or ‘history’ or ‘composition’ to gather quality information and data to prepare a comprehensive review paper. Titles, abstracts, and keywords which included any of the primary, or primary and secondary keywords, were studied, and 186 papers were found.

The search process was followed by a selection and screening process to select papers most relevant to the scope of this study. The screening methodology is described as follows:

i. The title and abstract were examined and checked whether the keywords ‘CB’ or ‘cigarette’ were consistent. If not, the paper was discarded.
ii. Reliable content that covered interesting technologies on CB recycling was examined and checked whether the methodological process utilized was provided by the authors. Content on the toxicity of cigarettes or toxicity of littered CBs were reviewed and only selected if reliable experiments were conducted. Papers that provided information on the history and composition of cigarettes were thoroughly read and analyzed to make sure data was provided.
iii. Publications that acknowledged the toxicity of CB waste or cigarettes were only selected.
iv. The references in the selected papers were examined and read.

Once all steps were considered and applied, a total of 101 papers were selected and reviewed.

2.1.1 History of Cigarette Butts

Tobacco was consumed in the form of cigars or through chewing tobacco in the late 19th and early 20th centuries. However, during the late 20th century, cigarettes became the preferred tobacco smoking method due to the low-cost and efficiency in nicotine absorption [28]. Now, tobacco is one of the most widely used addictive substances in the world, growing in at least 124 countries and occupying more than 4.3 million hectares of land, which is equivalent to the entire land footprint of Switzerland [29].

By 1950, major concerns began escalating regarding the health effects of smoking. Researchers found a statistical connection between smoking and lung cancer, and a significant number of articles were published in medical and academic journals addressing the negative health effects. Following these findings, the tobacco industry responded by manufacturing low-tar and filtered cigarettes [30]. Tar is one of the toxic chemical particles remaining after burning tobacco and constitutes the leading health risks associated with smoking, including cancer, while nicotine is the chemical that causes addiction [12]. The filter provides a mouthpiece for smokers and prevents the release of tobacco flakes into the smoker’s mouth. Subsequent to the modifications made to cigarettes, the nicotine yield per smoked

cigarette decreased from 2.7 to 0.85 mg, and the tar yields decreased from 38 to 12 mg during the period 1953 to 1996. The market share of filtered cigarettes jumped from 5% in 1953 to 20% in 1955, and, by 1975, 90% of the manufactured cigarettes contained filters [31]. However, it should be noted that the reduced yields described here are for machine smoked cigarettes, and not measured in humans.

According to the Washington DC-based Ocean Conservancy's global beach clean-ups, since the development of cigarette filters, CBs have been reported as the most collected waste each year since the beginning of the initiative in 1986 [32]. A single machine can make up to 20,000 or more filter cigarettes per minute, which is equivalent to 333 cigarettes per second; consequently, 67 cartons of cigarettes are produced every 40 sec. In a year if 5.7 trillion cigarettes are manufactured, and about 97% of them contain filters, more than one million tonnes of plastic is being littered into the environment. At present, China is the largest producer and consumer of cigarettes, having consumed 2.35 trillion cigarettes in 2016 (Fig. 2.1). If current trends continue, cigarette production is expected to increase and, subsequently, CB litter.

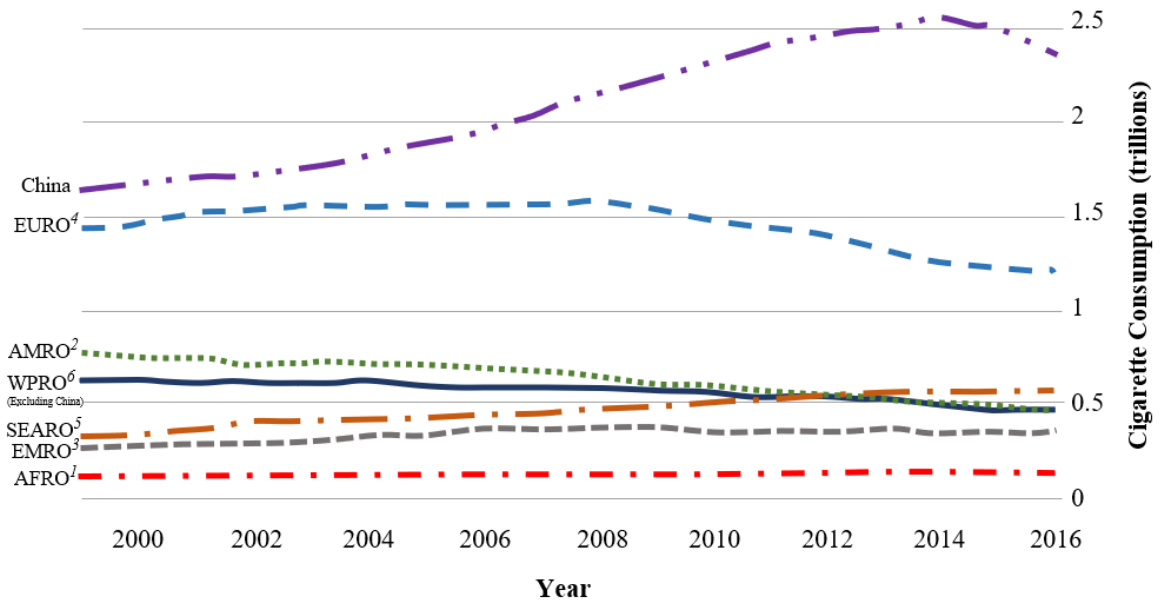


Fig. 2.1 Global cigarette consumption by WHO region, 1999 – 2016, in trillions (adapted from The Tobacco Atlas [4]) . [1]Africa. [2]Americas. [3]Eastern Mediterranean. [4]European. [5]Southeast Asia. [6]Western Pacific

2.2 Composition of CBs

The modern composition of a cigarette consists of a filter, cigarette paper, tobacco, tipping paper, and additives (Fig. 2).

2.2.1 Paper

The tipping paper, which stabilizes the mouthpiece from saliva, is wrapped around the filter and contains small ventilation holes that allow fresh air to mix with the smoke, thereby diluting the toxic mix of chemicals inhaled [33]; the cigarette paper is below the tipping paper and holds the tobacco blend; as shown Fig. 2.2. However, once the CB is discarded into the environment, cigarette and tipping paper will still be remaining. The paper may take up to 3 - 5 months to biodegrade under ambient conditions [34].

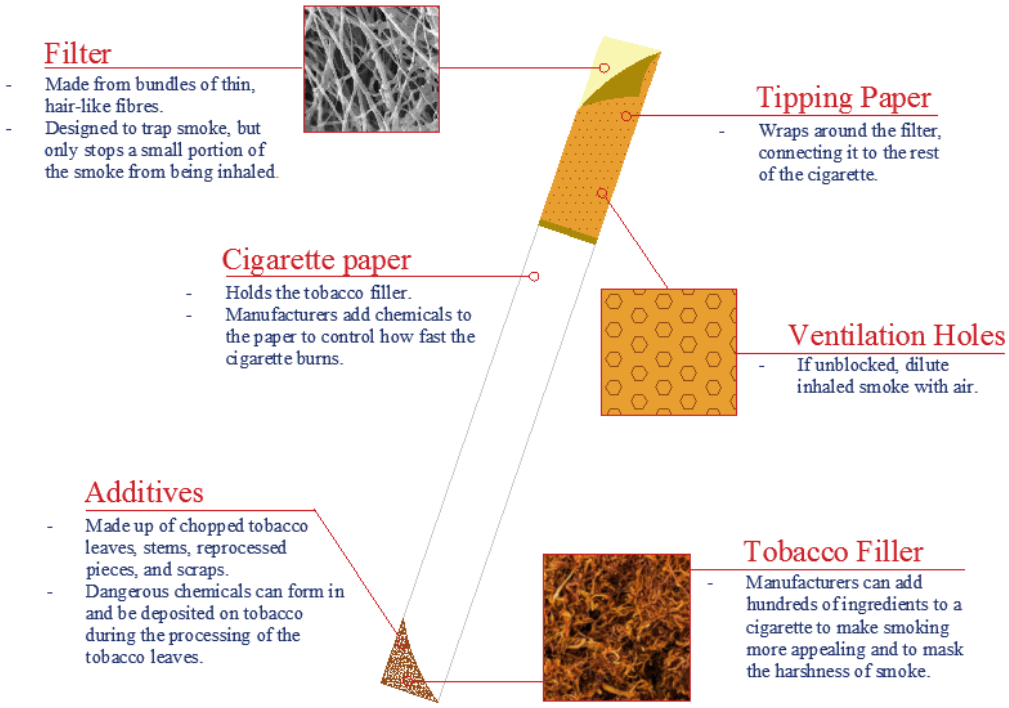


Fig. 2.2 Physical structure of a cigarette with each component identified and described.

2.2.2 Tobacco and Additives

The two major subgenera of the botanical genus Nicotiana are N. Rustica and N. Tabacum, which are the main commercial tobaccos [11]. The N. Tabacum is grown as the primary tobacco crop, which comes as flue-cured (Virginia tobacco), air-cured (burley tobacco), light air-cured, dark air-cured, and sun-cured tobacco (oriental tobacco) [35]. The composition of the tobacco blend varies according to the preferences of the country [11]. The manufacturing process of the tobacco consist of many stages, including growing the tobacco, harvesting the tobacco, curing the leaf, moistening, stripping, sorting, conditioning, aging, and, finally, blending the leaves [36]. The processing of the leaves produces several by-products, which are processed separately and then added back into the cigarette blend. Additionally, additives are incorporated into the tobacco mixture, including humectants and flavor additives. The

primary humectants in cigarette tobacco are glycerol and propylene glycol; their function is to preserve the plasticity and moisture in the cigarette and prevent or delay the drying of tobacco. While common flavor additives include licorice, cocoa solids, and various sugars, these additives act to mask the harsh flavors of smoke [11]. When a cigarette is discarded into the environment, unburned tobacco residue may remain in the cigarette butt.

2.2.3 Filter - Cellulose Acetate

Cellulose is a naturally occurring polymer of glucose ($C_6H_7O_6$) that is derived primarily from wood pulp and cotton linters. Each glucose unit consists of three hydroxyl groups (OH), which can hydrogen bond to adjacent chains; the result is a tough material [37]. Due to the extended chains of cellulose, it is almost impossible for solvents to penetrate in. The only way to process cellulose is to block the OH groups (hydrogen bonding) with acetate groups. Therefore, by esterifying natural cellulose with acetic acid, cellulose acetate is produced (Fig. 2.3) [38].

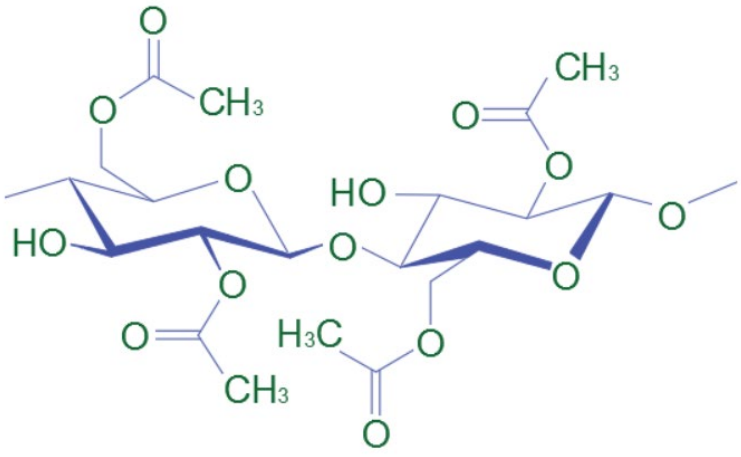


Fig. 2.3 Molecular structure of cellulose acetate, which is used in the filter of a CB.

Cellulose acetate is the primary component in the filter of a CB. Therefore the high degree of acetate substitution prevents the cellulose in the littered CB to access the microorganisms (bacterium, fungal, etc.) in the environment for biological decomposition [39, 40]. Ultraviolet (UV) radiation also promotes cellulose acetate degradation, however, the remaining tipping paper on the CB inhibits the sunlight UV radiation reaching the cellulose acetate fibers [40]. Ultimately the paper biodegrades, and the exposure of the UV radiation to the cellulose acetate surges. However, the composition of the cellulose acetate restricts the exposure of the sunlight to the surface area of the fibers. Additionally, the microorganisms within the environment do not accept the cellulose acetate as a food source [41]. Therefore, de-acetylation proceeds through hydrolysis, however cellulose acetate hydrolysis is a slow course at

ambient conditions. Hence why, under anaerobic conditions, it may take up to 1-2 months in soil, 6-9 months in freshwater, and 12 months or more under poor conditions for the CBs to degrade [14, 42].

2.3 Toxicity of Cigarette Butts

When a cigarette is smoked, it is converted to particulate and volatile phases. The particulate phase consists of non-volatile and semi-volatile chemicals, and the vapor phase consists of volatile chemicals. The vapor phase is the portion of the smoke that passes through the filter, and 95% of its weight is made up of 400-500 individual gaseous compounds including nitrogen, oxygen, carbon dioxide, carbon monoxide, water, argon, hydrogen, acetone, nitrogen oxides, and volatile sulfur compounds [30, 43]. The particulate phase is the portion of the smoke that is trapped in the filter and is made up of the remainder of the smoke weight with more than 3500 individual components; these include nicotine, nicotiana alkaloids, compounds specific to solanaceae, catechols, aromatic hydrocarbons, and polycyclic aromatic hydrocarbon (PAH) [11, 30].

PAH was the first carcinogen identified in a cigarette [44], and, in 2000, the number of carcinogens identified in tobacco smoking increased to 69 [45]. According to the International Agency for Research on Cancer [45], among the carcinogens, 69 are animal carcinogens, 48 are possibly carcinogenic to humans, 8 are probably carcinogenic to humans, and 11 are proven to be carcinogenic to humans [11, 30]. Hence, cigarette smoking is a serious threat to humans, for every cigarette burned, a smoker receives roughly 1.4 - 2.2 mg of dangerous compounds [46]. It is estimated a standard cigarette contains between 9 to 30 mg nicotine, of which 0.5 – 2 mg is inhaled by the smoker. While, 4 - 8 mg is the estimated toxic dose of nicotine intake for non-smokers [47]. The development of many organ systems can be detrimentally affected by nicotine due to the CBs toxicity, including the respiratory, cardiovascular, nervous, and reproductive systems [48, 49].

Smokers and non-smokers may be uninformed regarding the toxicity level of CBs, and the potential hazard the waste may pose to the environment and wildlife [50]. When CBs are littered into the environment, significant amounts of toxic chemicals are leached and potentially accumulated in water channels, streets, nature strips, and footpaths, thereby posing a serious threat to local organisms and aquatic species [50]. Children are more likely to be exposed to CBs through oral contact as they often like to explore their environments, and 1 - 2 mg of nicotine is sufficient to cause vomiting and nausea due to the CBs toxicity [47]. CBs are reported as the most common toxic substance causing symptoms in children [51].

In 2011, Moerman and Potts [52] investigated the leaching behavior of heavy metals in CBs. The main purpose of the study was to determine the concentration of metals leached and the effect on the surrounding environment. To understand how the pH level affects leaching behavior, a pH range of

6.00 - 4.00 was selected, which is typically observed in natural rainwater. It was found the metals studied have different leaching behaviors over time. A direct relationship was found between the concentration of leached metals and the soaking period. An increase in metal concentration leached for Ba, Fe, Mn, and Sr was seen over time, while no significant change in metal concentration was seen for Ti, Pb, Zn, and Ni. In contrast, a decrease in metal concentration for Cu, Al, Cr, and Cd was observed over time. The variable relationships for the selected metals Ba, Zn, and Cu can be seen in Fig. 2.4. This indicates that a piece of cigarette litter is a point source of metal contamination for a least a month. Moreover, it suggests that the longer the waste remains in the environment, the greater the contamination of Ba, Fe, Mn, and Sr [52].

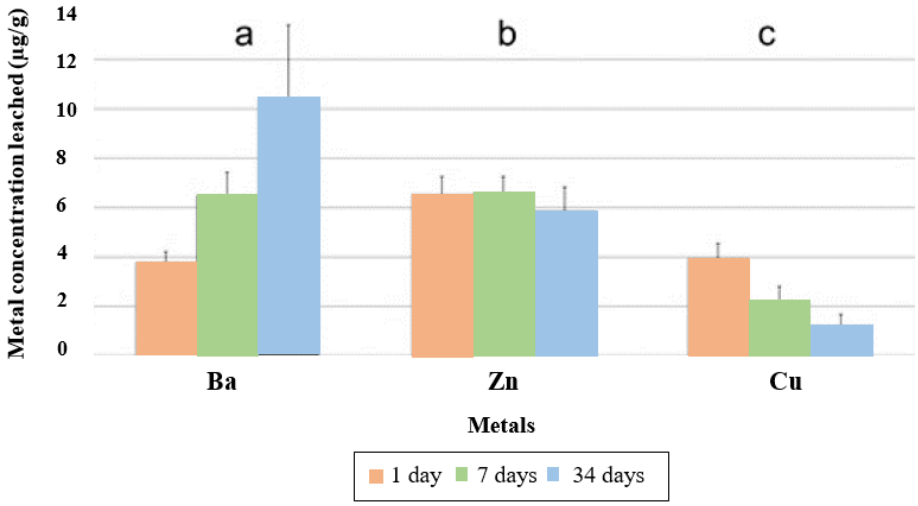


Fig. 2.4 Relationships observed between metal concentration leached from smoked cigarettes and soaking time in aqueous solution (adapted from Moerman and Potts [52])

To analyze the toxicity of CBs, the metal concentrations of whole unsmoked cigarettes were measured. The experimental studies exposed more than one-fifth of Mn, Zn, and Ba, one quarter of Cu and half of Sr and Pb may potentially be leached into the environment [53, 54]. When placed in contrast to the estimated amounts of CBs littered annually (4.95 trillion) [55], the results indicate 9.26 ± 0.5 kg (Al), 8.76 ± 2.52 kg (Ba) and 38.1 ± 8.2 kg (Fe) may enter the environment each year from CB litter [52].

While Slaughter et al. [17] conducted a comprehensive study to identify the LC50 (lethal concentration 50) of CB leachate to freshwater and marine ecosystems. The LC50 for leachate from smoked CBs was 0.97 and 1.1 CB/l for both the freshwater fathead minnow and marine topsmelt. While the leachate from smoked cigarette filters exhibited reduced values of 4.3 and 1.8 CBs/l for both fish species. For unsmoked cigarette filters, LC50 values of 13.5 and 5.1 CBs/l were found for fish species [17]. Therefore, the results indicate the toxicity of CB leachate increase from unsmoked to smoked filter.

Chevalier et al. [56] investigated the possible release of nanoscale particles from CB waste by leaching the waste with rainwater. The results revealed the release of 4.12 +/- 0.24% organic carbon in the spectrum of 10 nm up to 400 nm. Moreover, the analysis showed that As, Pb, Cd Cu, Ni, Cr, Co, Al, Mn, Zn, and Fe metallic species were principally attached to the nanoscale particles per gram of carbon released [56]. Hence, the results indicate an evolving global contamination of toxic chemicals, which requires urgent attention and investigation.

To evaluate the genotoxic potential of CBs , Di Giacomo et al. [57] studied the mutagenicity of CBs in the bacterial reverse mutation test on Escherichia coli WP2uvrA and Salmonella typhimurium TA98 and TA100 strains. The experiments were both conducted in the presence and absence of the S9 exogenous metabolic activator. The findings confirmed the mutagenicity of CBs in the presence of the exogenous metabolic activator S9, thereby verifying that CBs contain pro-carcinogenic substances. As a possible preventative strategy, the natural sesquiterpenes β-caryophyllene oxide (CRYO) and β-caryophyllene (CRY) compounds were studied to investigate their ability to inhibit the mutagenicity of CBs. This strategy was based on previous studies proposing the chemopreventative properties of CRYO and CRY [58-62]. The experimental results confirmed the antimutagenicity properties of CRY and CRYO on CBs, and, hence, the possibility of using the compound as an environmental decontaminator. Furthermore, the paper suggests utilizing the compound as a maceration solution to apply during disposal for smokers and for use during degradation procedures for CB waste [57].

2.4 Research on Valorisation of Cigarette Butts

2.4.1 Mixing in Fired Clay Bricks

Due to the durable nature of the material, bricks have been used in construction for many years. Fired clay bricks are mainly composed of clay and sand, which are bound together under high temperatures in a kiln, making a remarkably strong construction material. Lately, incorporating waste materials in fired clay bricks has been investigated in the hope of offering a sustainable solution to a growing pollution catastrophe. Common waste materials include rubber [63], polystyrene [64], fly ash [65], sludge [66], sawdust and limestone [67], and processed waste tea [68]. Moreover, in 2016, Ukwatta et al. [69] conducted a study based on incorporating biosolids in fired clay bricks. The results indicated that by adding up to 50% biosolids in fired clay bricks, acceptable physical and mechanical properties can be achieved, and up to 25% energy can be saved during the firing process [69]. Although recycling waste material in fired clay bricks may improve porosity, strength, and shrinkage, further comprehensive testing is required in the field of toxic leaching.

In a recent research, the application of recycling CBs in fired clay bricks were comprehensively investigated [1, 22, 24, 70]. The experimental procedure involved preparing clay bricks with 2.5%, 5%,

7.5%, and 10% CB content by weight. CBs of varying sizes and brands were provided by Buttout Australia Pty Ltd (Fig. 2.5). The mixtures were prepared using a Hobart mechanical mixer with mixing times of 5, 10, and 15 min. The samples were then compacted manually in molds using predetermined masses, and the specimens were dried at 105 ºC for 24 h before being fired in a furnace at 1050 ºC.



Fig. 2.5 Prepared CBs of varying sizes and brands for recycling [1]

The samples were tested based on their mechanical and physical properties and compared to control bricks with 0% CB content (Fig. 2.6). In addition, thermal conductivity, the effects of mixing time, and heating rate on brick properties, energy savings during firing, emission of gaseous compounds, and leachability were tested [1].

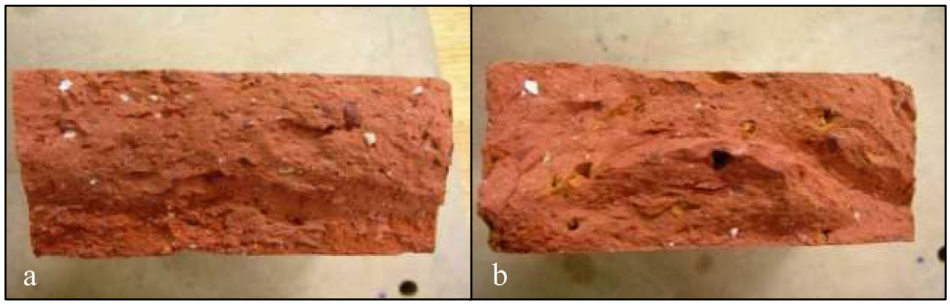


Fig. 2.6 Cross-sectional view of bricks with 0% (a) and 2.5% (b) CBs [1]

The property test results show a dramatic decrease in compressive strength from 25.65 MPa with 0% CB content to 3 MPa with 10% CB content (Fig. 2.7). The dry density decreased from 2118 kg m^{-3} for the control sample to 1482 kg m^{-3} with 10% CB content, producing lightweight bricks. The water absorption increased from 5% with 0% CB content to 18% with 7.5% CB content. Moreover, the results show a firing energy saving of 30.8% with the incorporation of 2.5% CB content, and a 58.4% energy

saving with a 5% CB content [1]. Additionally, an increase in CB content resulted in a reduction of thermal conductivity significantly. Therefore, incorporating CBs in the manufacturing of bricks may solve a growing pollution problem, however, further research is essential, including a life cycle assessment.

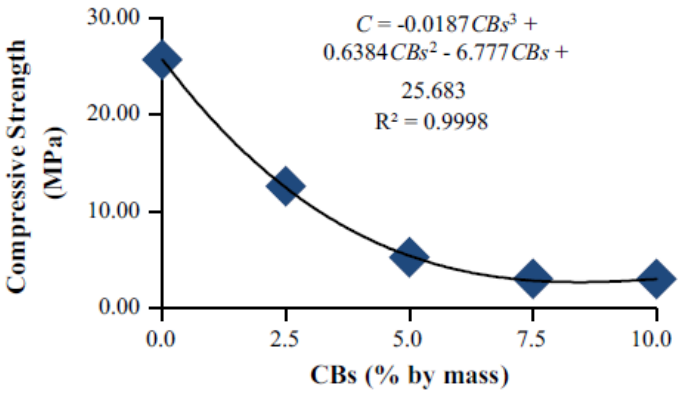


Fig. 2.7 Compressive strength versus CB% by mass [1]

2.4.1.1 Leachate test on fired clay bricks incorporated with CBs

The main environmental concern regarding recycling CBs in construction material is the possible leaching of contaminants, such as heavy metals. Therefore, leachate tests are necessary to assess the risk of contaminated leaching. Kadir and Mohajerani [70] investigated the possible levels of leachability of manufactured fired clay bricks incorporating 0%, 2.5%, 5%, and 10% CB content. Leachate tests were carried out based on the Australian Bottle Leaching Procedure (ABLP), Toxicity Characteristics Leaching Procedure (TCLP), and Static Leachate Test (SLT). The objective of the study was to determine whether the leaching of heavy metals including Arsenic (As), selenium (Se), Mercury (Hg), Barium (Ba), Cadmium (Cd), Chromium (Cr), Lead (Pb), Silver (Ag), Zinc (Zn), Copper (Cu), and Nickel (Ni) from the modified bricks exceeded the regulatory standards. The results exhibited a positive correlation between the leachate of heavy metals and leaching time; by day 134, a decrease in leachate was noted [70]. Low concentrations in leaching were found, which can be attributed to the high firing temperature during the firing process of the clay bricks incorporating CBs. The tests confirm that all the heavy metal leachate values were insignificant and below the standard acceptable limits.

2.4.2 Mixing with Asphalt Concrete

Asphalt concrete is a composite material consisting of coarse and fine aggregates, bituminous binder, and filler [71]. The material plays a fundamental role in flexible pavement structures, due to its ability to withstand high volumes of traffic load in various environmental conditions. Due to its composite nature, mixing asphalt concrete with various waste materials has gained widespread attention. The

application of waste materials in asphalt concrete mixtures may solve major environmental issues, reduce the demand for natural aggregates, and improve the performance of asphalt concrete [72]. Common modes of failure for asphalt concrete include rutting due to excessive deformation, cracking due to fatigue, raveling due to oxidation, and hardening of the binder.

Therefore, methods to improve the properties of asphalt concrete are constantly being investigated. Experiments have confirmed that the incorporation of car tires [73], glass [74], pulverized computer circuit boards [75], and nuclear and hazardous wastes [76] have improved the chemical and mechanical properties of asphalt concrete. Paranavithana and Mohajerani [77] investigated the possible application of recycled concrete aggregates in asphalt concrete. The study found lower resilient modulus, creep, and volumetric property values (except for the percentage of air voids) when compared with the values for fresh aggregate specimens [77].

In a recent study, a group of researchers investigated the application of CBs in asphalt concrete [19]. The experiment involved preparing 10 kg m^{-3}, 15 kg m^{-3}, and 25 kg m^{-3} CBs encapsulated with different bitumen classes (C170, C320, and C600) (Fig. 2.8) and incorporating them into an asphalt concrete mixture with Class 170 Bitumen (Fig. 2.9). CBs of varying sizes and brands were provided by Buttout Australia Pty Ltd. The initial step of the experimental procedure involved drying CBs in an oven to eliminate any trapped moisture. Once dried, the CBs were encapsulated with different classes of hot bitumen (C170, C320, and C600) and left to dry (Fig. 2.8). The aggregates were placed in the oven for 24 h at 105º, once dried, the asphalt mix, the aggregates, and bitumen were placed in the oven and heated for 1 h at 150 ºC. The aggregates were then mixed with the bitumen for 3 min, then the asphalt mix and mold were conditioned for 1 h at 150 ºC in the oven and mixed together with the encapsulated CBs. Once conditioned, the samples were compacted using a gyratory compactor. Following a similar procedure to the first investigation, the second investigation involved assessing the effects of incorporating 10 kg m^{-3} of CBs encapsulated with paraffin wax into asphalt concrete mixes manufactured with Classes 170 and 320 bitumen.



Fig. 2.8 CBs encapsulated in bitumen [19]



Fig. 2.9 Asphalt concrete samples manufactured with CBs of 10 kg m^{-3}, 15 kg m^{-3}, 25 kg m^{-3}, and no CBs (control) [19]

Several tests were carried out for both investigations, and the experimental results demonstrated encouraging results. Asphalt mixes with 10 kg m^{-3} and 15 kg m^{-3} CB content encapsulated with bitumen demonstrated acceptable bulk density, percentage of air voids, and resilient properties. The stability and flow values satisfied the [71] light, medium, and heavy traffic condition requirements of 5.5 kN for light traffic and 6.5 kN for medium and heavy traffic conditions (Fig. 10). Asphalt mixes with 10 kg m^{-3} CB content encapsulated with paraffin wax exhibited a decrease in the percentage of air voids and voids in mineral aggregate for C170 samples and an increase for C320 samples [19]. Both samples satisfied the standard requirements of 3-7% for air voids and a minimum of 15% for voids in mineral aggregate, while the stability values only satisfied light traffic conditions for road pavements. In addition, the results suggest a reduction in thermal conductivity, and, subsequently, on the Urban Heat Island effect due to the increase in porosity [19]. However, further investigations are necessary for the application of CBs encapsulated with paraffin wax and bitumen in asphalt concrete for widespread industry adoption.

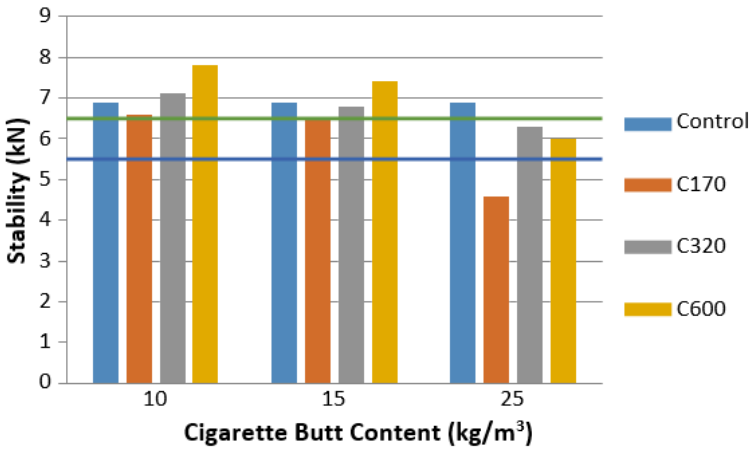


Fig. 2.10. The effect of different amounts of CBs on the Marshall Stability for CBs encapsulated with different classes of bitumen (C170, C320, C600) [19]

2.4.3 Production of Cellulose Pulp

Pulp is a lignocellulosic fibrous product manufactured by mechanically or chemically extracting cellulose fibers from renewable wood raw material. A group of researchers in Brazil developed a technology to produce cellulose pulp by recycling CBs [78]. The main idea of the technology is based on separating all the cellulose content in the CB by hydrolyzing the cellulose acetate in the presence of strong alkali (Fig. 2.11). The experiment involved four different processes, including dissolving all chemicals and ash absorbed in the filter, separating the cellulose acetate into cellulose, eliminating lignin from the residual tobacco and mitigating its cellulose, and, finally, disaggregating the remaining paper [78].

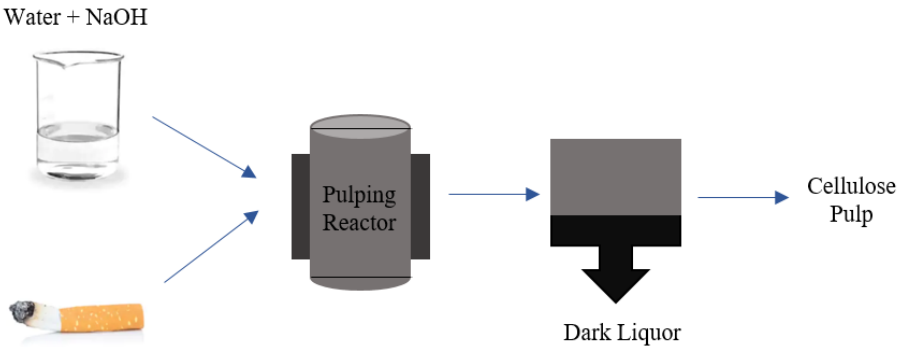


Fig. 2.11 Cellulose pulp production flow chart from CBs

At the end of the filtering process, cellulose pulp and a dark viscous liquor were produced. The experimental results confirmed that the cellulose pulp could, in fact, be used in the paper industry.

However, the dark liquor produced during the pulping process contained very high resistant organic content, including toxic compounds. Several treatment processes were conducted on the effluent, including precipitation by acidification, coagulation, and ozonation to reduce the chemical oxygen demand (COD) value, however, all methods were unsuccessful. Therefore, after precipitating the effluent with a pH of 3, Teixeira et al. [78] proposed that the clarified effluent should be collected and used in a new pulping process. However, the proposed progression for the clarified effluent fails to offer a sustainable method. Constant production of effluent will be produced, and the effluent produced will be more than what is required for the alkaline pulping, therefore, a practical solution is essential.

2.4.4 Utilization as Corrosion Inhibitor

Many investigations have been conducted to determine the corrosion inhibitive efficiency of nicotine [79], with one study presenting promising results involving applying CBs as corrosion inhibitors for N80 steel at 90º in a hydrochloric acid solution [80]. The investigation involved performing liquid chromatography (LC)/mass spectrometry (MS) and an infrared (IR) reaction analysis system to detect the chemical compositions of CB water extracts to ultimately determine the best aging time. The aging process involved adding five CBs in 100 mL distilled water and continuous monitoring by an LC/MS and an IR reaction analysis system to determine the highest inhibition efficiency of the aging solution. The experimental results recorded 24 h as being the best aging time. To assess the corrosion inhibitive effect of the aged solution on N80 steel at 90 ºC in hydrogen chloride (HCI), electrochemical and weight loss techniques were used. During the investigation, various CB water extracts and HCI concentrations were used to determine their influence on corrosion inhibition. The results presented a maximum corrosion inhibition efficiency of 94.6% in 10% HCI solution with 5% inhibitor. However, it was determined that the increase in HCI solution increased the corrosion attack [80].

Following this study, Zhao et al. [81] compared the corrosion inhibitive effect of anaerobic and aerobic CB water extracts on N80 steel at 90º in HCI to determine the effect of oxidation. Mass spectrometry/liquid chromatography and the infrared reaction analysis system test method were adopted to contrast the chemical compositions of the anaerobic and aerobic extracts. Fewer types of compounds were found in the anaerobic extracts than in the aerobic extracts, thereby indicating that the aerobic condition may favor CBs water extraction over the anaerobic condition. Additionally, to study the corrosion inhibitive mechanism and assess the corrosion inhibitive effect of both anaerobic and aerobic CB water extracts, potentiodynamic polarization, X-ray diffraction, weight loss, electrochemical noise, and impedance were used. The results revealed that the aerobic extracts consisted of more compounds and had greater corrosion inhibitive effects than the anaerobic extracts [81].

To further enhance the inhibition efficiency, Zhang et al. [82] suggested incorporating cuprous chloride (CuCI) to inhibit the corrosion of N80 steel in HCI solution. The study involved comparing the corrosion inhibition of water extracts from CBs in the presence and absence of CuCI on N80 steel at 90º in 15% HCI. In addition to the tests performed in the previous two papers, an X-ray photoelectron spectroscopy test was conducted to examine the composition of the protective coat formed by the CuCI. The results revealed an increase in inhibition efficiency of 95.3% with the addition of CuCI and 9% CB water extract [82]. In addition, the X-ray photoelectron spectroscopy results confirm that the copper added water extract from CBs improved the N80 steel surface with a protective coat. Although the results seem promising, the paper requires further research. The experimental procedure requires further clarification, and a comprehensive study is required on a leachate analysis as the paper fails to mention the prospect of toxic leaching during the lifetime of the modified material. A summary of all research conducted on the application of CBs as a corrosion inhibitor on N80 steel can be seen in Table 2.1.

Table 2.1 Applying CBs as a Corrosion Inhibitor for N80 Steel

Preparation of Inhibitors from CBs	CBs	HCI	Modifier	Corrosion Inhibitive Evaluation Techniques	Corrosion Inhibition Efficiency	Reference
5 CBs are added into 100 mL distilled water and aged for 24 h	5%	10%	-	• Weight loss • Potentiodynamic polarization • Impedance measurement	94.6%	[80]
5 CBs are added into 100 mL of distilled water and aged under bubbling air conditions for 24 h	5%	15%	-	• Weight loss • Electrochemical noise • Potentiodynamic polarization • Impedence • X-ray diffraction	92.9%	[81]
5 CBs are added into 100 mL of distilled water and aged for 24 h and reacted with 0.5 g of CuCI at 60 °C for 2 h.	5%	15%	9% CuCI	• Weight loss • Electrochemical noise • Potentiodynamic polarization • Impedence • X-ray photoelecetron spectroscopy	95.3%	[82]

2.4.5 Preparation of Porous Carbon

Porous carbon is a versatile material and can be used for several applications due to the material's high porosity, relative easy surface chemical tuning, large surface area, and excellent electron conductivity [83]. However, the preparation involves expensive and/or toxic chemicals and sophisticated techniques. Polarz et al. [84] found a simple one-step procedure for preparing porous carbons, which involves inexpensive precursors, cellulose acetate in the form of CBs. In conducting extensive experiments, it was found that complex biomolecules like cellulose acetate can produce porous carbons that exhibit well-defined positions on different length scales ranging from the nanometre to the micrometre [84]. Based on these findings, Lee et al. [85] and Yu et al. [86] investigated recycling used CBs as a proper carbon source for use in super-capacitors. The process involved heat treating used CBs under nitrogen (N) containing atmosphere to prepare N-doped mesoporous and microporous hybrid carbon material (NCF) through a one-step method; pyrolysis (carbonization) (Fig. 2.12). The combination of an enlarged surface area, the pseudocapacitive behavior, and well-developed mesopores and micropores resulted in achieving the desired supercapacitive performance [85].

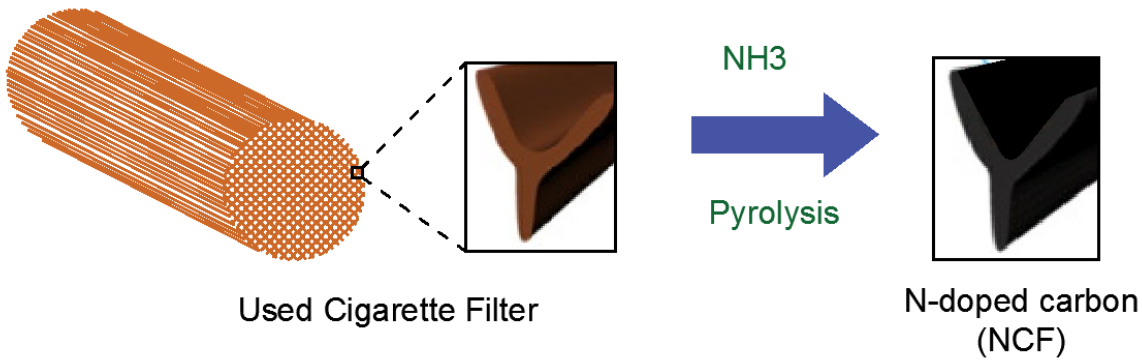


Fig. 2.12 Fabrication procedure used for preparing NCF by pyrolysis of a used cigarette filter

Similarly, Ghosha et al. [87], and Yazdi et al. [88] prepared a conducting material from used CBs through heat treatment; the study required no pre-treatment for the CBs prior to the carbonization process. The process involved characterizing and investigating the electrochemical properties of the pyrolyzed material through dynamic light scattering, Raman spectroscopic, X-ray diffraction, ultra-violet visible spectroscopy, Fourier transform infrared spectroscopy, zeta potential measurements, and field emission scanning electron microscopy [88]. The results confirmed the prospect of using the pyrolyzed material in electronic applications as the material demonstrated good conductivity [87]. However, further investigations are required in this field of study for widespread industry adoption.

2.4.6 Preparation of Activated and Charred Carbon

Hamzah and Umar [89] investigated the potential of preparing active carbon from CB waste through microwave induced KOH activation. The microwave heating method was adopted in preparing active carbons due to the method's rapid and uniform heating capacity. The process involved an activation time of 20 min at 630 Watts and a frequency of 2.45 GHz. The paper proposed that the resultant active carbon can be used as an alternative absorbent material for oil-water separation, supercapacitors, and methylene blue due to its structural diversity and large surface area [89].

Masoudi Soltani et al. [90] investigated producing charred carbon through the pyrolysis of CBs. The charred carbon was modified with NHO_3, and a decrease in surface area and micropores was observed. The study analyzed the efficiency of using non-modified and nitric acid-modified charred carbon to remove lead from the aqueous solution [90].

2.4.7 Utilization as Sound Absorbing Material

In a recent research, Escobar and Maderuelo-Sanz [91] proposed using CBs as an absorbing material due to the porous nature of the waste [83]. The investigation involved analyzing the acoustical performance of a new kind of material using used and unused CBs. In addition to studying the effects of compression, the influence of the length of butts, and a comparison with other materials. The results revealed a high absorption coefficient of < 0.90 from mid-frequencies (> 2000 Hz), indicating that CBs can be used as a substitute for artificial based commercial products [91]. The samples used for this study comprised intact CBs, however, CBs contain toxic chemicals, and, therefore, require pre-treatment. Maderuelo-Sanz et al. [92] investigated the acoustical performance of treated CBs. The CBs were initially dried at 80º and sealed in plastic bags; this process was conducted to disinfect the waste from any toxic chemicals and remove humidity. The physical properties, including fiber diameter, porosity, skeletal density, and flow resistivity, were measured. The acoustical performance of the new material was assessed and compared to other fibre materials, the results revealed that the performance of both materials were rather similar, however, the CB material demonstrated better performance [92].

2.4.8 Production of Superhydrophobic Sorbent

Ou et al. [93] found that by hydrolyzing cellulose acetate fibers from CBs in a sodium hydroxide (NaOH) solution, and, subsequently, in an ethanolic solution of hexadecyltrimethoxysilane, superhydrophobic/ superoleophilic fibers (SFs) can be produced. The produced SFs can be used for oil spill clean-up due to its wetting behaviors towards kerosene and water. The study involved testing the clean-up efficiency of the SFs in a staged kerosene spill, and it was found that the SFs had a 96% efficiency in absorbing the kerosene. Additionally, the moisture content in the collected SFs was found

to be lower than the kerosene poured onto the water. Hence, the SFs may be used in the clean-up of oil spills, and, simultaneously, overcome the detrimental effects of CBs [93].

Similarly, Ifelebuegu et al. [94] and Xiong et al. [95] investigated a two-step process to convert waste cigarette filters into a oleophilic and superhydrophobic sorbent for the application in spill clean-up and water/oil separation. The process involved cleaning the filters ultrasonically, followed by modifying the surface through chemical vapor deposition using methyltrichlorosilane. Ifelebuegu et al. [94] found the functionalized waste filters can exhibit superhydrophobic properties, absorbing 16 to 26 times their weights in numerous oils with up to 20 cycles of reuse capacity. While Xiong et al. [95] discovered an absorption capacity of 80% and 82% for pump and silicone oil after 10 cycles.

A similar investigation on recycling cigarette filters for oil-water separation was conducted by Liu et al. [96] and Liu et al. [97], which involved preparing cigarette filter coated meshes (CFCMs) through a facile electrospinning approach. The CFCMs achieved a performance of underwater superoleophobicity without the need for any chemical modification. An extraordinary separation efficiency larger than 99.9% for water/oil mixtures was achieved after many cycles [96, 97].

2.4.9 Use as Biofilm Carrier

A study was carried out to evaluate waste cigarette filters in tobacco factories as a biofilm carrier in the integrated fixed film activated sludge (IFAS) process [98]. Biofilm reactors are used for the removal of nitrogen species and organic carbon in biological wastewater treatment. Hence, this study aimed to develop a new, efficient, and inexpensive carrier with the use of cigarette filters. The investigation confirmed the effectiveness and feasibility of using cigarette filter rods as a biofilm carrier in an IFAS system to simultaneously remove organic matters and nutrients from wastewater [98]. Following this study, Sabzali et al. [99] investigated the use of cigarette filter rods as a biofilm carrier in an anaerobic moving bed biofilm reactor (AMBBR). The study revealed that cigarette filter rods are feasible and effective as Kaldnes media for the removal of phosphate and organic matter from wastewater. Additionally, the new bi-carrier is cost effective and durable and can increase the effective mixed liquor suspended solids in an anaerobic reaction basin due to the modified materials roughness, high surface area, and high porosity properties [99].

2.4.10 Preparation of Cellulose Nano-crystals

In a recent research, Abu-Danso et al. [100] investigated the use of CBs for the removal of diclofenac from water. To obtain phosphate ions (HPO) - cellulose nano-fibers (CNFs), acetyl groups were cleaved through alkali treatment and etched with HPO from phosphoric acid. While sulfuric acid was used under high temperatures to cleave phenol moieties from CNFs to obtain cellulose nano-crystals (CNCs). The

experiments revealed a maximum removal capacity of 107.9 mg g^{-1} of Dfc through the synthesis of HPO-CNFs, suggesting that celluloses extracted from CBs can potentially be used as an absorbent for the removal of Dfc from water [100].

2.4.11 Use as Vector Control

A number of strategies have been developed to control vectors in dengue epidemic areas with insecticides [101]. However, the species resistance has resulted in little success. CBs abound in chemical products, which are known to have pesticidal, fungicidal, insecticidal, and herbicidal properties [102]. Therefore, in search of an alternative control method, Dieng et al. [103] and Mondal et al. [104] found CB waste to be a potential source to manage mosquito-borne diseases. The study involved examining the developmental alterations of Aedes albopictus in the presence of CB waste [103]. The results exhibited an increase in the mortality of Aedes albopictus when subjected to CB waste solutions. Subsequently, Dieng et al. [105] assessed the vulnerability of Aedes aegypti to CBs during their development and found an interruption of larval development when subjected to CB waste. They tested modifications in egg hatching, longevity, reproductive capacity, fecundity, and ovipositional behaviors. Based on this study, Dieng et al. [106] further investigated the effects of CB exposure on the dengue vector Aedes aegypti to examine whether the waste may alter any biological parameters of parents and their progeny. The results indicate that CB waste can be developed as a novel vector control strategy as it has an indirect effect on adult life traits [106]. However, there are many unknown factors within this area of study, and, therefore, further research is essential. The valorization of CBs into these 11 novel materials has been investigated and summarized in Table 2.2.

Table 2.2 Summary of some of the significant research conducted on the valorization of CBs

CB Recycling Methods	Processing Used	Remarks	Ref.
Mixing in fired clay bricks	No Processing	• The addition of CBs demonstrated a decrease in dry density producing lightweight bricks. The CBs completely burnt off during the firing process. • A firing energy saving of 30.8% with the incorporation of 2.5% CB content was found due to the high heating value of CBs. • An increase in CB content resulted in a reduction of thermal conductivity significantly due to the increase in porosity. • The paper proposes that the incorporation of 1% CBs in the manufacturing process of fired clay	[1, 22, 24, 70]

		bricks can completely offset the CBs produced in the world. • Further research is recommended, including a life cycle assessment.	
Mixing in asphalt concrete	No Processing	• Asphalt mixes with CB encapsulated bitumen, and paraffin wax demonstrated acceptable bulk density, stability, percentage of air voids, and resilient properties. • Results suggested a reduction in thermal conductivity and subsequently on the Urban Heat Island Effect. • Substituting CB waste in the production of asphalt concrete will result in a decrease in demand for natural aggregates. • A leachate analysis and life cycle assessment are essential.	[19]
Production od cellulose pulp	Hydrolyzing the CBs for 3 hr in the presence of NaOH	• The extracted cellulose pulp from the CBs can be used instead of renewable wood raw material in the paper industry. • A dark liquor is produced containing very high resistant organic content, including toxic compounds. • A solution is required for the disposal or reuse of the dark liquor.	[78]
Utilization as corrosion inhibitor	CBs aged for 24 h	• A corrosion inhibition efficiency of 94.6% and 92.9% for 10% and 15% HCI • A corrosion inhibition efficiency of 95.3% for 15% HCI and 9% CuCI • The corrosion inhibitor can be used on N80 steel in construction. • Leachate analysis is required for the CB extracted metal surface.	[80-82]
Preparation of porous carbon	Hydrolyzing the CBs in nitrogen under heat	• Porous carbons that exhibit well-defined positions on different length scales ranging from the nanometre to the micrometer was produced. • The CBs can be used as a proper carbon source in super-capacitors and in electronic applications as it demonstrated good conductivity.	[84-88]

		• The experimental method involves the removal of the coated paper to obtain the filter. However, on a large scale, this method is not practical. Moreover, the disposal method of the coated paper is not discussed. • Another experimental method consists of utilizing only half of the filter to exclude the effect of pre-contained activated carbon, however the treatment/disposal of the other half is not considered.	
Preparation of activated and charred carbon	Microwaving the CBs Pyrolysis of CBs in NHO_3	• Active carbon can be used as an alternative absorbent for oil-water separation, supercapacitors, and methylene blue due to its structural diversity and large surface area. • Charred carbon can be used to remove lead from aqueous solution • The research proposes to wash CBs in hot water and unwrap from the cigarette paper, on a larger scale, this method is impractical to achieve.	[89, 90]
Utilization as sound absorbing material	Manual removal of cellulose acetate from the CB	• The product can be used as an absorbing material due to the modified materials elevated acoustical performance. • The results revealed a high absorption coefficient of < 0.90 from mid-frequencies (> 2000 Hz). • The filter can be used as a substitute for artificial based commercial products. • The toxicity of the filter is not considered, therefore further research is essential to investigate the release of heavy metals and toxic chemicals during the use of this new material.	[91, 92]
Production of superhydrophobic sorbent	Hydrolyzing the CBs in NaOH and ethanolic solution of hexadecyltrimethoxysilane. Chemical vapor deposition using methyltrichlorosilane on CBs.	• The produced superhydrophobic fibers can be used for the clean-up of water/oil spills • The processed cellulose acetate fibers demonstrated acceptable wetting behaviors towards kerosene and water. • The filters absorbed oil 16 to 26 times their weight.	[93-97]

	Facile electrospinning on the CBs.	• A 99.9% water/oil mixture separation efficiency was achieved after many cycles. • A life cycle assessment is necessary to assess the chemical characteristics of the superhydrophobic fibers once it deteriorates its sorption capacity and needs to be disposed of.	
Use as biofilm carrier	Integrated fixed-film activated sludge process used on the CBs	• The cigarette filter rods can be used as effective Kaldnes media for the removal of phosphate and organic matter from wastewater. • Further research is essential including a leachate analysis and life cycle assessment.	[98, 99]
Preparation of cellulose nano-crystals	Acetyl groups of CBs cleaved through alkali treatment and etched with HPO from phosphoric acid	• The celluloses extracted from CBs can potentially be used as an absorbent for the removal of diclofenac from water. • A maximum removal capacity of 107.9 mg g^{-1} of Dfc through the synthesis of HPO-CNFs was achieved. • Further research is essential including a leachate analysis and life cycle assessment.	[100]
Use as vector control	Soaking of CBs in distilled water	• An increase in mortality of Aedes albopictus was found in the presence of CBs. • CB waste can be developed as a novel vector control strategy as it has an indirect effect on adult life traits of Aedes aegypti. • To specifically target dengue vectors in an open environment is a complex process that requires extensive research.	[103-106]

2.5 Discussion and conclusion

The estimated volume of produced cigarettes worldwide, the persistency, as well as the toxic content after usage, support the necessity for efficient disposal and recycling strategies to solve a global CB waste catastrophe. Existing disposal strategies for collected CB waste include incineration and landfilling. However, both techniques are considered unsustainable and inefficient due to environmental and economic concerns. Therefore, this study has reviewed possible technology on the valorization of CB waste and examined the viability of the research conducted. Eleven CB valorization proposals and practices have been studied, including; (1) mixing in fired clay bricks, (2) mixing in asphalt concrete,

(3) production of cellulose pulp, (4) utilization as corrosion inhibitor, (5) preparation of porous carbon, (6) preparation of activated and charred carbon, (7) utilization as sound absorbing material, (8) productions of superhydrophobic sorbent, (9) use as biofilm carrier, (10) preparation of cellulose nano-crystals and (11) use as vector control.

Based on the extensive research conducted on the toxicity of CBs, it is clear the chemical and physical characteristics of CBs can potentially be utilized. However, this does not endorse the practicability and sustainability of the recycling method adopted. Hence why it is crucial to conduct a life cycle assessment of any new material recycled with CBs to prevent the beginning of a new waste catastrophe for the environment.

Throughout the study, various CB processing methods were reviewed, and a vast majority of the recycling methods made use of the chemicals trapped in the CB or the cellulose acetate filter. In contrast, to a minority of studies utilising the CB as a whole, without employing a processing technique. In summary, the following conclusions have been derived from this investigation:

- A practical recycling method involved incorporating CBs in the production of bricks. Whereby, the literature reviewed proposes that all CBs can sustainably be recycled in fired clay bricks by incorporating 1% CBs in 2.5% of the world's brick production. Additionally, the experimental results were found to be within the standard limits for various properties, including strength, water absorption, efflorescence, and shrinkage, which are important parameters when manufacturing bricks. Moreover, the leaching of contaminants from CBs in bricks was found to be negligible. However, further research is recommended, predominantly a life cycle assessment.
- CBs have the potential to be used as an aggregate in asphalt concrete. Asphalt concrete is a combination of aggregates and asphalt cement, whereby the aggregate constitutes 90 – 95% of the total mixture. Therefore the replacement of CBs in the production of asphalt concrete for natural aggregates can offer a sustainable alternative to reduce the demand for natural aggregates. Due to the toxic nature of CBs, the encapsulation process will act to prevent the leaching of those chemicals by inhibiting water from reaching the filters. Moreover, the physical and mechanical properties generally demonstrated acceptable bulk density, percentage of air voids, and resilient properties, while the stability values only satisfied the light traffic conditions for road pavements. However, research on CBs in asphalt concrete is limited, therefore further experimental work, including a leachate analysis and life cycle assessment, is essential.

- Extracting the cellulose fibers from the cellulose acetate for use in the pulp and paper industry offers a sustainable solution to deforestation. However, the process consequently leads to the production of wastewater containing toxic chemicals from the cellulose acetate. Despite the inconclusive nature of the results, the extracted cellulose pulp from the CBs can decrease the harvesting of trees for paper manufacturing. Moreover, the porous nature of CBs allows it to be a perfect absorbing material with exceptional acoustical performance. Which can be used as a substitute for artificial based commercial products. However, the literature reviewed requires clarification on the pre-treatment process of the CBs due to the highly toxic nature of the waste.
- The cellulose acetate in the CB is an important target for CB recycling. A vast majority of researches has utilized the porous nature of the waste for water and wastewater treatment applications. Including the fabrication of superhyrophobic fibers for clean-up of water/oil spills; biofilm carrier for the removal of phosphate and organic matter from wastewater; cellulose nanocrystals and nanofibres for the removal of diclofenac from water; active carbon for oil-water separation, and charred carbon for the removal of lead from aqueous solution. Although there is various equipment available for wastewater treatments, the utilization of CBs offers an innovative and sustainable approach. It is important all components of the CB is considered and not discarded as waste or require further management.
- Porous carbon derived from CBs are of high surface area with excellent electron conductivity and can be used for supercapacitors and electronic applications. Porous carbon can be used in several applications, however, is considered expensive and consists of sophisticated techniques. Therefore recycling CBs to produce porous carbon is a practical technique, however, the research conducted requires further experimentation and clarification. The methods adopted fail to recycle the whole CB, but rather attempt to make use of only one component of the waste and discard the remaining.
- The nicotine content trapped in the CB can be extracted to be used as an effective corrosion inhibitor for metals. However, applying the extract on metal raises environmental concerns for the possible leachate of toxic chemicals. Similarly, utilizing CBs grounded on its pesticidal, fungicidal, insecticidal, and herbicidal properties for vector control may result in the release of pollutants during the recycling process.

In conclusion, valorization of CBs may be an environmentally friendly way to manage a continuously growing pollution calamity. In some cases, it even implies an enhancement to the modified material. The aim of this review was to systematically evaluate the toxic nature of CBs, comprehensively review all major research studies on recycling CB waste, and investigate the feasibility and sustainability of

the recycling methods adopted. Based on the review, the following research and developments are recommended:

- A life cycle assessment to evaluate the positive and negative environmental impacts of the proposed CB recycling procedures from the collection of the waste through to the production process and, finally, the packaging and storage.
- The preparation and implementation of new policies for different recycling options through the affiliation of government agencies, the environmental protection authority (EPA), and waste recycling industries. In relation to the collection, safety, and storage of CBs; processing procedures of the various applications; toxicity and leachability; long term conditions and disposal.
- The evaluation of employee exposure, transport, collection processes, and consumer exposures for all recycling schemes mentioned.

Chapter 3: Recycling of Cigarette Butts in Fired Clay Bricks: A New Laboratory Investigation

Abstract

Cigarette butts (CBs) are the most commonly littered waste material in the world. It is estimated that over 5.7 trillion cigarettes are consumed worldwide each year. Consequently, millions of tonnes of highly toxic waste are contaminating the environment. CBs are composed of cellulose acetate filters—a polymer with poor biodegradability—and which, depending upon the environmental conditions, can take many years to decompose. In this study, fired clay bricks were manufactured with 0.5%, 1%, 1.5%, and 2% CBs by mass and tested against control bricks with 0% CBs. The results revealed a decrease in compressive strength from 48.6 MPa for 0% CB content bricks to 30.8 MPa for 1% CB content bricks, and a decrease in dry density with the increase in CB content, from 2114 kg m^{-3} for the control bricks to 1983 kg m^{-3} and 1969 kg m^{-3} for 1% and 2% CB content bricks. The highest value of water absorption appeared for 2% CB content bricks, which reached an absorption rate of 13.1% compared to 9% for the control bricks. The energy required during the firing process was calculated with a saving of up to 10.20%, for bricks incorporating 1% CBs. The thermal conductivity of the samples showed a reduction of 17% from 1.078 to 0.898 W $m^{-1} \cdot K^{-1}$ with the addition of 1% CBs. In addition, the manufactured bricks were tested for efflorescence, an initial rate of absorption (IRA), microstructural analysis, and shrinkage. A life-cycle assessment (LCA) is recommended to analyze the environmental impacts of bricks incorporating CBs.

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3.1 Introduction

Due to the increasing requirements of environmentally-friendly, lightweight, and low-cost construction materials, many studies have been devoted to finding ways to optimize the performance of the materials used in the building industry, such that the effect on the environment is minimized [1,2]. Ultimately, one-third of the world's energy consumption is consumed in buildings [3], while 50% of this is lost due to the poor thermal conductivity of the walls [4]. Fired clay bricks are the most conventionally used masonry material around the globe [5]. Their interesting physical, thermal, and mechanical properties allow them to be the ultimate material for the construction of buildings, especially their durability, compactness, and strength [6].

Buildings consume great amounts of thermal energy. During the winter, the internal warmth generated by heating is lost due to the high thermal conductivity of traditional bricks, while, in the summer, the coolness can escape due to the same reasoning. Therefore, it is essential to develop a method to produce thermally-resistant walls, while remaining in the realms of practicality and maintaining the aesthetic appeal of the construction. Hence, enhancing the insulation effectiveness of construction material will significantly lower the impact on the environment.

Many studies have been conducted to find ways to improve the thermal conductivity of fired clay bricks, with one being the recycling of waste materials composed of high organic content. It was found that the greater the energy consumption during the firing process, the greater the thermal conductivity of the bricks [7,8]. Therefore, the incorporation of additive materials with high organic content may create voids within the structure during the firing process and reduce energy consumption [9]. Wastes have higher heating values (HHV), which are added through self-combustion within the clay mixture, and, hence, less energy is required during the firing process [10]. In doing so, the porosity will increase due to the creation of pores, and the thermal conductivity and density of the fired clay bricks will decrease [11,12]. However, these modifications to the microstructure and composition of the brick will have a negative impact on the mechanical resistance of the material.

The correlation between density and thermal conductivity is evident [9]. However, it is conceivable to discover situations where there is a variation between thermal conductivity and density [7,13]. The sources of these variations may be the nature and distribution of the pores or the presence of certain mineralogical components [13,14]. A study was completed on 29 samples of clay bricks in which the thermal conductivity was determined. Furthermore, the relationship with their physical, microstructural, and compositional features was evaluated. It was found that mineralogical components, like Ca-rich silicates, quartz, and amorphous phase, play a significant role in decreasing the insulation properties of fired clay bricks, while open pores have a positive effect on thermal conductivity [13].

A number of studies have been conducted on recycling wastes in fired clay bricks to investigate the relationship between the use of additives in fired clay bricks and the micro-pores and nano-pores formed as a result. Such additives include waste sawdust [15], desulfurization slag and basic oxygen furnace aggregates [16], marble powder [17], biomass incineration [18], recycled paper processing residues [4], bio-briquette ash [19], sugarcane bagasse ash [20], and cigarette butts [21–27]. A recent study revealed the substructures produced by the additive olive pomace (10% weight) that resulted in a 14.4% decrease in thermal conductivity [28]. Comparing this to expanded vermiculite (10% content), a pore-forming material that expands when heated declined from 0.96 to 0.65 W mK^{-1}, which is a 32% drop in thermal conductivity [3]. While olive mill solid residue (OMSD) demonstrated the highest decrease in thermal conductivity of 70% with the use of 15% OMSD in fired clay bricks [29].

3.1.1 Cigarette Butts

Cigarette butts (CBs) are the most commonly littered waste material in the world. The Tobacco Atlas [30] reported that 5.7 trillion cigarettes were consumed worldwide in the year 2016 and that about 97% of the cigarette filters were made up of cellulose acetate filters, which is a modified natural polymer [31]. In Australia, every year, more than 20 billion cigarettes are consumed, and 7 billion are littered into the environment (Fig. 3.1) [32]. Clean Up Australia [33] reported that CBs were the most common source of rubbish collected under the miscellaneous category, which represented 91.5%.



Fig. 3.1. Littered CBs.

When CBs are littered, harmful chemicals are leached into the environment, which poseleading concerns for the global environment and the quality of urban and aquatic life. In cigarette smoke, it is possible to classify more than 4000 chemical components that are generated during burning or that are distilled from tobacco, of which 69 are carcinogenic [34,35]. Cigarettes generally incorporate cellulose acetate filters, which are designed to fully or partially retain particulate smoke components, including tar and toxic chemicals [36]. Cellulose acetate filters have poor biodegradability and can take up to 18

months to decompose under normal environmental conditions, and, hence, pose a severe dilemma in terms of toxic waste and litter disposal [37,38]. CBs littered on streets are commonly washed away into stormwater drains and end up on beaches, or in rivers and harbors where they leach toxic chemicals [39].

Landfilling and incineration are possible disposal methods for CBs. However, both methods are costly and universally unsustainable. The incineration of CBs can result in the emission of various hazardous substances, and landfilling of CBs will continue to be an environmental hazard regardless of the location of the landfill and can be severely detrimental to the environment and human health [40,41]. Therefore, it is becoming vital to find a practical and effective way to recycle CB waste. An alternative to the growing pollution catastrophe may be the method of recycling CBs into composite building materials such as fired clay bricks.

Mohajerani et al. [27] propose that all CBs can be sustainably recycled in fired clay bricks by incorporating 1% CBs in 2.5% of the world's brick production. The study involves incorporating various percentages of CBs in clay bricks, including 2.5%, 5%, 7.5%, and 10% [23,24,27]. Experiments have confirmed a steady decrease in dry density and compressive strength with an increase in the CB content of the clay bricks. This decrease in density increases the porosity, which results in an exponential drop in thermal conductivity. Thus, this results in improved thermal performance and greater energy efficiency [26]. The lower energy consumption during the firing process is contributed through the high calorific value of the CBs. CBs have higher heating values (HHV), which are added through self-combustion within the clay mixture, and, hence, less energy is required during the firing process [10]. Moreover, the high temperatures during the firing stage will result in the volatilization of dangerous components and eliminate any toxic compounds [27].

This paper is the continuation of an ongoing study on the incorporation of CBs in fired bricks. In this study, the properties of 0.5%, 1%, 1.5%, and 2% CB bricks by mass (equivalent to approximately 10 kg m^{-3}, 20 kg m^{-3}, 30 kg m^{-3}, and 40 kg m^{-3} CBs) were investigated and compared with controlled fired clay bricks (0 kg/m^3 CBs). The energy saving percentage during the firing process of bricks and the reduction in thermal conductivity are presented as a function of the CBs present in the mixtures. Additionally, numerous tests were carried out, including water absorption, compressive strength, efflorescence, an initial rate of absorption (IRA), density, shrinkage, and microstructural analysis.

3.2 Materials and Methods

3.2.1 Materials

The soil used in preparing the fired clay bricks was sandy silty clay (MC), which was provided by PHG Bricks Victoria (*Victoria, Australia*). Geotechnical laboratory tests were conducted on the soil,

including the particle size distribution test (PSD), liquid limit, and plastic limit, according to the Australian Standards [42–44].

CBs of varying sizes and brands were provided by Buttout Australia Pty Ltd. (Melbourne, Australia), which were collected from dry receptacles. On arrival, the CBs were dried and sanitized at 105 °C for 24 hr in the oven (Fig. 3.2), and then stored in airtight plastic bags.

Chemical and mineral analyses were conducted to determine the major chemical and mineral constituents of the experimental soil using X-ray fluorescence (XRF) (S4 Pioneer, Bruker, MA, USA) and an X-ray diffraction analyzer (XRD) (D8 Endeavor, Bruker, MA, USA). Thermo-gravimetric analysis (TGA) was carried out using the Perkin-Elmer TGA 8000 instrument (Waltham, MA, USA) to investigate the weight loss of the 0% and 1% CB content clay mixtures.



Fig. 3.2. CBs used in this study.

The geotechnical properties of the soil used in this study were determined according to the Australian Standards [43,44]. By testing the Atterberg limits, it was found that the liquid limit for the brick soil was 34%, while the plastic limit was 25%, respectively. These values are the average results from three replicate tests.

The particle size distribution for the soil sample was determined through sieve analysis and is shown in Fig. 3.3. The brick soil comprised 77.5% fine particles (<75 μm). According to these results, the Unified soil classification of the brick soils is sandy silty clay (MC) [42].

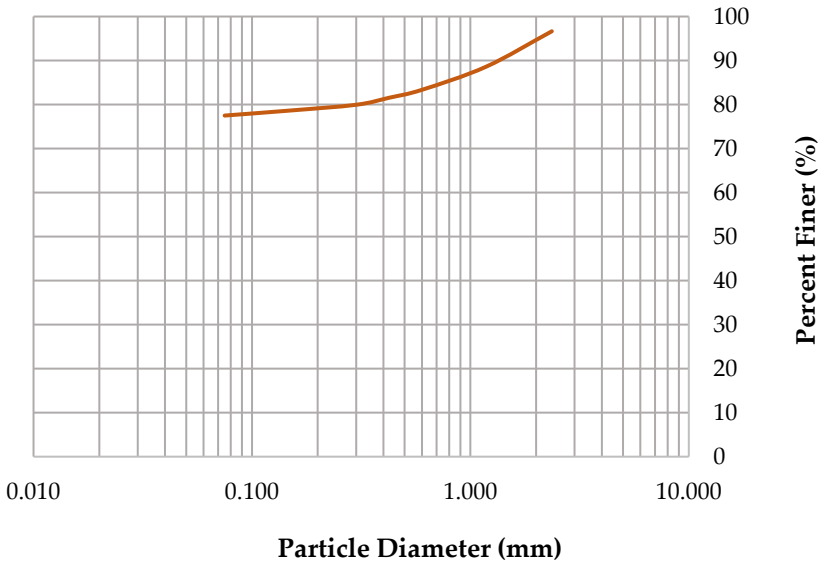


Fig. 3.3. Particle-size distribution curve of brick soil used in this study.

Chemical analyses were conducted to obtain the chemical composition of the brick soil using XRF. As shown in Table 3.1, the main compound found in the soil was silicon dioxide, which is followed by aluminium oxide and ferric oxide. XRD was utilized to identify the main crystalline phases on a <75-µm sample of the brick soil (Fig. 3.4). The brick soil displayed Quartz (SiO_2) as the primary crystalline phase while Muscovite ($KAI_2(AISi_3O_{10})(F,OH)_2$), Clinochlore ($Mg_5AI(AlSi_3O_{10})(OH)_8$), Mikasaite ($Fe_2(SO_4)_3$), and Albite ($Na(AlSi_3O_8)$ presented a lower intensity.

Table 3.1. Soil chemical properties.

Compound Formula	Brick Soil
SiO_2	62.84%
Al_2O_3	19.95%
Fe_2O_3	6.15%
K_2O	4.87%
Na_2O	2.46%
MgO	1.62%
TiO_2	1.19%
CaO	0.27%
Loss on ignition	9.60%

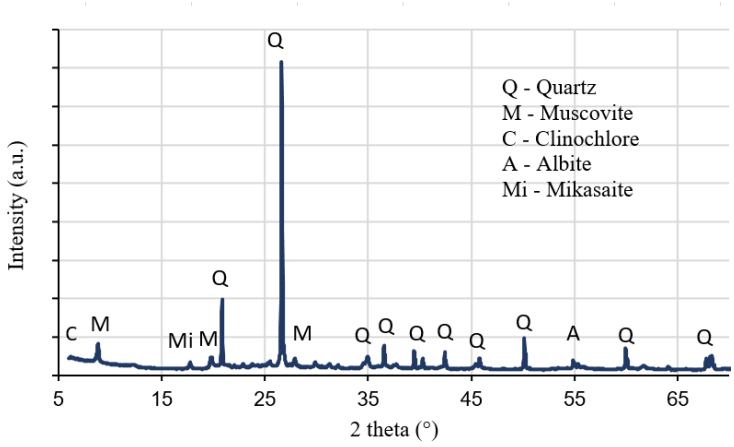


Fig. 3.4. XRD patterns of brick soil.

TGA was performed to monitor and compare the weight loss of the 0% and 1% CB content clay mixtures while the samples were exposed to a constant rate of heat. The samples were heated from 30 °C to 850 °C at 20 °C min^{-1} in an air atmosphere with a purge rate of 15 mL min^{-1}. Fig. 3.5 presents the thermogravimetric (TG) and differential thermal (DT) curves for 0% and 1% CB content clay mixtures, respectively. The 0% and 1% CB content clay mixtures demonstrated mass loss characteristics of 6.1% (0% CB content clay mixture) and 7.3% (1% CB content clay mixture). The initial decrease occurred at 150 °C. This is due to the evaporation of water within the mixture. While a majority of the mass loss occurred at 150–400 °C and 500–600 °C, during these ranges, organic matter is burned and water molecules of the chemical compounds are released. The final mass loss occurred between 600–800 °C. This can be related to the decomposition of carbonates and sulphides. The 1.2% difference in mass loss between the two samples could be due to the high volatile organic content present in the 1% CB content clay mixture that burnt off during the high temperatures.

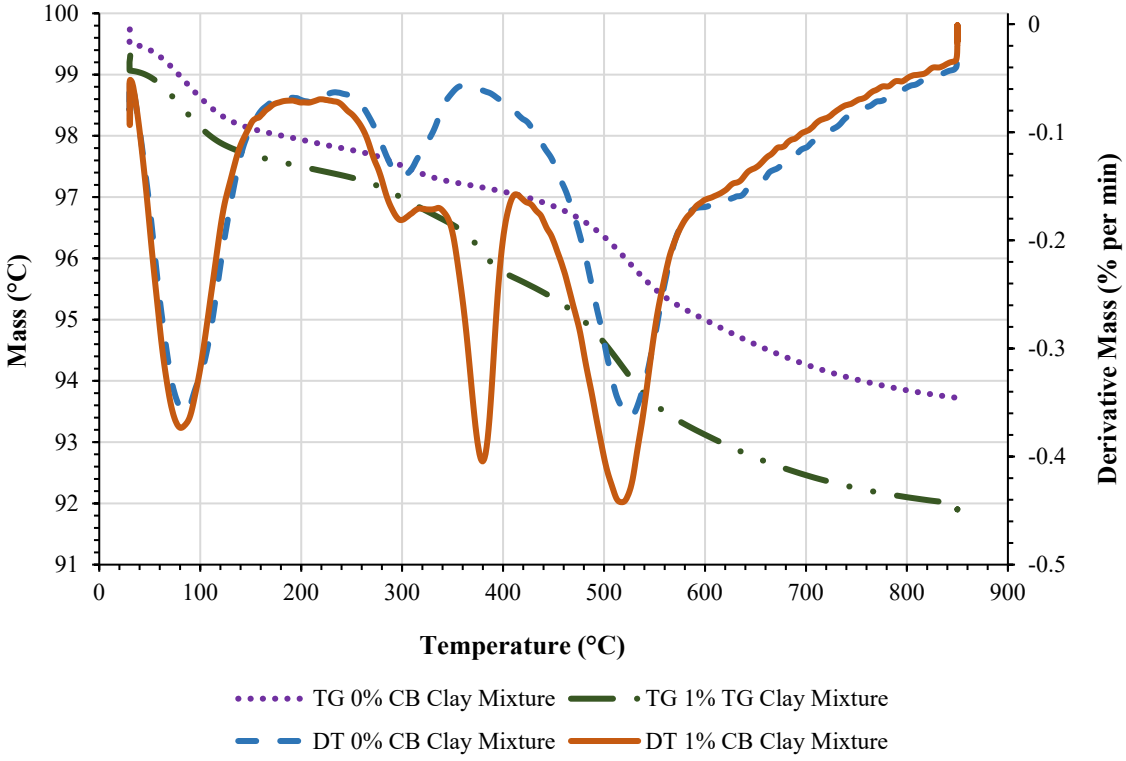


Fig. 3.5. TG and DTA curves of 0% CB content clay mixture and 1% CB content clay mixture.

3.2.2 Brick Sample Preparation

The bricks were prepared incorporating varying percentages of cigarette butts and a control sample with 0% CBs to investigate the mechanical and physical properties of the bricks. The mixes were categorized by the percentage of CBs by the mass of the brick. CB (0.0) was the control sample and consisted of no CBs. CB (0.5) consisted of 0.5% CBs by mass. CB (1.0) consisted of 1% CBs by mass, CB (1.5) consisted of 1.5% CBs by mass, and CB (2.0) consisted of 2% CBs by mass.

The soil and CB mixtures were prepared using a Hobart mechanical mixer for a duration of 25 min with a 15.5% water content. Then it was placed in airtight plastic bags for 48 h to uniformly distribute the moisture throughout the mix. The mixtures were then compacted in molds of a 100-mm diameter to yield bricks with heights of approximately 50 mm with a compaction pressure of 240 kPa. The specimens were air dried for 24 h before being placed in an oven for 24 h at 105 °C. Once removed from the oven, the samples were then fired in an electric furnace at 1050 °C for 3 h. During each stage of the preparation process, the height, diameter, and weight of the samples were measured using an electric scale and digital caliper. The fired samples were tested for water absorption, compressive strength, efflorescence, initial rate of absorption (IRA), density, shrinkage, thermal conductivity, microstructural analysis, and energy savings, according to the Australian Standards [45] (Fig. 3.6). All the results presented are the average of three replicates.

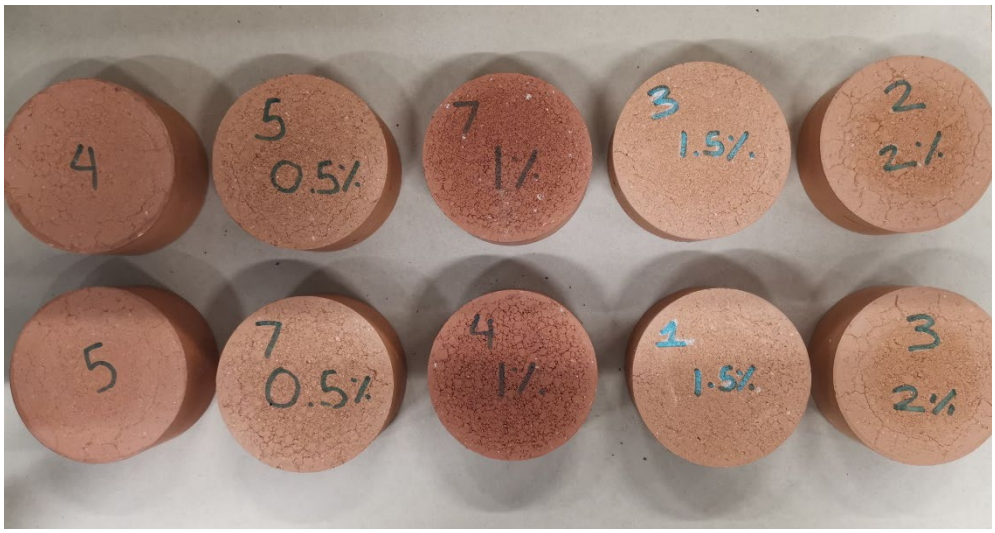


Fig. 3.6. Some of the samples prepared in this study.

3.2.3 Energy Saving

The calorific value of a material determines the amount of heat (energy) that is produced by the complete combustion of that material. Therefore, utilizing a material with a high calorific value as a partial replacement material in a mixture with a low calorific value will result in a reduction in the overall required energy for bricks, since additional energy is produced by the constituents of the material when combusted, as compared to a standard mixture.

In the calculation method [27], the firing energy for the fired clay bricks incorporating CBs was calculated, assuming that 2 MJ kg^{-1} energy was used for firing the bricks. This value was based on a survey conducted in 1993–1994, which shows that the firing energy consumption is in a range of 2 to 3 MJ kg^{-1} in Association of Southeast Asian Nations (ASEAN) countries [46]. A laboratory experiment was conducted to determine the calorific value of the CBs. Samples of used CBs were tested, and the calorific value for the CBs was found to be 16.53 MJ kg^{-1}.

The calculations were conducted for the CB bricks by considering the weight of CBs incorporated as well as the calorific value of the cellulose acetate. Therefore, the estimated firing energy saved through the addition of CBs into bricks and compared to the control brick was calculated as follows.

Energy Used for Control Clay Brick (MJ)	$Q_1 = q \times m_1$	(3.1)
Energy Used for Clay—B Brick (MJ)	$Q_2 = q \times m_2 - CV \times m_3$	(3.2)
Energy Saved (MJ)	$Q_1 - Q_2 = (q \times m_1) - (q \times m_2 - CV \times m_3)$	(3.3)
Energy Saved (%)	$\Delta E(\%) = \frac{Q_1 - Q_2}{Q_1} \times 100\%$	(3.4)

$$= \frac{(q \times m1) - (q \times m2 - CV \times m3)}{q \times m1} \times 100\%$$

Where:

q = energy used for brick firing = 2 MJ kg^{-1}

m_1 = mass of clay in control clay brick (kg)

m_2 = mass of clay in clay-CB brick (kg)

m_3 = mass of CBs in clay-CB brick (kg)

CV = calorific value of cellulose acetate = 16.53 MJ kg^{-1}

3.2.4 Thermal Conductivity

Due to the increasing requirements for energy savings, the thermal insulation properties of building materials, such as bricks, have become significant. The heat loss from a building depends on the thermal conductivity of the materials in the roof and wall. While the thermal conductivity of a brick is the quantity of heat transmitted through the brick. Components that influence the effective thermal conductivity of bricks are generally the geometrical design [47], microstructure, and mineralogical composition [48].

As part of these studies, Kadir and Mohajerani [22] developed Eq. (3.5) to estimate the thermal conductivity of brick samples. The thermal conductivity of 256 results of various types of concrete, aggregate, and bricks was obtained to develop an equation using regression analysis as a function of density. Whereby, it was found that K (thermal conductivity) can be represented using its relationship with Dry Density.

$$K = 0.0559e^{(0.0014)(Dd1)} \tag{3.5}$$

Utilizing Eq. (3.6) – (3.8), Eq. (3.9) was developed to estimate the energy transfer of the fired clay brick samples. According to Fourier's Law, we find that the Heat Flow Rate (W) can be expressed as:

$$q = k \times A \times \frac{\Delta T}{L} \tag{3.6}$$

Since the Heat flow rate is measured in Watts, we can also determine that, through reconfiguring, the Current Intensity formula can be found, which is shown below.

$$Q = \frac{Q}{\triangle T} \tag{3.7}$$

Hence,

$$\frac{Q}{t} = k \times A \times \frac{\Delta T}{L} \tag{3.8}$$

Re-arranging the above equations, we see a new formula for energy transfer.

$$Q = \frac{\mathrm{k} \cdot A \cdot \Delta T}{L} \times t \tag{3.9}$$

where:

q = heat flow rate (W)

$\Delta T = T_{hot} - T_{cold}$ = change in temperature (Kelvin)

Q = energy transfer (Wh)

T = time (h)

k = thermal conductivity (W mK^{-1})

L = thickness of the brick

A = surface area of the material transferring heat

3.3 Results and Discussion

3.3.1 Properties of the Fired Clay Bricks

Property tests were conducted for five types of the sample: 0% CBs, 0.5%, 1%, 1.5%, and 2% CBs. The results can be seen in Figs. 3.7 – 3.12. The dry density of the prepared bricks decreased from 2114 kg m^{-3} for controlled bricks to 1983 kg m^{-3} for bricks with 1% CBs by mass. The addition of CBs resulted in the bricks becoming more porous. The development of pores within the structure of the brick can be associated with the added CBs that burnt off during the firing process as a consequence of the volatile organic content in the CBs.

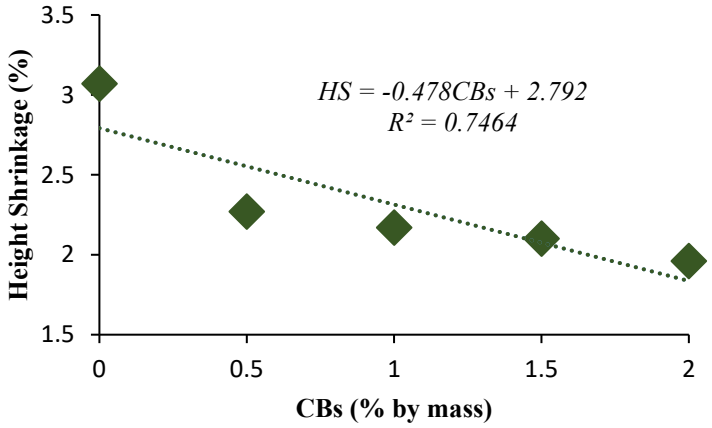


Fig. 3.7. Height shrinkage versus CB percentage by mass.

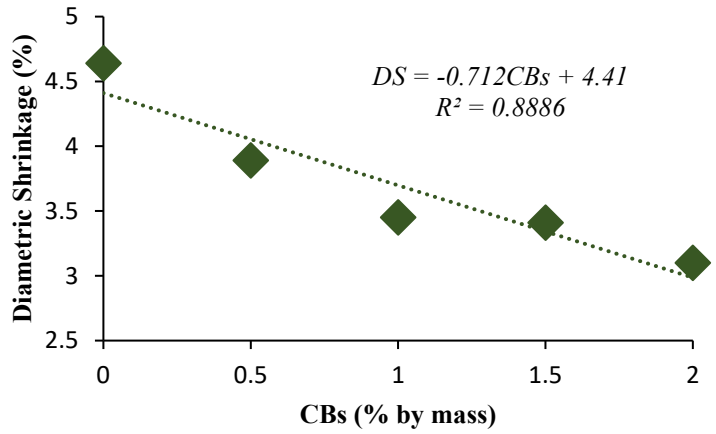


Fig. 3.8. Diametric shrinkage versus CB percentage by mass.

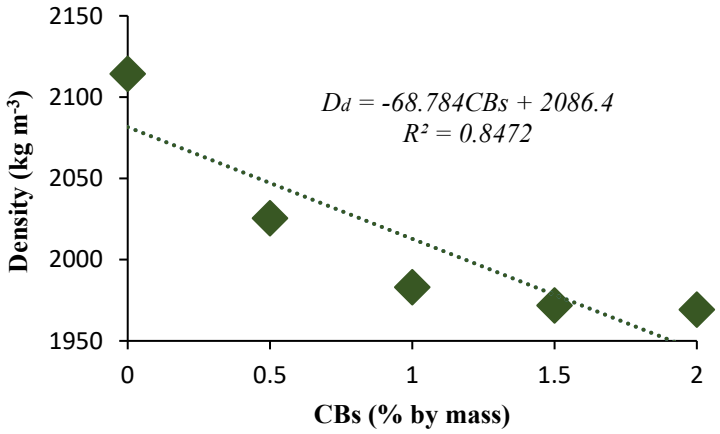


Fig. 3.9. Dry density versus CB percentage by mass.

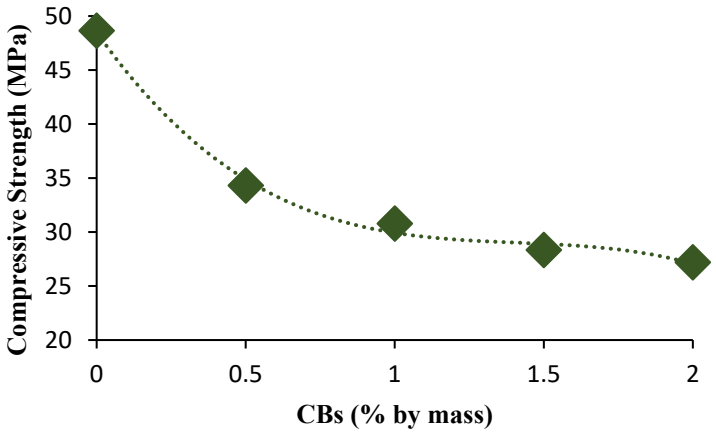


Fig. 3.10. Compressive strength versus CB percentage by mass.

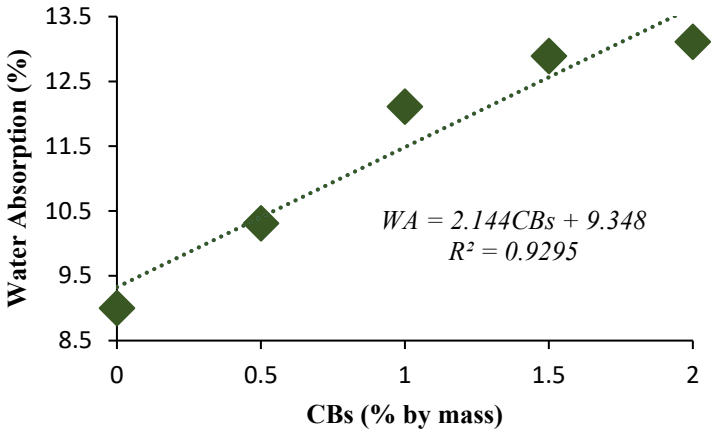


Fig. 3.11. Water absorption versus CB percentage by mass.

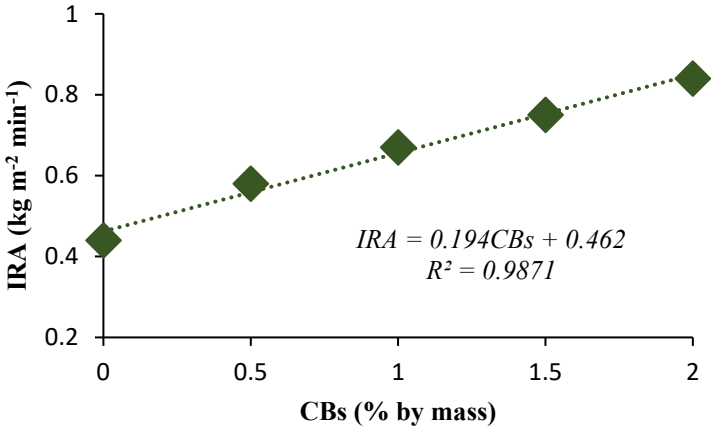


Fig. 3.12. IRA versus CB percentage by mass.

The compressive strength of the bricks tested decreased from 48.6 MPa for 0% CB content to 30.8 MPa for 1% CB content, and 27.2 MPa for 2% CB content. The minimum recommended compressive strength is 3–5 MPa, and 5–10 MPa for non-load and load-bearing fired clay bricks, respectively [49–51]. The reduction in strength was associated with the loss of organic content in the CBs, which were exposed to firing in the furnace. This resulted in the formation of bigger pores due to the CBs that burnt off, which lowered the strength of the CB-amended fired clay bricks. Similar results have been observed with the incorporation of hemp shiv fibers in fly ash-based alkaline mortars. The addition of hemp fibers led to a reduction in compressive strength due to the increase in porosity [52].

The higher percentage of CB content resulted in an increase in water absorption and IRA. The water absorption was measured at 12.1% for 1% CBs by mass compared to 9% for 0% CBs by mass (Fig. 3.13). The greatest value of water absorption was measured at 13.1% for 2% CBs by mass. The IRA

was found to be 0.67 kg m^{-2} min^{-1} for 1% CBs by mass and 0.44 kg m^{-2} min^{-1} for 0% CBs by mass, which falls within the IRA range of 0.2 and 5 kg m^{-2} min^{-1} [53]. A similar trend was observed with the incorporation of biosolids in fired clay bricks. The increase in the percentage of biosolids resulted in an increase in IRA due to the creation of pores during the firing process [54]. Water absorption and IRA values signify the long-term durability performance of the bricks. Therefore, excessively high values can lead to cracking as a consequence of volume changes.



Fig. 3.13. 24-h cold water and 5-h boiling water test.

Drying and firing shrinkage are essential indicators for evaluating the suitability of the added CBs from a durability point of view. The bricks are exposed to high temperatures in the furnace. Therefore, any cracks that occur during drying can result in brick failure during firing. The results revealed a decreasing trend in height and diametric shrinkage with the increase in CB content. The diametric shrinkage values obtained were 4.64%, 3.45%, and 3.1% for 0%, 1%, and 2% CBs by mass bricks, respectively. The decreasing trend may be the consequence of the volatile content present in the CBs that burnt off, which results in a change in size. A similar trend in the variation of shrinkage in fired clay bricks with the increase in waste glass has been reported by Phonphuak et al. [55].

Analysis of efflorescence was carried out for the clay bricks and the clay-CB bricks. Efflorescence is the formation of salt deposits on or near the surface of the brick that causes a change in appearance [56]. For comparison purposes, 10 samples were prepared and sorted into two pairs so that both specimens of each pair had the same appearance. All external faces of each pair of specimens were examined (Fig. 3.14). The results for the potential efflorescence of the control, 0.5%, and 1.5% CB bricks displayed “Nil” observable efflorescence, while there was “Slight” efflorescence for 1%, and “Moderate” efflorescence for the 2% CB bricks, according to the Australian Standards [57].

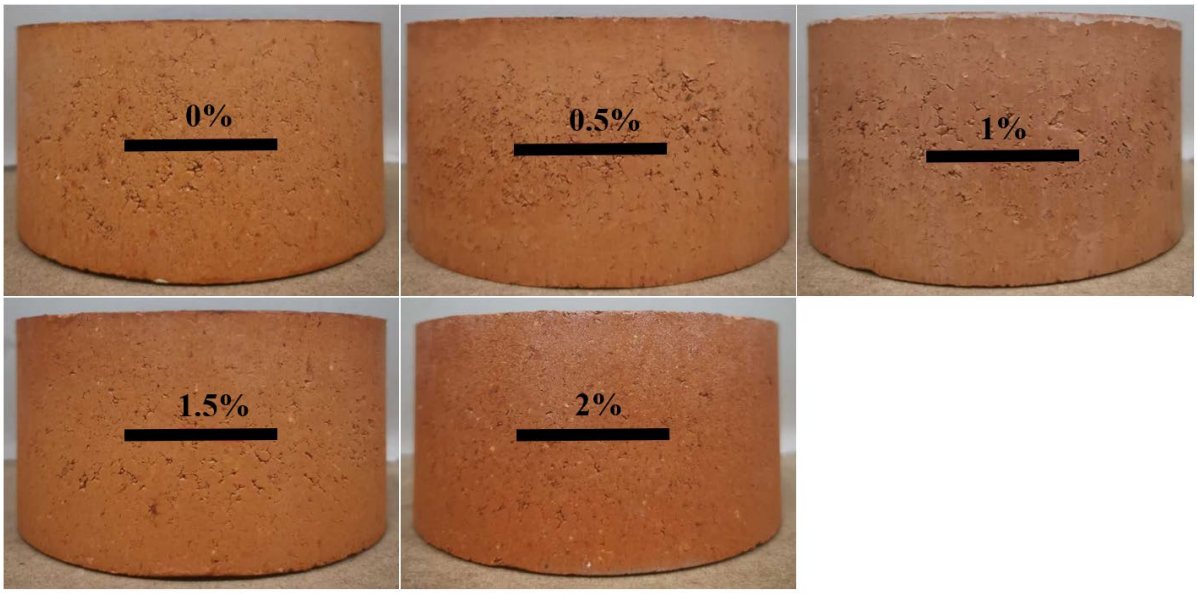


Fig. 3.14. Efflorescence on bricks for mixes with 0%, 0.5%, 1.0%, 1.5%, and 2.0% CBs.

3.3.2 Energy Savings

Using Eq. (3.4), the estimated amounts of energy saved during the firing process of 0.5%, 1%, 1.5%, and 2% CBs by mass bricks were calculated and are given in Table 3.2 and Fig. 3.15. The results show that, by firing bricks with 1% CBs by mass, there could be a possible 10.20% energy saving when compared to the control bricks. Furthermore, a 14.71% energy saving can be achieved with an increase in the CB content (2% CB bricks). This is due to the waste having a higher heating value, which is added through self-combustion within the clay mixture during the firing process [9].

Table 3.2. Calculated value of energy saved during the firing of brick samples.

Mixture Identification %	Mass of Clay Brick, m_1 Kg	Mass of Clay-CB Brick m_2 Kg	Mass of CBs in Clay-CB Brick m_3 Kg	Energy Used for Brick Firing q MJ kg^{-1}	Calorific Value CV MJ kg^{-1}	Energy Used for Control Brick Q_1 MJ	Energy Used for Clay-CB Brick Q_2 MJ	Energy Saved $Q_1 - Q_2$ MJ	Percentage of Energy Saved $(Q_1 - Q_2)/Q_1 \times 100$ %
CB (0.5)	0.758	0.749	0.004	2	16.53	1.52	1.44	0.080	5.27
CB (1.0)	0.758	0.742	0.007	2	16.53	1.52	1.36	0.155	10.20
CB (1.5)	0.758	0.741	0.011	2	16.53	1.52	1.30	0.218	14.36
CB (2.0)	0.758	0.738	0.011	2	16.53	1.52	1.29	0.223	14.71

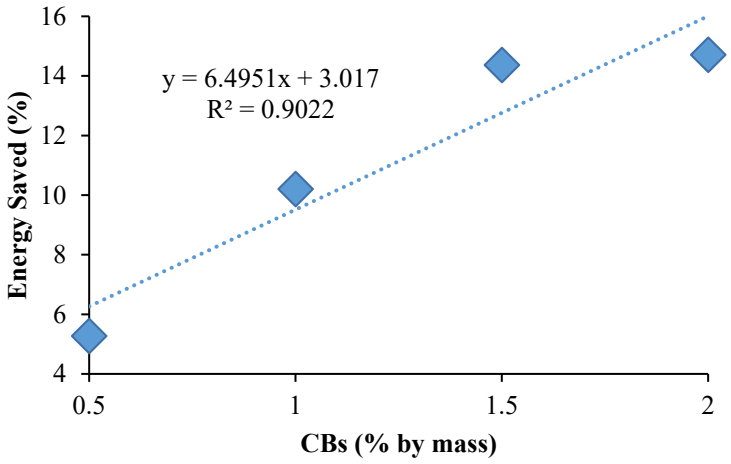


Fig. 3.15. Energy saving versus CB percentage by mass.

3.3.3 Thermal Conductivity

Using Eq. (3.5), the estimated thermal conductivities for five different brick samples were calculated and are given in Table 3.3 and Fig. 3.16. As shown, the increase in CB content results in a reduction in density and thermal conductivity. The incorporation of 1% CBs reduces thermal conductivity by 17%. Lower thermal conductivity is beneficial, as it will decrease the energy consumed for cooling and heating.

Table 3.3. Thermal conductivity of all estimated samples.

Mixture Identification %	Dry Density Kg m^{-3}	Thermal Conductivity W mK^{-1}	Reduction of Thermal Conductivity %
CB (0)	2114	1.078	0
CB (0.5)	2026	0.953	11.66
CB (1.0)	1983	0.898	16.76
CB (1.5)	1972	0.884	18.06
CB (2.0)	1969	0.880	18.35

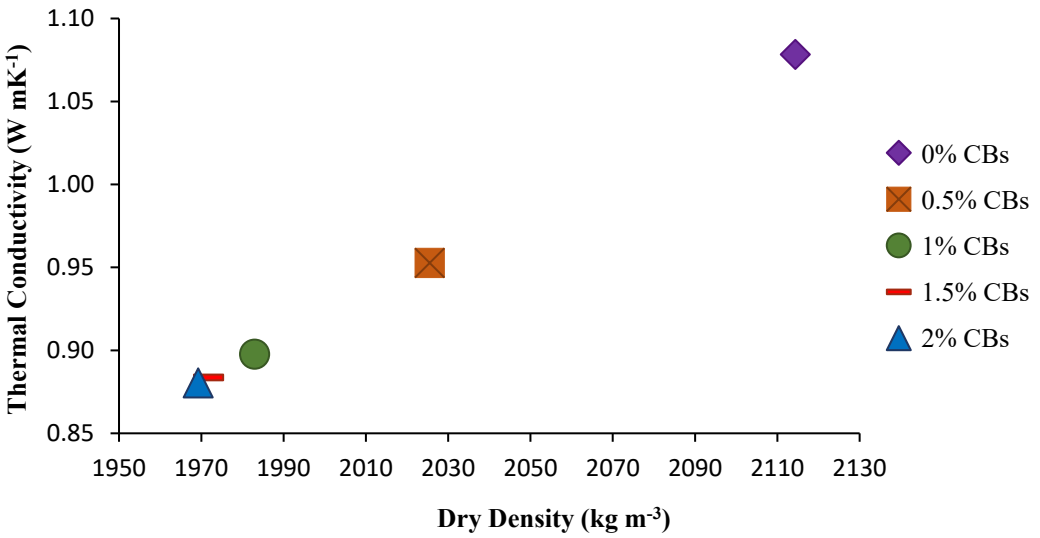


Fig. 3.16. Thermal conductivity of each brick sample.

Based on the new energy transfer, Eq. (3.9), the energy savings between a fired clay brick wall with 0% CB content and a fired clay brick wall with 0.5%, 1%, 1.5%, and 2% CB content was calculated (Table 3.4). For the purpose of this example, a 3-m high and 3-m wide wall was considered over a period of 1 year (Fig. 3.17). The thermal conductivity values are taken from Table 3.3. The temperature of the inner and outer surfaces of the wall is assumed to be 283.15 k and 273.15 k, respectively. Heat flows from the warmer object to the cooler object until they are at the same temperature. The depth of the brick is taken as 0.1 m, and it is assumed that the heat transfer through the wall is steady and that thermal conductivity is constant.

Table 3.4. Energy savings between a fired clay brick wall with 0% CB content and 0.5%, 1%, 1.5%, and 2% CB content.

Mixture Identification %	Thermal Conductivity k W/mK	Area of Wall A m^2	Change in Temperature $\Delta T = T_{hot} - T_{cold}$ kelvin	Time t h	Thickness of Brick L m	Energy Transfer Q kwh	Energy Saved %
CB (0)	1.078	9	10	8760	0.1	8.50	-
CB (0.5)	0.953	9	10	8760	0.1	7.51	11.60%
CB (1.0)	0.898	9	10	8760	0.1	7.08	16.70%
CB (1.5)	0.884	9	10	8760	0.1	6.97	18.00%
CB (2.0)	0.880	9	10	8760	0.1	6.94	18.37%

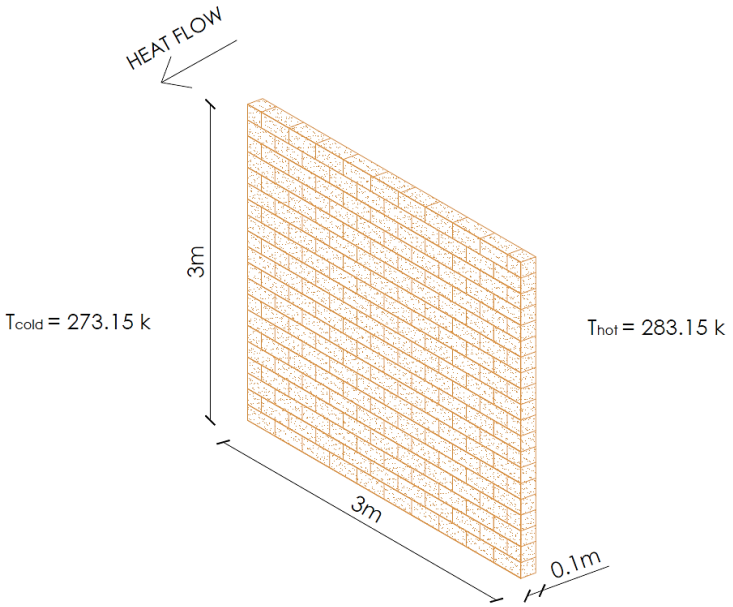


Fig. 3.17. Fired clay brick wall.

In Table 3.4 it can be seen that the increase in CB content corresponds to the reduction in thermal conductivity, and, consequently, a reduction in the energy transfer between the fired clay brick walls. An energy savings of 16.7% was achieved for a brick wall with 1% CB content compared to a 0% CB content wall. This reduction in energy transfer, and, therefore, the energy savings will decrease the costs of heating and cooling. A life-cycle assessment (LCA) is recommended to analyze the environmental impacts of fired clay bricks by incorporating CBs [58].

3.4 Conclusions

This study investigated the properties of 0.5%, 1%, 1.5%, and 2% CB bricks by mass and compared them with those of standard fired clay bricks. Numerous tests were conducted, including water absorption, efflorescence, initial rate of absorption (IRA), density, shrinkage, microstructural analyses, thermal conductivity, and energy savings based on the Australian Standards.

The compressive strength of the bricks tested decreased from 48.6 MPa for 0% CB content to 30.8 MPa for 1% CB mass. The lowest compressive strength value was obtained for the 2% CB content bricks (27.2 MPa). The minimum recommended compressive strength is 3–5 MPa, and 5–10 MPa for non-load and load-bearing fired clay bricks, respectively.

However, the dry density decreased with the increase in CB content from 2114 kg m^{-3} for control bricks to 1983 kg m^{-3} and 1969 kg m^{-3} for 1% and 2% CB content bricks, which produces light-weight bricks. The advantages of light-weight construction material are cheaper to produce and have a lower overall life cycle energy usage and design flexibility. The diametric shrinkage values obtained were 4.64% and

3.45% for the 0% and 1% CBs by mass bricks, respectively, which rises fairly linearly as the CB content increased.

The highest value of water absorption occurred for the 2% CB content bricks, which reached an absorption rate of 13.1% compared to 9% for the control bricks. The initial rate of absorption was found to be 0.67 kg m^{-2} min^{-1} for 1% CBs by mass bricks and 0.44 kg m^{-2} min^{-1} for the control bricks. The increase in CB content resulted in the increase in pores both in size and occurrence, which increased the water absorption rate of the bricks.

The thermal conductivity was estimated by using a prediction model. The estimations showed that an increase in CB content could result in a significant decrease in the thermal conductivity, and, consequently, result in lower energy costs for cooling and heating.

The estimated amounts of energy saved during the firing process of 0.5%, 1.5%, 1% CB, and 2% CB content bricks were found using calculations. Approximate savings of 10.20% and 14.71% were seen for 1% and 2% CB content bricks, respectively.

Consequently, bricks incorporating CBs have the potential to provide substantial energy savings and a unique, sustainable recycling solution to a growing global waste catastrophe. However, it is essential to adhere to international regulations and standards regarding the handling and recycling of toxic waste materials.

The results found in this study complete and confirm the previous findings on recycling CBs in fired clay bricks at RMIT University and the published proposal that all CBs can be sustainably recycled in fired clay bricks by incorporating 1% CBs in 2.5% of the world's brick production [27].

Chapter 4: Energy Savings, Thermal Conductivity, Micro and Macro Structural Analysis of Fired Clay Bricks Incorporating Cigarette Butts

Abstract

In this study, lightweight structural bricks incorporating various percentages of cigarette butts (CBs) are produced to investigate the thermal properties, energy savings, porosity and micro- and macro-structure. A novel method was developed to monitor the energy consumption of the furnace during the firing process, and Scanning Electron Microscopy (SEM) and X-ray micro-Computed Tomography (micro-CT) imaging were used to characterize the quantity, distribution, size, and shape of the micro- and macro-scale pores of the brick samples incorporating CBs. The energy savings of the microporous bricks with 1% CBs by weight (wt.) varied between 8% (theoretical value) and 10.2% (measured value). Clay bricks showed an increase in porosity and a decrease in density with the increase in CB content, which reduced thermal conductivity. Thermal conductivity was measured to reduce from 0.463 to 0.347 W m^{-1} K^{-1} with the addition of 2% CBs by wt. compared to the control bricks. The 3D models of the bricks incorporating various percentages of CBs demonstrated a homogenous distribution of pores within the brick structures. On a micrometer scale, the size of the micro- and macro- pores varied within a range of 0.035 to 1.365 mm.

The research presented in this chapter has been published in the following:

Kurmus, H. and Mohajerani, A. 2021. Energy Savings, Thermal Conductivity, Micro and Macro Structural Analysis of Fired Clay Bricks Incorporating Cigarette Butts. Construction and Building Materials, 283. https://doi.org/10.1016/j.conbuildmat.2021.122755.

4.1 Introduction

The demand for bricks is growing, particularly in economically developing countries, where infrastructure development projects are expanding. Establishing energy-efficient buildings is the method of improving the efficiency of buildings through the sustainable use of resources. The construction industry alone consumes between 30 to 40% of the total energy available in the world, while 41% of the total energy consumed in the construction industry is employed for heating and cooling [107, 108]. Although the production of bricks is expanding, the technology used in firing bricks remains the same, whereby energy is consumed inefficiently. In general, 1 MJ kg^{-1} of energy is required to transform a clay mixture into a brick [109]. However, the specific firing energy consumption ranges from 2 to 3 MJ kg^{-1}, which is roughly 25 to 35% of the total costs associated with producing bricks [110]. Therefore, producing energy-efficient residential blocks can potentially decrease the usage of energy, lower costs, and offer sustainable alternatives to boost energy conservation [111, 112].

To reduce the energy demand in the building sector, and, consequently, the costs associated with energy usage, it is essential to manufacture lightweight, low-cost, and environmentally friendly building materials. Producing high insulation capacity bricks can potentially diminish the building's energy consumption. According to the European standard EN832, walls should be produced from materials with a heat transfer coefficient of approximately 0.4 to 0.7 W m^{-2} K^{-1} [113]. The standard brick with a density of 2400 kg m^{-3} is equivalent to a thermal conductivity value of 1.4 W m^{-1} K^{-1} [114]. The thermal conductivity of a brick is mainly based on the density; therefore, increasing the thermal transmittance implies producing high porosity bricks, which can be achieved with the addition of waste material. Organic wastes contribute to auto-thermal combustion due to their high calorific value. Therefore, the energy consumed during the firing of bricks incorporating wastes will essentially be lower. As such, there is a growing interest among researchers to investigate potential waste products to incorporate in the production of fired clay bricks to improve their thermal properties and attain energy savings [115-119]. Some of the organic wastes that have been studied include glass powder and palm oil fly ash [120], iron tailings [121], biosolids [122, 123], waste marble sludge [124], and cigarette butts (CBs) [1].

Each year, globally, over 1.2 million tons of toxic CB waste is being littered into the environment, which, potentially, can take up to 10 years to decompose under normal environmental conditions [4]. Littered CBs consist of arsenic, tar, CO, nicotine, and many other hazardous chemicals [125]. The release of toxic chemicals and heavy metals to the environment results in the contamination of plants, animals, water bodies, and soil [51, 52]. Being a major environmental issue, this has led researchers to explore feasible and practical recycling techniques. The incorporation of CBs in construction materials,

particularly in fired clay bricks [1] and asphalt concrete [19-21], has gained pervasive interest.

4.1.1 Recycling Cigarette Butts in Fired Clay Bricks

In 2016, after extensive research, Mohajerani et al. [1] proposed that by incorporating 1% CBs into 2.5% of the bricks produced worldwide, the CB dilemma could conceivably be eradicated [1, 22-24]. With the addition of 2.5% CBs in fired clay bricks, density decreased by 8.3%, while bricks incorporating 1% CBs, the compressive strength was determined to be 19.53 MPa. All the mechanical and physical property values were within the acceptable range of the Australian Standards [1].

Through the addition of CBs, porous bricks were produced, as shown in Fig. 4.1, and a drastic decrease in thermal conductivity was observed with the increase in CB content [1]. Fired clay bricks incorporating 1% CBs resulted in an energy saving of 9%. For the energy savings, the specific firing energy was assumed to be 2 MJ kg^{-1}, and the calorific value of the cellulose acetate was presumed to be 19 MJ kg^{-1} for the calculation method. Therefore, it should be mentioned, that changes in the assumed values (specifically the firing energy and calorific value) will lead to variations in the energy savings for bricks incorporating CBs.

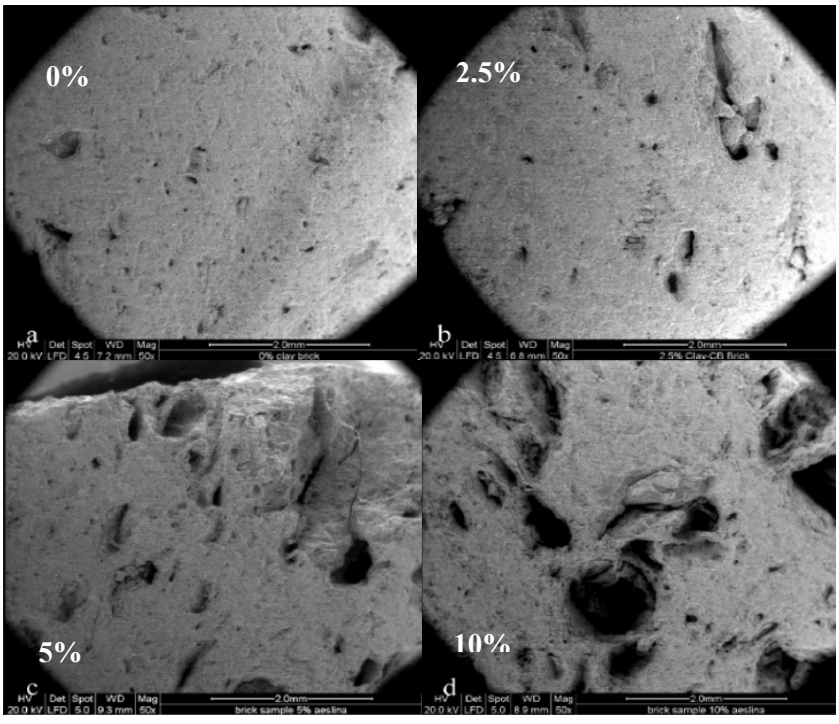


Fig. 4.1. ESEM images of FCBs with 0%, 2.5%, 5%, and 10% CBs (adapted from Mohajerani et al. [1]).

The implementation of recycling CBs in bricks on an industrial scale was explored extensively by Mohajerani et al. [2]. The paper suggested the use of naphthalene containing mothballs during the

collection and storing phase of CBs to reduce the bacteria content. While, to eliminate the odor from Volatile Organic Compound (VOC) emissions in CBs, the use of UV light was recommended. The paper further discussed CB collection systems, energy savings, properties of bricks incorporating CBs, and an implementation procedure (Fig. 4.2). All the results confirmed that the addition of 1% CBs in 2.5% of the brick production is adequate to free the world from CBs [2].



Fig. 4.2. Green and fired clay bricks (adapted from Mohajerani et al. [2]).

In 2020, Kurmus and Mohajerani [126] conducted a comprehensive leachate analysis of heavy metals for used, unused, and fired CBs, and unfired and fired clay bricks incorporating various percentages of CBs. The leaching results were compared to the EPA Victoria solid industrial waste guidelines and were found to be below the regulatory limits, and, therefore, were categorized as non-hazardous waste [126].

This paper is a continuation study on recycling CBs in fired clay bricks. The objective of this study is to investigate and analyse the effects of incorporating CBs on energy savings, porosity, thermal conductivity, and the micro- and macro-structure of fired clay bricks. Scanning Electron Microscopy (SEM) and X-ray micro-Computed Tomography (micro-CT) was used to quantify the porosity and characterize the size, shape, and distribution of the pores within the brick structure. The energy consumption of the furnace during the firing process of the clay brick specimens was monitored via the development of a novel method. The utilization of CBs in the production of clay bricks can be valuable in enhancing the thermal insulation properties of construction buildings, reduce the energy consumption of conventional furnaces while lowering costs and promoting eco-friendly and sustainable building materials. In this study, clay bricks incorporating various percentages of CBs (0%, 0.5%, 1%, 1.5%, and 2% by wt.) are investigated.

4.2 Materials and Methods

4.2.1 Preparation of brick samples

For brick manufacturing, CBs and clay (soil) were used as the raw materials. The raw clay material was obtained by PGH Bricks and Pavers (VIC, Australia), and the CBs were provided by Butt Out Australia Pty Ltd. But Out is a CB collection company in Melbourne that collects CBs from dry receptacles that are installed in outdoor areas. The CBs and soil were oven dried for 24 h at 105 ºC before being transferred into plastic bags. Various tests on the soil and brick specimens were performed to determine their fundamental properties. The clay powder was characterized by mineralogical and chemical analysis using the X-ray diffraction pattern (XRD) (D8 Endeavor, Bruker, MA, USA) and X-ray fluorescence (XRF) (S4Pioneer, Bruker, MA, USA). Particle size distribution was determined using sieve analysis according to the Australian Standard [127]. The washing method procedure was followed, which involved mixing the clay material with an antifocculation solution (sodium hexametaphosphate) in a large tray. The mixture was then washed on a 75 μm sieve, the sample retained on the 75 μm sieve was dried, and the mass was determined. The dry sample was then passed through sieves from 2.36 mm to 150 μm, and the masses retained on each sieve were recorded.

Selected combinations of soil containing 0%, 0.5%, 1%, 1.5%, and 2% CBs by wt. were prepared. The raw mixtures were mixed and homogenized using a Hobart mechanical mixer for 25 min with a 15.5% water content. Cylindrical samples were prepared with the use of a compactor with a compaction pressure of 240 kpa and dried at 105 ºC for 24 h. After drying, the green brick samples were fired at 1050 ºC for 3 h in the Thermoconcept electrical chamber furnace (Model KC 220/13, Hylec Controls Pty Ltd, NSW, Australia), which has a volume capacity of 0.216 m^3. The ramp rate of the furnace was set to 0.7 °C/min, and when the temperature reached 1050 °C, it was held for 3h and then cooled down at a rate of 0.7 °C/min to a safe temperature.

4.2.2 Energy Saving

4.2.2.1 Calculation Method

In order to determine the calorific value of CBs, a laboratory investigation was undertaken by a group of researchers for various CB sample types. Based on the study, an average value of 16.53 MJ kg^{-1} was determined to be the calorific value of CBs [2]. Partially replacing clay with a high calorific value material (CBs) during the brick preparation will allow the CBs to act as energy carriers in the brick firing phase due to the high percentage of organic and carbonaceous substances contained in the waste. Utilizing a calorific value of 16.53 MJ kg^{-1} and assuming that 2 MJ kg^{-1} energy is used for the firing of bricks, the energy-saving through the addition of CBs compared to the conventional brick making was

determined [110]. In the calculation method [1], the calculations were conducted by considering the mass of CBs (m_3) and clay (m_2) in the clay-CB brick, mass of the controlled sample (m_1), energy used for brick firing (q), and the calorific value of the cellulose acetate (CV).

$$\text{Energy Saved, } \Delta E(\%) = \frac{(q \times m_2) - (q \times m_2 - CV \times m_3)}{q \times m_1} \times 100\% \quad (4.1)$$

4.2.2.2 Measurement Method

Energy-saving tests were performed on 0%, 0.5%, 1%, 1.5%, and 2% CBs by wt. samples to compare the energy usage by the furnace during firing. For the purpose of this investigation, the environmental and internal conditions were intended to be kept the same for all mixtures, including the initial temperature within the furnace, the external temperature of the furnace, the time when the furnace was operated, moisture content of the mixture, operation of the ventilation system, and program setup of the furnace. Therefore, to manage all the conditions during the operation of each mixture, each firing session was reduced to the minimum temperature required for brick making to decrease the time needed for the firing process, while roughly two-thirds of the furnace capacity was filled with samples. It is known that silica changes its crystal form at 573 °C and is a prevalent component of the quartz component of the clay used for making bricks, consequently the minimum temperature range for brick firing should be between 600 and 700 °C [110]. Therefore, a ramp rate of 2 °C min^{-1} and a maximum temperature of 700 °C was selected for the furnace set up for each mixture. This particular furnace setup was only applied to the energy savings measurement method.

The energy usage of each mixture during the firing process was evaluated by means of a Fluke 1736 Three-Phase Power Energy Logger, which contains a power box and a magnetic sensor clamp. The experimental set up to measure the energy consumption of the furnace can be seen in Fig. 4.3. The main plug is connected to the power box, which contains a magnetic sensor clamp fixed around a red cable to measure the current flowing to the furnace. The current is then transmitted to the energy logger, and the voltage is transmitted through a black cable that is connected to both the main cable and the logger. The captured voltage and current are then automatically converted to energy demand, and the energy consumption of the furnace is given for the period desired. All data are the mean of three replicate tests.

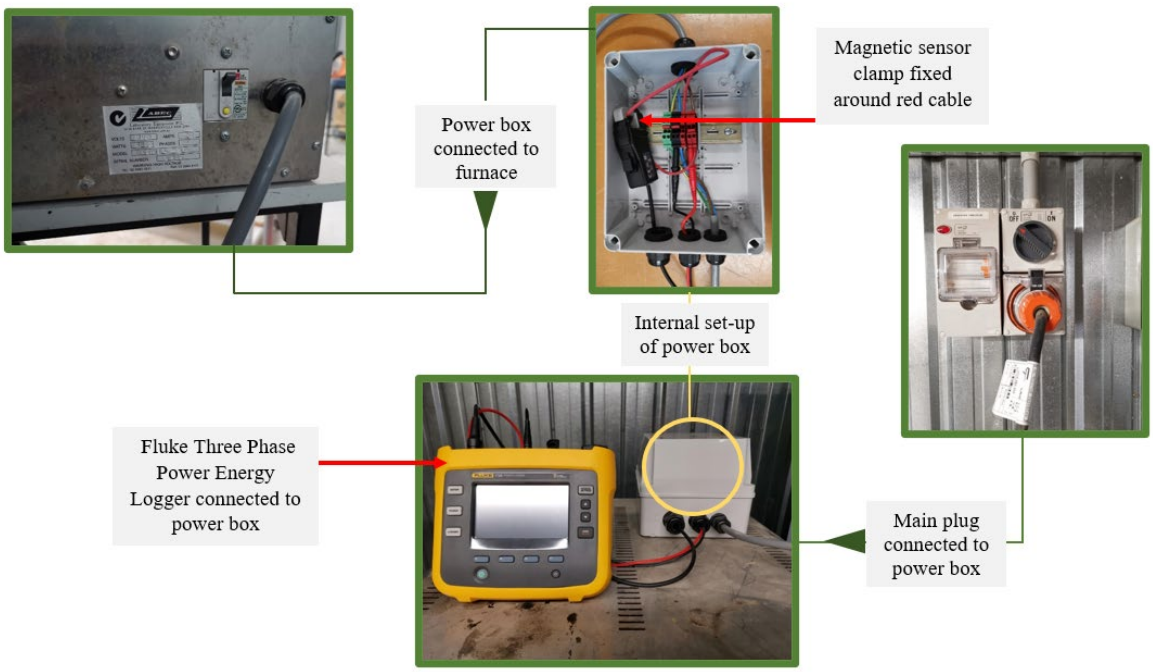


Fig. 4.3. Experimental setup used to measure energy consumption during the firing of bricks.

4.2.3 Micro- and Macro-Structural Analysis

The distribution and texture of pores in the FCBs incorporating CBs were analysed using SEM (Quanta 200 SEM) and micro-CT (Bruker SkyScan 1275) imaging. Micro-CT provides data related to the connectivity and orientation of the pores within the structure. In addition, the micro-CT permits the visualization of the internal structure of the samples and will explain how the addition of CBs influences the physical-mechanical characteristics of fired clay bricks. Brick samples with a diameter of 2.5 cm and 5 cm in height were placed in the micro-CT. The samples were scanned at 100 kV and 100 µA exposure, producing an axial sequence of X-ray attenuation imagery. A 1 mm copper filter was employed between the brick and tube to reduce the hardening of the artifacts and amplify the contrast between the sample phases. The subsequent 3D imagery consisted of a voxel size of 35 µm resolution, and data on internal porosity, pore distribution, and pore thickness were collected.

4.2.4 Thermal Conductivity

The addition of waste material in the production of fired clay bricks can drastically improve the thermal conductivity and resistance properties of the construction material, so, effectively, the building will not lose too much heat in the winter or gain too much heat in the summer [128]. When wastes are introduced into the clay brick mixture, during the firing phase, the organic components burn off, and micro- and

macro-scale pores are created within the brick structure. Essentially the bricks become less dense, and the thermal conductivity value decreases [129].

The determination of thermal conductivity of FCBs incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by wt. using the Transient Line Source technique was carried out. The apparatus adopted in this study is the Transient Line Source (TLS) meter (TLS-100, Poly Technologies, Australia), which complies with standard ASTM D5334-14 [130]. The TLS-100 measures thermal conductivities between 0.1 to 5 W m^{-1} K^{-1}, with an accuracy of 5% and reproducibility of 2%. The line source method is based on a constant current source (q) being delivered to heat the sample, whereby the temperature rise is recorded over a defined period of time. The slope (a) from a plot of the logarithm of time versus temperature rise is used in the calculation of the thermal conductivity (k) and is described by Eq. (4.2):

$$k = \frac{q}{4\pi a} \tag{4.2}$$

A sensor needle (4 mm in diameter and 50 mm in length) comprising a thin heating wire and temperature sensor sealed in a steel tube (to protect the sensor) was completely inserted in the brick sample (Fig. 4.4). A hole 4 mm in diameter was drilled into the brick sample using the drill bit provided to host the probe, and thermal contact grease was applied to the probe to minimise the sample/sensor contact resistance. The measurements were performed with an average temperature of 20 ± 2 °C and a measurement time of 5 min each. A time interval of 15 min for each test was adopted to allow for the thermal equilibration between the sample and the probe. Each sample was measured three times, and the mean of the three measurements was retained.



Fig. 4.4. Thermal conductivity testing of fired clay brick using the TLS-100.

4.3 Results and Discussion

4.3.1 Characterization of raw materials

The chemical composition measured by XRF of the clay soil is provided in Fig. 4.5. According to the data, the soil comprises a significant amount of silica (SiO_2) (62.59%), followed by alumina (Al_2O_3) (19.95%) and, to a reduced degree, iron (Fe_2O_3) (6.15%) and potassium oxide (K_2O) (4.87%). The recommended SiO_2 and Al_2O_3 content the clay should be composed of when used for brick making should be within the range of 20% to 50% and 10% to 20%, respectively [131]. The XRD patterns of the clay soil present the mineral phases (Fig. 4.6). The soil was mainly composed of Quartz (SiO_2), but also comprised Muscovite ($KAI_2(AISi_3O_{10})(FOH)_2$), Albite ($Na(AlSi_3O_8)$, Clinochlore ($Mg_5AI(AlSi_3O_{10})(OH)_8$), and Mikasaite ($Fe_2(SO_4)_3$) minerals.

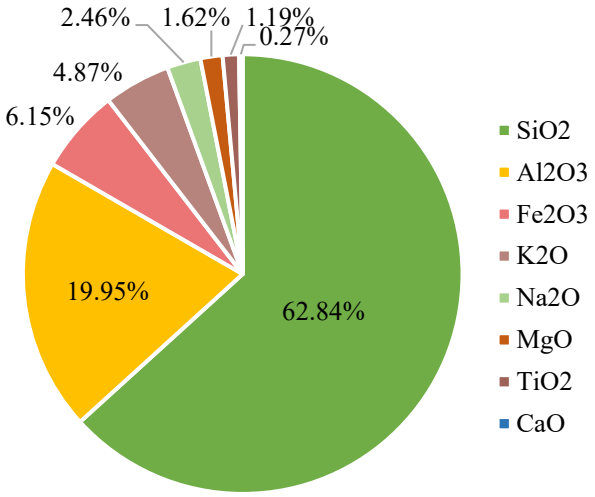


Fig. 4.5. Chemical composition of the clay soil sample.

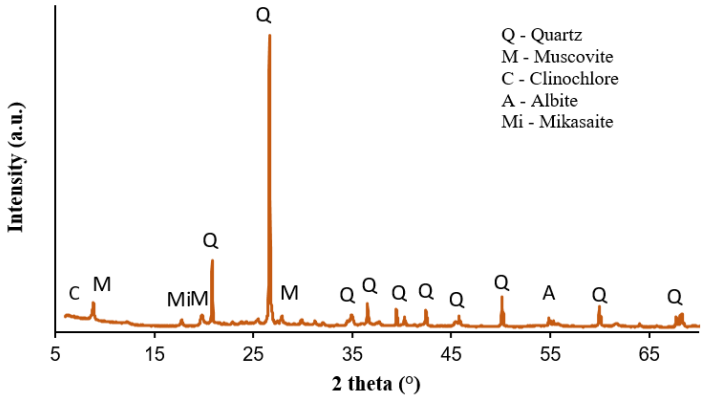


Fig. 4.6. X-ray diffraction pattern of the clay soil sample (adapted from Kurmus and Mohajerani [115].

The particle size distribution curve of the clay soil is displayed in Fig. 4.7 and was determined through sieve analysis [127]. The results revealed that 77.5% of the sample was <75 µm, which was in conformity with the high silica content in the clay soil. The soil was classified as sandy silty clay (MC) in accordance with the unified soil classification system [127], having a liquid limit of 34% and a plastic limit of 25%. A microscope image at 6000× magnification of the main raw material clay and the CB fibers under the optical microscope can be viewed in Fig. 8. The image of raw clay material depicts a variety of particle sizes in accordance with the particle size distibution curve, with clay particle sizes smaller than 20 µm (Fig. 4.8(a)).

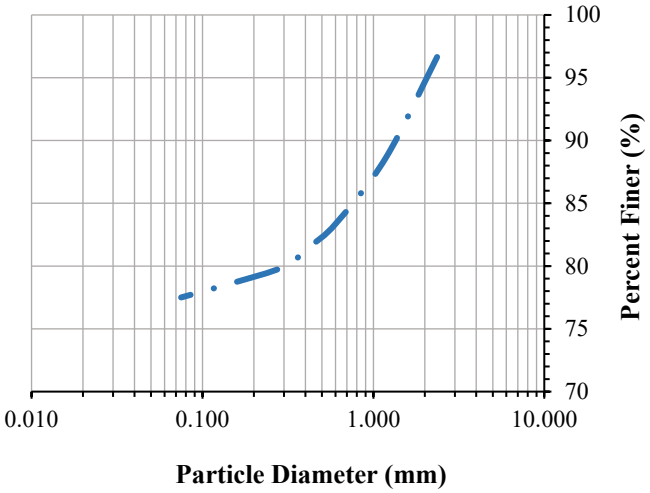
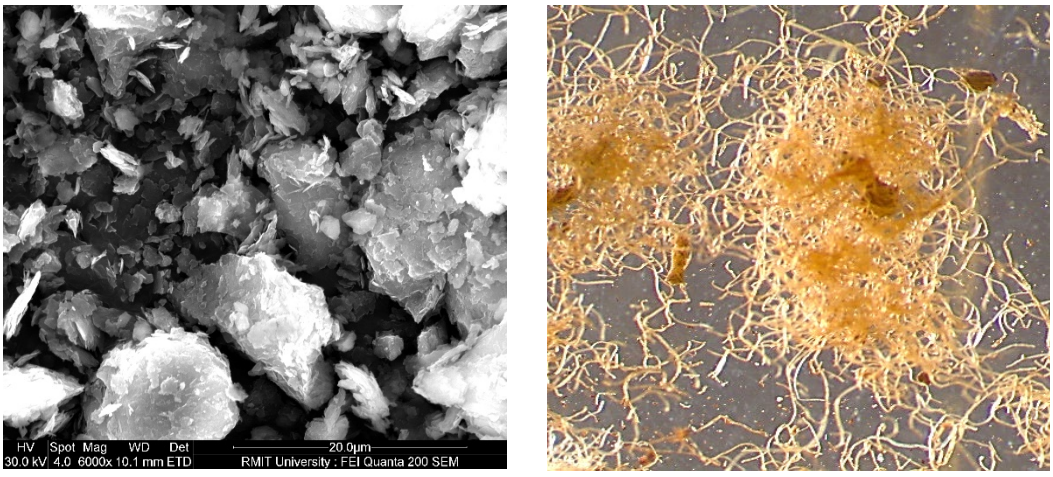


Fig. 4.7. Particle size distribution of the brick soil.



(a) (b)

Fig. 4.8. (a) SEM image of raw clay soil and (b) CB fibers under the optical microscope.

4.3.2 Energy Savings

4.3.2.1 Calculation method

Using Eq. (4.1), the estimated amounts of energy saved during the firing process of 0.5%, 1%, 1.5%, and 2% CBs by wt. bricks were determined and are presented in Fig. 4.9. The results indicate that by firing bricks with 1% and 2% CBs by wt., potential energy savings of 10.2% and 14.71% can be achieved in comparison to bricks containing no CBs. CBs have a higher heating value compared to clay. Therefore, the partial replacement of clay with CBs results in additional heat being released from the combustion of the waste during the firing process [132]. The higher the content of CBs, the greater the energy saved during firing. Similarly, Boldyrev [133] observed 20% to 40% energy savings by recycling coal-mining waste in fired clay bricks, with the highest energy saving being 60%. The range of coal-mining content incorporated into the clay brick varied between 5% and 15% by mass.

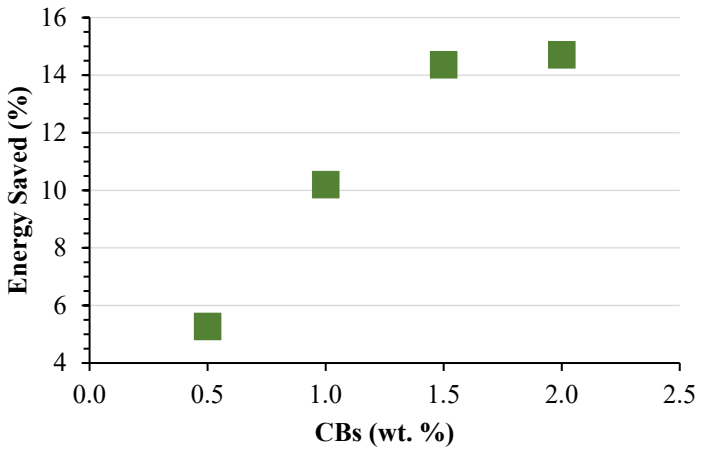


Fig. 4.9. Calculated energy-saving versus CB%.

4.3.2.2 Measurement Method

The calorific contribution provided by wastes during the firing process of fired clay bricks is regarded as a vital phenomenon in terms of energy conservation. CBs are composed of cellulose acetate (polymer) filters, and the melting point of cellulose acetate ranges between 164 and 170 °C [134]. Therefore, once the furnace reaches a temperature between 164 and 170 °C, the CBs combust, and additional energy is available for the furnace [134]. As shown in Fig. 4.10, a relationship has been derived between the CB content of the brick mixture and the energy saved during the firing process. The total energy-used values were captured by the energy logger for each session and can be seen in Table 4.1.

The findings demonstrate that by firing bricks with 1% CBs by wt., an 8% energy saving can be achieved in comparison to the conventional brick (0% CBs) and a maximum 13.3% energy saving with the addition of 2% CBs by wt. This shows that the percentage of energy savings is significantly contingent on the percentage of CBs present within the clay mixture. Similarly, Dondi et al. [135] reported a 15% energy saving during the firing process of fired clay bricks incorporating 4 to 5% sawdust by mass. The calorific value of the sawdust waste ranged between 7 and 19 MJ kg^{-1}, which effectively contributed to the firing process.

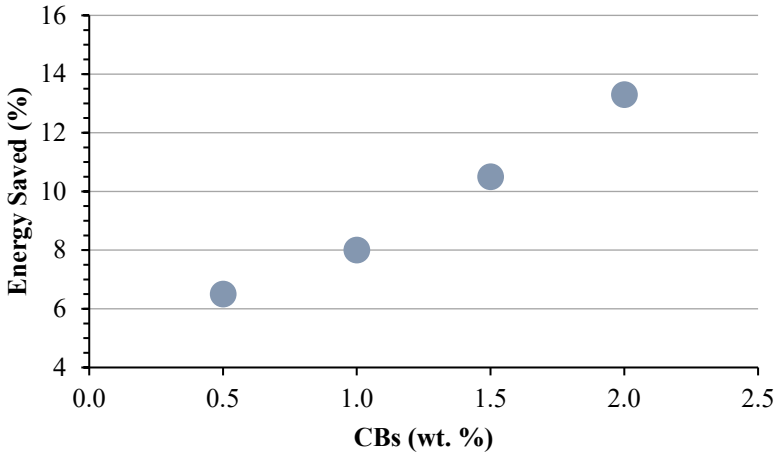


Fig. 4.10. Measured energy-saving versus CB%.

Currently, the total production of bricks in Australia per year is 1.6 billion [136], and the average brick weighs 3.2 kg. Consequently, the total mass of bricks produced in Australia per year can be estimated (5,120 million kg). Assuming the energy consumption for firing bricks is 2 MJ kg^{-1} [110], and the average electricity usage rate in Victoria is 23.272 c kWh^{-1} [137], the average cost of firing bricks in Australia is roughly $661.96 million per year. Based on the derived assumptions and calculations, the estimated cost savings in the energy consumption of firing bricks incorporating CBs are computed in Table 4.1. As shown in Table 4.1, with an energy saving of 8%, an estimated $53.15 million can be saved during the firing process of bricks incorporating 1% CBs by wt. compared to controlled bricks.

Table 4.1. Measured value of energy consumption during the firing of bricks incorporating CBs and estimated cost savings.

		Mixture Identification				
		CB (0.0)	CB (0.5)	CB (1.0)	CB (1.5)	CB (2.0)
Total Measured Energy Consumption of Furnace by Logger	kWh	6.85	6.40	6.30	6.13	5.94

Energy Saved by Incorporating CBs in Fired Clay Bricks	%	-	6.50	8.00	10.50	13.30
Estimated Cost of Firing Bricks in Australia per year (million)	$	-	661.96	661.96	661.96	661.96
Estimated Cost Savings in Energy Consumption of Firing Bricks Incorporating CBs (million)	$	-	43.20	53.15	69.58	88.04

The deviation between the theoretical value and actual measured value of the energy savings may be the cause of several factors, including the assumed specific firing energy value in the calculation method, the maximum temperature reached in the furnace during the firing process; moisture contents of the mixtures; program setup of the furnace, and the calorific value of the CBs. However, the major source of discrepancy is essentially the assumed specific firing energy value of 2 MJ kg^{-1}. In reality, based on a survey performed in ASEAN countries during the years 1993 to 1994, the specific firing energy consumption spans from 2 to 3 MJ kg^{-1} [110]. In this investigation, a lower firing energy value was selected to achieve a higher energy saving. However, to investigate whether a change in the specific firing energy will result in a digression in energy savings, Fig. 4.11 was produced.

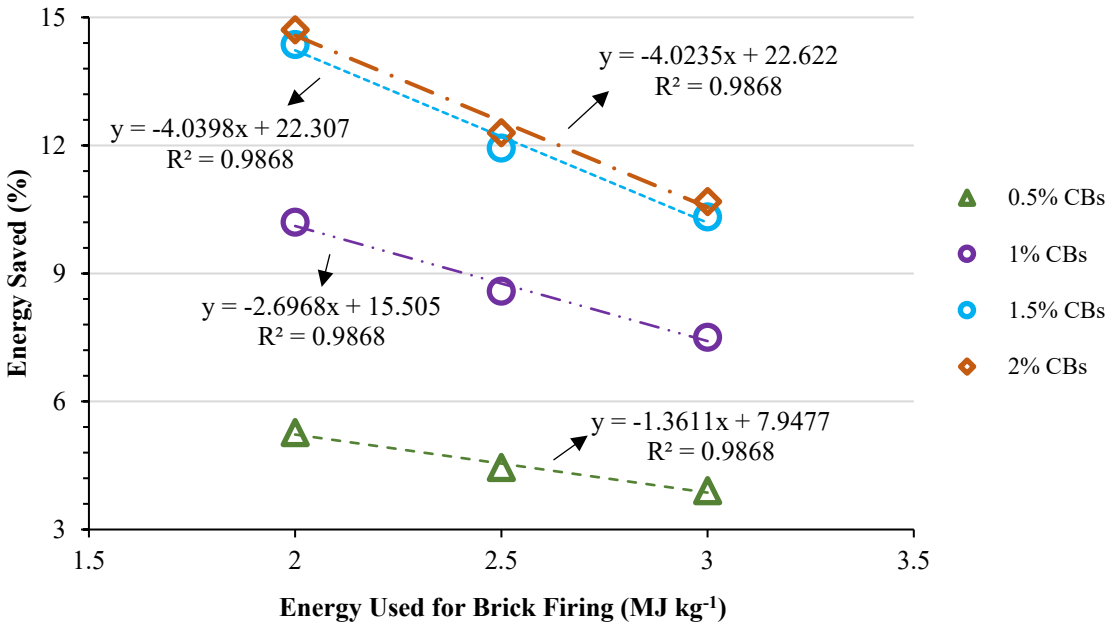


Fig. 4.11. Calculated energy savings for brick specimens incorporating CBs for different specific firing energy values.

As observed in Fig. 4.11, an increase in specific firing energy results in a decrease in the percentage of energy saved for all four brick-CB combinations. For firing energy values of 2, 2.5, and 3 MJ kg^{-1}, an

energy saving of 10.2%, 8.58%, and 7.5% was achieved for bricks incorporating 1% CBs. A specific firing energy value between 2.5 and 3 MJ kg^{-1} is consistent with the actual measured value of 8% energy savings. Therefore, based on the calculation and measurement methods, we can conclude that for bricks incorporating 1% CBs by wt. a minimum energy saving of 8% is achievable.

It is critical to mention the energy savings investigation is based in a laboratory environment, and when compared to an industrial scale, the results may vary due to many factors. The conventional kiln operation in a large scale brick production involves a tunnel kiln, where the loading, firing, and unloading process is a continuous process and can operate over a range of months [2]. In comparison to a laboratory environment, bricks are fired in batches of 20 in an electric furnace for a duration of 2 days. The heat loss and efficiency of both processes may vary, and therefore the energy savings results may differ. However, if we were to compare the firing of controlled bricks to bricks incorporating CBs on an industrial scale, we can assume both settings and conditions are similar, including the heat loss, energy consumption, efficiency, firing time, and brick quantity. The only variance in both operations is the amount of heat added to the firing process through the combustion of the CBs, which will potentially decrease the energy consumption during the firing process. Therefore, although we cannot assume a certain 8% of energy savings is achievable, we can confirm an energy savings of up to 8% can be achieved. However, further research on an industrial scale is recommended.

4.3.3 Micro and Macro-Structural Analysis

As discussed above, thermal conductivity is significantly affected by density and porosity. However, it is not only the porosity and density but the size, shape, and distribution of the pores that play an important role in thermal conductivity. This is supported by the heat transfer mechanism, which can be depressed or enhanced by the distinct mean path of air molecules entrapped in the pores, and, subsequently, by the diverse pore size [138]. Therefore, to investigate this phenomenon, micro-CT imaging was used to characterize the distribution, size, and shape of the micro- and macro-scale pores within the brick structure formed from the addition of CBs.

A total of 1581 2D images of cylindrical brick samples of 2.5 cm in diameter and 5 cm in height with a 35 µm thickness was obtained. The 2D scanning images depict the size, shape, and abundance of the micropores and macropores at particular sections of the sample. Fig. 4.12 represents 2D slices of random sections selected from the various brick samples incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by wt. with the thresholding operation. The thresholding operation is a manual interface, which has an inherent error margin based on the human perception of the edge of the void region [139]. The black regions represent pores, and the white and light grey regions represent coarse aggregates. As

shown in Fig. 5.12 the pores vary in size, shape, and quantity with the various samples. An increase in pore frequency is observed with the increase in CB content, which is due to the creation of voids from the burnt CBs during the firing stage.

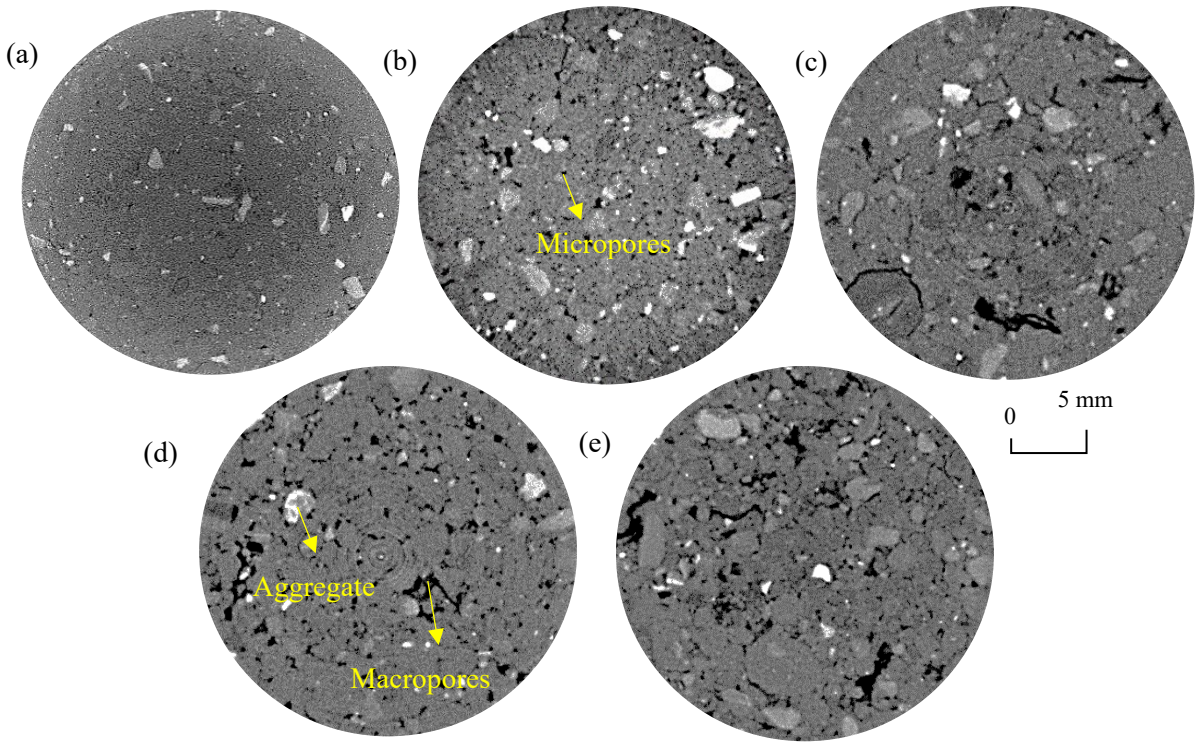


Fig 4.12. 2D CT scanning images at random positions of (a) 0%, (b) 0.5%, (c) 1%, (d) 1.5%, (e) 2% CBs by wt. bricks.

To explore the 3D structural features of the pores, including the distribution, interconnectivity, and size, 2D images were used in the 3D numerical simulation. (1) Pore distribution: layered micro- and macro-pores are homogeneously distributed in a vertical direction. (2) Pore interconnectivity: in most cases, the micropores are not connected with each other in an isolated state. However, those micropores distributed in strips connect with each other and the macropores to some extent. (3) Pore size: macro- and micropores have various sizes. Macropores are generally greater than 0.08 mm in diameter, and micropores are less than 0.08 mm in diameter. Micropores occur in isolated points and in strips [140].

The determination of the structural separation resulted in a relative volume distribution over discrete size intervals. The relative volume is plotted against the midpoint of the class. To obtain better knowledge regarding the size and quantity of pores for each brick sample, Fig. 4.13 was developed. The average pore size interval for bricks incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by wt., range from 0.035 – 0.455 mm, 0.035 – 1.015 mm, 0.035 – 1.085 mm, 0.035 – 1.365 mm, and 0.035 – 1.365

mm, respectively. The main pore size interval for all the samples was between 0.175 and 0.315 mm, while the largest pore size of 1.33 mm was found in bricks incorporating 1.5% and 2% CBs by wt.

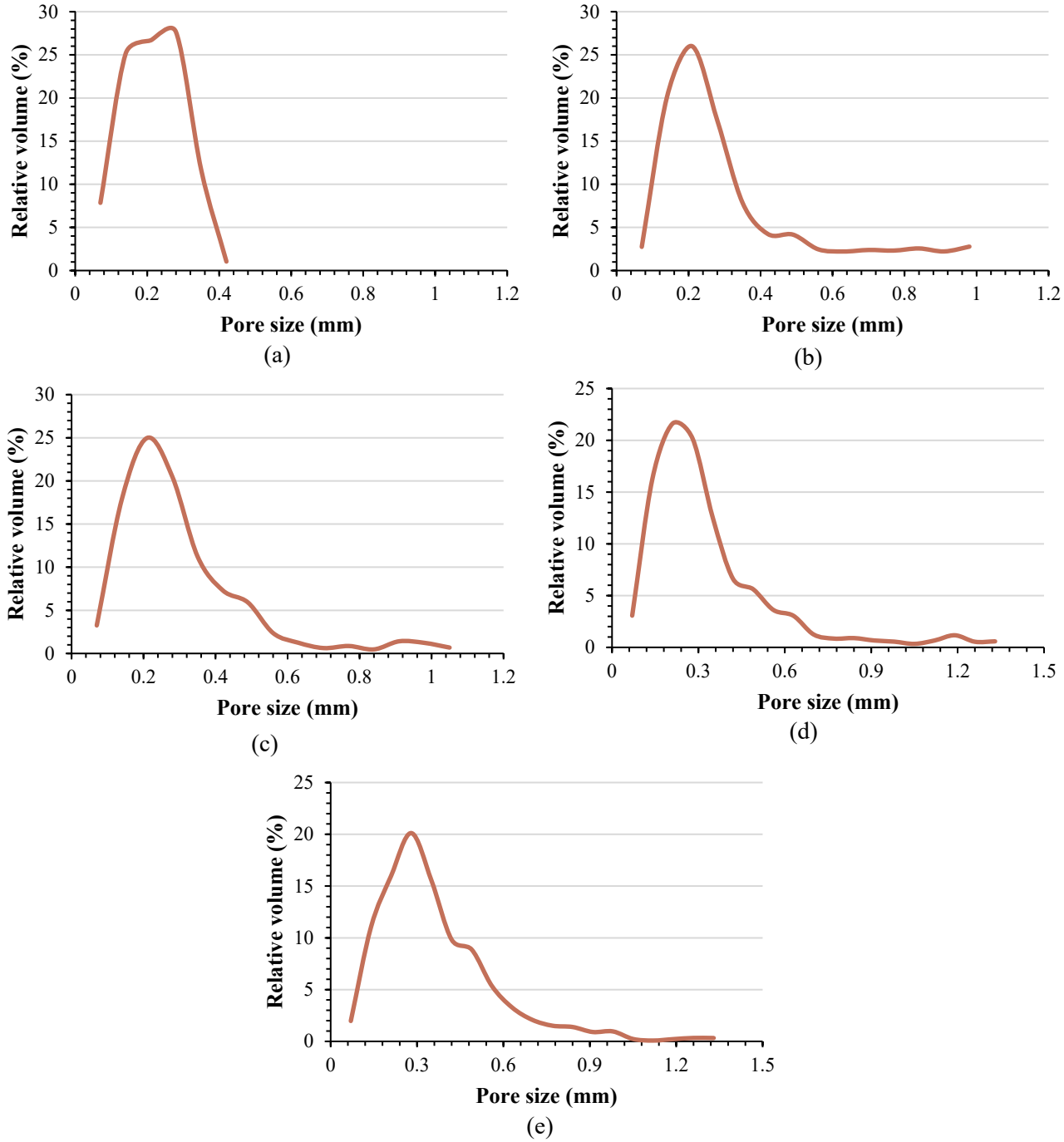
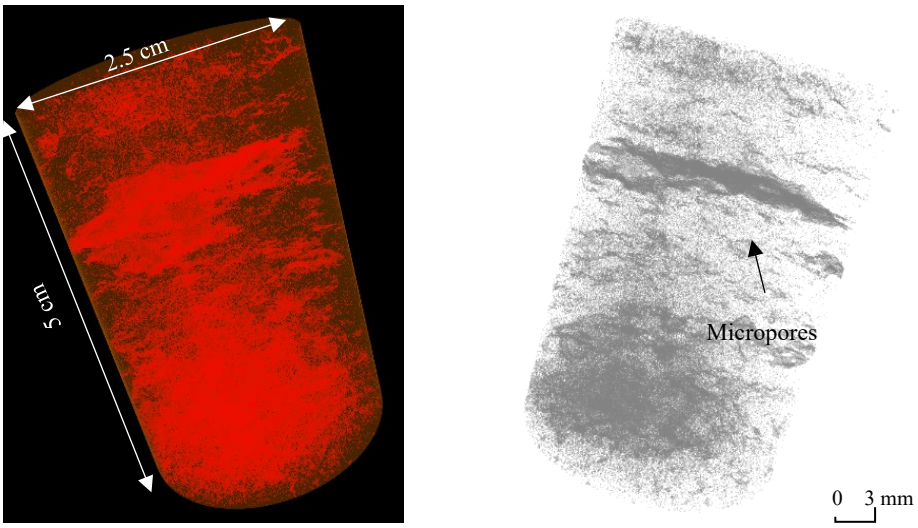


Fig. 4.13. Relative volume distribution over pore size intervals for bricks incorporating (a) 0%, (b) 0.5%, (c) 1% (d) 1.5%, (e) 2% CBs by wt.

The micro CT detected a high quantity of micropores (8%) of size 0.07 mm for the control bricks

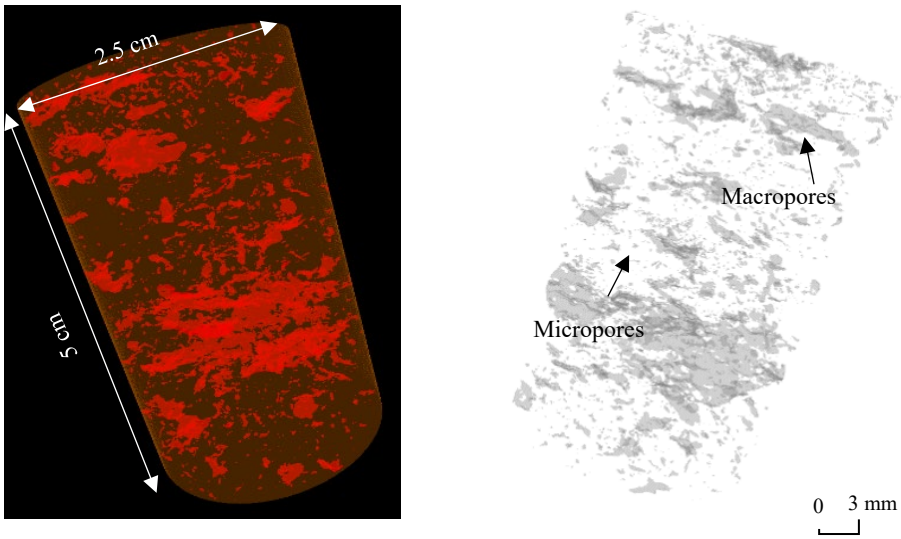
compared to the bricks containing CBs. This is likely due to the air bubbles that were formed during the compaction process as a result of the air entrapped in the soil mixture (Fig. 4.14). As sintering progresses, pores start to round and become smaller when compared to the solid-state and are completely isolated from the surface at the end of the sintering phase [141]. Sintering accelerates over 950 °C because of the vitreous phase formations. This phase penetrates into the pores and closes them, and then separates them from the neighboring pores; this is why the pores are evenly distributed within the control brick [142] (Figs. 4.14 – 4.18).

In contrast, in the bricks incorporating CBs, the CBs burn during the sintering process, and the pores are formed in the structure by CO_2 emissions [142]. Because the liquid phase will decrease with the reduction in clay quantity, the formed pores cannot be filled completely [141]. This results in the interconnection of micro-pores and the pores formed due to the burnt CBs, which produces macro-pores. As shown in Figs. 4.15 – 4.18, the increase in CB content results in an increase in pores within the brick structure. The image on the left of each figure represents the whole brick sample with the pore structure and the image on the right exclusively represents the pore network.



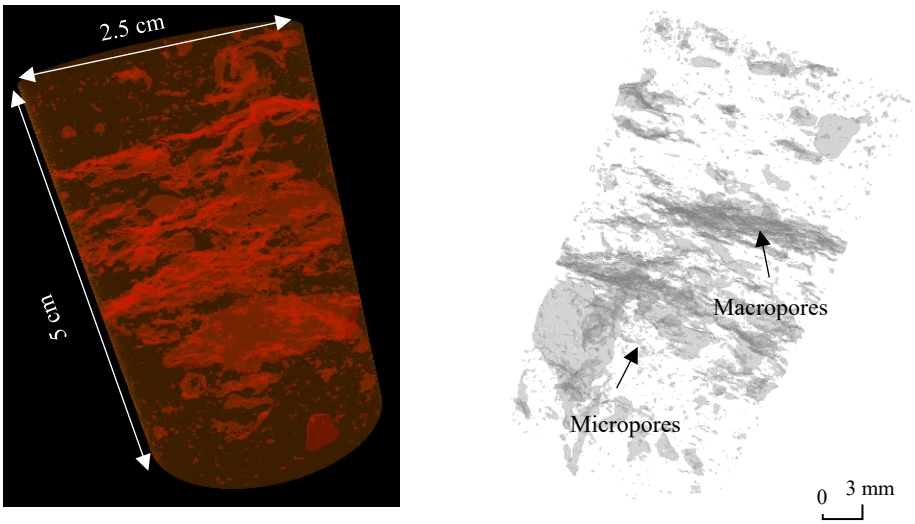
(a) 3D image of whole sample (b) 3D image of pores

Fig. 4.14. Reconstructed 3D CT image at microscale of brick incorporating 0% CBs by wt.



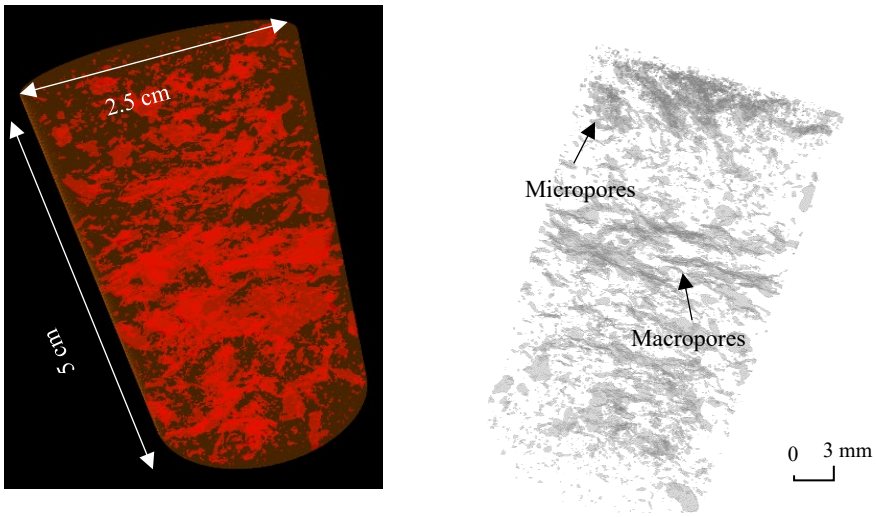
(a) 3D image of whole sample (b) 3D image of pores

Fig. 4.15. **Reconstructed 3D CT image at microscale of brick incorporating 0.5% CBs by wt.**



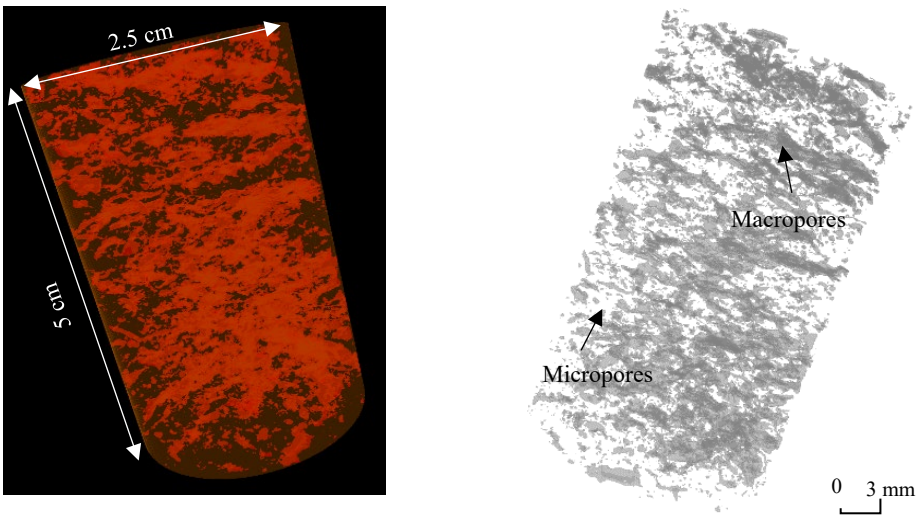
(a) 3D image of whole sample (b) 3D image of pores

Fig. 4.16. Reconstructed 3D CT image at microscale of brick incorporating 1.0% CBs by wt.



(a) 3D image of whole sample (b) 3D image of pores

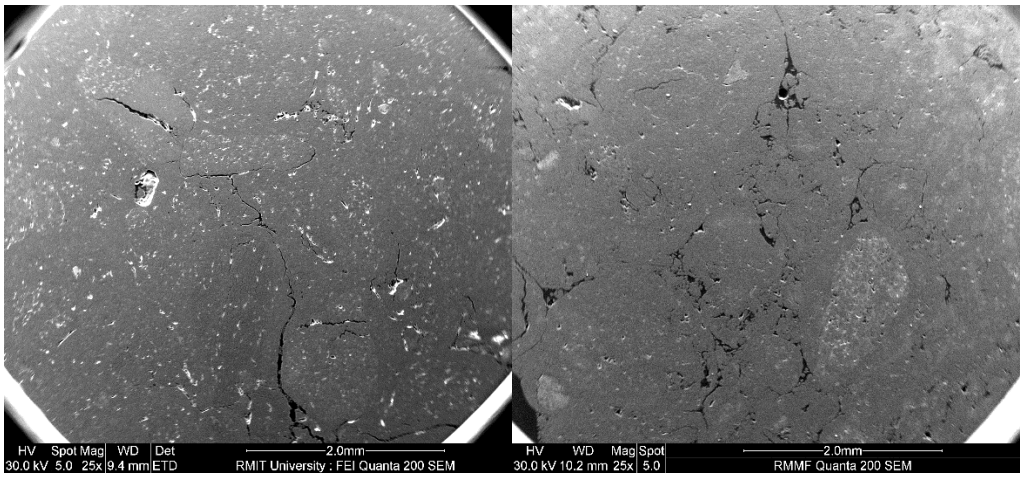
Fig. 4.17. Reconstructed 3D CT image at microscale of brick incorporating 1.5% CBs by wt.



(a) 3D image of whole sample (b) 3D image of pores

Fig. 4.18. Reconstructed 3D CT image at microscale of brick incorporating 2.0% CBs by wt.

The 3D models of the bricks incorporating CBs show a homogenous distribution of pores within the brick structures. This indicates that the CBs within the clay mixtures were evenly distributed and homogenized during mixing. These changes were also seen in the SEM images, as shown in Fig. 4.19 for FCBs incorporating 0.5% and 1.5% CBs at 25× magnification. The number of cracks and pores increased with the increase in CB content, which corresponds with the increase in porosity values from the ct-scan analysis.



(a) (b)

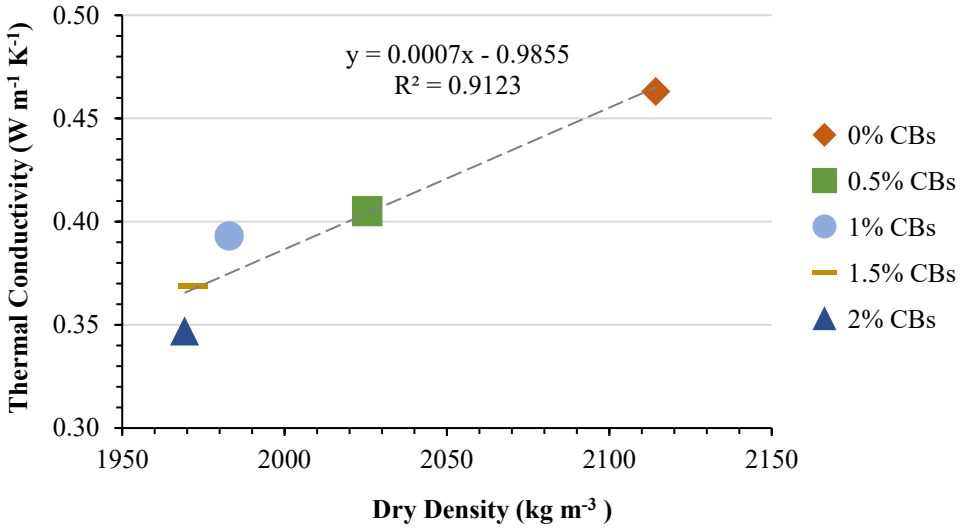
Fig. 4.19. SEM images of FCBs incorporating (a) 0.5% and (b) 1.5% CBs.

4.3.4 Thermal Conductivity

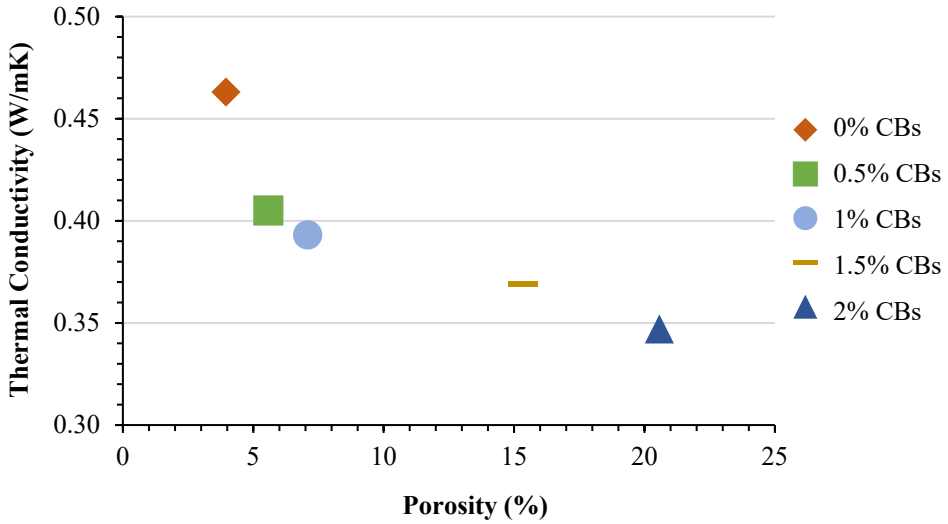
The recent trend in the brick manufacturing industry has revealed an increase in the use of combustible organic pore formers to improve the thermal insulation in fired clay bricks [143]. The bulk density is a key component that governs the overall thermal conductivity of solids. Images from micro-CT were used, and solids and voids were separated to obtain the porosity following the method described in the experimental method section. In the micro-CT analysis, the total porosity of the brick specimens was calculated as the volume of all pores as a percentage of the total volume. The attained porosity values were then analyzed to examine the relationship with the thermal conductivity values obtained from the TLS-100. The correlation between the thermal conductivity, density, and porosity of the brick samples incorporating various percentages of CBs is presented in Fig. 4.20.

A lower thermal conductivity value is desirable as it represents a lower heat loss and better insulating properties. The obtained thermal conductivity and density values for control bricks (0% CB content) was 0.463 W m^{-1} K^{-1} and 2114 kg m^{-3} and reduced to 0.393 W m^{-1} K^{-1} and 1983 kg m^{-3} with the addition of 1% CBs by wt. (Fig. 4.20(a)). The thermal conductivity was further reduced to a value of 0.347 W m^{-1} K^{-1} with the addition of 2% CBs by wt. Aouba et al. [144] similarly reported a decrease in the thermal conductivity from 0.51 W m^{-1} K^{-1} for the control specimens to 0.39 W m^{-1} K^{-1} with the addition of 8% olive stone flour to fired clay bricks. Likewise, Munir et al. [124] observed a drop in thermal conductivity from 0.56 to 0.47 W m^{-1} K^{-1} by incorporating 15% waste marble sludge in clay bricks. The results obtained in this study are in agreement with Al-Hadhrami and Ahmad [145], thereby affirming that lower density results in better thermal insulation bricks.

The general increase in CB content resulted in a decrease in thermal conductivity and an increase in porosity, as shown in Fig. 4.20(b). Porosity values of 7.09% and 20.57% were obtained for bricks incorporating 1% and 2% CBs by wt. with corresponding density values of 1983 and 1969 kg m^{-3}. When both parameters are compared, we see a variance in the increase in porosity. Although the density values are quite reliable, the porosity values obtained from the micro-CT analysis may be influenced by subjectivity. Despite the technique (micro-CT analysis) possessing many advantages, it also presents limitations and difficulties. During CT analysis, the partial volume effect may result in changes in surfaces, higher degree of interconnection between pores, reduction of volume, and increase in pore size [146]. Moreover, image noise largely influences the calculation of certain parameters of pore size and, therefore, may lead to a substantial decrease of the noise pixel size inside the pore space [147]. In consideration of the mentioned factors, which may result in calculation errors, the density values present a more reliable data set in comparison to the micro-CT pore analysis results.



(a)



(b)

Fig. 4.20. Relationship between thermal conductivity and (a) density and (b) porosity of brick samples incorporating CBs.

4.4 Conclusion

In this study, bricks were manufactured incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by wt. and were tested for energy savings, thermal conductivity, porosity, micro- and macro-structural analysis. The current study has confirmed that it is possible to obtain light bricks from CBs as low-cost materials. The utilization of CB waste as a raw material in manufacturing fired clay bricks can be an important way of recycling for the final disposal of this abundant waste, leading to conservation of the environment. The key findings of the present work are as follows:

- Based on the novel method developed to monitor the energy consumption of the furnace during the firing of brick specimens, for bricks incorporating 1% CBs by wt., a minimum energy saving of 8% is achievable. This means that an estimated $53.15 million in terms of electricity usage could be saved in Australia every year, thereby leading to economic construction.
- The porosity, and micro- and macro-scale pore size, and distribution of bricks incorporating CBs by wt. were assessed by reconstructing a 3D section of samples from the micro-CT images. Based on the analysis, the following was observed:
  - The porosity ranged from 3.96% to 20.57% for bricks incorporating 0% to 2% CBs by wt. The loss of organic content in the CBs during the firing process resulted in the formation of macropores within the brick structures.
  - On a micrometer scale, the size of the micro- and macro- pores varied, and their diameters

ranged between 0.035 and 1.365 mm. The largest pore size of 1.365 mm was observed for bricks incorporating 1.5% and 2% CBs by wt.

- The 3D models of the bricks incorporating CBs demonstrated a homogenous distribution of pores within the brick structures. In most cases, the micro- and macro-pores were not connected to each other. This indicates that the CBs within the clay mixtures were evenly distributed and homogenized during mixing.

- A reduction of up to 15.12% in thermal conductivity was achieved with the addition of 1% CBs by wt. in clay bricks compared to traditional bricks. That is a decrease from 0.463 W m^{-1} K^{-1} to 0.393 W m^{-1} K^{-1} in thermal conductivity due to the formation of pores from the burnt CBs, thereby improving the thermal performance. Consequently, this will decrease the energy required for cooling and heating in households.
- Density decreased with the addition of CBs, due to the burning of CBs in the brick body, producing lightweight bricks, which, in turn, may potentially reduce transportation and labor costs.

Finally, recycling CBs in fired clay bricks offers a sustainable solution to a growing pollution problem, and, by recycling CBs in bricks, it is possible to completely offset the total littered CBs in Australia, and, possibly, from around the world. Moreover, it is cheaper to produce in terms of energy requirements, the more CBs are added in the manufacturing of fired clay bricks, the further the costs associated with energy decrease.

Chapter 5: Leachate Analysis of Heavy Metals

Part 5.1: Leachate Analysis of Heavy Metals in Cigarette Butts and Bricks Incorporated with Cigarette Butts

Abstract

Billions of cigarette butts (CBs) are discarded as litter in the environment every year worldwide. As CBs have poor biodegradability, it can take several years for them to break down while leaching toxic chemicals and heavy metals. Mohajerani et al. (2016), based on long-term research, developed a method for the recycling of CBs in fired clay bricks with promising results. This paper presents and discusses the leaching behavior of potentially hazardous metals from used, unused, and shredded used CBs, and unfired and fired clay bricks incorporating CBs. The leachate analysis was conducted according to the Australian Bottle Leaching Procedure (ABLP) for pH values 2.9, 5.0, and 9.2. The aim was to quantify the amount of heavy metals leached, determine the relationship between the metal concentration leachate, pH of the solution and condition of the sample, and examine the effect of firing on the leaching capability of bricks. The leachate results were then compared to the concentration limits for heavy metals set by the USEPA national primary drinking water and the EPA solid industrial waste hazard categorization thresholds to assess the suitability of fired clay bricks incorporating CBs. Metals Cu, Zn, Mn, Al, Fe, Ti, and Ba demonstrated the highest leachate concentrations for pH 2.9 and pH 5.0 for used CBs. This suggests that used CBs are more prone to leaching heavy metals in areas with highly acidic rain compared to the natural range of precipitation. The leaching behavior of fired bricks incorporating CBs was considerably lower than that for the unfired bricks due to the immobilization of heavy metals during the firing process. However, the leaching of Cr and Ni was almost completely impeded after the firing of the bricks, and more than 50% of all the tested heavy metals were hindered.

The research presented in this chapter has been published in the following:

Kurmus, H. and Mohajerani, A., 2020. Leachate Analysis of Heavy Metals in Cigarette Butts and Bricks Incorporated with Cigarette Butts. Materials, 13(12), 2843. 10.3390/ma13122843.

5.1.1 Introduction

The littering of cigarette butts (CBs) has become a critical environmental issue around the world. In Australia, 20 billion filtered cigarettes are consumed annually, and it is predicted that over 7 billion are thrown into the environment [6]. In 2016, 5.7 trillion cigarettes were consumed worldwide, and it is estimated that by 2025, cigarette production will increase by 50% [4].

Over 97% of cigarettes have cellulose acetate filters [4], which are designed to partially preserve particulate smoke components. The major toxic agents include nicotine, polycyclic aromatic hydrocarbons (PAHs), compounds specific to Solanaceae, nicotiana alkaloids, and catechols [148, 149]. Cellulose acetate filters have poor biodegradability, and, under normal environmental conditions, they can take up to 18 months to decompose; thus, becoming a source of toxic waste, polluting waterways and groundwater, and harming aquatic life and animals [50]. Several companies in Australia have developed CB disposal systems consisting of receptacles, the service includes the removal and final disposal of the CB waste. The common methods implemented for the disposal of collected CBs include landfilling and incineration. However, neither technique is economically viable or environmentally friendly [10]. Therefore, it is possible for companies to directly deliver CB waste to brick manufacturing facilities to be used for recycling.

The prevalent heavy metals found in CBs that prompt a likely risk to human health, aquatic organisms, and the environment, are Na, As, Mg, Sc, Pb, La, Bi, Cr, Mn, Cs, Co, Ni, Cu, K, Ag, Zn, Cd, Fe, Hg, Al, Sb, Au, Se, and Te [150, 151]. To investigate the transfer of toxic metals in the tobacco to the smoke component, several studies have been conducted [152-154]. Benson et al. [155] found that the concentrations of Cu, Mn, Pb, and Zn are relatively higher in the filler tobacco compared to the concentration in the smoked cigarette filter, except for Cd. While Dobaradaran et al. [156] found that Hg and Pb may be potential sources of prolonged contamination for aquatic organisms in coastal areas based on their response to heavy metals [156]. From a general point of view, Ar and Cr are causally associated with cancer in humans, while Ni and Cd are probably carcinogenic to humans [153]. Heavy smokers accumulate more than twice the body burden of Cd compared to a non-smoker. The inhaled heavy metals may prompt disorders of mineral metabolism, such as osteoporosis [150].

A promising approach to a growing pollution calamity may be through the recycling of CBs in fired clay bricks. Mohajerani et al. [1] carried out a comprehensive study on recycling 0%, 2.5%, 5% 7.5%, and 10% CBs in fired clay bricks with promising results. The tests carried out include density, compressive strength, water absorption, initial rate of absorption, shrinkage, modulus of rupture, thermal conductivity, and energy savings. A decrease in compressive strength was observed from 25.65 MPa to 12.57 MPa for bricks incorporating 0% and 2.5% CBs. While water absorption decreased from 18% to 5% for bricks incorporating 7.5% and 0% CBs. The study found an increase in CB content

resulted in less dense clay bricks with greater porosity and consequently improved thermal performance and energy savings. A comprehensive study was completed for 1% CBs in fired clay bricks for the confirmation of the results. The compressive strength for controlled brick decreased from 25.65 MPa to 19.52 MPa for bricks incorporating 1% CBs, and water absorption increased from 5% to 7.4% with the addition of 1% CBs. The paper proposed that by incorporating 1% CBs in 2.5% of the world's brick production, all CBs can sustainably be recycled [22, 23, 157].

Several other investigations have been conducted to recycle wastes in the production of fired clay bricks. Such wastes include waste marble powder [158], recycled paper processing residue [143], sugarcane bagasse ash [159], olive pomace [160], bio-briquette ash [161], expanded vermiculite [162], and olive mill solid residue [163]. However, placing toxic wastes in construction materials poses serious exposure risks to the environment and human health. Consequently, it is essential to conduct comprehensive environmental valuations of innovative building materials incorporating wastes prior to their use as commercial products.

The main environmental issues raised from the use and disposal of construction materials incorporating various hazardous wastes are the possible leachate of toxic chemicals and heavy metals to the surrounding environment. Therefore, it is essential to conduct a comprehensive leachate analysis to classify and quantify those harmful constituents. Currently, there are over 50 leaching tests available for various sample conditions, pH values, and solid to liquid ratios to mimic a range of possible environmental conditions. A leachate test determines the liquid that may liquify or dissolve toxic chemicals or heavy metals and potentially enter the environment [164]. The methods vary in terms of the mass and particle size of the sample, the type and volume of the leachate fluid, the leachate delivery method, and the leachate contact time. The main leachate methods have been summarized in Table 5.1.1, including their applications and limitations.

Table 5.1.1 Applications and limitations of various leachate testing methods.

Method	Applications	Limitations	Ref.
Australian Bottle Leaching Procedure (ABLP) (AS 4439.3)	• Inorganic and semi-volatile organic constituents • Range of disposal-to-land scenarios • Applicable for liquid and solid wastes	• Only suitable for particle sizes less than 2.4 mm • Not applicable for encapsulated wastes	[165]

Toxicity Characteristics Leaching Procedure (TCLP) (Method 1311)	• Aerobic conditions • Organic and inorganic analytes • Applicable for liquid, solid and multiphasic wastes	• Not suitable for anaerobic conditions • Suitable only for short-term leaching	[166]
Synthetic Precipitations Leaching Procedure (SPLP) (Method 1312)	• Inorganic and organic constituents • Volatile organic material leachate • Aerobic conditions	• Suitable only for aerobic conditions • Suitable only for short-term leaching • Only applicable for particle size samples	[167]
Liquid-Solid Partitioning as a Function of Extract pH using a Parallel Batch Extraction Procedure (Method 1313)	• Inorganic and semi-volatile organic constituents • Non-volatile organic constituents • Evaluation of disposal, beneficial use, treatment effectiveness and site remedies • Suitable for a wide range of solid materials and pH values	• Not applicable for characterizing the release of volatile organic analytes	[168]
Liquid-Solid Partitioning as a Function of Liquid-Solid Ratio for Constituents in Solid Materials Using an Up-Flow Percolation Column Procedure (Method 1314)	• Inorganic and non-volatile organic constituents • Evaluation of disposal, beneficial use, treatment effectiveness and site remedies • Suitable for a wide range of granular materials • Provides options for the preparation of analytical samples	• Not suitable for monolithic materials without particle size reduction • Not applicable for characterizing the release of volatile organic analytes	[169]
Mass Transfer Rates of Constituents in Monolithic or Compacted Granular Materials Using a Semi-dynamic Tank Leaching Procedure (Method 1315)	• Inorganic analytes • Monolithic or compacted granular material • Suitable for a wide range of solid materials	• Not applicable for the release of organic analytes • Does not provide a solution considered under field conditions • Minimum sample size 5 cm	[170]
Method 1316: Liquid-Solid Partitioning as a Function of Liquid-to-	• Inorganic and non-volatile organic constituents • Natural pH of the solid material	• Not applicable to characterize the release of volatile organic analytes	[171]

Solid Ratio in Solid Materials Using a Parallel Batch Procedure (Method 1316)	• Suitable for a wide range of solid materials	• Does not provide a solution considered under field conditions	
Characterization of Waste—Leaching—Compliance Test for Leaching of Granular Waste Materials and Sludges—Part 1	• Inorganic constituents • Monolithic or granular material	• Not applicable for non-polar organic constituents • Only suitable for particle sizes less than 4 mm	[172]

By exploring the various leachate testing methods and comparing their applications and limitations, it is apparent that each method is limited to a specific material or environment. For particularly construction materials that are exposed to harsh weather conditions, the pH of the solution is very critical. Therefore the Australian Bottle Leaching Procedure (ABLP) [165] or Method 1313 [168] testing methods are recommended as both are pH dependence tests and are suitable for fired clay bricks. The leachates from CB bricks represent heavy metals, which can possibly leach into the surrounding environment, and pose significant concerns for the quality of aquatic and urban life. The Leaching Environmental Assessment Framework (LEAF) is the most comprehensive method due to its extensive scope and comprises of Methods 1313, 1314, 1315, and 1316 [168-171]. However, the four methods in LEAF are recommended to be used in conjunction when conducting a leachate analysis and can be time consuming and unnecessary.

Therefore, in reviewing the various leaching test methods, the ABLP [165] was determined to be the most suitable leaching procedure for this particular study based on the sample size, pH values, disposal to land scenarios, and condition of the sample in comparison to the USEPA methods. Previous studies concerning the leachate of bricks incorporating CBs have found the concentrations of the metals to be in trace amounts and less than the regulatory thresholds [1]. However, the framework and methodologies followed did not account for the leaching behavior of bricks incorporating CBs over the scope of potential environmental situations expected in various disposal scenarios. In addition, the procedure followed was based on the Australian Standards edition published in 1997.

This paper is the continuation of an ongoing study on the incorporation of CBs in fired clay bricks. The objective of this study was, therefore, to conduct two comprehensive leachate analysis investigations. In the first study, the leachate characteristics of used, unused, and shredded used CBs for various heavy metals were assessed. The aim was to quantify the amount of heavy metals leached from littered CBs and determine the correlation between the metal concentration leached and the pH of the solution. The second part of the study involved evaluating the leachate characteristics of unfired and fired clay bricks

incorporating 0%, 0.5%, 1%, 1.5%, and 2% CB content by mass. The leachate concentrations of various heavy metals were examined for unfired and fired bricks and compared to study the effect of firing on the immobilization of heavy metals of fired clay bricks. The leachate tests were performed in accordance to the ABLP method [165] for three pH values; 2.9, 5.0, and 9.2, to investigate how the pH affects the leaching behavior of the metals studied over a wide range of possible conditions. To evaluate the suitability of the fired clay bricks incorporating CBs, the leachate results were compared to the concentration limits for heavy metals set by the solid industrial waste hazard categorization published by the EPA Vic Industrial Waste Resource Guidelines [173] and the United States national primary drinking water thresholds [174].

5.1.2 Materials and Methods

5.1.2.1 Materials

Sandy silty clay (MC) and CBs were selected as the raw materials. The CBs of varying brands and sizes were supplied by Buttout Australia Pty Ltd. (Melbourne, Australia), while the soil was provided by PHG Bricks (Victoria, Australia). On arrival, the CBs were oven-dried at 105 °C (24 h) and disinfected. The oven-dried CBs were then placed in airtight plastic bags ready for use for the preparation of bricks samples. Particle size distribution (PSD) test was conducted on the brick soil using sieve analysis [127]. The brick soil was characterized by determining the mineralogical composition through an X-ray diffraction analyzer (XRD) (D8 Endeavor, Bruker, Billerica, MA, USA) and the chemical composition through X-ray fluorescence (XRF) (S4 Pioneer, Bruker, Billerica, MA, USA).

5.1.2.2 Preparation of Brick Samples

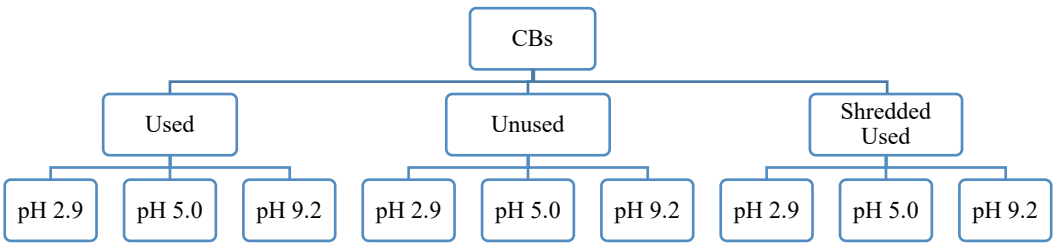
Five different mixtures were prepared and used in this experiment: clay bricks with 0% CBs by mass and bricks made of clay and 0.5%, 1%, 1.5%, and 2% CBs by mass. The mechanical and physical properties of the prepared brick samples were assessed and prepared to be used for the leachate analysis test. Two batches of specimens were prepared and mixed in a Hobart mechanical mixer for 25 min with a water content of 15.5% and then compacted in cylindrical molds of 50 mm x 100 mm. The clay brick specimens were oven-dried at 105 ºC. The oven dried bricks were named as ‘unfired bricks.’ The second batch of unfired specimens was then fired at a temperature of 1050 ºC for 3 h. The fired brick was named as ‘fired bricks.’ The fired brick samples were tested for the initial rate of absorption (IRA), compressive strength, shrinkage, water absorption, efflorescence, microstructural analysis, and density according to the Australian Standards [175] (Fig. 5.1.1). All the results presented are the average of three replicates.



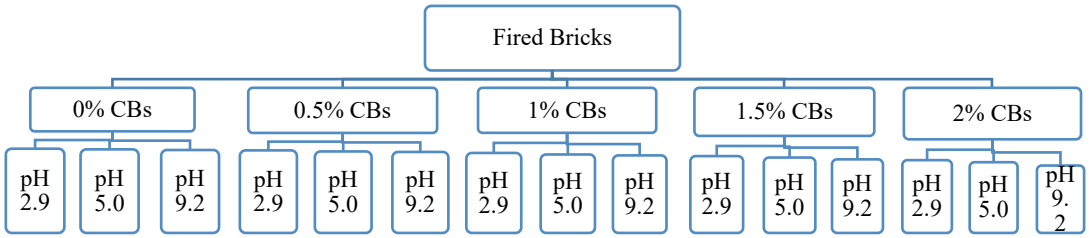
Fig. 5.1.1. Some of the prepared brick samples with different percentages of cigarette butts (CBs).

5.1.2.3 Sample Preparation and Leachate Analysis

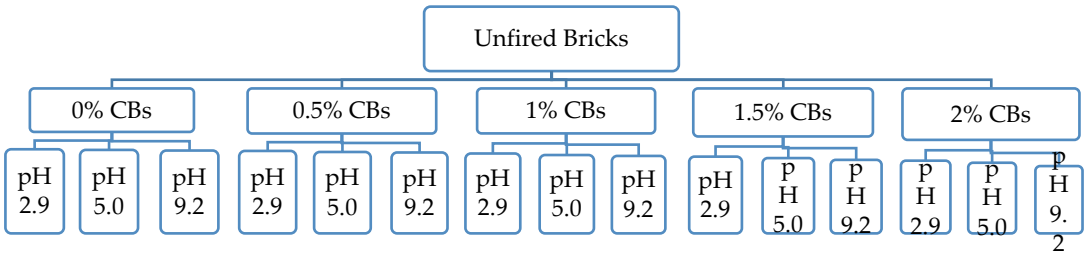
To evaluate the relationship between the metal concentration leached and the pH of the leaching solution, and to simulate various environmental conditions, three pH values were selected, 2.9, 5.0, and 9.2 (Fig. 5.1.2). The metals selected for both investigations were based on their toxicity levels to the local and aquatic organisms, their presence in CBs, and their ability to easily be analyzed by the Inductively Coupled Plasma Mass Spectrometry (ICP-MS). The leaching of heavy metals from the prepared brick samples was determined based on the ABLP [165]. Heavy metals, including Cadmium (Cd), Iron (Fe), Aluminum (Al), Lead (Pb), Arsenic (As), Mercury (Hg), Barium (Ba), Nickel (Ni), Copper (Cu), Titanium (Ti), Chromium (Cr), and Zinc (Zn), were analyzed for used, unused, and shredded used CBs. While Chromium (Cr), Copper (Cu), Silver (Ag), Barium (Ba), Mercury (Hg), Selenium (Se), Nickel (Ni), Cadmium (Cd), Arsenic (As), Lead (Pb), and Zinc (Zn) were examined for unfired and fired clay bricks incorporating CBs. In the ABLP method as presented in Fig. 5.1.3, the brick samples were crushed with a hammer and sieved using a 2.4 mm sieve.



(a)



(**b**)



(**c**)

Fig. 5.1.2. Leaching procedure for leachates derived from (a) used, unused, and shredded used CBs and (b) fired clay bricks and (c) unfired clay bricks incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by mass.

The fluid extract of pH 2.9 was prepared by adding glacial acetic acid (5.7 mL) to 900 mL of water, which was later diluted to 1 L with water. pH 5.0 was prepared by adding glacial acetic acid (5.7 mL) to approximately 900 mL of water to which sodium hydroxide solution (64.3 mL of 1 mol L^{-1}) was added; the solution was then diluted to 1 L with water and mixed well. pH 9.2 was achieved by dissolving sodium tetraborate decahydrate (borax) (38.2 g) in approximately 900 mL of water and then diluting this to 1 L with water. The pH of this fluid was to be between 9.1 and 9.3, and where the pH was not between these levels, the fluid was discarded.

Samples were prepared based on a 1:20 ratio in screw cap bottles. The mass of the crushed fired clay brick samples used in this study was 10 g, while the volume of the extraction fluid was 200 mL. The bottles were then placed in an agitator and rotated at 30 ± 2 rpm for 18 ± 2 h. The fluid extract was attained by filtering the samples through a 0.45-microfilter to separate it from the solid phase. The ICP-MS was utilized to analyze the extracted samples for heavy metals. The results exhibited in this study are the average of three replicates.

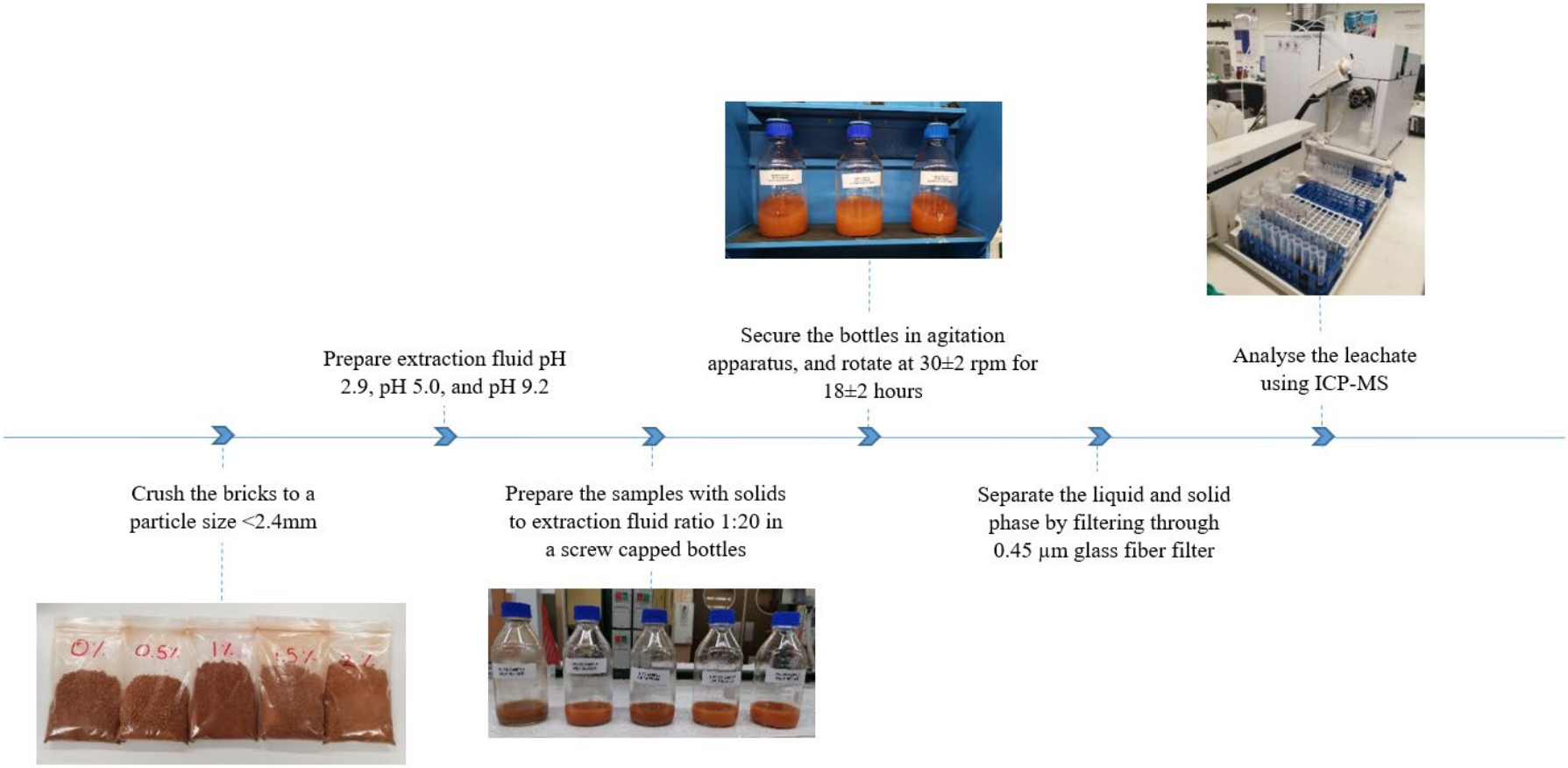


Fig. 5.1.3. Flow chart of the leachate analysis procedure followed in this study.

5.1.3 Results and Discussion

5.1.3.1 Characterization of Raw Materials

The brick soil used in this study was sandy silty clay (MC) according to the Unified soil classifications [127]. The results of the PSD are shown in Fig. 5.1.4, which was carried out using sieve size analysis [127]. For the brick soil, the plastic limit was found to be 25% and 34% for the liquid limit. The XRF results of the chemical composition of the brick soil are given in Fig. 5.1.5, while the XRD results of the mineralogical composition of the brick soil can be seen in Fig. 5.1.6. It was observed that the main crystalline phase on a <75 µm soil sample of the brick soil was Quartz (SiO_2).

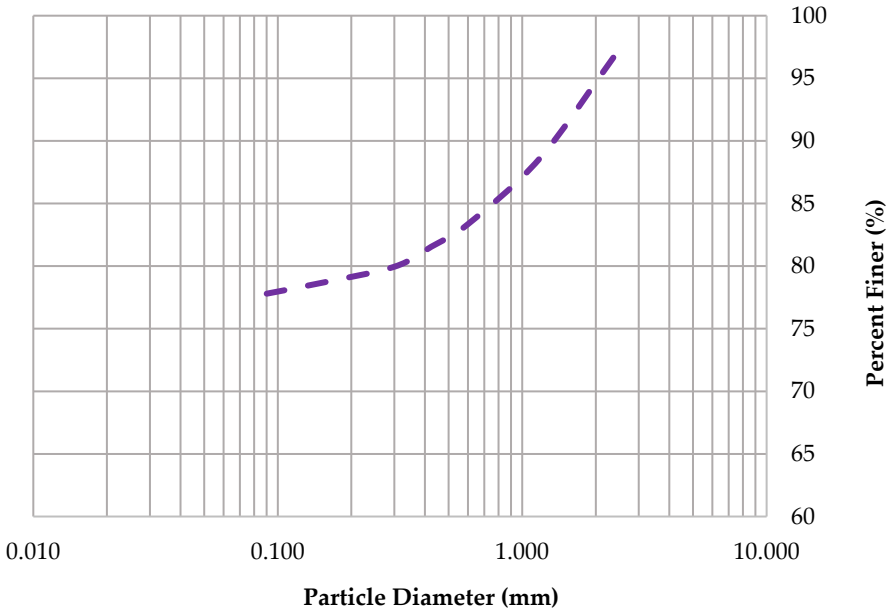


Fig. 5.1.4. Particle size distribution (PSD) curve of brick soil used in this study.

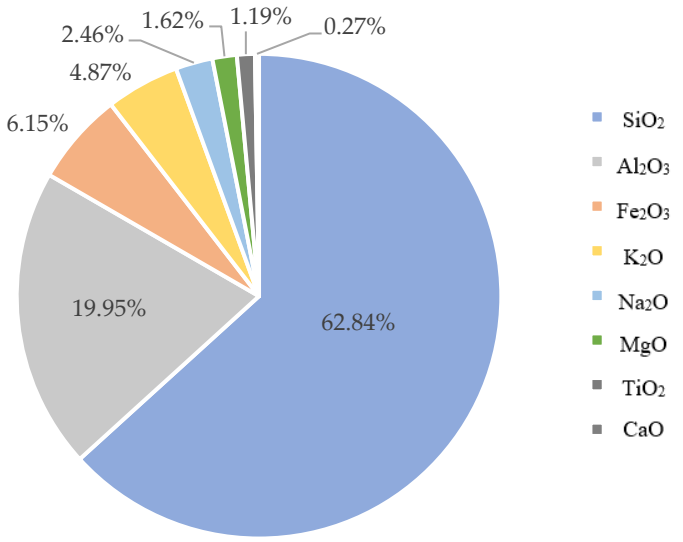


Fig. 5.1.5. Brick soil chemical properties (adapted from Kurmus and Mohajerani [176]).

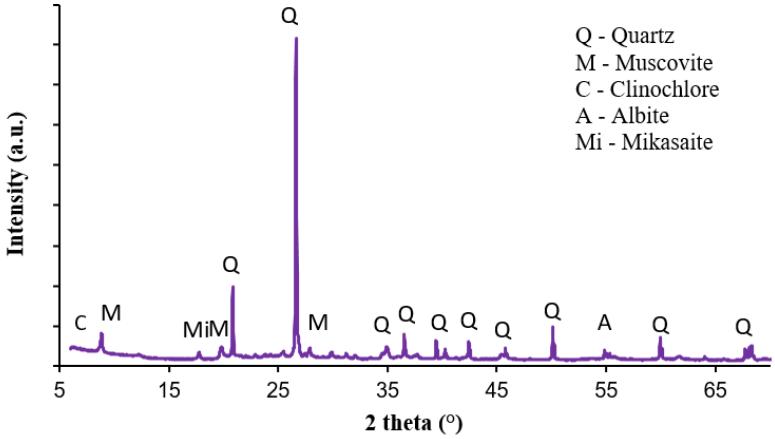


Fig. 5.1.6. X-ray diffraction (XRD) patterns of brick soil (adapted from Kurmus and Mohajerani [115].

5.1.3.2 Properties of Fired Clay Bricks

The properties of 0%, 0.5%, 1%, 1.5%, and 2% CB content by mass bricks are shown in Table 5.1.2 [116, 177]. The compressive strength of a brick is a crucial quality for construction materials. In this study, the results indicate that the strength of the clay brick was significantly dependent on the quantity of CBs added. A decrease in compressive strength was found with the decrease in density and the increase in CB content. The results reveal that the compressive strength was in the range of 27.2 – 48.64 MPa, with a decrease in CB content from 2% to 0% by mass. Although there is a substantial decrease in strength with the increase in CB content, the values are considerably above the standard minimum compressive strength of 17.2 MPa for Grade MW (moderate weathering) bricks [178]. The density of the fired bricks was in the range of 1969 - 2114 kg $m^{-3,}$ with a decrease in CB content from 2% to 0%

by mass. Light-weight bricks have several benefits in construction and production, including the reduction in transportation and labor costs and improving the thermal insulation properties of bricks.

Table 5.1.2. Properties of brick samples with 0%, 0.5%, 1%, 1.5%, and 2% CB content.

Mixture Identification (%)	Height Shrinkage (%)	Diametric Shrinkage (%)	Average Density (kg m^{-3})	Compressive Strength (MPa)	Average Water Absorption (%)	Initial Rate of Absorption (kg m^{-2} min^{-1})
CB (0)	3.07	4.64	2114.33	48.64	9.0	0.44
CB (0.5)	2.27	3.89	2025.5	34.31	10.3	0.58
CB (1.0)	2.17	3.45	1983	30.78	12.1	0.67
CB (1.5)	2.1	3.41	1971.8	28.34	12.9	0.75
CB (2.0)	1.96	3.1	1969.22	27.2	13.1	0.84

The loss of water in the brick between the clay particles, which results in the proximate packing of those particles, is called firing shrinkage. The results show that the firing diametric and height shrinkage values decrease with the increase in CB content. It was found that the diametric shrinkage values were between 3.1% and 4.64%, with the decrease in CB content in the bricks from 2% to 0% by mass, which is below the maximum standard shrinkage value of 8% [179].

Water absorption determines the durability of a brick. When a high content of water can infiltrate a brick, the durability decreases, which can give rise to frost damage, salt crystallization, and change in volume, which can eventually result in structural damage to the building, therefore, a brick must be adequately dense to avoid the encroachment of water [22, 180]. According to Table 5.1.2, water absorption and IRA increase with the increase in CB content. This is due to the decrease in density and the creation of pores from the CBs that were burnt off during the firing process. Bricks with a CB content of 0% to 2% by mass presented water absorption values between 9% to 13.1%. The water absorption for standard fired clay bricks should range between 5% to 20% [65]. The IRA values ranged between 0.44 to 0.84 kg m^{-2} min^{-1} for bricks incorporating between 0% to 2% CBs by mass. The values presented for IRA fall within the acceptable range of 0.2 to 5 kg m^{-2} min^{-1} [65]. Nevertheless, the water absorption and IRA results comply with the typical requirement range.

5.1.3.3 Leachate Analysis of Cigarette Butt (CB) Samples

Cigarettes contain a distinctive quantity of trace elements known to have toxicological effects on living organisms, and the smoking process drastically increases the concentrations of those elements in the cellulose acetate filter [153]. To identify and quantify the metals leached from discarded CBs and

determine the relationship between the metal concentration leached, the pH of the solution and the condition of the sample, a comprehensive leachate analysis was conducted based on the ABLP [165].

To simulate natural conditions, the leaching procedures used in this study were designed as closely as possible to an open environment where CBs can be found. To investigate how the pH affects the leaching behavior of metals from CBs, three pH values were selected: 2.9, 5.0, and 9.2. The selected metals for this study include Ni, Zn, As, Hg, Al, Cu, Ba, Cr, Fe, Ti, Pb, and Cd and are based on their high degree of toxicity and presence in CBs [150-154].

As illustrated in Fig. 5.1.7, the findings suggest that differences in pH over a wide range of precipitation have an effect on the behavior of the metal concentrations leached from unused CBs. A direct relationship between the concentrations of Cu, Zn, Mn, Al, and Ba leached, and the corresponding pH value was observed. Cu, Zn, Mn, Al, and Ba demonstrated the highest leachate concentration for pH 2.9, followed by pH 5.0. While pH 9.2 exhibited the lowest metal leachate concentration. The concentration level for Ba was 7.610, 4.572, and 3.1 mg L^{-1} for pH 2.9, 5.0, and 9.2. This implies that unused CBs are more likely to leach heavy metals in areas with highly acidic rain compared to the natural range of precipitation. This is due to the enhancement of metal mobilization in acid rain. The hydrogen (H^+) ion in the acidic rain disturbs the cations (positively charged ion) from their connection sites, this results in the decrement of cation exchange capacity and increase in the concentration of cations in CBs, therefore allowing the cations to be leached from the waste [181]. Likewise, the toxic ions (toxic metals) usually bound to the negatively charged surface and are displaced by H^+ ion, too [182].

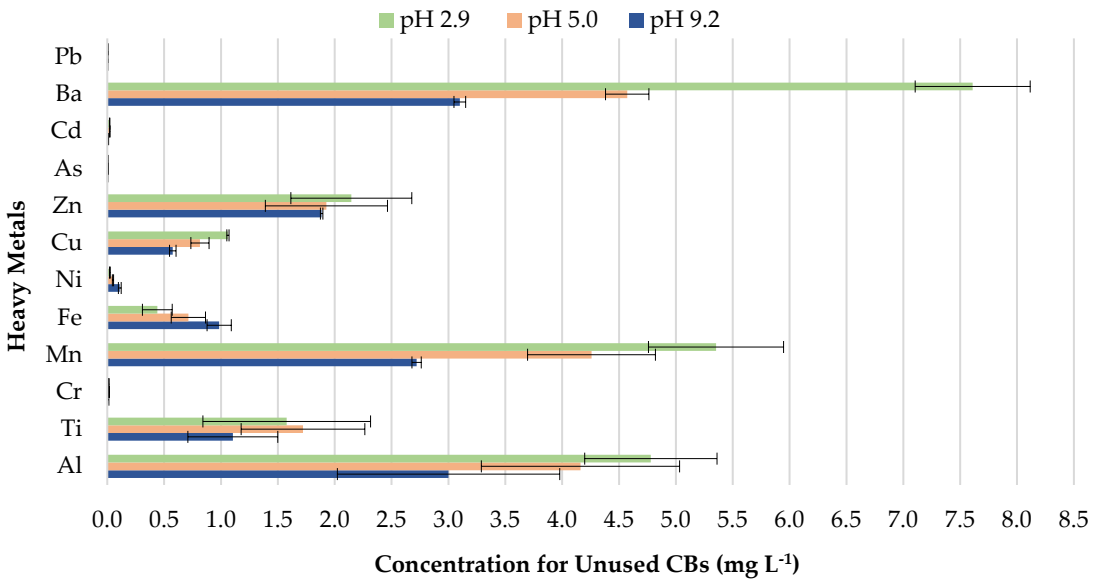


Fig. 5.1.7. Leachate concentrations of unused CBs.

The concentration of metals leached for used CBs for pH values 2.9, 5.0, and 9.2 can be seen in Fig. 5.1.8. The results indicate that changes to the pH of precipitation will likely enhance or reduce the magnitude of metal contamination from used CBs. The metal concentration leachate for Zn, Mn, and Al for pH 5.0 was found to be much more significant compared to pH 2.9 and 9.2. In comparison, Ba, Cu, Ti and Fe depicted greater metal contamination for pH 2.9. Similarly, to unused CBs, used CBs are more prone to leaching in highly acidic areas due to the enhancement of metal mobilization in acidic water.

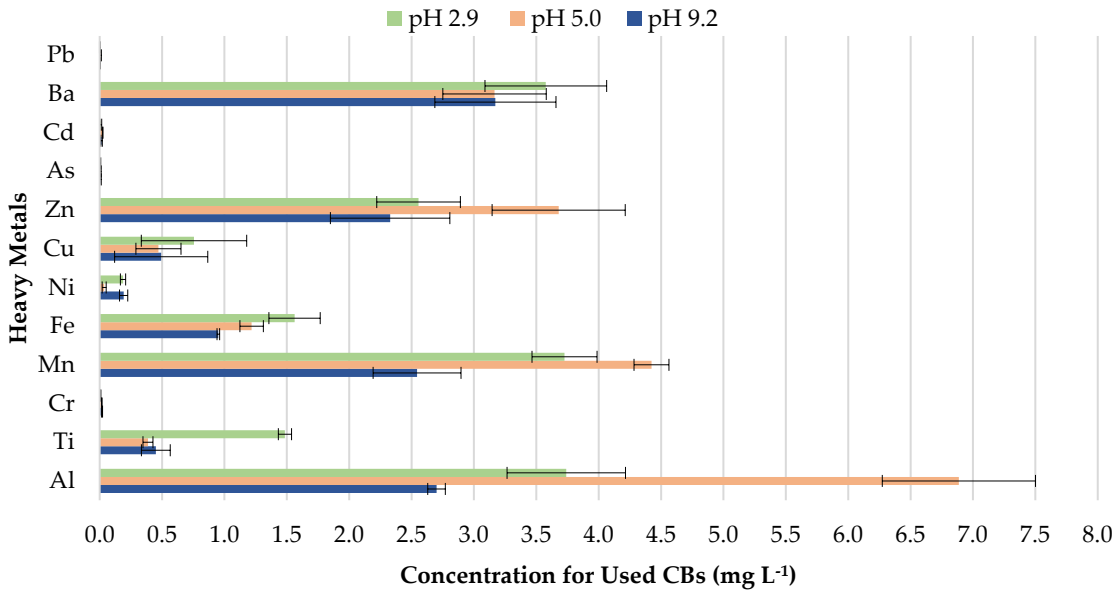


Fig. 5.1.8. Leachate concentrations of used CBs.

A direct relationship between the metal concentration leached and pH value was found for shredded used CBs. The concentration of the metals leached for pH values, 2.9, 5.0, and 9.2, can be seen in Fig. 4.9. pH 5.0 resulted in the highest magnitude of metal contamination for Ba, Zn, Mn, and Al, followed by pH 2.9, while pH 9.2 demonstrated the least possible metal contamination. The relationship between metal leachate and pH value supports the results attained for both unused and used CBs.

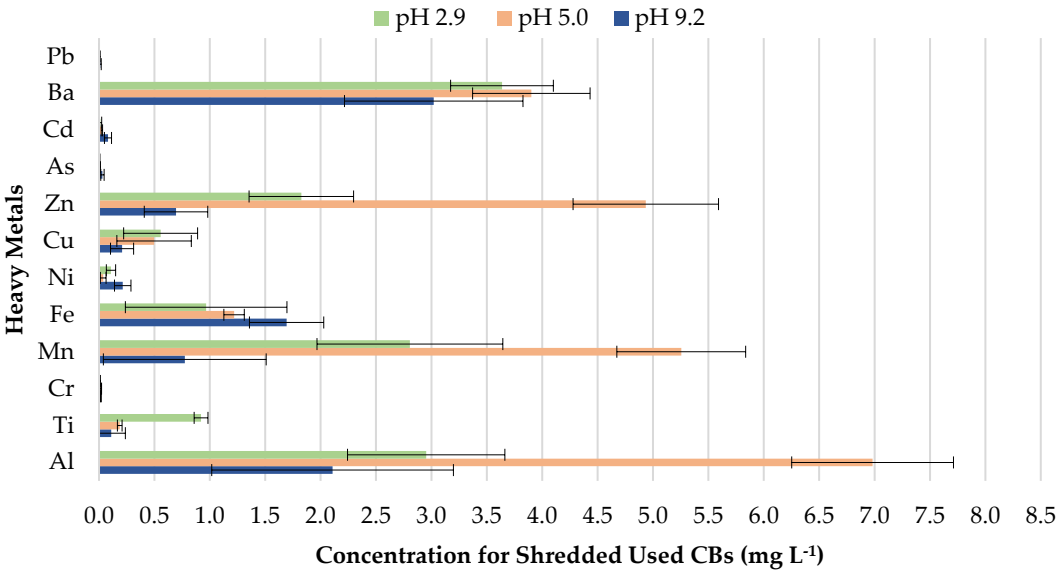


Fig. 5.1.9. Leachate concentrations of shredded used CBs.

To determine whether the smoking process influences the quantity of heavy metals found within CBs, the metal leachate concentrations for pH 5.0 for unused, used, and shredded used CB samples was analyzed. A linear relationship between the concentration of metals leached and the condition of the sample was found. Cr, Cd, Zn, Mn, Al and Fe contamination was greater for shredded used CBs compared to used and unused CBs (Fig. 5.10). For shredded used CBs, 6.982 mg L^{-1} leachate was recorded for Al, compared to 6.887 mg L^{-1} leachate for used CBs and 4.161 mg L^{-1} for unused CBs. Generally, Al is one of the most abundant elements found in CBs due to its presence in tobacco. Al toxicity can lead to tissue-damaging effects, alteration in calcium metabolism in several organ systems, and Alzheimer's disease [183]. The results suggest that used and shredded used CBs are more prone to leaching heavy metals Cr, Cd, Zn, Mn, Al and Fe compared to unused CBs. One explanation for this observation is that an enhanced level of Cr, Cd, Zn, Mn, Al and Fe is absorbed and retained by the filter during combustion. The remnant tobacco, which comprised of burnt and unburnt tobacco, effectively exacerbated the toxicity [17]. Other metals, including Pb and As are of insignificant concern for all sample conditions as their leachate concentrations are below 0.08 mg L^{-1}.

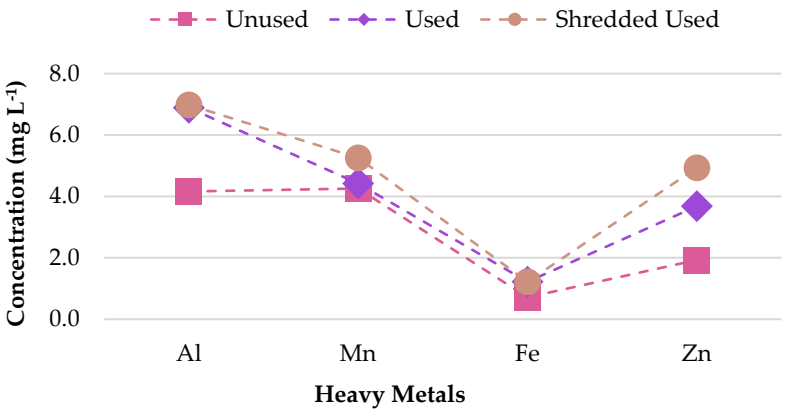


Fig. 5.1.10. Al, Mn, Fe, and Zn leachate concentrations for pH 5.0.

In contrast, the leaching of Cu, Ba, and Ni at pH 5.0 demonstrated a distinctive trend. Unused CBs exhibited the highest leachate concentration for all three metals, followed by shredded CBs, while used CBs displayed the least metal contamination. One explanation for this observation is that majority of the weight of Cu, Ba, and Ni pass through the vapor phase, contrary to being trapped in the filter.

5.1.3.4 Australian Bottle Leaching Procedure (ABLP) Leachate Analysis of Fired Clay Bricks

The toxicity of CBs is a concern when incorporated in construction materials like fired clay bricks. Therefore, it is essential to conduct extensive environmental valuations of innovative building materials modified with waste materials prior to being used as a commercial product. The transfer of heavy metals from bricks incorporating CBs to the environment is determined through a leachate analysis using the ABLP [165] and compared with the threshold regulatory limits for heavy metals set by the USEPA [174] and Industrial Waste Resource Guidelines [173]. Table 5.1.3 and Figs. 5.1.11 – 5.1.15 summarize the ICP-MS results for heavy metals using the ABLP method for pH 2.9, pH 5.0, and pH 9.2. All data are means of three values relative to standard deviation. The standard deviation is relatively small for most of the results, confirming the efficiency of the method.

Table 5.1.3. Australian Bottle Leaching Procedure (ABLP) (<2.4 mm) leachate results for heavy metals in crushed brick samples using inductively coupled plasma mass spectrometry (ICP-MS).

Heavy Metals	Concentration Level (mg L^{-1})		Percentages of CBs by Mass (mg L^{-1})															
			0%			0.5%			1%			1.5%			2%			
	USEPA Drinking Water [a]	EPA Vic Solid Waste [b]	pH 2.9	pH 5.0	pH 9.2	pH 2.9	pH 5.0	pH 9.2	pH 2.9	pH 5.0	pH 9.2	pH 2.9	pH 5.0	pH 9.2	pH 2.9	pH 5.0	pH 9.2
Cr	0.1	2.5	0.004 ± 0.004	0.014 ± 0.003	0.009 ± 0.003	0.005 ± 0.001	0.015 ± 0.003	0.007 ± 0.004	0.008 ± 0.001	0.013 ± 0.003	0.010 ± 0.002	0.007 ± 0.001	0.012 ± 0.003	0.009 ± 0.001	0.011 ± 0.002	0.016 ± 0.001	0.016 ± 0.005
Ni	-	1	0.007 ± 0.006	0.044 ± 0.004	0.004 ± 0.004	0.010 ± 0.005	0.043 ± 0.012	0.003 ± 0.002	0.056 ± 0.005	0.041 ± 0.013	0.025 ± 0.003	0.008 ± 0.004	0.039 ± 0.013	0.003 ± 0.001	0.203 ± 0.163	0.043 ± 0.015	0.010 ± 0.004
Cu	1.3	100	1.559 ± 0.331	0.523 ± 0.128	0.035 ± 0.009	1.536 ± 0.295	0.557 ± 0.082	0.043 ± 0.009	1.134 ± 0.472	0.543 ± 0.099	0.053 ± 0.009	1.496 ± 0.136	0.480 ± 0.208	0.042 ± 0.027	0.860 ± 0.076	0.499 ± 0.122	0.035 ± 0.004
Zn	-	150	1.792 ± 0.181	0.628 ± 0.016	0.104 ± 0.092	1.909 ± 0.174	0.645 ± 0.195	0.140 ± 0.014	1.876 ± 0.291	0.406 ± 0.058	0.267 ± 0.081	1.867 ± 0.337	0.335 ± 0.095	0.045 ± 0.027	2.251 ± 0.487	0.543 ± 0.076	0.222 ± 0.163
As	0.01	0.35	0.005 ± 0.001	0.010 ± 0.009	0.003 ± 0.001	0.011 ± 0.002	0.011 ± 0.006	0.005 ± 0.001	0.004 ± 0.003	0.014 ± 0.001	0.006 ± 0.002	0.044 ± 0.012	0.038 ± 0.025	0.015 ± 0.002	0.005 ± 0.001	0.015 ± 0.003	0.001 ± 0.000
Se	0.05	0.5	-	0.002 ± 0.001	-	0.002 ± 0.001	0.002 ± 0.001	-	0.013 ± 0.001	0.002 ± 0.001	0.003 ± 0.002	0.003 ± 0.001	0.003 ± 0.001	0.002 ± 0.001	0.015 ± 0.001	0.003 ± 0.001	-
Ag	-	5	-	-	-	-	0.009 ± 0.004	-	-	-	-	-	-	-	-	0.001 ± 0.000	-
Cd	0.005	0.1	0.001 ± 0.001	-	-	-	0.001 ± 0.000	-	-	0.001 ± 0.000	-	-	-	-	-	0.001 ± 0.001	-
Ba	2	35	1.450 ± 0.168	1.648 ± 0.138	1.447 ± 0.435	1.473 ± 0.221	1.770 ± 0.182	1.100 ± 0.118	1.513 ± 0.268	1.481 ± 0.075	1.432 ± 0.083	1.530 ± 0.14	1.526 ± 0.222	1.312 ± 0.287	2.747 ± 0.984	2.098 ± 0.566	1.032 ± 0.417
Hg	0.002	0.05	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Pb	0.015	0.5	0.012 ± 0.003	0.010 ± 0.007	0.001 ± 0.001	0.005 ± 0.001	0.008 ± 0.005	0.001 ± 0.000	0.050 ± 0.028	0.006 ± 0.002	0.006 ± 0.001	0.003 ± 0.003	0.007 ± 0.001	0.001 ± 0.001	0.011 ± 0.004	0.010 ± 0.008	0.002 ± 0.001

[a] USEPA [174]
[b] Industrial Waste Resource Guidelines [173]
Note: All data are means of three value

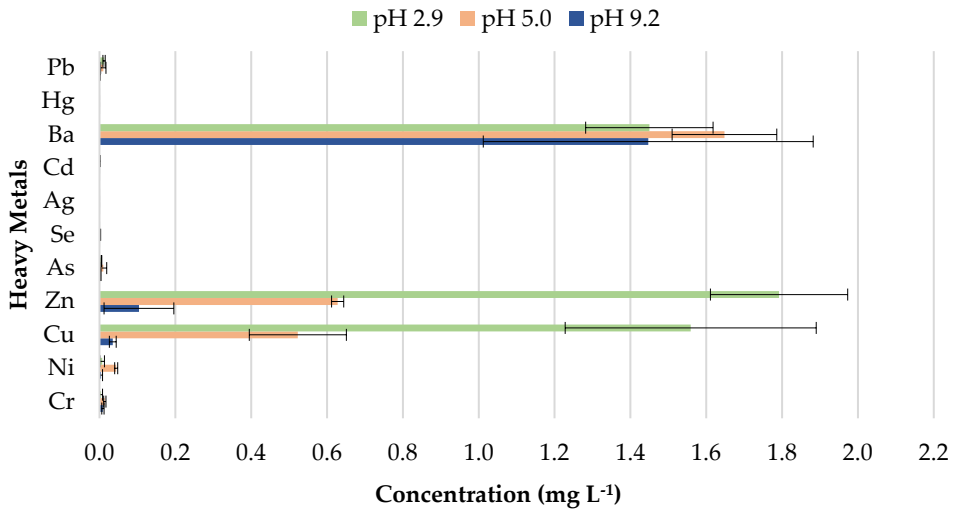


Fig. 5.1.11. Leachate concentrations for crushed bricks with 0% CB content.

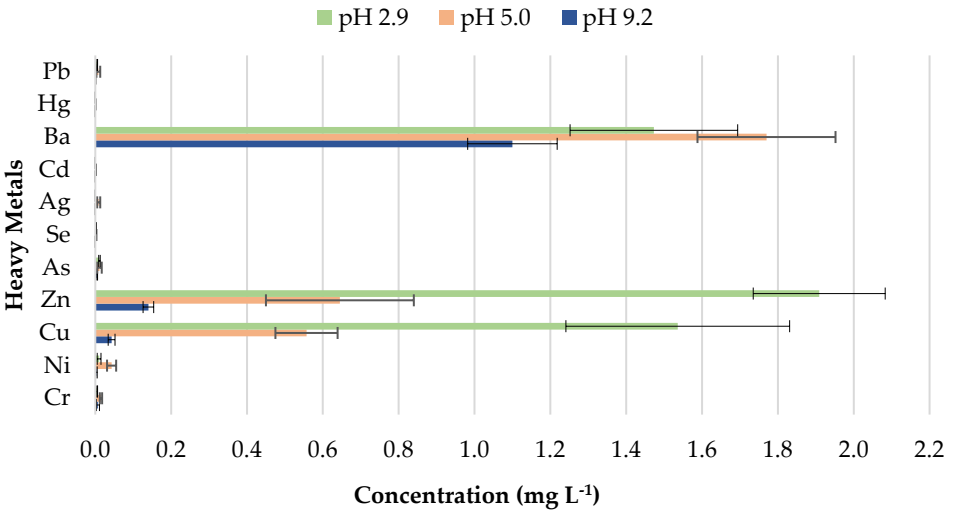


Fig. 5.1.12. Leachate concentrations for crushed bricks with 0.5% CB content.

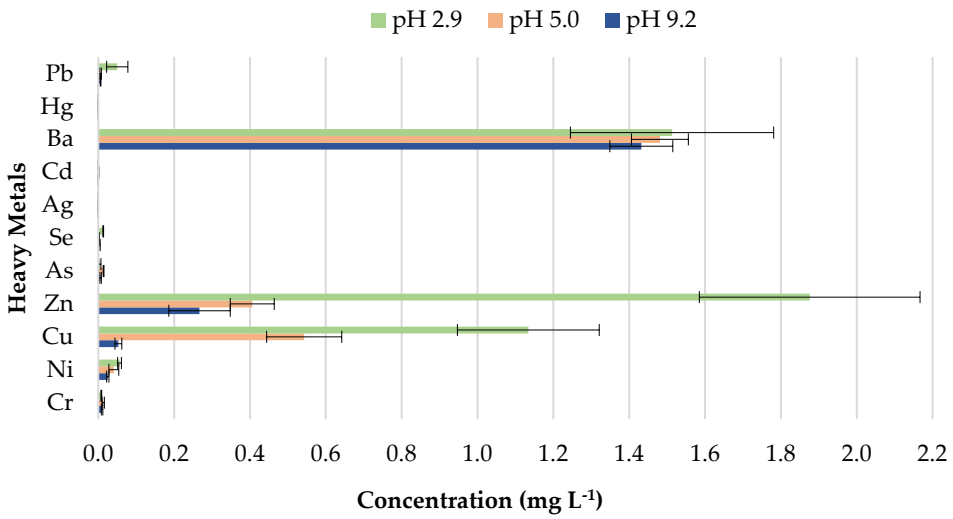


Fig. 5.1.13. Leachate concentrations for crushed bricks with 1% CB content.

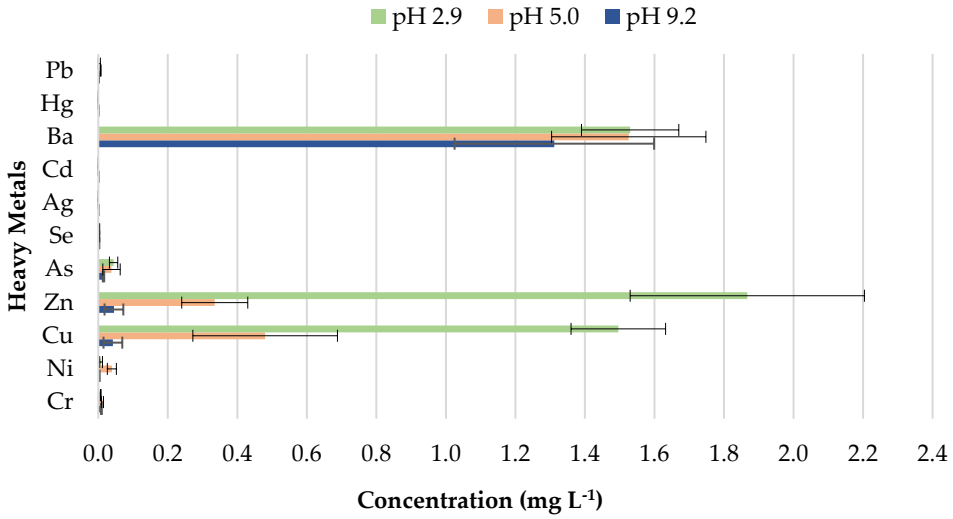


Fig. 5.1.14. Leachate concentrations for crushed bricks with 1.5% CB content.

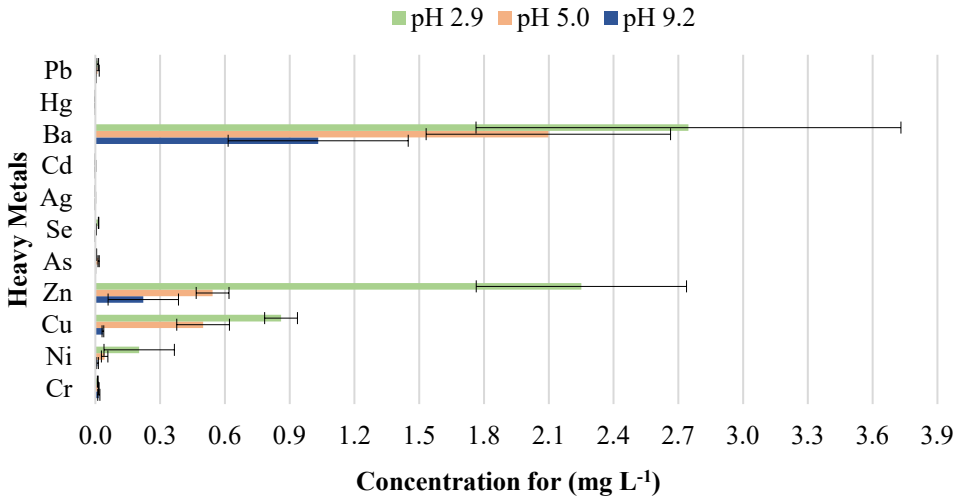


Fig. 5.1.15. Leachate concentrations for crushed bricks with 2% CB content.

The findings of this study indicate that differences in pH have an effect on the quantity of metal concentration leached for fired clay bricks incorporating CBs. The results imply that changes to the pH will likely increase or decrease the magnitude of metal contamination from fired clay bricks. In this study, an increasing trend in the metal concentration leachate was observed for fired clay bricks with the increase in CB content from 0% to 2% by mass. This is due to the increase in CB content, as discussed above, CBs are more prone to leaching heavy metals in areas with a pH of 2.9, compared to a pH of 9.2. The effect of the pH on the convergence of heavy metals plays a crucial role in the migration and transformation of metals. For instance, Yang et al. [184] found that the speciation of Cd in sediment increases with the increase in pH from 4.5 – 9.5. While Zhang et al. [185] found the release tendency of Ni and Cu to be lower in a pH range of 5 - 10, compared to pH 0 – 4, probably because of the low concentration of dissolved organic carbon (DOC) in deionized water.

A direct relationship between the pH and the metal concentration leachate was found. A decrease in pH from 9.2 - 2.9 resulted in an increase in the metal concentration leached for Zn, Cu, Ba, and Ni for bricks incorporating 1% and 2% CBs by mass (Figs. 5.1.16 & 5.1.17). This implies that pH values, such as those tested in areas with highly acidic rain (pH 2.9) outside the natural scope of precipitation, will result in the metal contamination of Zn, Cu, Ba, and Ni. This is because acidic water composes of a high number of H^+ ions. Therefore the surface of the particles is positively charged, which results in weak metal sorption and the release of toxic ions such as Zn, Cu, Ba, and Ni [185]. Other metals, including As, Cr, Se, Ag, Cd, Hg, and Pb are of insignificant concern as the leachate concentrations are below 0.03 mg L^{-1}.

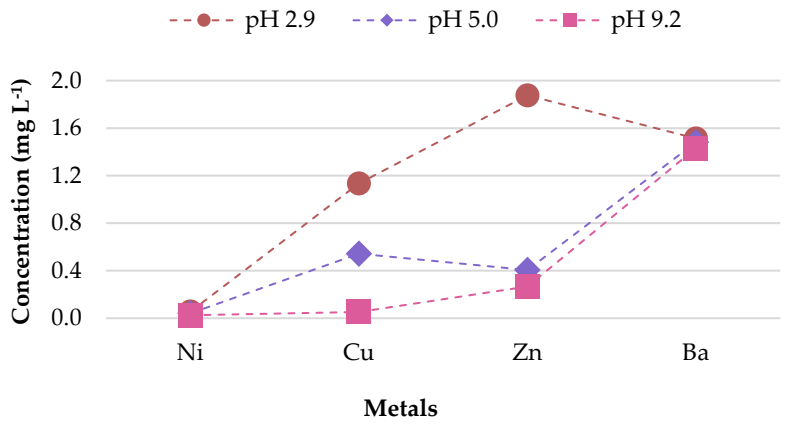


Fig. 5.1.16. Variable relationship observed between metal concentrations leached from 1% CB bricks and pH value.

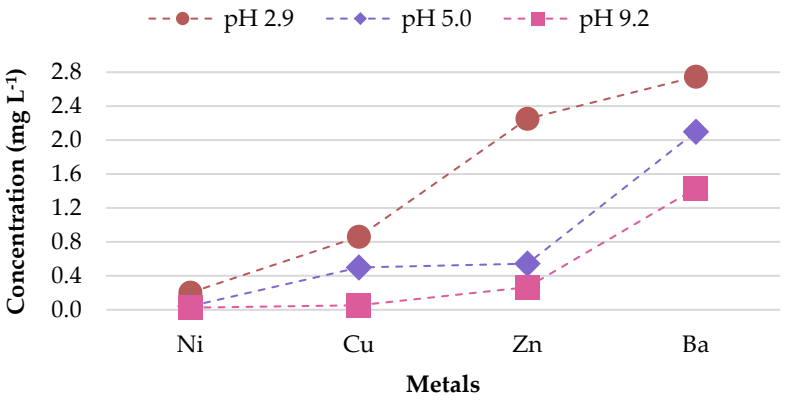


Fig. 5.1.17. Variable relationship observed between metal concentrations leached from 2% CB bricks and pH value.

It can be deduced from Table 5.1.3 that the metal concentrations of the leachate are trivial when compared with the regulatory threshold limits set by the Industrial Waste Resource Guidelines [173]. The results confirm that fired clay bricks incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by mass can be categorized as non-hazardous waste under various pH values as they are clearly below the regulatory limits for heavy metals. To better interpret the insignificance of the results, the leachate concentrations were compared to the United States national primary drinking water thresholds set by the USEPA [174]. Cu and Ba leachate concentrations for 0%, 0.5%, 1%, 1.5%, and 2% CB bricks were the only metals moderately above the threshold limits, while the remaining leachate concentrations were considerably below the limits.

5.1.3.5 Unfired Bricks and Fired Bricks Comparative Leachate Analysis

Concerns about heavy metal contamination to the environment that might be initiated through the incorporation of CBs, which cannot be combusted during the firing process, represent a major issue. Therefore, to investigate the variations in concentrations of 0%, 0.5%, 1%, 1.5%, and 2% CB bricks before and after firing, pH value 9.2 was selected. The results are presented in Figs. 5.1.18 – 5.1.21 and Table 5.1.4. All data are means of three values relative to standard deviation. The findings show that the leaching behavior of the fired clay bricks is notably lower after the firing process. This is the result of the suppression of metals that are locked in the bricks when fired to high temperatures. This behavior can be associated with the silicate-based physio chemical containment process of fired clay bricks [135]. The main compounds found in the brick soil were silica (SiO_2) and iron oxide (Fe_2O_3). Silica enhances the interconnection and compactness of the material, which can result in metal immobilization [186], while iron oxide can assist in the formation and stability of the material, because of the crystallization-dehydration of less crystallized free iron oxides and the removal of organic materials [187]. Hence decreasing the leaching potential of heavy metals from the fired clay bricks incorporating CBs.

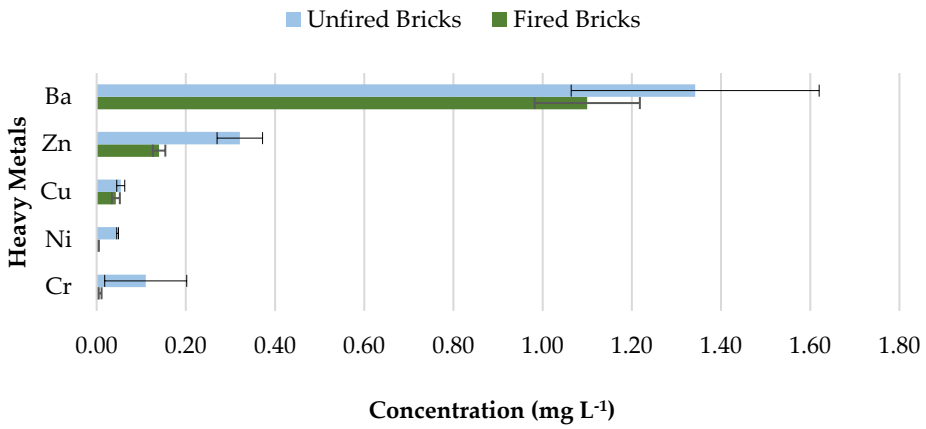


Fig. 5.1.18. Heavy metal concentration of 0.5% CB content bricks before and after firing.

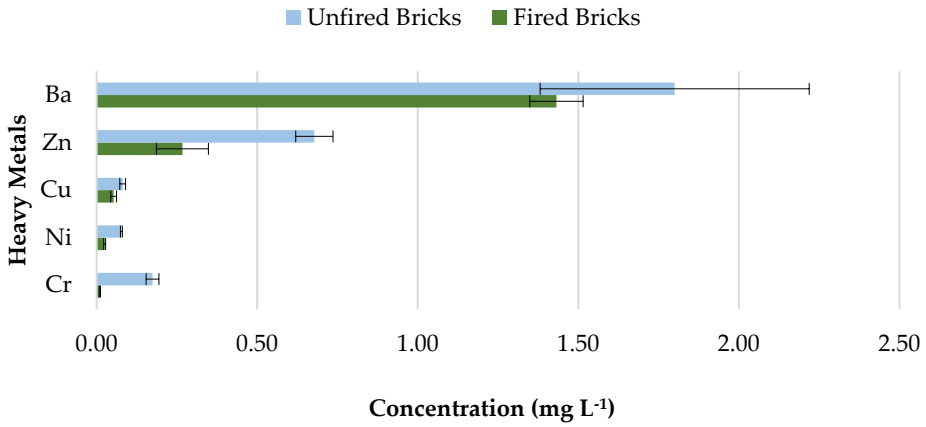


Fig. 5.1.19. Heavy metal concentration of 1% CB content bricks before and after firing.

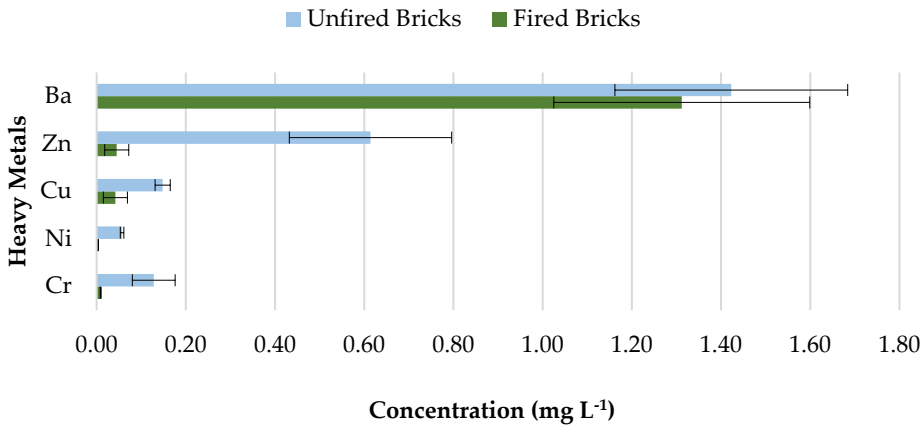


Fig. 5.1.20. Heavy metal concentration of 1.5% CB content bricks before and after firing.

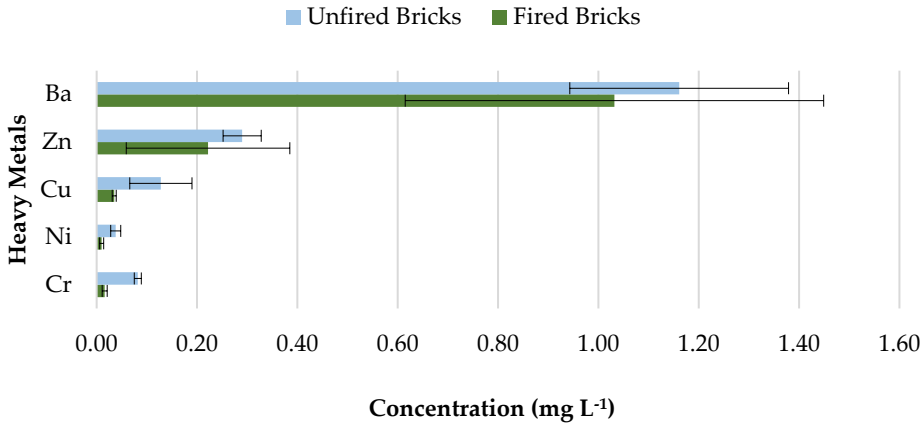


Fig. 5.1.21. Heavy metal concentration of 2% CB content bricks before and after firing.

The fluctuations in the leachate concentrations for unfired and fired bricks incorporating 0.5%, 1%, 1.5%, and 2% CBs by mass are summarized in Table 5.1.4. The results indicate that the addition of CBs in fired clay bricks considerably reduces the leaching ability of heavy metals. For example, the leaching of Cr and Ni was almost completely impeded after firing for bricks incorporating 0.5% and 1.5% CBs by mass. In addition, over 50% of all tested heavy metals were immobilized after firing for 0.5%, 1%, 1.5%, and 2% CB bricks. However, Ba demonstrated the least change in metal contamination after the firing process. More than 90% of Cr was immobilized in CB fired clay bricks compared to CB unfired bricks. Incorporating CBs in fired clay bricks resulted in a decrease in the metal contamination of the toxic waste, which can be an environmental advantage compared to the littering of CBs. The high firing temperature immobilized the heavy metals within the brick due to the crystalline structure of the material.

To further study the leaching performance of toxic metals at various liquid to solid ratios and for compacted granular material, leaching tests can be conducted according to Method 1314 [169] and Method 1315 [170]. Although the leaching procedure followed in this study considers a wide range of pH values subject to changing environmental conditions, it does not consider leaching subject to various liquid to solid ratios and compacted granular material. However, it can be stated that fired clay bricks amended with CBs are safe under various conditions and environments as the results are significantly below the threshold limits.

Table 5.1.4. Leaching concentrations of unfired and fired bricks containing 0.5%, 0.5%, 1%, 1.5%, and 2% CBs by mass for pH 9.2.

Heavy Metals	Percentage of Leaching Drop before and after Firing (%)											
	0.5%			1%			1.5%			2%		
	Unfired Bricks	Fired Bricks	% Leaching Drop	Unfired Bricks	Fired Bricks	% Leaching Drop	Unfired Bricks	Fired Bricks	% Leaching Drop	Unfired Bricks	Fired Bricks	% Leaching Drop
Cr	0.110 ± 0.092	0.007 ± 0.004	94	0.174 ± 0.020	0.010 ± 0.002	94	0.128 ± 0.048	0.009 ± 0.001	93	0.082 ± 0.007	0.016 ± 0.005	80
Ni	0.047 ± 0.002	0.003 ± 0.002	94	0.077 ± 0.003	0.025 ± 0.003	68	0.057 ± 0.004	0.003 ± 0.001	95	0.038 ± 0.010	0.010 ± 0.004	74
Cu	0.054 ± 0.009	0.043 ± 0.009	20	0.081 ± 0.009	0.053 ± 0.009	35	0.148 ± 0.017	0.042 ± 0.027	72	0.128 ± 0.062	0.035 ± 0.004	73
Zn	0.321 ± 0.051	0.140 ± 0.014	56	0.678 ± 0.058	0.267 ± 0.081	61	0.614 ± 0.182	0.045 ± 0.027	93	0.290 ± 0.038	0.222 ± 0.163	23
Ba	1.342 ± 0.278	1.100 ± 0.118	18	1.800 ± 0.419	1.432 ± 0.083	20	1.423 ± 0.261	1.312 ± 0.287	8	1.161 ± 0.218	1.032 ± 0.417	11

Note: All data are means of three values.

5.1.4 Conclusions

The incorporation of CBs into fired clay bricks could provide a viable resolution to an escalating waste pollution tragedy, with acceptable mechanical and physical properties and a new cost-effective construction product. Therefore, the results presented here may clarify issues raised regarding the possible leachate of heavy metals from the use and disposal of bricks incorporating CBs. The following conclusion can be drawn from this study:

- Various leachate testing methods were compared, including the ABLP and USEPA methods. In reviewing the various leaching test procedures, the ABLP was determined to be the most suitable leaching procedure for this study based on the sample size, pH values, disposal-to-land scenarios, and condition of the sample.
- Metals Cu, Zn, Mn, Al, Fe, Ti, and Ba demonstrated the highest leachate concentration for pH 2.9 and 5.0 for used CBs. This implies that used CBs are more likely to leach heavy metals in areas with highly acidic rain compared to the natural range of precipitation. This is due to the enhancement of metal mobilization in acidic water, whereby the toxic ions (toxic metals) bound to the negatively charged surface and are displaced by H^+ (hydrogen) ion.
- It was found that used and shredded used CBs are more prone to leaching heavy metals Cr, Cd, Zn, Mn, Al, and Fe compared to unused CBs. This is because an enhanced level of Cr, Cd, Zn, Mn, Al, and Fe is absorbed and retained by the filter during combustion.
- The leachate results for fired clay bricks incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by mass were found to be below the regulatory limits set by the EPA Victoria solid industrial waste guidelines. Therefore, bricks modified with CBs can be categorized as non-hazardous waste.
- The leaching of Cr and Ni was almost completely hindered after the firing process of clay bricks incorporating 0.5% and 1.5% CBs by mass. In addition, more than 50% of all the tested heavy metals were immobilized after firing for 0.5%, 1%, 1.5%, and 2% CBs by mass bricks. Hence the addition of CBs in fired clay bricks may be a feasible option to hinder the heavy metals present in CBs.
- The compressive strength of the CB brick samples varied considerably with the percentage of organic content within the mixture. However, the values were considerably above the minimum requirement of 17.2 MPa for compressive strength. The addition of CBs produced less dense and porous bricks, with better thermal insulation properties.
- The water absorption and IRA results were within the acceptable range, indicating CB brick samples are competent to endure harsh weather conditions.

It is recommended to investigate the leaching performance of toxic metals at the various liquid to solid ratios and for compacted granular material, as described in Method 1314 [169] and Method 1315 [170],

to consider leaching of bricks subjected to various disposal or use scenarios. Moreover, acceptable limits for heavy metals for various applications, particularly for construction material manufactured with waste material, should be developed. However, it can be stated fired clay bricks incorporating CBs are safe under various conditions and environments as the leachate concentrations are adequately below the regulatory limits. Moreover, the addition of CB waste in brick manufacturing offers a sustainable recycling solution to a growing pollution issue.

Part 5.2: Physical Properties and Leachate Analysis of Bitumen and Paraffin Wax Encapsulated Cigarette Butts and its use in Road Construction

Abstract

Littered cigarette butts (CBs) are a form of poorly biodegradable waste. Attempts have been made to prevent the pollution caused by the waste through the recycling of CBs in construction materials. A novel method involving the encapsulation of CBs with paraffin wax and bitumen was developed, which investigated its use in hot-dense asphalt concrete with promising results. The objective of this study was to carry out a comprehensive leachate analysis to examine the effectiveness and efficiency of the encapsulation method for the prevention of leachates from CBs and to investigate the physical properties of wax and bitumen encapsulated CBs. The leachate analysis was conducted using the Australian Bottle Leaching Procedure (ABLP) for pH values 2.9, 5.0, and 9.2. The leaching behavior of encapsulated CBs was insignificant compared to fresh CBs due to the immobilization of metals after being encapsulated in bitumen and wax. The findings suggest that the variance in pH influences the behavior of the metal concentration leached. Generally, pH 2.9 resulted in the highest metal contamination for all sample types, followed by pH 5.0 and pH 9.2. The water absorption decreased significantly from 383% for fresh CBs to 4% and 5% for bitumen and wax encapsulated CBs. Micro-X-ray CT imaging was used to study and analyze the porosity and pore structure of CBs, and a substantial decrease in porosity from 66% to 0.5% was determined when CBs were encapsulated with bitumen.

The research presented in this chapter has been accepted in the following:

Mohajerani, A., Kurmus, H., Rahman, M.T., Smith, J.V., Woo, S.S., Jones, D., Dastjerdi, M.G., Nguyen, B.T. and Calderon, C., 2021. Bitumen and Paraffin Wax Encapsulated Cigarette Butt; Physical Properties and Leachate Analysis. Journal of Pavement Research and Technology. https://doi.org/10.1007/s42947-021-00063-9

5.2.1 Introduction

5.2.1.1 Cigarette butt pollution

Cigarette butts (CBs) are one of the most frequently littered wastes in the world [188]. Worldwide, smokers roughly consume 6.5 trillion cigarettes each year, and an estimated one-third are disposed of correctly while the remaining 4 trillion CBs are discarded into the environment [188]. In Australia, 7 billion CBs are carelessly thrown into the environment each year [189]. It costs the Victorian economy $25.7 million for tobacco-related litter control and roughly $7.5 million in San Francisco every year [188, 189]. Furthermore, the environmental impacts of the continuous littering of CBs are of greater concern for the health of humans, wildlife, and marine and freshwater organisms [125].

The abundance and slow decomposition rate of CBs exacerbate the detrimental effects on the environment. Approximately 95% of CBs are comprised of plastic filaments that are fabricated from synthetic fibers (cellulose acetate) derived from wood pulp [190]. The conventional composition of a cigarette can be seen in Fig. 5.2.1. The decomposition of cellulose acetate varies between a few months to several years, depending on various environmental factors [40, 191-194], and only dissipates 37.8% of their initial mass after two years of decomposition [195].

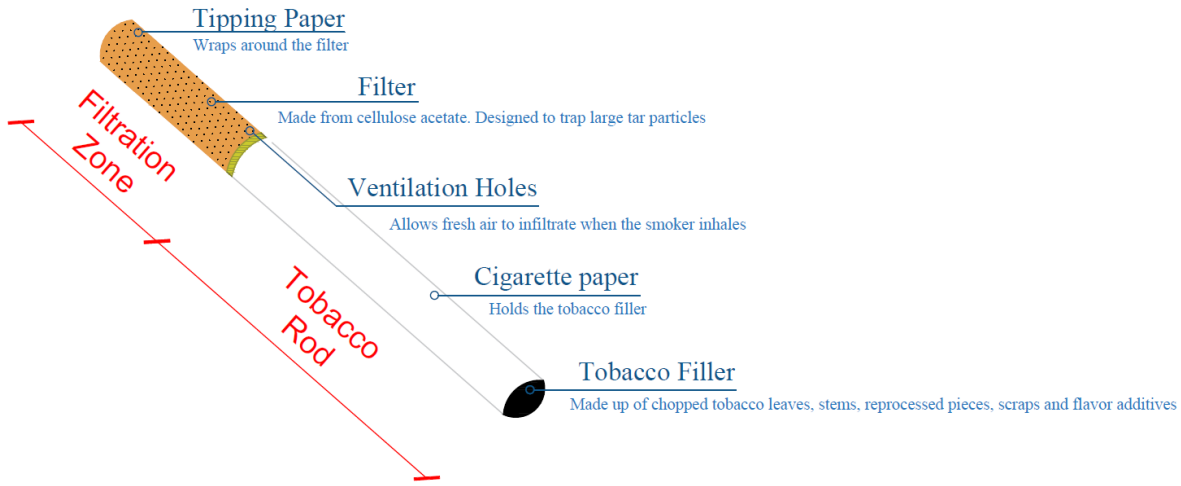


Fig. 5.2.1. Physical structure of a cigarette with each component described and identified.

Cigarette filters are designed to absorb and trap smoke components, including tar and toxic chemicals [196]. Cigarette smoke contains up to 4,000 chemical components, approximately 3,000 in the gas phase and 1,000 in the tar phase. The compounds include N-nitrosamines, aromatic amines, hydrogen cyanide, ammonia, formaldehyde, benzene, acetaldehyde, phenols, pyridines, and toxic heavy metals, including nickel and cadmium that can combine to form over 60 chemicals that are known to be carcinogenic [11, 196]. Smoked CBs contain a substantial quantity of nicotine and other compounds produced during the combustion of tobacco and, essentially, pose a higher toxic effect in contrast to

unsmoked cigarettes [125]. Therefore, when CBs are littered, toxic chemicals and heavy metals are leached into the environment, resulting in a significant impact on living organisms and the health of the ecosystem [50, 196-198].

The possible disposal methods for CBs include incineration and landfilling. However, neither method is sustainable or economically feasible. CBs remain an environmental hazard when sent to landfills, and the continuous disposal of hazardous wastes to landfills is becoming costly and difficult [125, 199, 200], while the process of incineration can result in the release of harmful emissions [10].

Therefore, it is becoming crucial to find an effective and practical method to recycle CB waste. One environmentally friendly way to solve this growing global pollution problem may be through the sustainable method of recycling CBs in construction materials. Indeed, several studies have been carried out on the incorporation of CBs in fired clay bricks [1, 23, 24, 116, 177, 201, 202]. Various mixes containing between 0% to 10% CBs by weight (wt) were prepared and tested for density, shrinkage, initial rate of absorption (IRA), water absorption, compressive strength, thermal conductivity, and energy savings. Bricks incorporating 1% CBs by wt demonstrated a 5.2% decrease in dry density and a 24% decrease in compressive strength compared to the control bricks. While water absorption and IRA increased from 9% to 12% and 0.44 kg m^{-2} min^{-1} to 0.67 kg m^{-2} min^{-1} for bricks incorporating 0% and 1% CBs by wt. By using the calculation method, an energy saving of 10.2% was estimated for fired clay bricks incorporating 1% CBs by wt. Furthermore, it was estimated that if 1% CBs were incorporated into 2.5% of the bricks produced worldwide, the CB problem could possibly be eradicated from the world [1, 203].

5.2.1.2 Encapsulation of CBs

Mohajerani et al. [19] developed a novel method involving the encapsulation of CBs with bitumen and paraffin wax and incorporating them into asphalt concrete (AC). Dense graded asphalt samples containing different classes of bitumen (C170, C320, C600) encapsulated CBs (10 kg m^{-3}, 15 kg m^{-3}, 25 kg m^{-3}) were manufactured according to the Austroads [204]. The stability values for asphalt mixes with 10 kg m^{-3} and 15 kg m^{-3} CB content encapsulated with C170, C320, and C600 bitumen satisfied the conditions for road pavements for light, medium, and heavy traffic, respectively. However, the AC mixes incorporating 25 kg/m^3 CBs encapsulated with C170, C320, and C600 bitumen did not fulfill the minimum conditions for medium and heavy traffic. The flow values were all within the aimed range of 2 to 4 mm other than the AC mix comprising 10 kg m^{-3} CB content encapsulated with C320 bitumen. A decline in the inflow was observed for the AC mixes from 2.5 mm for the controlled sample to 1.8 mm for the C320 bitumen encapsulated CB sample. Moreover, the increase in CB content ensued in an increase in the inflow. For C170 bitumen, the resilient modulus decreased from 4703 MPa for the AC mix containing no CBs to 3387 MPa for the AC mix containing a CB content of 15 kg m^{-3}. It is apparent

that the increase in CB content in the AC mix results in a decrease in the resilient modulus. However, it can be noted that the resilient modulus for the 15 kg m^{-3} and 25 kg m^{-3} CB content AC mixes was within the 2500 to 4000 MPa resilient modulus target range.

The bulk density and maximum density results for the AC bitumen encapsulated CB samples ranged from 2262 to 2421 kg m^{-3} and 2352 to 2546 kg m^{-3}. Two trends were observed, the increase in CB content resulted in a decrease in the maximum density and bulk density, and the maximum density was found to be greater than the bulk density. Lower density is desirable as it will decrease the thermal conductivity.

As a continuation of the parent research, Rahman et al. (2020) investigated the application of bitumen encapsulated CBs in stone mastic asphalt (SMA). The research has revealed mechanical properties like Marshall stability, flow, and resilient modulus of modified SMA [205]. SMA samples were compacted (by maintaining density) using 1%, 2%, and 3% (by weight of asphalt mix) bitumen encapsulated CBs, and results from laboratory investigation indicated up to 1% use of bitumen encapsulated CBs met industry requirements [205]. Later as an expansion of the original research, rutting, fatigue, and moisture damage characteristics of dense graded asphalt prepared with bitumen encapsulated CBs were investigated, and the use of up to 1% bitumen encapsulated CBs met industry requirements and supported the application of bitumen encapsulated CBs in asphalt [206].

The process of encapsulation involves the coating of hazardous materials in matrices to prevent the escape of toxic chemicals [207]. The use of bitumen to encapsulate radioactive waste started in the 1960s and was used due to its insolubility and impermeability to water [207, 208]. Water-soluble radioactive wastes, such as spent organic ion exchangers and bottom residues from evaporative treatment, are exceptionally suitable for bituminization [208]. Prefabricated bitumen membranes and asphalt concrete are used to seal the soil surface where water pollution is likely to occur to prevent the subsoil from being polluted by industrial waste [209].

Another effective mixture for encapsulation is paraffin wax. Paraffin wax is obtained from petroleum and was first commercially produced in 1867, less than ten years after the initial petroleum well was drilled [210]. Throughout the 20th century, it was one of the most popular and earliest water repellent finishes [211]. A highly efficient way to create water repellent fabric was to coat it with paraffin wax. This coated the fibers with a thin layer of wax, especially those exposed to water, thereby giving the fabric exceptional water repellent properties. Therefore, the use of paraffin wax as an encapsulation material can potentially avert the escape of the toxic chemicals and heavy metals found in CBs.

5.2.1.3 Objective

The objective of this study was to carry out a comprehensive leachate analysis to examine the effectiveness and efficiency of the encapsulation method for the prevention of leachates of heavy metals

from CBs and investigate the physical properties of wax and bitumen encapsulated CBs. The leachate analysis was performed using the Australian Bottle Leaching Procedure (ABLP) [212] for pH values 2.9, 5.0, and 9.2. Method 1313 [168] is a similar leachate testing method to the ABLP, both being pH-dependent. However, Method 1313 is part of the four procedures in the Leaching Environmental Assessment Framework (LEAF), and it is recommended all four methods be used in conjunction when examining a sample. In the ABLP, the metals studied include Be, Mg, Al, Se, As, Pb, V, Mn, Fe, Co, Ni, Cu, Cr, Ag, Zn, Mo, Cd, Ti, Th, and U. To evaluate the suitability of bitumen and wax encapsulated CBs and their possible use in road construction, the leachate results were compared to the concentration limits for heavy metals set by the Australian drinking water guidelines [213] and solid industrial waste hazard categorization thresholds [173]. To determine the effectiveness and efficiency of the developed method, the encapsulated CBs underwent microstructural analysis and were tested for weight, density, bulk density, water absorption, and strength.

5.2.2 Materials and methods

5.2.2.1 Bitumen

Bitumen, also known as asphalt binder, is a highly viscous petroleum-based hydrocarbon that can be found in natural deposits or refined from crude oil. The primary use of bitumen is in road construction, where it is used as a binder between aggregates, however, it is also commonly used to create bituminous waterproofing products and to encapsulate radioactive waste [15]. Approximately 102 million tons of bitumen are used per year, and of that, about 85% is used as the binder in asphalt concrete for roads. There are different grades of bitumen available that all have different properties. The different grades in Australia are detailed in Standards Australia [214], including the method of testing and classification. In this research, Bitumen PMB A10E was used to encapsulate CBs. The properties of the bitumen used according to the supplier specifications are given in Table 5.2.1 [3].

Table 5.2.1. Properties of bitumen PMB A10E (adopted from Puma Bitumen [3]).

Properties	Typical value
Softening Point	95°C
Torsional Recovery at 25 °C	73%
Viscosity at 165 °C	0.9 Pa·s
Stiffness at 2 5°C	25 kPa
Segregation	0.5%
Consistency at 60 °C	16,000 Pa·s

5.2.2.2 Paraffin wax

Paraffin wax is a crystalline mixture of solid straight-chain hydrocarbons, characterized by its solid state at room temperature. It is a white and colorless soft, solid wax. Paraffin wax is obtained from petroleum, and the common uses of paraffin wax include candles, polishes, electrical insulators, wax paper, and cosmetics [210]. The main properties of the type of paraffin wax used in this paper are summarized in Table 5.2.2.

Table 5.2.2. Wax properties provided by Australian wax company SDS

Appearance and odor	Solid at room temperature Slight odor May be handled and transported as hot, molten liquid > 100 °C
Color	At ambient temperature: white powder When melted yellow liquid
Melting point	80°C*
Boiling point	Not applicable, decomposes
Vapor pressure	< 0.1 kPa (250° C)
Specific gravity	0.92 approximately at 20 °C
Solubility	Insoluble in water
Percent volatile by water	Nil at 20 °C
Corrosiveness	Not corrosive

*Mohajerani et al. [19]

5.2.2.3 CB Encapsulation process

5.2.2.3.1 Pre-processing of CBs

Used CBs of varying sizes and brands were supplied by Buttout Australia Pty Ltd and dried in the oven for 24 h at 110 °C to remove the moisture content. It is critical that the CBs are completely dried out to ensure that the bitumen and wax completely fill the voids within the CBs during the encapsulation process. After drying, the excess paper and tobacco were removed from the CBs.

5.2.2.3.2 Bitumen encapsulation of CBs

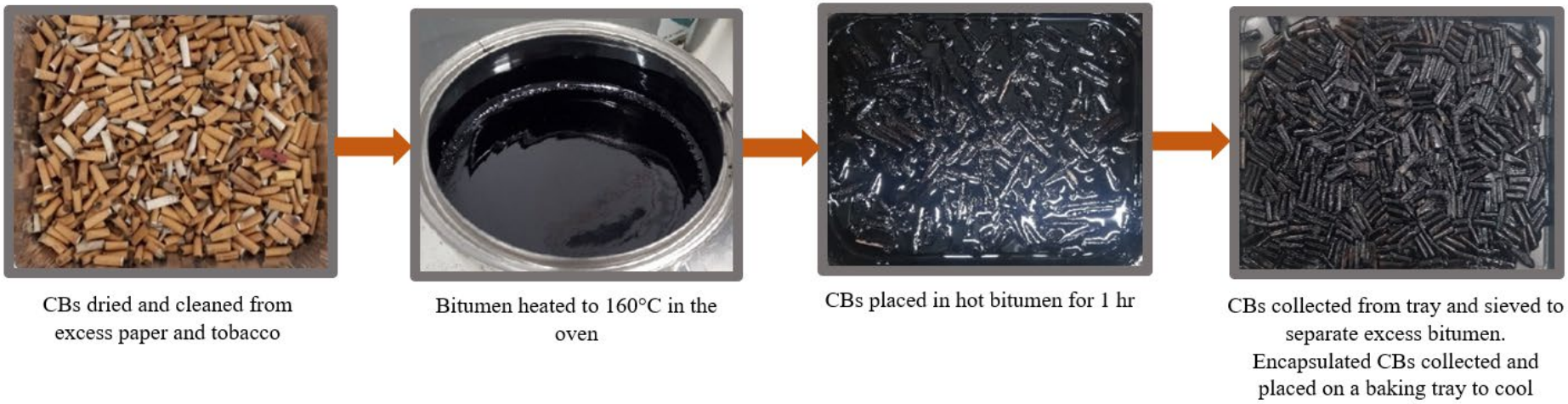
The procedure followed to encapsulate CBs with bitumen is shown in Fig. 5.2.2(a). To increase the workability of the bitumen as a liquid form to perform an effective encapsulation, the bitumen was placed in the oven in a container for 2 h at 160 ºC. In a thin aluminum tray, the liquid bitumen was poured, and CBs were placed in the tray and then placed in the oven for 1 hr at 160 ºC. Throughout the oven drying process, the bitumen-CB mixture was stirred with a steel rod every 10 min. During this process, the air voids within the CB filter were replaced with bitumen. Once removed from the oven,

the encapsulated CBs were collected and placed on a 6.7 mm sieve to break up the excess bitumen from individual bitumen encapsulated CBs. The collected encapsulated CBs were stored at room temperature in a tray.

5.2.2.3.3 Wax encapsulation of CBs

The procedure followed to encapsulate CBs with wax is demonstrated in Fig. 5.2.2(b). A saucepan was used to heat the wax to 140 ℃ until it was completely melted. The CBs were placed in the mixture and gently pressed to the bottom of the pot to ensure the wax completely infiltrated into the voids of the CBs. The CBs were then left to soak in the hot wax for approximately 10 min. Once the CBs were soaked in wax for 10 min, the CBs were carefully removed from the hot wax and placed in a sieve to remove the excess wax. The CBs were then placed on a baking tray and left to dry and harden.

(a)



(b)

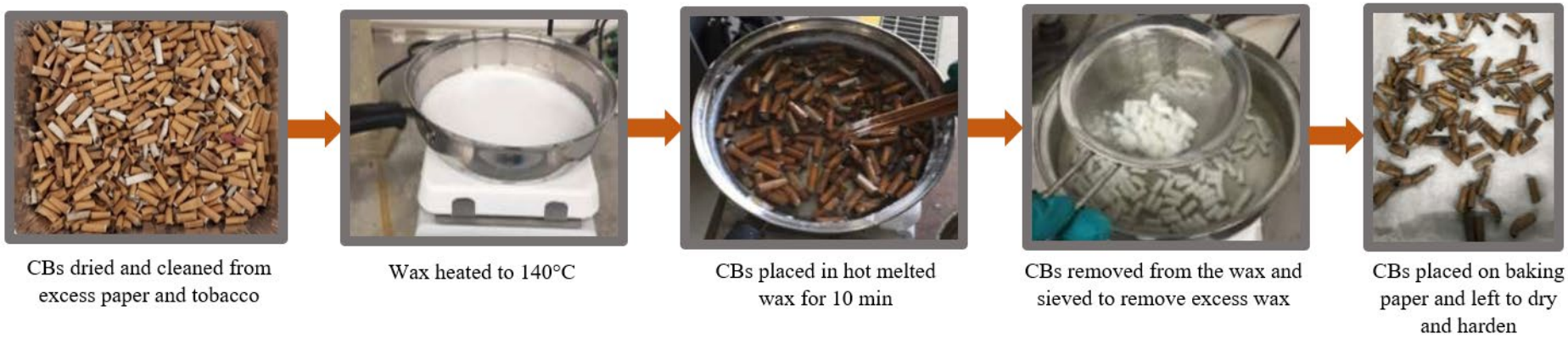


Fig. 5.2.2. Flow chart of the process followed to encapsulate CBs with (a) bitumen and (b) paraffin wax.

5.2.2.4 Leachate analysis

To determine whether the encapsulation process of CBs prevents or reduces the leaching of metals found within CBs, a comprehensive leachate analysis was conducted. The ABLP [212] was followed in this study on oven-dried CBs, fresh CBs, bitumen encapsulated CBs (broken), bitumen encapsulated CBs (unbroken), wax encapsulated CBs (broken), and wax encapsulated CBs (unbroken). To simulate various disposal scenarios, pH values 2.9, 5.0, and 9.2 were selected. Metals including Beryllium (Be), Magnesium (Mg), Selenium (Se), Aluminum (Al), Thallium (Ti), Vanadium (V), Molybdenum (Mo), Chromium (Cr), Lead (Pb), Manganese (Mn), Copper (Cu), Iron (Fe), Silver (Ag), Cobalt (Co), Arsenic (As), Nickel (Ni), Zinc (Zn), Cadmium (Cd), Thorium (Th), and Uranium (U) were examined for all six sample types. The experimental procedure followed in this study to prepare the leachates is shown in Fig. 5.2.3.

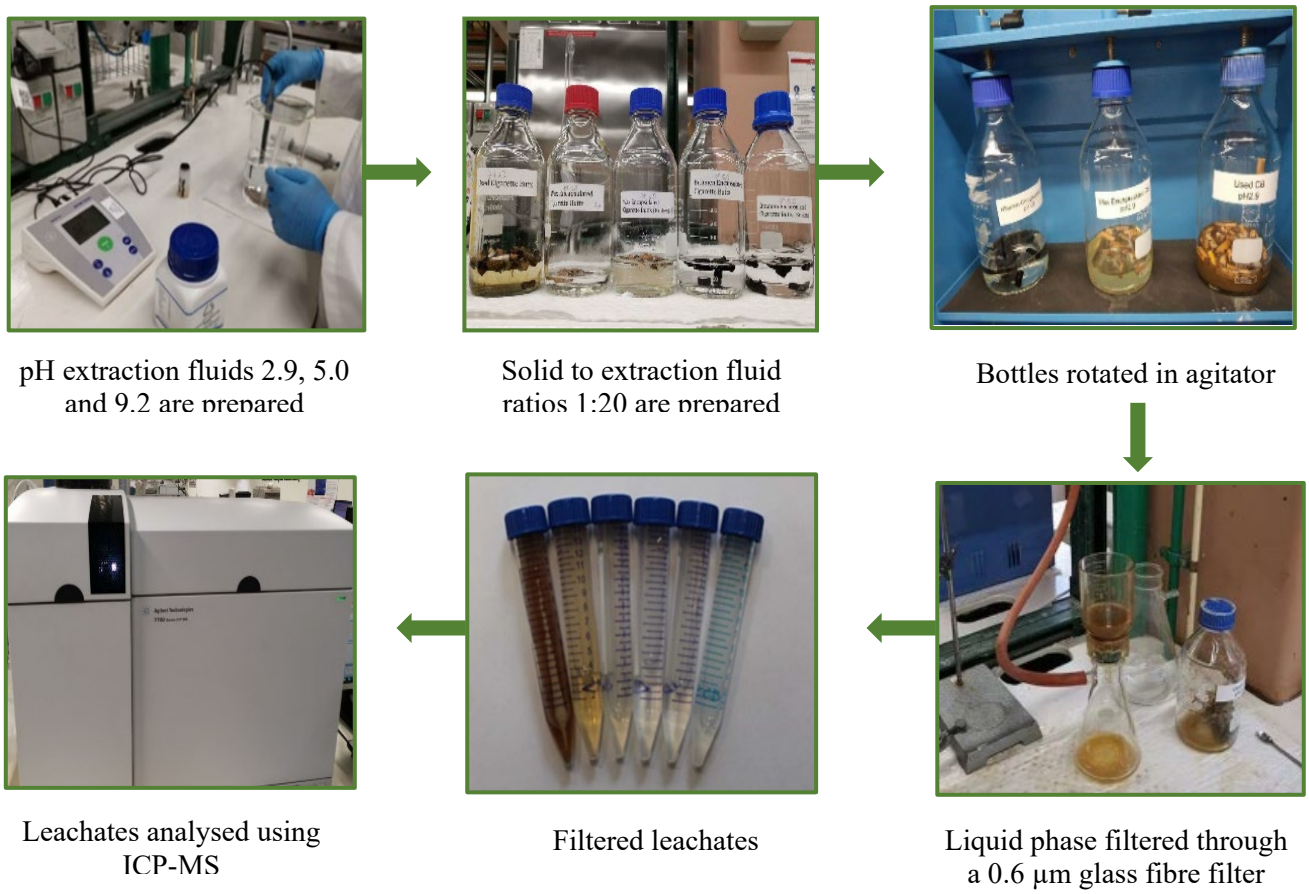


Fig. 5.2.3. Leachate analysis process followed in this study.

Initially, three different pH leaching fluids were prepared. For leaching fluid pH 2.9, 5.7 ml of glacial acetic acid was added to 900 ml of deionized water. The solution was then diluted with deionized water to a volume of 1L. The pH of the solution was then determined using a pH meter, and if the value was not in the range of pH 2.8 - 3.0, the solution was discarded, and the process started again. Leaching fluid pH 5.0 was prepared by adding 5.7 ml of glacial acetic acid to 900 ml of deionized water, then 64.3 ml of 1 mol L^{-1} sodium hydroxide solution was added to the solution. The fluid was then diluted

with deionized water to a volume of 1L and mixed well. The pH of the solution was then determined using a pH meter; if the pH value was 5.0, the solution was finished, while if it was in the range of 4.9 to 5.1, the pH was adjusted by the addition of 1 mol L^{-1} acetic acid or 1 mol L^{-1} sodium hydroxide solution. The final solution was then stored in an airtight container. Leaching fluid pH 9.2 was prepared by dissolving 38.2 g of sodium tetraborate decahydrate into 900 ml of water. The solution was then diluted with water to 1 L. The pH of the solution was then determined using a pH meter; if the pH was not 9.2, the solution was discarded, and the process started again.

Samples were prepared for various CB types with a solid to fluid ratio of 1:20 inside screw cap extractor bottles. The extraction bottles were rotated at 30 rpm for 24 h by means of a VELP Scientific agitation apparatus in an end over end manner. The liquid and solid phases were separated by filtering through a 0.6 μm glass fiber filter, and the leachates were transferred to centrifugal tubes for analysis. The leachates were then analyzed using the ICP-MS for metals.

5.2.3 Results and discussion

5.2.3.1 Mass and density of encapsulated CBs

To determine the mass of unused and used CBs, batches of 50, 100, and 200 unused CBs and a batch of 50 used CBs were weighed. The average mass was determined from each batch, and, therefore, the approximate mass of one unused and one used CB was calculated, as shown in Table 5.2.3. The mass of one unused CB was found to be approximately 0.420 g, while the mass of one used CB is approximately 0.830 g. There is a reasonable variance between the two samples, which is due to the excess tobacco and paper present in used CBs.

The Australian standards [215] and ASTM5 [216] were followed to determine the bulk density for unused, used, bitumen encapsulated, and wax encapsulated CB samples. The bulk density for bitumen encapsulated CBs was found to be 708.33 kg m^{-3} in contrast to 781.46 kg m^{-3} for wax encapsulated CBs. In the case of unused CBs, bulk density was 150.16 kg m^{-3} and 154.286 kg m^{-3} for used CBs. The obtained values for bulk density are essential when encapsulated CBs are intended for use as aggregate in construction material.

Table 5.2.3. Calculation of material content of CBs encapsulated with bitumen and wax.

Sample type	Weight before encapsulation (g)	Weight after encapsulation (g)	Weight of materials in filter (g)	Material content
Bitumen encapsulated CB	8.7	28.2	19.5	69.15%
Wax encapsulated CB	0.19	0.83	0.64	77.1%

5.2.3.2 Encapsulation material content

The average bitumen content of a CB encapsulated with bitumen was found to be 69.15%. In contrast, the wax content of a CB encapsulated with wax was found to be 77.1% (Table 5.2.3). There is a difference of 7.95% between the penetration percentage of the wax and bitumen encapsulated CBs. This indicates that the bitumen does not penetrate through the paper and/or the filter as effectively as the wax. This could be due to the much higher viscosity of bitumen than wax that makes it harder to penetrate, as one study found that the viscosity of bitumen had a large effect on its absorption into crumb rubber [217].

5.2.3.3 Water absorption

One of the vital properties of encapsulated CBs is water absorption, as the results will determine whether toxic chemicals and metals will escape if the material is penetrated by water. Dried, wax-encapsulated, and bitumen-encapsulated CB samples were completely submerged in water for 24 h and 48 hr to determine the degree of water absorption. To ensure the sample surfaces were completely dried after the soaking process, the samples were dried using a paper towel and then weighed. The water absorption values obtained could also be used to indicate the resistance to freeze-thaw action. A highly absorptive aggregate is unacceptable in hot mix asphalt, and if aggregate has a water absorption value of 5% or higher, it could be defined as a highly absorptive aggregate [218]. Table 5.2.4 presents the water absorption results for dried, bitumen encapsulated, and wax encapsulated CBs for 24 h and 48 h testing.

Table 5.2.4. Water absorption of dried, bitumen encapsulated and wax encapsulated CBs for 24-hr and 48-hr testing.

Properties		Dried CB		Bitumen encapsulated CB		Wax encapsulated CB	
		Test 1 24 h	Test 2 48 h	Test 1 24 h	Test 2 48 h	Test 1 24 h	Test 2 48 h
Dry mass	(g)	4.7	8.7	25.1	27.2	21.9	23.6
Mass, saturated surface dry	(g)	22.7	27.9	26	28.2	23	24.6
Water absorption	(%)	382.98	220.69	3.59	3.68	5.02	4.24
Water absorption drop	(%)	-	-	99	98	99	99

The results show that the encapsulation process of CBs significantly reduces the water absorption of dried CBs. For 24 h testing, the water absorption decreased from 382.98% for dried CBs to 3.59% and 5.02% for bitumen and wax encapsulated CBs, respectively. While for 48 h testing, the water absorption decreased from 220.69% for dried CBs to 3.68% and 4.24% for bitumen and wax encapsulated CBs, respectively. The results show that the encapsulated CBs behave as hydrophobic materials, meaning

water cannot be absorbed by the encapsulated CB filter. Hence, it can be deduced that the 3.59% and 5.02% (24 h) water absorption found for bitumen and wax encapsulated CBs is truly the surface water content. To place this in perspective, the water absorption drop was determined for both wax and bitumen encapsulated CBs relative to the water absorption for dried CBs (Table 5.2.4), and a 99% water absorption drop was discovered for CBs when encapsulated with bitumen and wax (24 h). The other 1% can be interpreted as the moisture content developed during testing.

5.2.3.4 Microstructural analysis

The penetration of the bitumen and wax during the encapsulation process is an important factor in achieving the minimum escape of toxic chemicals from within the encapsulated CBs. In this investigation, encapsulated CB samples were cut in half, and images were taken at three different magnifications to allow observation of the internal structure of dried, bitumen, and wax encapsulated CBs, as shown in Fig. 5.2.4.

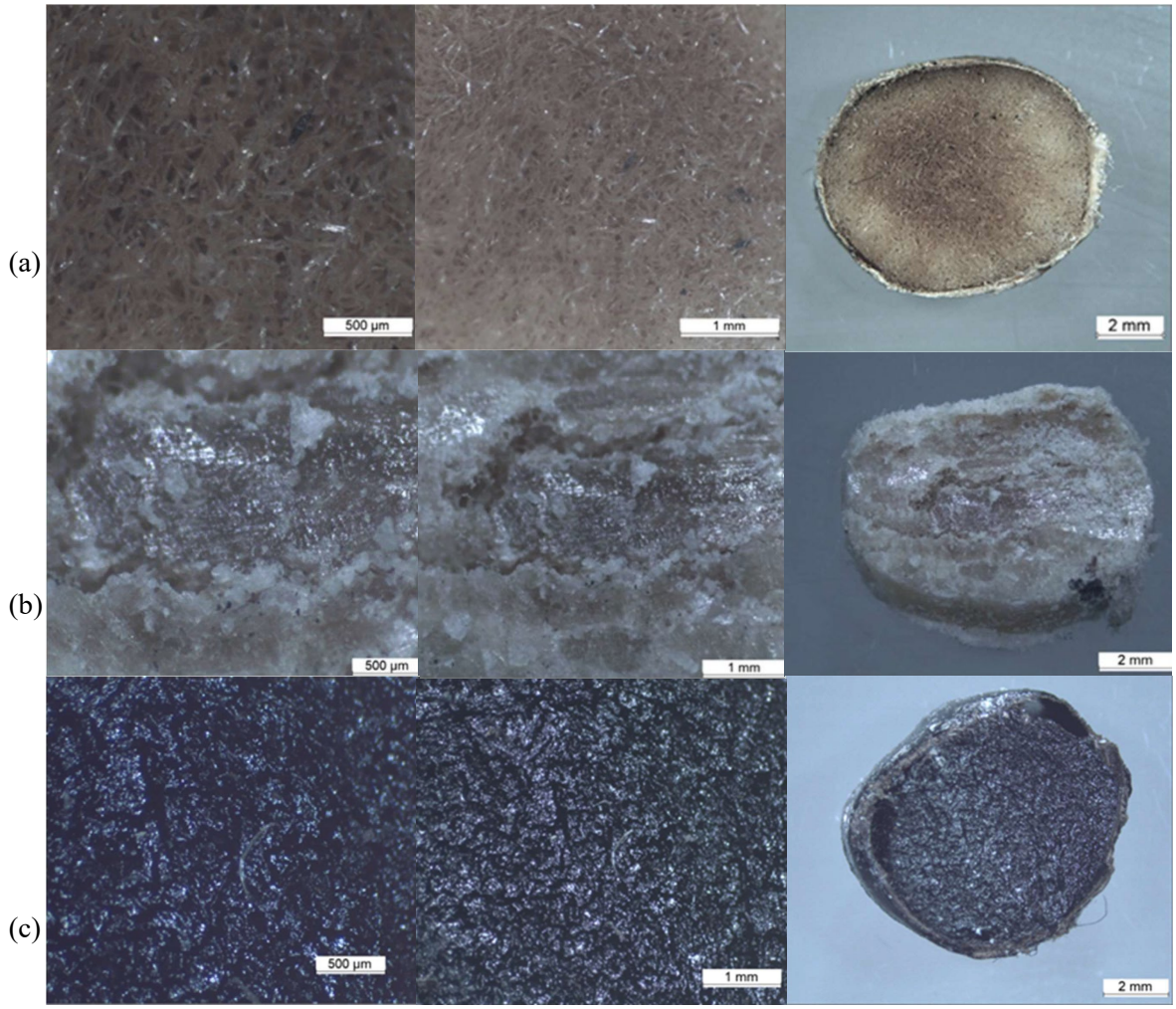


Fig. 5.2.4. CBs in an optical microscope (a) dried, (b) wax encapsulated, and (c) bitumen encapsulated.

The images taken by the microscope of the dried CBs show the fibrous nature of the cellulose acetate filter. There is also discoloration of the middle part of the filter due to being smoked. The internal surface of the bitumen and wax encapsulated CBs is characterized by an uneven and undulating surface with many small peaks and valleys. There are no breaks or discontinuities in the internal distribution of the bitumen except for around the edges of the CBs where the paper surrounds the filter. The 500 microns and 1 mm photos show that the bitumen and wax are very evenly spread throughout the entire filter and that the bitumen and wax have successfully penetrated the CB filter.

In addition to the microscopic images, Bruker SkyScan 1275 micro-X-ray Computed Tomography (CT) images were taken of a dried CB and bitumen encapsulated CB sample (Fig. 5.2.5). Fig. 5.2.6 represents random 2D section slices of a dried CB and bitumen encapsulated CB in which the black pixels represent pores. As shown in Fig. 5.2.6, the pores within the dried CB drastically decrease when encapsulated with bitumen. The encapsulation process effectively and completely penetrates the bitumen inside the filter and creates a dense and water-repellent material. Micro-X-ray CT provides data related to the quantity and orientation of pores within the structure. Porosity analysis is performed on white and black pixels within the object. The program can classify pores as being a number of black pixels encased by white pixels and differentiate between closed pores (bubbles) and open pores (connecting channels) [139]. Images from the micro-CT were used, and the solids and voids were separated to obtain the porosity; the porosity of selected samples was estimated in which a 66% porosity was found for the dried CB, and a 0.5% porosity was found for the CB encapsulated with bitumen. Hence, it can be seen that the encapsulation process significantly reduces the porosity.



Fig. 5.2.5. Bruker SkyScan 1275 X-ray micro-Computed Tomography

(a) (b)

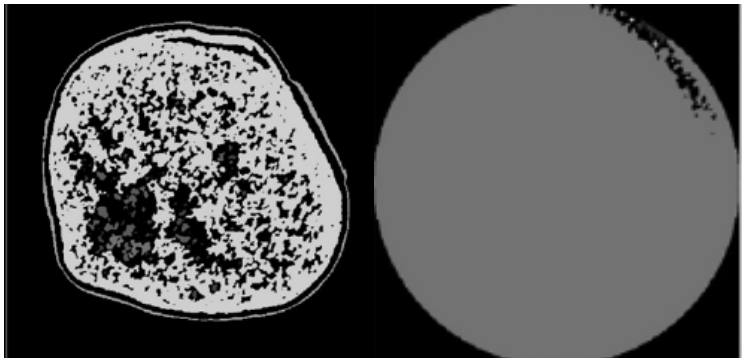


Fig. 5.2.6. CT-Scan model of (a) dried CB and (b) bitumen encapsulated CB.

5.2.3.5 Leachate analysis of CB samples

A leachate analysis was performed following the ABLP method to identify and quantify the metals leached from wax and bitumen encapsulated CBs, and analyse the relationship between the pH of the solution and the metal concentration leached, as well as to investigate whether the encapsulation process of CBs prevents or reduces the leachate of metals found within CBs. The selected metals for this study include Be, Mg, Al, Se, As, Pb, V, Mn, Fe, Co, Ni, Cu, Cr, Ag, Zn, Mo, Cd, Ti, Th, and U for wax and bitumen encapsulated CBs, and are insignificant.

Metals Al, Cr, Fe, Ni, and Cu, and metals Mg and K are presented separately due to the high variance in metal concentration in leachate. While the leachate concentrations for Al, Cr, Fe, Ni, and Cu are in the range of 0.001 to 0.954 mg L^{-1}, the leachate concentrations for Mg and K are between 3.442 and 105.316 mg L^{-1}. As demonstrated in Figs. 5.2.7 and 5.2.8, the findings imply that the variance in pH influences the behavior of the metal concentration leached from the bitumen and wax encapsulated CBs for both the broken and unbroken samples. Generally, pH 2.9 resulted in the highest metal contamination for all sample types, followed by pH 5.0 and pH 9.2. This implies that wax and bitumen encapsulated CBs are more susceptible to leaching metals in the presence of highly acidic water. This shows that different environmental conditions play a major role in the leaching behavior of CBs and validates that all leachate tests should be performed for various environmental scenarios.

Comparing the leachate concentrations of metals from wax and bitumen encapsulated CBs, wax encapsulated CBs depict higher leachate concentrations. For Al (pH 2.9), the leachate of bitumen encapsulated CBs for broken and unbroken is 0.263 mg L^{-1} and 0.230 mg L^{-1}, respectively, whereas, for wax encapsulated CBs, it is 0.954 and 0.943 mg L^{-1} for broken and unbroken, respectively. This is due to bitumen having a higher cohesiveness compared to paraffin wax, which assists in trapping metals more effectively.

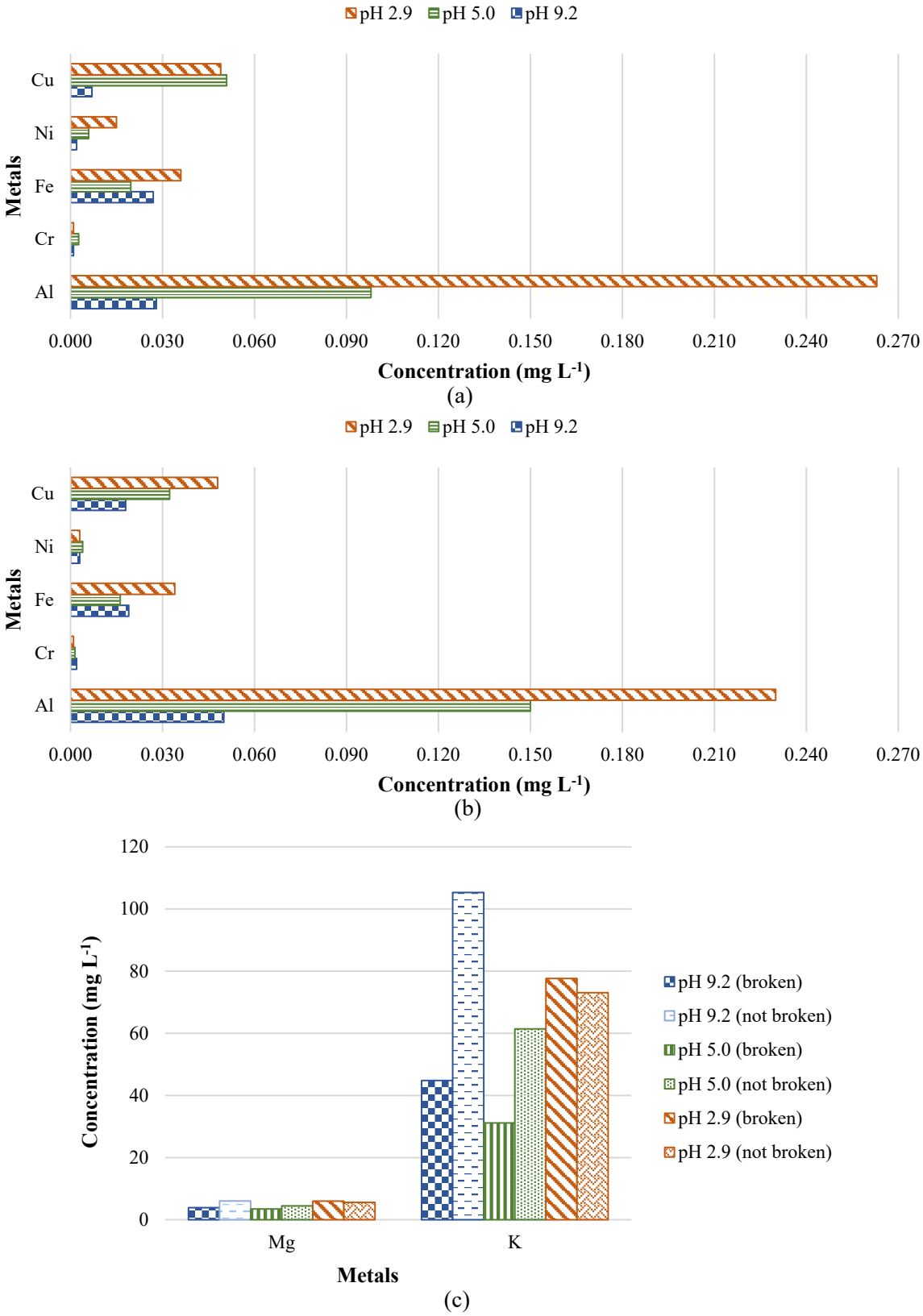
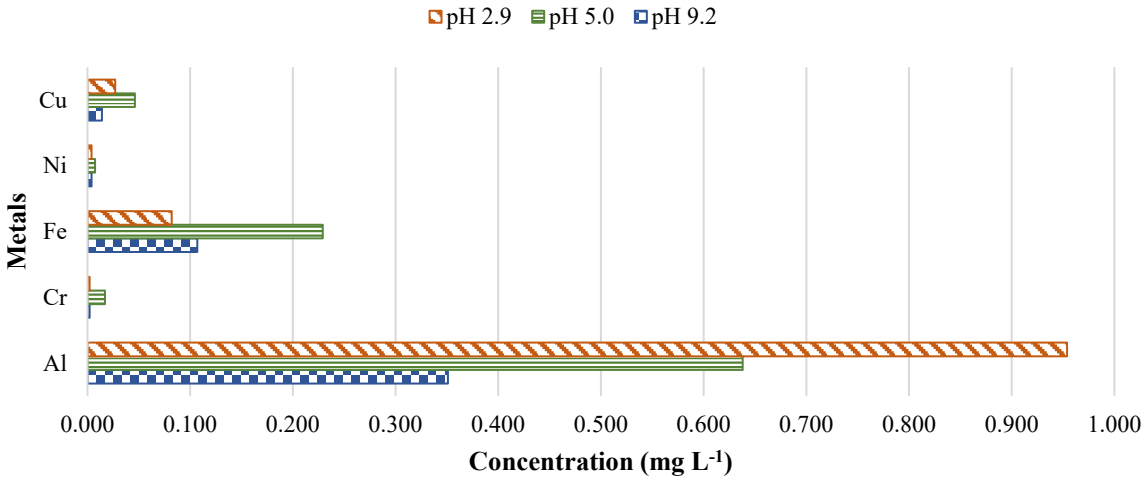
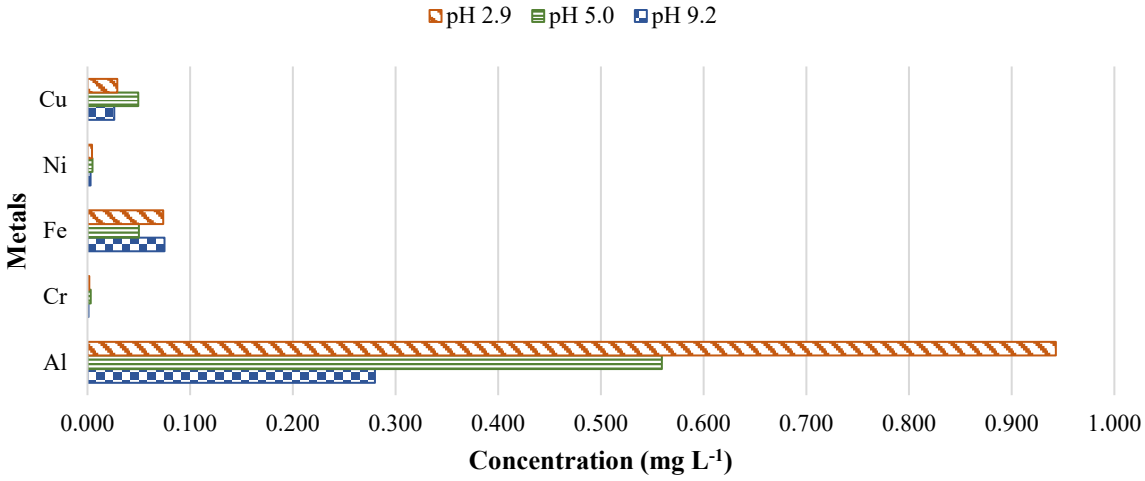


Fig. 5.2.7 Metal concentration of bitumen encapsulated CBs (a) broken, (b) unbroken, and (c) broken and unbroken samples.

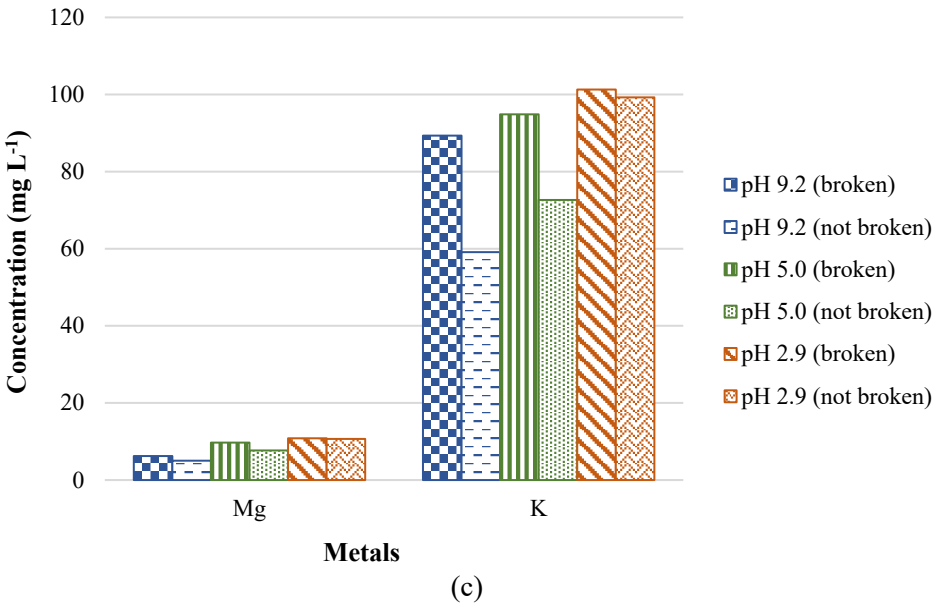
No relationship between the leachate concentrations of wax and bitumen encapsulated CBs for broken and unbroken was found. For pH 5.0, bitumen encapsulated CBs (broken) leached 0.098 mg L^{-1} for Al and 0.051 mg L^{-1} for Cu, whereas, for bitumen encapsulated CBs (unbroken), Al leached 0.150 mg L^{-1} and 0.032 mg L^{-1} for Cu. The results show that the damaged samples (broken) generally result in a higher leachate concentration than the undamaged samples (unbroken). This shows that the damage reduces the effectiveness of immobilizing the leachate of metals. However, it is to be noted that there is not a significant variance to the intact samples, with the largest drop in performance being only 2.63%. This shows that if the encapsulated CBs were to be damaged, they would still be able to contain the metals within the CB.



(a)



(b)



(c)

Fig. 5.2.8. Metal concentration of wax encapsulated CBs (a) broken, (b) unbroken, and (c) broken and unbroken samples.

Tables 5.2.5, 5.2.6, and 5.2.7 depict the leachate results for metals in bitumen and wax encapsulated CBs for pH values 2.9, 5.0, and 9.2, respectively. The leachate concentrations of wax and bitumen encapsulated CBs were compared to the Industrial Waste Resource Guidelines [173] and were found to be below the regulatory threshold limits for heavy metals for distinct pH values. To further highlight the triviality of the leachate concentrations, the results were compared to the Australian drinking water heavy metal guideline thresholds [213].

To investigate the variations in concentrations of dried CBs before encapsulation and after bitumen and paraffin wax encapsulation, the leaching drop was determined. The findings in Tables 5.2.5, 5.2.6, and 5.2.7 show that the leaching behavior of CBs is considerably lower after being encapsulated with wax and bitumen. This is the result of sealing the CBs with viscous and insoluble materials, which prevents the release of metals or the intrusion of any liquid. For bitumen and wax encapsulated CBs, the leaching of metals Be, Cd, Se, As, Ti, Mo, Pb, Th, U, and Ag were immobilized by 100% while metals Mg, K, Fe, Cr, Co, Al were hindered between 70% to 98% for pH 2.9. Cu demonstrated a leaching drop of 2% for bitumen encapsulated CBs, and, although the leaching drop is trivial, it must be noted that the leachate concentrations are less than 0.05 mg/L. The samples for pH 5.0 demonstrated similar leaching drops. Metals Be, Se, Cd, As, Ti, Mo, Th, U, and Ag were immobilized by 100% for both wax and bitumen encapsulated CBs. While pH 9.2 resulted in the most effective immobilization of metals, whereby all metals were hindered by more than 80% for both sample types. Encapsulating CBs resulted in a significant decrease of 100% immobilization of toxic metal contamination, which can be an environmental advantage compared to the littering of CBs. To investigate whether the condition of a

bitumen and wax encapsulated CB affects the leaching capability of the material, leachate tests were conducted for broken bitumen and wax encapsulated CB samples for pH values 2.9, 5.0, and 9.2. Similar results to those tabulated in Tables 5.2.5, 5.2.6, and 5.2.7 were found for broken bitumen and wax encapsulated CB samples. This indicates that when encapsulated CBs are incorporated in the construction of asphalt concrete, the possible damaged state (broken) will not affect the leaching behavior of the toxic waste.

Table 5.2.5. Leaching concentrations of dried, bitumen, and wax encapsulated cigarette butts for pH 2.9.

Metals	Concentration Level (mg L^{-1})		Percentage of leaching drop before and after encapsulation				
	Australian drinking water guidelines[1]	EPA Victoria solid waste thresholds[2]	Dried CBs (mg L^{-1})	Bitumen encapsulated CBs		Wax encapsulated CBs	
				Concentration (mg L^{-1})	Leaching Drop	Concentration (mg L^{-1})	Leaching Drop
Be	0.06	35	0.000	<0.001	100%	<0.001	100%
Mg	-	-	58.741	5.578	91%	10.658	82%
Cd	0.002	0.1	0.011	<0.001	100%	<0.001	100%
Se	0.01	0.5	0.001	<0.001	100%	<0.001	100%
K	-	-	563.256	73.057	87%	99.257	82%
As	0.01	0.35	0.003	<0.001	100%	<0.001	100%
V	-	-	0.004	<0.001	100%	0.001	70%
Cr	0.05	2.5	0.010	0.001	90%	0.002	82%
Ti	-	-	0.001	<0.001	100%	<0.001	100%
Cu	2	100	0.049	0.048	2%	0.029	41%
Fe	-	-	2.032	0.034	98%	0.074	96%
Mo	0.05	2.5	0.000	<0.001	100%	<0.001	100%
Co	-	-	0.009	0.001	89%	0.002	75%
Pb	0.01	0.5	0.004	<0.001	100%	<0.001	100%
Ni	0.02	1	0.016	0.003	81%	0.004	72%
Th	-	-	0.000	<0.001	100%	<0.001	100%
U	0.017	-	0.001	<0.001	100%	<0.001	100%
Al	-	-	3.643	0.230	94%	0.943	74%
Ag	0.1	5	0.000	<0.001	100%	<0.001	100%

[1]Australian Government [213]
[2]Industrial Waste Resource Guidelines [173]
Note: All data are means of three values

Table 5.2.6. Leaching concentrations and leaching drops of dried, bitumen, and wax encapsulated cigarette butts for pH 5.0.

Metals	Concentration Level (mg L^{-1})		Percentage of leaching drop before and after encapsulation (%)				
	Australian drinking water guidelines[1]	EPA Victoria solid waste thresholds[2]	Dried CBs (mg L^{-1})	Bitumen encapsulated CBs		Wax encapsulated CBs	
				Concentration (mg L^{-1})	Leaching Drop	Concentration (mg L^{-1})	Leaching Drop
Be	0.06	35	0.000	<0.001	100%	<0.001	100%
Mg	-	-	82.460	4.448	95%	7.661	91%
Cd	0.002	0.1	0.006	<0.001	100%	0.001	84%
Se	0.01	0.5	0.002	<0.001	100%	<0.001	100%
K	-	-	701.741	61.422	91%	72.667	90%
As	0.01	0.35	0.003	<0.001	100%	<0.001	100%
V	-	-	0.005	<0.001	100%	0.001	90%
Cr	0.05	2.5	0.011	0.001	87%	0.003	71%
Ti	-	-	0.000	<0.001	100%	<0.001	100%
Cu	2	100	0.080	0.032	60%	0.049	38%
Fe	-	-	1.358	0.016	99%	0.050	96%
Mo	0.05	2.5	0.001	<0.001	100%	<0.001	100%
Co	-	-	0.010	0.001	94%	0.001	87%
Pb	0.01	0.5	0.005	<0.001	100%	0.001	79%
Ni	0.02	1	0.019	0.004	80%	0.005	75%
Th	-	-	0.000	<0.001	100%	<0.001	100%
U	0.017	-	0.001	<0.001	100%	<0.001	100%
Al	-	-	2.775	0.150	95%	0.559	80%
Ag	0.1	5	0.000	<0.001	100%	<0.001	100%

[1]Australian Government [213]
[2]Industrial Waste Resource Guidelines [173]
Note: All data are means of three values

Table 5.2.7. Leaching concentrations and leaching drops of dried, bitumen, and wax encapsulated cigarette butts for pH 9.2.

Metals	Concentration Level (mg L^{-1})		Percentage of leaching drop before and after encapsulation (%)				
	Australian drinking water guidelines[1]	EPA Victoria solid waste thresholds[2]	Dried CBs (mg L^{-1})	Bitumen encapsulated CBs		Wax encapsulated CBs	
				Concentration (mg L^{-1})	Leaching Drop	Concentration (mg L^{-1})	Leaching Drop
Be	0.06	35	0.075	<0.001	100%	<0.001	100%
Mg	-	-	65.536	6.020	91%	5.025	92%
Cd	0.002	0.1	0.011	<0.001	100%	0.001	91%

Se	0.01	0.5	0.002	<0.001	100%	<0.001	100%
K	-	-	770.329	105.316	86%	59.106	92%
As	0.01	0.35	0.004	<0.001	100%	<0.001	100%
V	-	-	0.024	0.001	96%	0.001	96%
Cr	0.05	2.5	0.033	0.002	94%	0.001	97%
Ti	-	-	0.000	<0.001	100%	<0.001	100%
Cu	2	100	0.209	0.018	91%	0.026	88%
Fe	-	-	30.340	0.019	100%	0.075	100%
Mo	0.05	2.5	0.010	<0.001	100%	<0.001	100%
Co	-	-	0.019	0.001	95%	0.001	95%
Pb	0.01	0.5	0.035	0.005	86%	0.001	97%
Ni	0.02	1	0.066	0.003	95%	0.003	95%
Th	-	-	0.010	<0.001	100%	<0.001	100%
U	0.017	-	2.842	<0.001	100%	<0.001	100%
Al	-	-	16.156	0.050	100%	0.280	98%
Ag	0.1	5	0.001	<0.001	100%	<0.001	100%

[1]Australian Government [213]
[2]Industrial Waste Resource Guidelines [173]
Note: All data are means of three values

5.2.4 Conclusions and recommendations

5.2.4.1 Conclusion

The objective of this paper was to carry out a comprehensive leachate analysis to examine the effectiveness and efficiency of the encapsulation method for the prevention of leachates from CBs and investigate the physical properties of wax and bitumen encapsulated CBs. The leachate analysis was conducted using the Australian Bottle Leaching Procedure (ABLP) [212] for pH values 2.9, 5.0, and 9.2. The metals studied included Be, Mg, Al, Se, As, Pb, V, Mn, Fe, Co, Ni, Cu, Cr, Ag, Zn, Mo, Cd, Ti, Th, and U. Numerous experiments were conducted in accordance with the appropriate Australian standards to determine the physical properties of wax and bitumen encapsulated CBs. Based on the experimental studies, the following conclusions can be drawn:

- The wax content of wax encapsulated CBs was found to be 77.1%, and the bitumen content of bitumen encapsulated CBs was found to be 69.15%, thereby demonstrating the effectiveness of the encapsulation process.
- For 24 h testing, water absorption decreased from 382.98% for used CBs to 3.59% and 5.02% for bitumen and wax encapsulated CBs. That is a water absorption drop of 99% for bitumen and wax encapsulated CBs. Therefore, it can be presumed that the water absorption value is relative to the surface water content (moisture content) and, essentially, not the absorption of the filter.

- Water penetration plays a vital role in spreading leachate containing toxic materials into the environment. Water absorption of only 3.59% for bitumen encapsulated CBs is exceptionally low, and when placed in an AC mixture, bitumen encapsulated CBs will be encapsulated with bitumen consecutively. Therefore, signifying an almost 0% leachability possibility of toxic chemicals and metals during the use and disposal of the waste.
- Micro-X-ray CT imaging was used to study and analyze the porosity and pore structure of CBs, and a substantial decrease in porosity from 66% to 0.5% was determined when CBs were encapsulated with bitumen.
- The leachate concentrations of metals Be, V, Co, As, Se, Mo Ag, Cd, Ti, Pb, Th, and U for bitumen and wax encapsulated CBs were found to be below 0.001 mg/L, and, therefore, are of insignificant concern.
- The findings suggest that the variance in pH influences the behavior of the metal concentration leached from bitumen and wax encapsulated CBs for both broken and unbroken samples. Generally, pH 2.9 resulted in the highest metal contamination for all sample types, followed by pH 5.0 and pH 9.2.
- In comparing the leachate concentrations of heavy metals from wax and bitumen encapsulated CBs, generally, wax encapsulated CBs depicted higher leachate concentrations. This is due to bitumen having a higher cohesiveness compared to paraffin wax, which assists in trapping heavy metals more effectively
- The results showed that damaged samples (broken) generally result in a higher leachate concentration than undamaged samples (unbroken). For pH 5.0, bitumen encapsulated CBs (broken) leached 0.098 mg L^{-1} for Al and 0.051 mg L^{-1} for Cu, whereas, for bitumen encapsulated CBs (unbroken), Al leached 0.150 mg L^{-1} and Cu leached 0.032 mg L^{-1}. This shows that the damage reduces the effectiveness of immobilizing the leachate of metals.
- The leachate results for wax and bitumen encapsulated CBs were found to be insignificant compared to the limits set by the EPA Victoria solid industrial waste guidelines and, therefore, can be categorized as non-hazardous.
- The findings show that the leaching behavior of CBs is considerably lower after being encapsulated with wax and bitumen. This is the result of sealing the CBs with viscous and insoluble materials, which prevents the release of metals or the intrusion of any liquid.
- Many health and safety issues arise from the storage of CBs, the process of encapsulating CBs, and their inclusion into manufactured materials. Due to their toxic nature, correct OH&S processes need to be pursued with all aspects of the encapsulation process.
- This research has revealed unknown physical properties like mass, density, encapsulation materials content, and water absorption of wax encapsulated CBs and bitumen encapsulated

CBs. Microscopic analysis has confirmed the uniformity of the encapsulation process. Comprehensive leachate analysis has proved the efficacy of the encapsulation method in reducing the leaching of heavy metals from encapsulated CBs. As heavy metal leaching is one of the key environmental issues worldwide, this research has provided a sustainable solution that will contribute to solving global cigarette butt pollution. This research has also reviewed a potential application of encapsulated CBs in asphalt concrete, and promising results from laboratory investigation have supported the recycling method of encapsulated CBs in asphalt concrete.

5.2.4.2 Recommendations

The following recommendations are made based on the current study:

- The leachate testing performed should be repeated to confirm the results and trends obtained. In addition, different damage scenarios could be tested to more accurately show the effect on leachate properties. Most importantly, when they become available, testing methods that are applicable to solid wastes should be used to repeat the testing in this paper to achieve more accurate leachate properties.
- Only one type of bitumen was used in this research, bituminous macadam base course A10E. This could be an important factor in the properties of the final encapsulated CB, so investigations need to be made into whether the grade of bitumen affects the penetration, water absorption, and, in turn, the leachate properties.
- The interaction between encapsulated CBs added to manufactured materials and how this affects the physical and chemical properties was found in this research. Changes to the encapsulation process, such as the removal of paper around the filter, are also important to investigate as such factors affect the water absorption and material content properties and could also affect the leachate properties. Testing whether the compression of the CB filters before encapsulation could affect different properties is another interesting factor.
- Finally, although a method of encapsulation that results in negligible water absorption for bitumen has been developed in laboratory conditions, further research is required into the feasibility and method required for a large-scale industrial approach to the recycling of CBs through the encapsulation process.

Chapter 6: Polycyclic Aromatic Hydrocarbons (PAHs) in Fired Clay Bricks Incorporating Cigarette Butts

Abstract

Cigarette butts (CBs) are the most common littered waste in the world and may contain polycyclic aromatic hydrocarbons (PAHs) from the incomplete combustion of tobacco during burning. Therefore, to investigate the potential PAH residual remaining in fired clay bricks (FCBs) incorporating CBs and examine the environmental impact of utilizing toxic waste in the production of FCBs, a comprehensive PAH extraction analysis was conducted. The Soxhlet extraction method was utilized to conduct a qualitative and quantitative analysis of sixteen toxic Environmental Protection Authority (EPA) Polycyclic Aromatic Hydrocarbons (PAHs) in FCB samples incorporating CBs using gas chromatography-mass spectrometry (GC–MS). The concentrations of the mean total (Σ)PAHs for FCBs incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by weight (wt) were found to be 0.183, 0.180, 0.242, 0.234, and 0.463 μg mL^{-1}. As expected, PAHs with higher water solubility and volatility, naphthalene, fluorene, anthracene, pyrene, fluoranthene, and chrysene were found at higher concentrations compared to lipophilic PAHs. The ΣPAH concentrations for all five FCB–CB mixes were well below the EPA Victoria solid waste hazard categorization threshold for industrial waste. Moreover, the samples were studied for their carbon content using the carbon, hydrogen, nitrogen, and sulfur (CHNS) analyzer and thermogravimetric analysis (TGA). The results confirm an almost 100% combustion process of CBs during the firing process. A content less than 0.3% suggests that all carbon within the FCB–CB mixture relatively disappeared during the firing process up to 1050 °C. However, further research regarding the emission of volatile organic compounds (VOCs) during the production of FCBs incorporating CBs should be conducted.

The research presented in this chapter has been published in the following:

Kurmus, H, Mohajerani, A and Grist, S. 2021. Polycyclic aromatic hydrocarbons in fired clay bricks incorporating cigarette butts. Materials, 14(8), 2032. https://doi.org/10.3390/ma14082032.

6.1 Introduction

Globally, over 5.4 trillion cigarettes are produced, and approximately one-third of cigarette butts (CBs) are littered into the nearby environment [125, 219, 220]. In Australia, CBs are the most common littered waste under the miscellaneous category, representing 91.5% [8, 125]. When littered, CBs persist in the environment for up to 10 years due to the composition of the filter. The filter of a CB is made from cellulose acetate, and, although cellulose is biodegradable, the acetate content prevents the cellulose from biological decomposition in an open environment [40, 125]. A study conducted in a controlled laboratory condition found that it took up to 720 days for a CB to lose 30% to 35% of its total weight while decomposing [221].

In cigarette smoke, it is possible to classify more than 4000 chemical components generated during burning or distilled from the tobacco [125]. Essentially, the role of the filter of the CB is to trap and absorb the particulate smoke components, consisting of more than 3500 non-volatile and semi-volatile chemical compounds. The main toxic compounds include polycyclic aromatic hydrocarbons (PAHs), nicotine, metals, catechols, carbonyls, alcohols, nicotine alkaloids, and compounds specific to Solanaceae [11, 30]. Therefore, when CBs are improperly disposed of, they pose a toxicant risk to the global environment and urban and aquatic life, whereby harmful chemicals are leached [50, 149]. The leachate of the toxic compounds may eventually contaminate surface water and groundwater [151, 222], and ultimately bioaccumulate in the human food chain and wildlife [17, 18].

Based on several years of research, Mohajerani et al. [1] proposed that the CB conundrum can potentially be eliminated by incorporating 1% CBs into 2.5% of the world's total FCB production. Various mechanical and physical tests were conducted to determine the practicality and competency of incorporating toxic waste in the production of FCBs, and the results were promising [22, 23, 115, 157, 176]. The tests included compressive strength, density, water absorption, initial rate of absorption, efflorescence, shrinkage, energy savings, thermal conductivity, and gas emissions. The research continued with the implementation phase of recycling CBs in FCBs, whereby several sterilization methods and odor elimination techniques were reviewed, CB collection systems were explored, and an implementation procedure was proposed on an industrial scale [2]. However, an environmental issue that may evolve through the addition of CBs in the construction of FCBs is the potential leachate of heavy metals and toxic chemicals during the use and disposal of the material. In 2020, Kurmus and Mohajerani Kurmus and Mohajerani [126] conducted a comprehensive leachate study of heavy metals from FCBs incorporating various percentages of CBs. The modified FCBs were found to be below the regulatory threshold limits and categorized as non-hazardous waste when compared to the EPA Victoria industrial waste guidelines. Regarding PAHs in FCBs modified with CBs, no studies have been conducted.

PAHs in CBs are dangerous compounds produced through the incomplete combustion and pyrolysis of tobacco. They consist of two or more aromatic rings and are hydrophobic in nature with very low water solubility. PAHs are a concern, considering that they bioaccumulate and linger in the environment, with carcinogenic, teratogenic, and mutagenic properties [223, 224]. In 2008, 28 PAHs were identified as priority pollutants by the National Waste Minimization Program, a project which was funded by the US Environmental Protection Agency [225]. In a study conducted by Dobaradaran et al. Dobaradaran et al. [222], it was found that PAHs, particularly fluorine, acenaphthylene, naphthalene, and acenaphthene, are released into the environment from freshly littered CBs. In addition to CBs, PAH contamination in soil is a common issue, because soil is less receptive to chemical and biological degradation and the PAHs tend to absorb tightly into the organic matter present in the soil. Moreover, the recalcitrance of PAHs towards treatment increases with prolonged aging time, promoting the sequestration of PAH molecules into micropores [226].

Thus, the aim of this study was to conduct a comprehensive evaluation of PAHs in FCBs incorporated with CBs to examine the potential environmental impact of utilizing this toxic waste in the production of FCBs. In this study, a direct Soxhlet extraction method was employed using dichloromethane as the solvent. The aim of the extraction process is to effectively remove the analyte (PAHs) from its matrix, with minimal solvent usage [227]. The extract was analyzed for sixteen US EPA priority PAH compounds using gas chromatography–mass spectrometry (GC–MS) in accordance with the US EPA Method 3540C [228]. The PAH concentrations were then compared to the Environmental Protection Authority (EPA) Victorian solid industrial waste hazard categorization and management thresholds [173] to assess the suitability of FCBs incorporating CBs. Furthermore, the FCB samples incorporating CBs were studied for their carbon content using the carbon, hydrogen, nitrogen, and sulfur (CHNS) analyzer and thermogravimetric analysis (TGA).

6.2 Materials and Methods

6.2.1 Materials and Brick Manufacturing Methodology

Soil and CBs are the raw materials that were used in this study. A sieve analysis was conducted to classify the soil, and, according to unified soil classification, the soil was categorized as sandy silty clay (MC) [127]. The soil material was obtained from PGH Bricks and Pavers (Melbourne, Australia) and the CBs, of distinct brands and sizes, were provided by Butt Out Australia Pty Ltd (Melbourne, Australia). The raw materials, CBs and soil, were initially oven-dried at 105 °C for 24 h. Various percentages (0%, 0.5%, 1%, 1.5%, and 2% CBs by weight (wt)) of CBs were then added and uniformly mixed with the soil for 25 min in a mechanical mixer at a moisture content of 15.5%. The fast-mixing process caused the CBs to shred and homogenously integrate into the soil mixture. Prior to being placed in the oven at 105 °C for 24 h, the samples were compacted in cylindrical molds at a compaction

pressure of 240 kPa. The final step involved firing the brick samples in the furnace at 1050 °C for 3 h. For each batch, three replicate samples were prepared, tested, and the mean value was reported. The compressive strength, water absorption, initial rate of absorption, shrinkage, and density property results for the FCBs incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs by wt. can be found in Kurmus and Mohajerani Kurmus and Mohajerani [115].

The chemical and mineralogical composition of the CB brick samples were determined through AXS D8 Endeavor wide-angle X-ray diffractometer (XRD) (Bruker, MA, USA) and AXS S4 Pioneer spectrometer X-ray fluorescence (XRF) (Bruker, MA, USA). Powder samples were prepared by crushing and sieving the FCB samples incorporating CBs to 10 μm in size. The XRD was operated at 45 kV and 35 mA in reflection scanning mode from 6° to 90° for a total scan time of 29 min.

6.2.2 Carbon Analysis

The CHNS-2400 Elemental Analyser (PerkinElmer, MA, USA) was used to analyze the total carbon, hydrogen, nitrogen, and sulfur content in FCBs incorporating 0%, 0.5%, 1%, 1.5%, 2%, and 5% CBs by wt. The purpose of this investigation was to determine whether carbon was trapped within the core of the brick structure from the potential incomplete combustion of CBs during the firing process. Therefore, for this experiment, only the extracted core samples were analyzed. The core of the 100-mm-diameter bricks was extracted using a 50-mm drill-saw, as shown in Fig. 6.1. The extracted core bricks were crushed and sieved through a 1.14-mm sieve, and the quadrat sampling method was employed to select a homogeneously distributed sample for testing.

CHNS analysis was conducted by combusting samples of 5 mg at a high temperature of 850 °C in a chamber in the presence of oxygen. The occurrence of oxygen allows individual elements to combine with oxygen and form gaseous byproducts and water vapor such as nitrogen (N_2), sulfur dioxide (SO_2), water (H_2O), and carbon dioxide (CO_2). The gases are then captured in the gas control zone and homogenized. The homogenized gases are depressurized through a column, and the separation approach is used to elute the gases, which are measured and detected by a thermal conductivity detector. In this study, three replicate samples were prepared, and the mean average was reported.

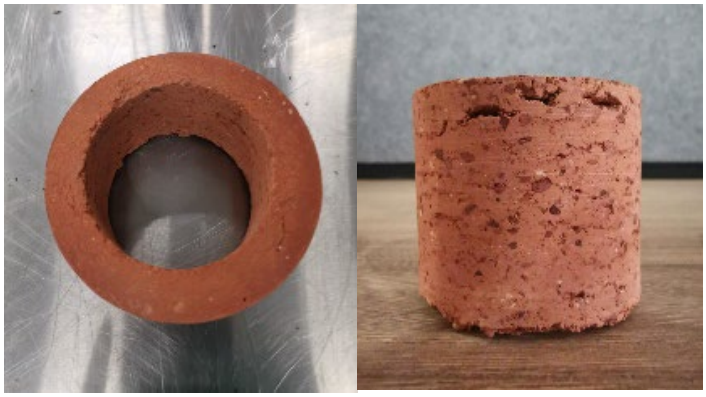


Fig. 6.1. The core of the brick extracted for CHNS analysis.

Thermogravimetric analysis (TGA) and differential thermal analysis (DTA) were performed using the TGA 8000 Autosampler (PerkinElmer, MA, USA) to monitor the mass loss of the brick samples incorporating CBs. For the thermal analysis, the samples were heated from 30 °C to 850 °C at a constant heating rate of 20 °C min^{-1} in a 15 mL-per-min flow of nitrogen. For the purpose of this investigation, the brick samples were crushed to a particle size of <75 µm. All samples were tested under the same conditions.

6.2.3 Polycyclic Aromatic Hydrocarbon (PAH) Analysis

6.2.3.1 Extraction Method

The Soxhlet extraction technique is a widely employed method to extract PAHs from sediments and soils [229]. Therefore, the brick samples were extracted in accordance with the USEPA Soxhlet Extraction Method 3540C [228]. In the basic Soxhlet extraction test, 15 g of crushed brick samples of particle size <1 mm is placed into a cellulose thimble, which is then extracted using 300 mL of solvent via the reflux cycle. In this case, dichloromethane was used as the solvent. The whole assembly was heated for 8 h using an isomantle. The extracts from the Soxhlet extractor were concentrated to 10 mL using the rotary evaporator and then diluted. The concentrated samples were mixed with five isotopically labeled internal standard solutions to make a final concentration of 10 µg mL^{-1}. The samples were stored at 4 °C for GC–MS measurement.

6.2.3.2 Calibration Standards

PAH calibration standards, including five types of isotopically labeled internal standards and 16 types of PAHs, were used for quantification. The PAH mix (2000 µg/mL) was purchased from Supelco and included acenaphthene, dibenz[*a*,*h*]anthracene, anthracene, chrysene, benz[*a*]anthracene, benzo[*b*]fluoranthene, fluorene, benzo[*k*]fluoranthene, naphthalene, benzo[*a*]pyrene, phenanthrene, benzo[*ghi*]perylene, pyrene, fluoranthene, acenaphthylene, and indeno [1,2,3-*cd*]pyrene. The internal

standards (2000 µg/mL) were also purchased from Supelco and included benz[*a*]anthracene-d12 solution, chrysene-d12 solution, naphthalene-d8 solution, perylene-d12 solution, and phenanthrene-d10 solution.

Five-point standard calibration solutions were prepared to ensure the accuracy and precision of the PAHs for quantitative and qualitative analysis. The standard calibration solutions were measured at levels ranging from 1.25 to 25 µg mL^{-1} and internal standards of 10 µg/mL. Each analytical series included one blank test. The 16 PAHs and five internal standards in the samples were defined by the precursor ions and retention time, as shown in Table 6.1.

Table 6.1. Physico-chemical characteristics and optimized MRM acquisition parameters of target PAHs.

Compound	Retention Time (min)	Precursor Ion (m/z)	Product Ion (m/z)	No. of Aromatic Rings	Water Solubility (mg L^{-1})
Naphthalene	5.51	128	128/128	2	31.69
Naphthalene-d8 (IS)	5.49	136	108/136		
Acenaphthylene	7.07	152	152/152	3	3.93
Acenaphthene	7.24	153	153/153	3	3.93
Fluorene	7.73	166	165/166	3	1.68–1.98
Anthracene	8.66	178	178/178	3	0.0446
Phenanthrene	8.71	178	178/178	3	1–1.6
Phenanthrene-d10 (IS)	8.64	188	160/188		
Pyrene	9.83	202	202/202	4	0.129–0.165
Fluoranthene	10.05	202	202/202	4	0.206
Chrysene	11.22	228	228/228	4	0.0015–0.0022
Chrysene-d12 (IS)	11.20	240	236/240		
Benz[*a*]anthracene	11.26	228	228/228	4	0.011
Benz[*a*]anthracene-d12 (IS)	11.23	240	236/240		
Perylene-d12 (IS)	11.75	241	241/242		
Benzo[*b*]fluoranthene	12.21	252	252/252	5	0.0012
Benzo[*a*]pyrene	12.24	252	252/252	5	0.0038
Benzo[*k*]fluoranthene	12.52	252	252/252	5	0.0008
Benzo[*ghi*]perylene	13.72	276	276/276	6	0.00026
Dibenz[*a,h*]anthracene	13.74	278	278/278	6	0.0005
Indeno[1,2,3-*cd*] pyrene	14.05	276	276/276	6	0.062

6.2.3.3 GC–MS Analysis

A GC–MS system connecting a gas chromatograph with a triple-quadrupole mass spectrometer and equipped with an auto-sampler and sample preparation robot (Agilent 7000 Series GC–MS triple quad system) was used to analyze the PAH compounds. Two Agilent DB-5MS columns 5 m long and 0.25 mm in diameter were used with mid-column backflush. The multi-mode injector was operated in solvent elimination mode with injection at 5 °C, ramping at 600 °C min^{-1} to 325 °C. The oven temperatures were between 5 and 325 °C after an initial holding time of 1 min at 25 °C min^{-1}. To attain an initial oven temperature of 5 °C, CO_2 was used as the carrier gas for cooling at a constant flow of 3 mL min^{-1}. The cryogenics mode offered the ultimate conditions for the solvent vent mode. The injection volume was set to 0.2 µL and the electronic energy to 70 eV. Each run took approximately 17 min.

6.3 Results and Discussion

6.3.1 Characterization of Brick Samples

The chemical composition of the raw clay material and bricks incorporating CBs is presented in Table 6.2. The data provided in Table 6.2 show that the major compounds in the raw clay are silicon dioxide (SiO_2), saluminum oxide (Al_2O_3), iron(III) oxide (Fe_2O_3), and potassium oxide (K_2O). Similarly, the chemical compounds found in the FCB samples incorporating various percentages of CBs were SiO_2, Al_2O_3, Fe_2O_3, and K_2O. The XRD analysis results for the raw clay soil and FCBs specimens incorporating CBs are presented in Fig. 6.2. The XRD peak at 26.7° indicates that the soil and brick samples incorporating CBs are mainly made of quartz, which is consistent with the XRF data presented in Table 6.2, demonstrating the presence of a large amount of silica. The second-highest peak corresponding to the raw clay soil is muscovite, a hydrated phyllosilicate mineral of aluminum and potassium, which are present in quantities of 19.95% and 2.87%, respectively, in the XRF results.

Table 6.2. Chemical composition of raw clay soil and brick samples incorporating CBs.

Component	Raw Clay Soil (%)	Percentages of CBs by wt.				
		0%	0.50%	1%	1.50%	2%
Na_2O	2.46	0.5	0.4	0.4	0.4	0.5
MgO	1.62	1.1	1.1	1.2	1.1	1.2
Al_2O_3	19.95	16.8	16	16.5	16.3	16.8
SiO_2	62.84	53.8	52.6	54.8	53.2	55.3
P_2O_5	-	0.2	0.2	0.2	0.3	0.3
K_2O	4.87	3.5	3.4	3.4	3.5	3.5
CaO	0.27	0.2	0.2	0.3	0.3	0.4
TiO_2	1.19	0.9	0.9	0.9	0.9	0.9

Fe_2O_3	6.15	6.9	6.9	6.8	6.8	6.7

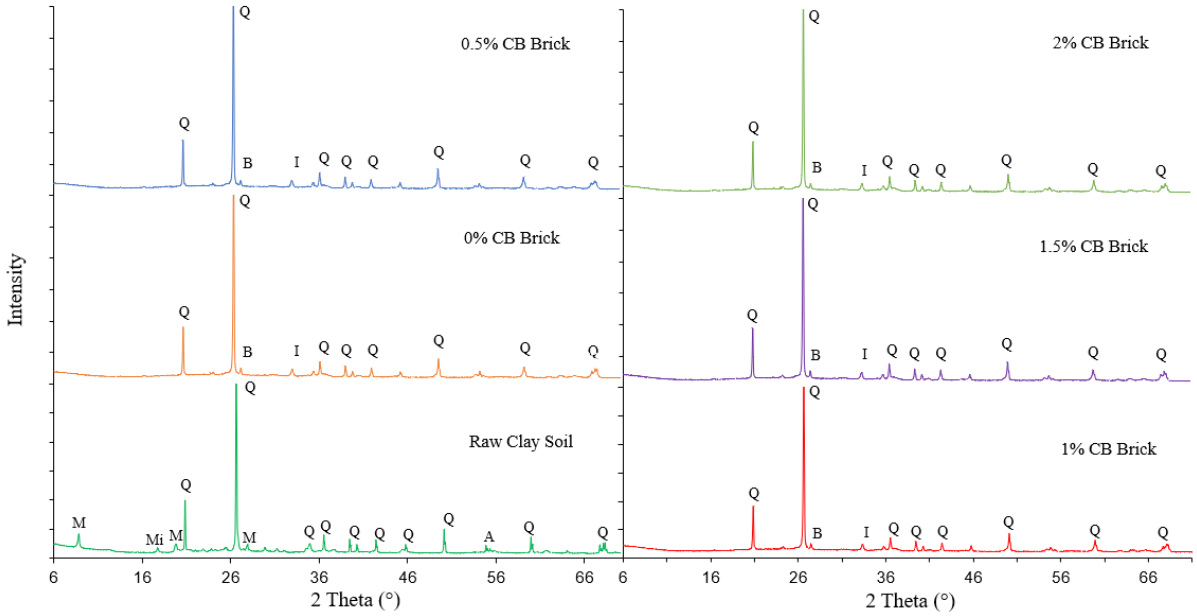


Fig. 6.2. XRD spectra of raw clay soil and FCBs incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs. Q: Quartz, M: Muscovite, Mi: Mikasite, A: Albite, B: Bismuth oxide, I: Iron chromium oxide.

6.3.2 Carbon Analysis

CBs contain a high content of carbon, which may stem from the unburnt tobacco, the wrapping paper, or the cellulose acetate filter, and when they are incorporated into the production of FCBs, carbon may be trapped within the brick structure. Therefore, to investigate whether the CBs within the brick structure completely burn off during firing, a CHNS and TG analysis was conducted. The results pertaining to organic CHNS contents and the mass loss from the TGA are shown in Table 6.3. As expected, FCBs incorporating 5% CBs contain the preeminent carbon content of 0.29% due to the high CB content in the FCB, and, although it is the highest, the effect will not be significant because of the very low percentage. A carbon content less than 0.3% suggests that all carbon within the FCB–CB mixture relatively disappeared during the firing process up to 1050 °C. As displayed in Table 6.3, the samples contained a high content of nitrogen, which varied from 3.74% to 6.36% in the FCB samples containing between 0% and 5% CBs. This can be explained by the fact that nitrogen is the most abundant element in the atmosphere, and it is present in organic materials—in this case, soil—which then become available to plants, such as tobacco [230]. For the hydrogen and sulfur contents, all samples were found to contain 0%.

The TGA results of the brick samples incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs demonstrated mass losses of 0.41%, 0.47%, 0.54%, 0.65%, and 0.68%. The weight loss between 50 and 120 °C is

mainly due to the removal of moisture, and the mass loss from 230 to 430 °C is primarily due to the decomposition and combustion of organic matter remaining from the incomplete combustion of the CBs or soil during the firing process at 1050 °C [231]. While the CHNS analyzer solely measures the carbon content, the mass loss from the TGA comprises both the carbon (combustion of organic matter) and moisture content, which is the source of the deviation between the results.

Table 6.3. Concentration of carbon, hydrogen, nitrogen, and sulfur content and mass loss in bricks incorporating 0%, 0.5%, 1%, 1.5%, 2%, and 5% CBs by wt.

Mixture Identification (%)	CHNS Results				TGA Results
	Carbon (%)	Hydrogen (%)	Nitrogen (%)	Sulfur (%)	Mass Loss (%)
CB (0)	0.02	Not detected	3.74	Not detected	0.41
CB (0.5)	0.03	Not detected	4.74	Not detected	0.47
CB (1.0)	0.02	Not detected	6.01	Not detected	0.54
CB (1.5)	0.01	Not detected	6.16	Not detected	0.65
CB (2.0)	0.04	Not detected	4.88	Not detected	0.68
CB (5.0)	0.29	Not detected	6.36	Not detected	-

6.3.3 Polycyclic Aromatic Hydrocarbon (PAH) Analysis

The selected GC–MS quantification ion chromatograms of 16 PAHs in a standard mixture can be seen in Fig. 6.3. The chromatogram displays excellent sensitivity, separation, and peak shape. The internal standards naphthalene-d8, phenanthrene-d10, perylene-d12, benz[*a*]anthracene-d12, and chrysene-d12 were used to quantify naphthalene, acenaphthene, acenaphthylene, anthracene, fluoranthene, fluorene, phenanthrene, pyrene, benz[*a*]anthracene, chrysene, benzo[*b*]fluoranthene, benzo[*k*]fluoranthene, benzo[*ghi*]perylene, benzo[*a*]pyrene, dibenz[*a*,*h*]anthracene, and indeno[1,2,3-*cd*]pyrene. The results behaved linearly over a concentration range from 1.25 µg/mL to 25 µg/mL for all 16 PAHs.

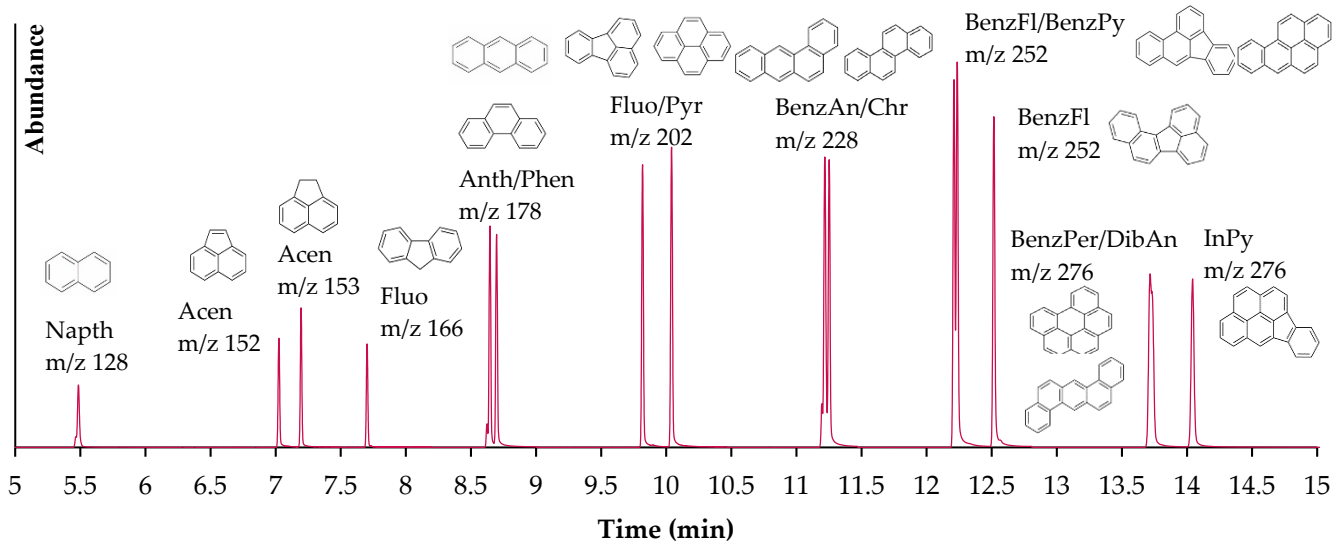


Fig. 6.3. GC–MS chromatogram at 25 µg mL^{-1} in DCM for 16 PAHs.

The results of the FCBs incorporating CBs for 16 PAHs are illustrated in Fig. 6.4. Currently, there are no similar studies involving the extraction of PAHs from FCBs or wastes recycled in FCBs to compare the results. However, research on the emission of PAHs from littered CB samples has been performed, whereby Dobaradaran et al. Dobaradaran et al. [222] found a mean ΣPAH concentration level of 24.6 µg mL^{-1} in freshly smoked CB samples and 20.4 µg mL^{-1} in city CB samples. When compared to the current study, the mean ΣPAH content in FCBs incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs was found to be at concentrations of 0.183, 0.180, 0.242, 0.234, and 0.463 µg mL^{-1}. It is apparent by incorporating CBs into the production of FCBs that a major decrease in PAH emissions into the environment is achievable. However, the major decrease in PAH concentrations may be due to several reasons, including the processes involved in manufacturing the FCB–CB samples. The brick manufacturing methods include oven drying, mixing (with water and soil), compacting, and firing, which may all result in the dissolution or evaporation of PAHs. This is further confirmed by the revised study [222], which found naphthalene to have the highest concentration level in both collected CBs from city streets and freshly smoked CB samples. In this current study, naphthalene was the second/third most prominent PAH for all sample types. The variations may be from the higher solubility of naphthalene, and thus the potential dissolution or evaporation of the PAH during the production processes, those factors being high temperatures or high-speed mixing.

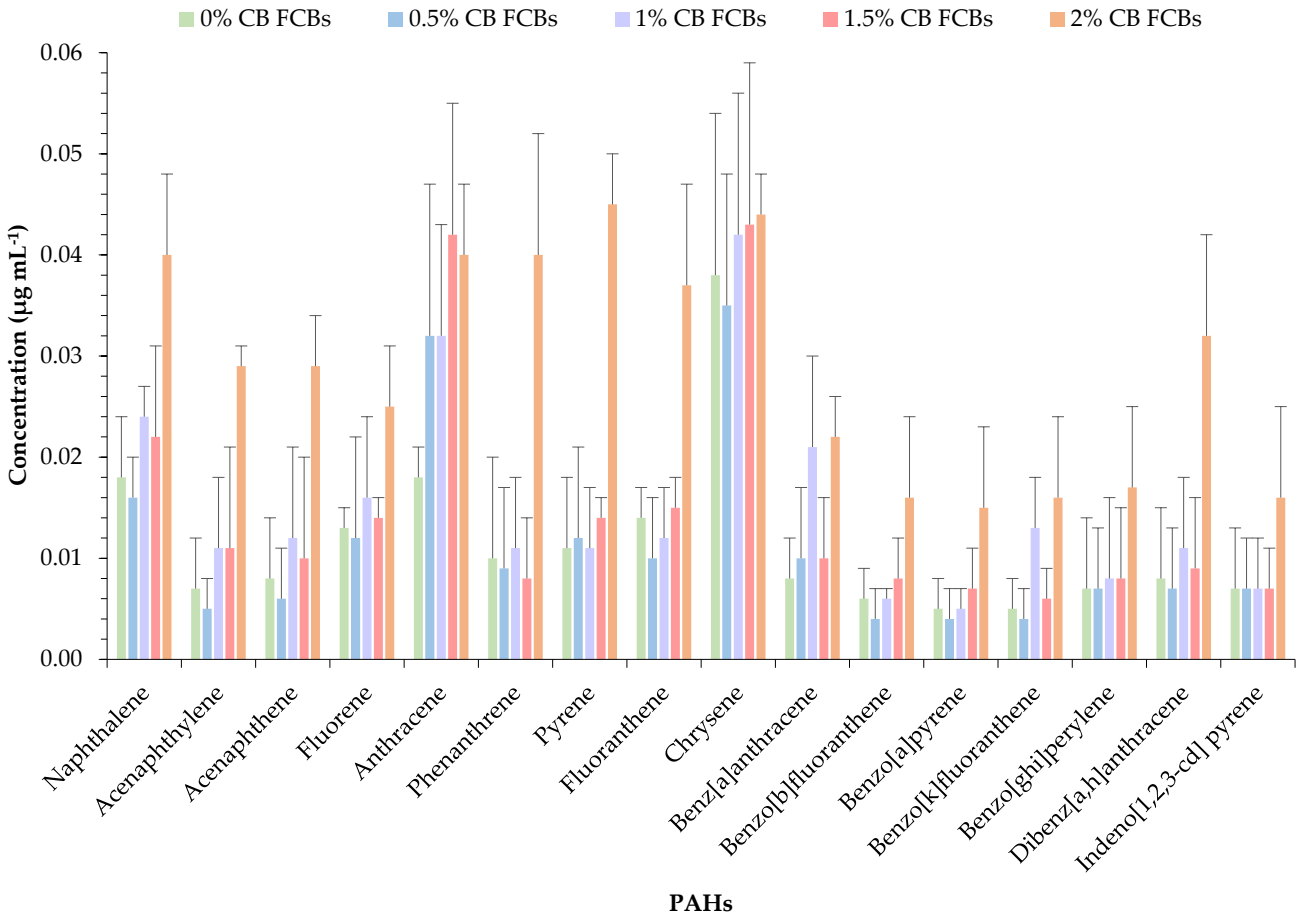
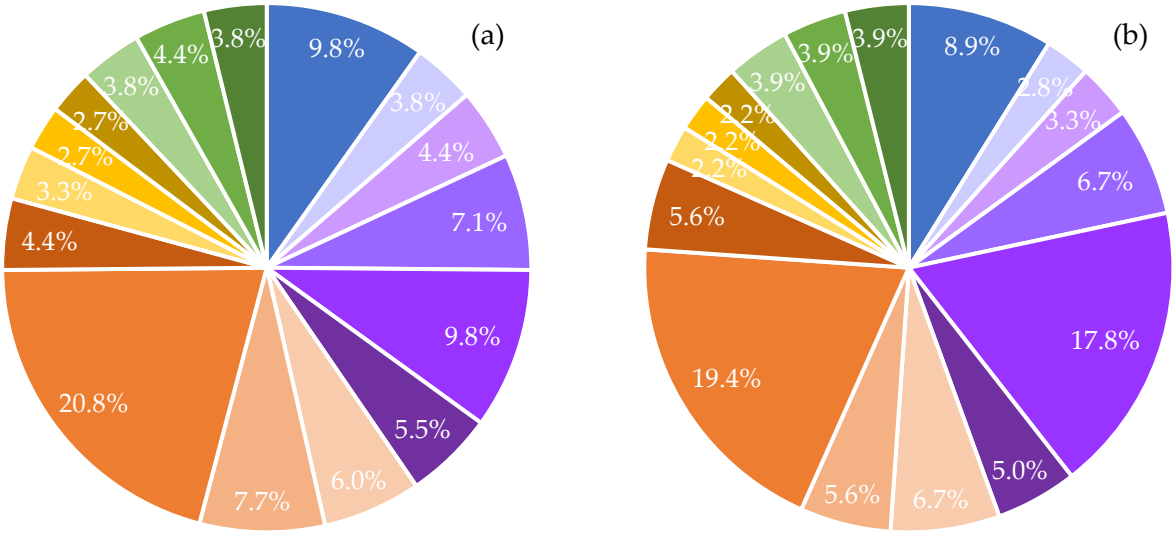


Fig 6.4. Results of PAHs found in FCBs incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs.

Another study that involved the leachate of ΣPAHs from CBs into various water types found concentration levels ranging from 3.0 to 5.0 µg mL^{-1}, 3.3 to 5.5 µg mL^{-1}, and 3.9 to 5.7 µg mL^{-1} for river, tap, and deionized waters, respectively [223]. The concentrations of this study are fairly low, as the study involved determining the leachate of PAHs from CBs in water samples through liquid–liquid extraction, compared to our study, which involved solid-phase extraction (SPE). The SPE method has a greater detection of sensitivity and is effective in extracting highly soluble substances. In addition to the method, the type of solvent used for extraction is critical. The initial solvent used for the Soxhlet extraction procedure was hexane–acetone (50:50). This decision was based on the theory of like dissolves like. Various studies found that dichloromethane (polar) resulted in low recoveries for all PAH (nonpolar) compounds when used as an extraction solvent, whereas hexane–acetone (nonpolar–polar) was primarily effective [232, 233]. However, in our study, hexane–acetone resulted in chromatograms with poor sensitivity, separation, and peak shape. For this reason, the solvent was changed to dichloromethane. Dichloromethane resulted in excellent chromatograms and high PAH recoveries, as shown in Fig. 6.3.

The sample with the highest ΣPAH concentration was in the FCB samples incorporating 2% CBs. The order of the ΣPAH concentrations in the FCBs according to the percentage of CBs incorporated was as follows: 2% CBs > 1% CBs > 1.5% CBs > 0% CBs > 0.5% CBs. The sample with the highest CB content exhibited the highest ΣPAH content as expected, the concentration of the 16 PAHs ranging from 0.015 to 0.044 µg/mL. However, it must be pointed out that the main raw material employed in preparing the brick samples was soil, and soil contamination is a common phenomenon. PAHs are known to absorb tightly to the organic matter in soil, rendering them less susceptible to biological and chemical degradation. For example, Saim et al. Saim et al. [227] extracted 1623 µg/mL of ΣPAHs from coal-derived contaminated land soil using the Soxhlet extraction method, while Yu et al. Yu et al. [234] extracted 582 µg kg^{-1} (0.582 µg mL^{-1}) from soils in the Guiyu area of China. Therefore, it is necessary to mention that the extracted PAH concentrations in the brick samples may be a result of the soil being contaminated, in addition to the CBs.

As shown in Fig. 6.5, the FCB–CB samples contained a high concentration of PAHs predominately with 2-3-4-rings compared to those with 5-6-rings. Chrysene and anthracene constituted 9.5% to 20.8% and 8.6% to 17.9%, respectively, of the mean ΣPAHs extracted from the brick samples. As expected, naphthalene, fluorene, anthracene, pyrene, fluoranthene, and chrysene, which are PAHs with higher water solubility and volatility, were found at higher concentrations compared to lipophilic PAHs, which include benzo[*b*]fluoranthene, benzo[*k*]fluoranthene, benzo[*ghi*]perylene, benzo[*a*]pyrene, dibenz[*a*,*h*]anthracene, and indeno[1,2,3-*cd*]pyrene. The lowest concentration was found at 0.004 µg/mL for PAHs with 5-rings. In general, the ΣPAH concentrations were the lowest in PAHs with 5-rings, followed by 6-rings, 2-rings, 3-rings, and 4-rings for all sample types.



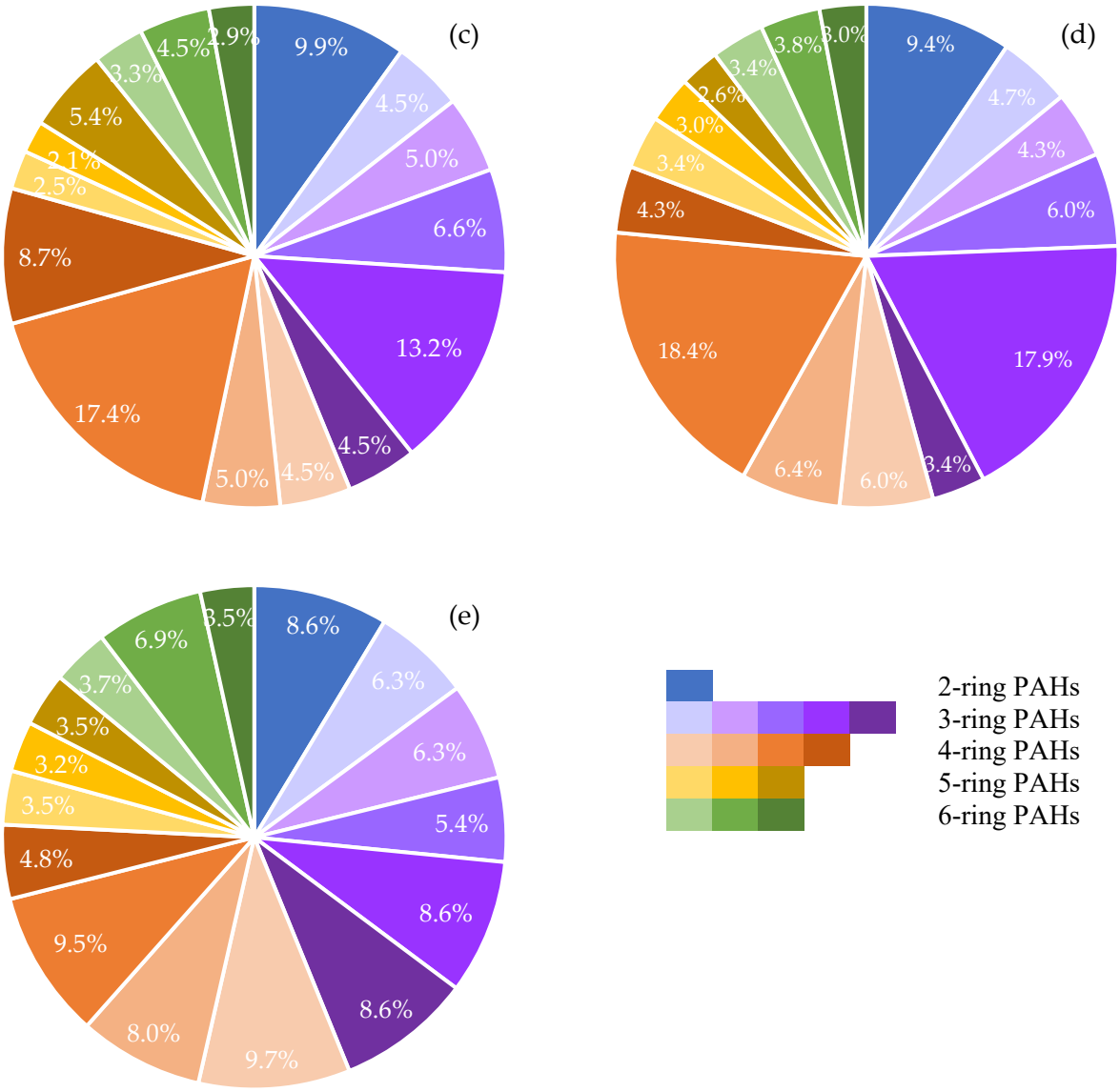


Fig. 6.5. Distribution of 2-3-4-5-6-ring PAHs (% of sum of 16 PAHs) in FCBs incorporating (a) 0%, (b) 0.5%, (c) 1%, (d) 1.5%, and (e) 2% CBs.

According to the Industrial Waste Resource Guidelines [173], if the total PAH concentration is below 50 μg mL^{-1}, the waste is considered industrial waste (IW). IWs are not considered prescribed wastes; therefore, they do not require control or management when disposed of to landfills, although it is recommended they undergo special management practices. In this case, the mean ΣPAHs for FCBs incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs range between 0.180 and 0.463 μg mL^{-1} and are significantly below the limit of 50 μg mL^{-1}.

6.4 Conclusions

In the present study, the concentration levels of 16 EPA-PAHs extracted from fired clay brick (FCB)

samples, incorporating various percentages of cigarette butts (CBs), were evaluated. The results show that lower-molecular-weight PAHs (light PAHs), including naphthalene, fluorene, anthracene, pyrene, fluoranthene, and chrysene, with higher volatility and lower lipophilicity, were released at very low concentrations compared to the higher-molecular-weight PAHs (heavy PAHs) from FCB–CB samples.

Among the detected PAHs, chrysene and anthracene dominated the emissions and constituted 9.5% to 20.8% and 8.6% to 17.9%, respectively, of the mean ΣPAHs extracted from the brick samples. Lipophilic PAHs including benzo[*b*]fluoranthene, benzo[*k*]fluoranthene, benzo[*ghi*]perylene, benzo[*a*]pyrene, dibenz[*a*,*h*]anthracene, and indeno[1,2,3-*cd*]pyrene demonstrated lower PAH concentrations due to their higher weights. The mean ΣPAH content in FCBs incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs was found in concentrations of 0.183, 0.180, 0.242, 0.234, and 0.463 μg mL^{-1}. As expected, the highest ΣPAH concentration was found in FCB samples incorporating 2% CBs. The concentrations of the mean ΣPAH for all samples were well below the EPA solid waste hazard categorization threshold limit for industrial waste.

Carbon, hydrogen, nitrogen, and sulfur (CHNS) and thermogravimetric (TG) analysis were conducted on the FCB–CB samples to investigate whether the CBs are completely combusted during the firing process. The results confirm an almost 100% combustion process during the firing process. A content less than 0.3% suggests that all carbon within the FCB–CB mixture relatively disappeared during the firing process up to 1050 °C.

It can be concluded that the results presented in this study clarify the issues raised regarding the possible release of toxic PAHs into the environment during the use and disposal of FCBs incorporating CBs or the potential storage of carbon in the FCB–CB mixture. However, further research regarding the emission of volatile organic compounds (VOCs) during the storage of CBs and during the mixing, oven-drying, and firing process of producing FCBs incorporating CBs should be conducted. In addition, potential deodorization and sterilization methods of CBs should be examined and implemented.

Chapter 7: Implementation of Recycling Cigarette Butts in Lightweight Bricks and a Proposal for Ending the Littering of Cigarette Butts in Our Cities

Abstract

Our cities, parks, beaches, and oceans have been contaminated for many years with millions of tonnes of unsightly and toxic cigarette butts (CBs). This study presents and discusses some of the results of an ongoing study on recycling in fired-clay bricks. *Energy Savings:* The energy value of CBs with remnant tobacco was found to be 16.5 MJ kg^{-1}. If just 2.5% of all bricks produced annually worldwide included 1% CB content, all the CBs currently produced could be recycled in bricks, and it is estimated that global firing energy consumption could be reduced approximately by 20 billion MJ (megajoules). This equates approximately to the power used by one million homes in Victoria, Australia, every year. *Bacteriological Study:* CBs were investigated for the presence of ten common bacteria in two pilot studies. *Staphylococcus* spp and *Pseudomonas aeruginosa* were detected in fresh used CB samples, and *Listeria* spp were detected in old used CB samples. All CB samples except the dried sample had significant counts of *Bacillus* spp. Some species of the detected bacteria in this study are pathogenic. Further confirmation and comprehensive microbiological study are needed in this area. Contact of naphthalene balls with CBs had a significant disinfecting effect on *Bacillus* spp. The implementation procedure for recycling CBs in bricks, odor from VOC emissions in CBs, sterilization methods, CB collection systems, and safety instructions were investigated and are discussed. *Proposal:* Considering the combined risks from many highly toxic chemicals and possible pathogens in cigarette butts, it is proposed that littering of this waste anywhere in cities and the environment be strictly prohibited and that offenders be heavily fined.

The research presented in this chapter has been published in the following:

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7.1 Introduction

Cigarette butts (CBs) are the most common type of waste material discarded in the world. In 2016, an estimated 5.7 trillion cigarettes were consumed around the globe (The Tobacco Atlas 2018; Carlozo 2018). While in Australia, 24 billion cigarettes were consumed, of which 7 billion were disposed of incorrectly [235]. That is, up to two-thirds of every smoked cigarette is littered in the environment (WHO 2017). Numerous countries have implemented strong tobacco control regulations, and a significant decline in cigarette consumption has been observed (Mackay et al. 2000). Despite this, it is estimated that the number of tobacco smokers is set to increase by 7 and 24 million in Nigeria and Indonesia, during the period 2015 and 2025 (The Tobacco Atlas 2018). This is due to further development in the market and population growth (Srbinoska et al. 2018).

There are over 4,000 chemicals present in a cigarette [236, 237], seventy-two of which are known to be cancer-causing carcinogens [238]. The main toxic agents include carbon monoxide, argon, aromatic hydrocarbons, hydrogen cyanide, phenol, nitrogen oxides, formaldehyde, acetaldehyde, acetone, benzene, ammonia, and pyridines (Hoffmann et al. 2001). The filter of a cigarette is made of cellulose acetate fibers, whereby the filter modifies the particulate smoke components through particle retention (Hoffmann et al. 1995). However, cellulose acetate filters have poor biodegradability and can take up to 10 years to decompose under normal environmental conditions (Hon 1977). Therefore when CBs are freely dispersed in the environment, they pose a critical problem in terms of toxic waste for the urban and aquatic life (Dieng et al. 2013; Marinello et al. 2019; Miscevska et al. 2006; Slaughter et al. 2006). Aquatic animals regularly consume CBs mistaking them for food, and have been found in the stomachs of fish, birds, sea turtles, and other creatures, leading to serious digestive issues (Rebischung et al. 2018; Kurmus and Mohajerani 2020a).

As the world's population increases, the number of CBs being littered is likely to rise drastically. In Australia, CBs were reported as the most common source of rubbish collected, representing 91.5% under the miscellaneous category (Clean Up Australia 2017). As a result, it is becoming more imperative that an effective solution for recycling CB waste is found and implemented. Marinello et al. (2019) and Torkashvand and Farzadkia (2019) comprehensively reviewed and investigated possible ways to manage CBs. Torkashvand and Farzadkia (2019) focused on developing control models for CB littering, while Marinello et al. (2019) analyzed possible recycling techniques for CBs and evaluated the disadvantages and advantages of those methods studied. In contrast, Kurmus and Mohajerani (2020b) reviewed key research studies on the recycling of CBs and investigated their toxicological properties. The paper explored the effectiveness, sustainability, and efficiency of those recycling methods. A major recycling method reviewed involved the recycling of CBs in the construction of asphalt concrete. Whereby it was disclosed through the addition of CBs in asphalt concrete, the demand for natural aggregates will decrease due to the partial replacement of aggregates with CBs. Moreover,

it was stated acceptable mechanical and physical properties, including air voids, resilient modulus, and bulk density, are achieved with the addition of CBs in asphalt concrete (Kurmus and Mohajerani 2020b). A current application being employed in Australia is the recycling of CBs to produce items such as pillows, plastic furniture, and shipping pallets. However, recycling CBs into objects on a small scale does not offer a practical and reliable solution in the long-term. The credibility and sustainability of the method must be investigated in terms of toxicity, leachability, and life cycle analysis.

This study is the continuation of ongoing research on the addition of CBs in lightweight bricks. In 2016, Mohajerani et al. (2016) tested the physical and mechanical properties, emissions, and energy savings of fired clay bricks incorporated with CB content and presented promising results. The proposal was to incorporate 1% CB content by weight in 2.5% of the world's brick production to solve a global waste catastrophe [1]. However, further testing and analysis are required for the brick manufacturing industry and government organizations to confirm that recycling CBs in fired clay bricks is a feasible and viable proposal. Therefore, the objective of this research is to investigate and develop practical and sustainable technologies for recycling all CBs (filters) in fired clay bricks on an industrial scale. The paper includes; (1) the calorific value of CBs and energy savings, (2) laboratory study on the manufacturing and properties of bricks containing CBs, (3) CB incorporation in the brick manufacturing process on an industrial scale, (4) CB odor issues and reduction methods, (5) CB collection systems, (6) bacteriological investigations on CBs and (6) possible sterilisation methods for CBs.

7.2 Calorific Value of Cigarette Butts and Energy Reductions

The incorporation of organic matter in clay mixtures assists in the firing process of clay bricks [239], and, hence, the recycling of waste materials in fired clay bricks is an area of wide ranging research [240]. To determine the energy saving benefits of the addition of CBs in bricks, the calorific value of CBs must be considered. The calorific value of a material determines the amount of heat (energy) that is produced by the complete combustion of that material. Materials with high calorific values (e.g. petrol) produce more energy when combusted than materials with low calorific values (e.g. wood). As a result, utilising a high calorific value material as a partial replacement in a mixture with a low calorific value will reduce the overall required firing energy for bricks, as, compared to a standard mixture, additional energy is produced by the materials' constituents when combusted.

An investigation into the calorific value of cigarette butts was undertaken. Samples of used CBs with remnant tobacco and used CB filters were tested with the aim of determining the calorific value of each. The results showed that the average gross calorific value of CBs with remnant tobacco (representing the typical used CBs found discarded on the ground) was 16.53 MJ kg^{-1}. A higher value of 16.99 MJ kg^{-1} was recorded for the cellulose acetate filter in isolation. However, the lower value of 16.53 MJ kg^{-1} has been used in this study. The cellulose acetate content from CB filters reduces the required firing

energy for clay bricks due to it possessing a higher calorific value than clay [1]. The energy required for the successful firing of bricks was taken as 2 MJ kg^{-1}, as it lies within the range of values stated by Prasetsan (1995) [110] and other investigators.

By utilising the tested CB calorific value and Eq. (7.1) to Eq. (7.4), the estimated energy savings from the incorporation of 1% CB content into clay bricks is 9.3%.

Energy required to fire control brick (MJ)	$Q_1 = q \times m_1$	(7.1)
Energy required to fire CB brick (MJ)	$Q_2 = q \times m_2 - CV \times m_3$	(7.2)
Energy saved (MJ),	$Q_1 - Q_2 = q \times m_2 - (q \times m_2 - CV \times m_3)$	(7.3)
Energy Saved (%)	$\Delta E = Q_1 - Q_2/Q_1 \times 100\%$	(7.4)

Where

m_1 = mass of control brick (no CB content)

m_2 = mass of clay in brick containing CBs

m_3 = mass of CBs in brick

$q = 2$ MJ/kg energy required for firing clay

CV = measured calorific value of used CBs, 16.53 MJ kg^{-1}

It is estimated that if 2.5% of all the bricks produced annually around the world include 1% CB content, the energy consumption of the process can be reduced by approximately 20 billion MJ (from the calorific value of 1% CBs with 9.3% energy saving) Mohajerani et al. (2016). This approximately equates to the power used by one million homes every year in the State of Victoria, Australia. Therefore adding 1% CBs (about 20 kg m^{-3}) to the brick clay results in significant energy savings, solves the CB pollution globally, and tangibly reduces the environmental impact of the brick manufacturing industry whilst providing financial savings to manufacturers through the reduction of energy for firing bricks.

7.3 Laboratory study on the manufacturing and properties of bricks containing CBs

7.3.1 Manufacture of samples

Cigarette butts (CBs) of varying sizes and brands were collected from dry bins and supplied by Butt-Out Australia Pty Ltd for this study. PGH Bricks and Pavers (Victoria) provided the brick soil, which is a brown silty sandy clay. The geotechnical parameters of the soil were determined according to AS/NZS 1289.5.1.1:2003 [241] and are presented in Table 7.1.

Table 7.1. Geotechnical Properties of the Brick Soil.

Test/Properties	Standard	Brick soil
Specific gravity	AS 1289.3.5.1	2.69
Liquid limit (%)	AS 1289.3.1.1	32
Plastic limit (%)	AS 1289.3.2.1	19
Plasticity index (%)	AS 1289.3.1.1	13
Australian soil classification	AS 1726-1993	CL
Optimum moisture content (%)	AS 1289.5.1.1	16
Maximum dry density (Mg/m^3)	AS 1289.5.1.1	1.78
Organic content (%)	BS 1377-3	1.23

The manufacturing process of the bricks first involved drying the soil and CBs (if any) in an oven for 24 h at 105 °C. The soil was taken out of the oven and crushed into finer particles to pass through a sieve size of 2.38 mm. The soil and oven dried CBs were then placed inside a 20 L capacity Hobart mechanical mixer and mixed for a total of 25 min. The calculated water content of 15.5% was gradually added during the mixing process. After the mixing was complete, the mixtures were sealed in airtight plastic bags and left to settle for a period of 48 h to uniformly distribute moisture throughout the mix.

The next step was to compact the mixes using a Servopac Gyratory compactor with a pressure of 240 kPa. A cylindrical mould was used during the compaction process. The moulds were 100 mm in diameter, in which a predetermined mass of soil was placed inside the mould to yield bricks with heights of approximately 50 mm.

After compaction, the samples were air dried for 24 h before being placed in an oven for 24 h at 105 °C. After being removed from the oven, the samples were fired in an electric furnace at 1020 °C for 3 h. The height, diameter, and weight of the bricks were measured using a digital caliper and electronic scale at each stage of the manufacturing process to compute the diametric and height shrinkage of the bricks.

After the bricks were removed from the furnace (Fig. 7.1) and allowed to cool to room temperature, they were tested according to AS/NZS 4456.1:2008 [242]. The compressive strength, initial rate of absorption (IRA), cold water absorption, shrinkage, and density were computed. Compressive strength tests were conducted on three samples from each mix using a Material Testing Machine (MTS), while the cold-water absorption and IRA testing were completed using a water bath.



Fig. 7.1. Bricks before and after firing.

7.3.2 Properties of the Bricks

The average properties of the brick samples tested are shown in Table 7.2. The control samples containing no CBs and 15.0% moisture content were manufactured as a benchmark for further results. An additional 0.5% moisture content was added to the first batch of 1% CB bricks to account for the effect of the CBs, which could absorb moisture from the soil and thereby result in an undesirably dry mix. The optimum moisture content was used to achieve the maximum density during the manufacturing of the samples.

As expected, the addition of 1% CBs resulted in a reduction of the compressive strength to 27.49 MPa for the 15.5% moisture content sample and 25.77 MPa for the 17.5% moisture content sample. The common compressive strength for standard bricks is between 10 and 20 MPa. Varying moisture contents were used to investigate what effects this would have on the compressive strength and other properties of the bricks. Interestingly, shrinkage was reduced for the 15.5% moisture content samples, which could be attributed to the absorption of moisture by the CBs prior to firing. The whole CBs present in this sample may have also provided a reinforcing effect to resist shrinkage. Shrinkage of the 17.5% moisture content bricks was very similar to that of the control, thereby indicating that, for small samples, shrinkage is not greatly affected by the inclusion of 1% CB content. Eliche-Quesada et al. (2012) similarly reported a minor decrease in shrinkage with the addition of biodiesel production residues in fired clay bricks.

Cold water absorption increased with a decrease in density from 2134 kg m^{-3} to 1964 kg m^{-3} (8%). The initial rate of absorption was also higher for the CB bricks. Once again, this is expected, as a reduction in density relates to an increase in pore volume and, hence, a more absorptive material. Similar

observations were reported in previous studies of bricks incorporating recycled paper processing residues (Sutcu and Akkurt 2009). The large initial rate of absorption for the 15.5% moisture content CB bricks can be explained by the voids left by the CBs after firing. As a result, for both aesthetic and physical reasons, manufacturers should ensure that the CBs are completely disintegrated and mixed evenly before firing. It is important to note that the water absorption and initial rate of absorption values obtained to satisfy the relevant masonry standards for low-medium rise structural bricks. Finally, the thermal conductivity of the bricks decreases with the corresponding reduction in density, 1.107 W m^{-1} K^{-1} to 0.873 W m^{-1} K^{-1}, a reduction of up to 21%. This is a fantastic result, as a lower thermal conductivity will reduce the amount of heat that the brick transfers. A house built with 1% CB bricks will use less energy for heating in winter, and less energy for cooling in summer. This reduction in operational energy will result in huge global energy savings above what has been estimated purely from the reduction in embodied energy due to the calorific value of CBs contributing to the firing process. The thermal conductivity was calculated using Eq. (7.5).

$$TC = 0.0558e^{0.0014D_d} \quad (7.1)$$

Where;

TC = thermal conductivity (W m^{-1} K^{-1})

D_d = dry density of clay bricks (kg m^{-3})

Table 7.2. Properties of manufactured brick samples.

Sample Identification	Moisture Content (%)	Compressive Strength (MPa)	Cold Water Absorption (%)	Initial Rate of Absorption (kg m^{-2} min^{-1})	Diametric Shrinkage (%)	Height Shrinkage (%)	Average Density (kg m^{-3})	Thermal Conductivity (W m^{-1} K^{-1})
CB (0%) (0 kg m^{-3})	15.0	43.17	8.15	0.31	5.38	6.16	2134.0	1.107
CB (1%) (20 kg m^{-3})	15.5	27.49	10.53	0.47	4.00	4.34	1991.0	0.906
CB (1%) (20 kg m^{-3})	17.5	25.77	11.51	0.39	5.39	5.98	1964.0	0.873

These results confirm the findings in Mohajerani et al. (2016) [1]. Investigations into the leachates from manufactured CB bricks have already been conducted [1]. Both the Australian Bottle Leaching Procedure (ABLP) and the Toxicity Characteristics Leaching Procedure (TCLP) were used on crushed bricks and solid bricks [1]. The testing was conducted on bricks containing 2.5, 5.0, 7.5, and 10.0% CB content. It was found that heavy metal concentrations were well below the levels stipulated by the USEPA [243] and the Environmental Protection Agency of Victoria [244] for concentrations of 10% and lower CB content. Also, a comprehensive leachate study on leachate analysis of heavy metals in

cigarette butts and bricks incorporated with cigarette butts has been carried out based on ABLP (2019) and USEP (2013), (Kurmus and Mohajerani 2020c).

7.4 Brick Manufacturing Processes—Industrial Scale

7.4.1 Automatic Processes

The manufacturing process for clay bricks can be summarised by the following sequential processes: mining of raw materials, grinding, screening, blending (where desired mixes are formed by adding admixtures/other types of soil), forming, shaping, drying, firing, cooling, and packaging/shipment of the material [245].

Stage 1. Winning

The raw materials used in the production of clay bricks include surface clays and shales, which are commonly acquired from open pit quarries [245]. Clay is the most commonly utilised material in the production of bricks. Approximately 96% of all bricks manufactured in the United Kingdom are produced with clay [246]. Typically, clay will be extracted in bulk quarries on a few occasions per year and stockpiled (bulk winning), which allows extraction to occur during good weather [247]. Sometimes quarries are located on-site alongside manufacturing facilities, whilst other facilities obtain raw materials via road or rail transportation [245]. Raw materials for brick making are stockpiled on-site, usually in the open, where rainwater washes out some salts that could cause efflorescence. These stockpiles may contain a sufficient volume of raw material to satisfy up to a year of brick production [246].

Stage 2. Primary Crushing

Raw materials are then fed into a primary crusher for initial size reduction, typically by a truck or front-end loader [245].

Stage 3. Blending

The minerals within the clay are responsible for the bricks' final properties, such as colour [246]. As a result, different clays and admixtures are combined at this early stage to produce a mix design that will achieve the desired properties of the final product [247]. Throughout this stage, water is added consistently, where the water, brick clay soil, and any additional additives are mixed together. The blending process also reduces the size of the materials within the mixture.

Stage 4. Grinding

The material, now initially reduced in size, is typically fed into a secondary crushing mechanism. Manufacturing plants typically utilise a pan mill for secondary crushing, which can be used with both dry and moist clay mixtures [248]. The plant used for grinding is generally heavy, creates a lot of

vibration, and, unlike blending, the mixture at this stage is extremely difficult to assess. The brick soil mix at this stage is very fine [247].

Stage 5. Screening

From here, the material is screened to ensure that the correct size reduction has been achieved. Oversized clay particles are returned to the secondary crushing system [248]. The material is then conveyed either to storage areas (such as stockpiles or silos) or directly into a mill room for forming.

Stage 6. Forming and Shaping

Extrusion, also known as the stiff-mud method, involves producing a mixture with the desired plasticity by adding 10-15% water content. Once the desired water content (WC) has been reached and the mixture is homogenous, it is passed through a vacuum chamber to de-air the mixture. De-airing removes pore air from the mixture, increases the plasticity, and contributes to the final product strength. Finally, the clay is extruded to produce a 'column' of clay, which is then coated if required and cut into bricks by a set of automatic cutting blades. Around 90% of all bricks produced in the United States of America are produced via extrusion [249].

Moulded,

The moulded brick process, otherwise known as the soft-mud process, is suitable for clay with water contents too high to be used in extrusion. Instead, a mixture with approximately 20-30% water content is formed. This mixture is placed and compacted in brick moulds to prevent the mix from sticking to sand or water. This method can be utilised by machine or hand [249].

Dry-Press,

This process is used for very low-plasticity clays. A mixture with up to 10% water content is created, which is then compressed by air or hydraulic rams into steel moulds [249].

Stage 7. Drying

Once the bricks have been formed, they are typically stacked onto kiln cars and sent into a holding area, where they are later loaded into a dryer. Most dryers utilise waste heat from the cooling zone of a kiln and are typically heated to 204 °C; however, some manufacturing plants may choose to use fuels like gas to heat up their driers [245]. It should be noted that, in some instances, air drying of bricks is utilised; however, this method is avoided where possible due to weather impacts, increased duration of production, and uneven drying. Two of the most common driers used are tunnel dryers, in which a kiln car will slowly move through a heated tunnel, and chamber dryers, where bricks are loaded onto pallets inside a large chamber [248].

Stage 8. Firing

The bricks are then fired in kilns. There are two main types of kiln – intermittent kilns and continuous kilns. Intermittent kilns, such as clamp and down-draught, must be allowed to cool before bricks can be unloaded and reloaded. They are typically only utilised where special colour effects are required [246]. Continuous kilns are used in large scale production and are capable of firing bricks at a steady rate due to the continuous loading, firing, and unloading process [247]. The most popular type of continuous kiln is a tunnel kiln, where bricks are loaded into kiln cars and pass through various carefully monitored temperature zones inside a tunnel. The advantage of this method is that the process is continuous. These tunnel kilns can range from 104 to 152 meters in length and can reach a maximum temperature of approximately 1090 °C in the firing zone.

Stage 9. Unloading, Packaging, and Shipping

After successful firing, bricks enter a cooling zone (if continuous kiln) where they can cool to near ambient temperatures. They are then removed from the kiln, stored, packaged, and shipped [245].

7.4.2. Manual Processes

Clay bricks have been utilised throughout history by many civilisations. Bricks were used in the construction of the city of Babylon about 6000 years ago [246]. Although the industrial revolution introduced machine processes into brick manufacturing, in some places around the world, bricks are still made by hand [246].

In some nations, such as India, brick manufacturing is predominately a manual process, where traditional processes of making bricks have changed little over time [250]. The typical process can be described as follows:

Stage 1. Winning

Clay is mined (either by hand or mechanical means) and stored in piles in the open. Rainwater is said to make the clay soft and removes unwanted chemical oxides from the clay.

Stage 2. Mixing

When ready, water is added and mixed with the clay until a mixture with enough plasticity for moulding is created. The mixing process is conducted with hands and feet.

Stage 3. Moulding

Once the desired consistency has been achieved, workers roll a large sample of the mixture in the sand, before being pushed into steel brick moulds. The sand acts as a lubricant that stops the mixture from sticking to the mould.

Stage 4. Drying

These moulds are placed into a drying area where the sun has access to the moulds. The moulds are occasionally turned over to prevent non-uniform drying and consequential warping of the bricks.

Stage 5. Firing

After a sufficient drying process (typically up to 2 weeks), the bricks are arranged in a kiln and fired. Once the firing is completed (decided at the discretion of the brick maker), bricks are sorted according to colour [250].

Stage 6. Cooling

Bricks are unloaded from the continuous kilns, where they can cool to ambient temperatures. In the case of intermittent kilns, the entire kiln and all the bricks it contains can cool in between firing cycles. Bricks are unloaded once the entire system has cooled.

Stage 7. Packaging and Transportation

The bricks are sorted depending upon the qualities, such as colour, and then packaged. These bricks may then be transported off-site, utilised in the construction of new kilns, or re-fired (if sufficient firing has not occurred) at the discretion of the plant manager.

It is important to note that whilst the general method of manual brick manufacture remains the same, regional variations are common, which accommodate factors, such as local geography, climate, etc.

In Africa and South America, large fires fuelled by wood are used to heat kilns, which are then insulated with grass or mud. In nations such as India and Mexico, large volumes of bricks are fired at once, and the thermal mass is used for insulation. Fuel for kilns can be wood or coal, but, in some cases, where these are not accessible, garbage and biomass can also be used [250].

7.5. CB Incorporation into the Automatic and Manual Process

7.5.1. Incorporation into Automatic Processes

To incorporate the CBs into the automatic manufacturing process, three methods are suggested: Method 1 – Addition of whole CBs with no modification, Method 2 – Addition of pre-shredded CBs, and Method 3 – Addition of a concentrated CB pre-mixed brick clay (with a high % of CB content).

Although three methods are suggested, for the most part, the methods share a common sequence. Once CBs have been delivered on-site, they will be stored, cleaned, and handled strictly according to OH&S requirements.

7.5.1.1. Method 1—Addition of Whole CBs with no Modification

Fig. 7.2 demonstrates the stages of the Automatic Manufacturing Process and the proposed stage to add the whole CBs with no modification. This paper suggests that whole CBs should be added during stage 3, the blending stage, with the addition of water. When CBs are introduced during this stage, the cohesive characteristics of the brick clay mixture and the addition of water reduce or even negate any potential airborne particles that may derive from the CBs. Adding CBs during stage 3 also reduces the initial size (disintegration) and allows CBs to be distributed better within the mix. Improvements in quality can be achieved with longer mixing times [1].

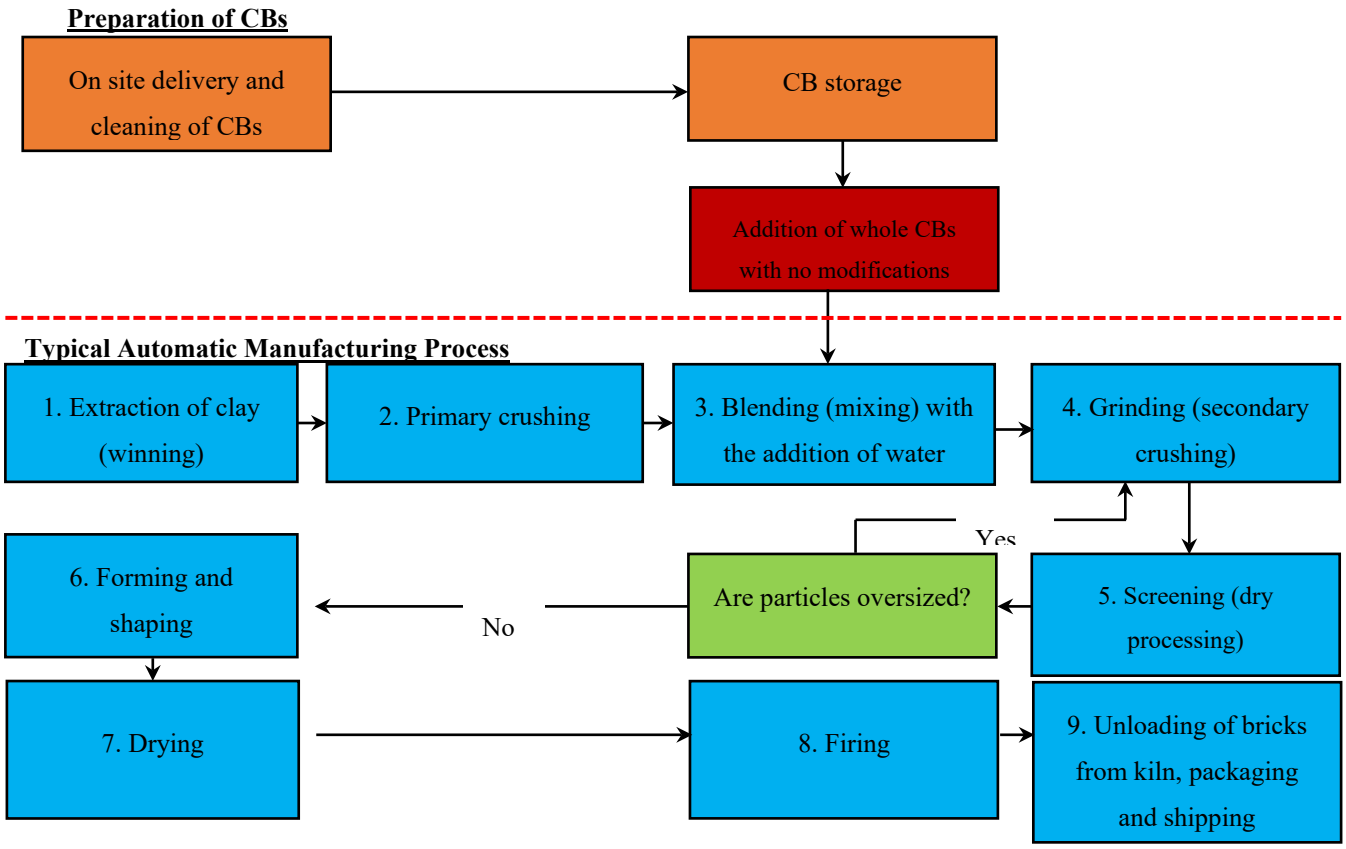


Fig. 7.2. Method 1—Addition of whole CBs into the Automatic Manufacturing Process.

7.5.1.2. Method 2—Addition of Pre-shredded CBs

Fig. 7.3 demonstrates the proposed stage to add the pre-shredded CBs. As the preparation process is different, refer to Fig. 7.2 for the common stages. This paper suggests that pre-shredded CBs should be added during stage 3, the blending stage, with the addition of water. When introducing the CBs to stage 3, it is important to minimise the exposure of the CBs to open air environments, as chemicals within the CBs may diffuse into the air with other dust particles. Exposure to this air could be a safety risk for nearby personnel. A pipeline pathway between the enclosed shredding system and blender is recommended to minimise environmental pollution as the shredded CBs are directly incorporated in

stage 3. Water is added throughout the blending stage as moisture within the mixture entraps chemicals and mitigates potential airborne particles.

It is important to understand that the actual shredding of the CBs must be performed in a fully enclosed environment, such as an enclosed industrial shredder. This is necessary as it keeps the dust and chemicals from spilling out into the environment. Enclosed shredding minimises environmental pollution and the exposure of personnel that could occur when compared to shredding in an open air environment. Compared to method 1, as the CBs are already pre-shredded, it is expected that there will be improved uniformity in particle size and distribution within the mix, leading to clay bricks with more uniform properties.

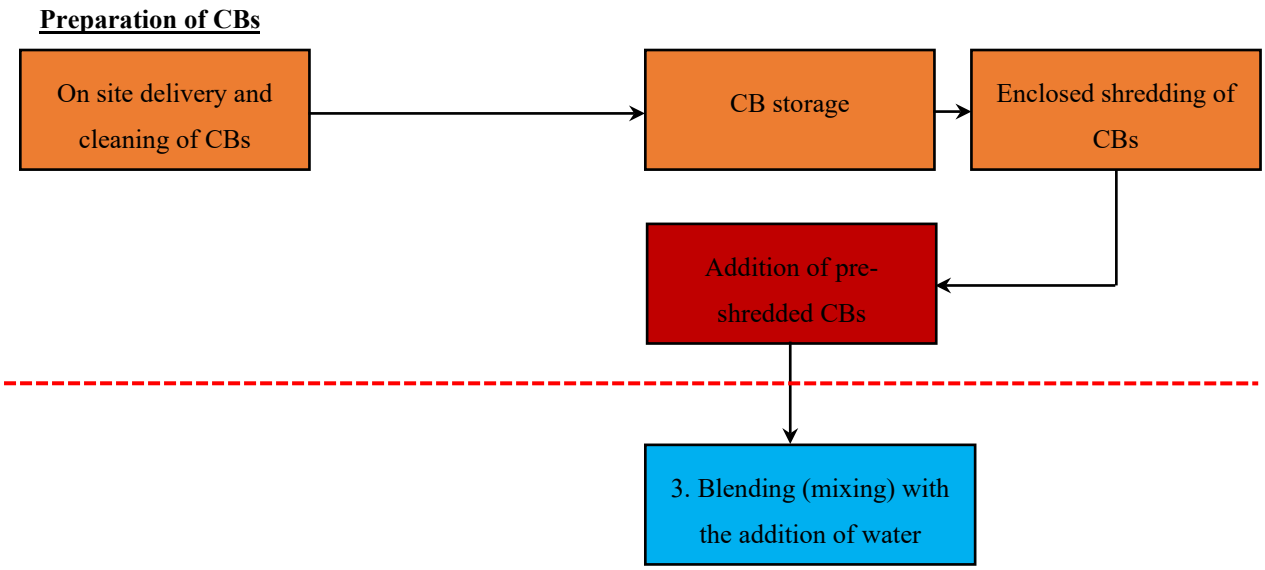


Fig. 7.3. Method 2—Addition of pre-shredded CBs into the Automatic Manufacturing Process.

7.5.1.3. Method 3—Addition of a Concentrated CB Pre-mix (With a High % Content of CB)

Fig. 7.4 demonstrates the proposed stage to make and add the concentrated CB pre-mix. As the preparation process is different, refer to Fig. 7.2 for the standard stages. This paper suggests that the concentrated CB pre-mix should be added during stage 3, the blending stage, with the addition of water. The concept of this method allows clay brick manufacturers to make a CB pre-mix and store it until needed for specialised brick orders. This means that manufactures can create different types of brick products that contain varying CB content, which can be made to order by the client. This pre-mix consists of a high content of CBs, water, and soil. The premix is then incorporated during the blending stage into the main line of bricks being produced. Manufacturers should take care when adding the premix into blending to ensure that the high concentration CB content mix is diluted to the desired CB content outcome.

An issue with this method is that it may require another conveyor/plant to transport the pre-mix into stage 3. This suggestion requires a capital investment, which should be considered before implementing it into the production line.

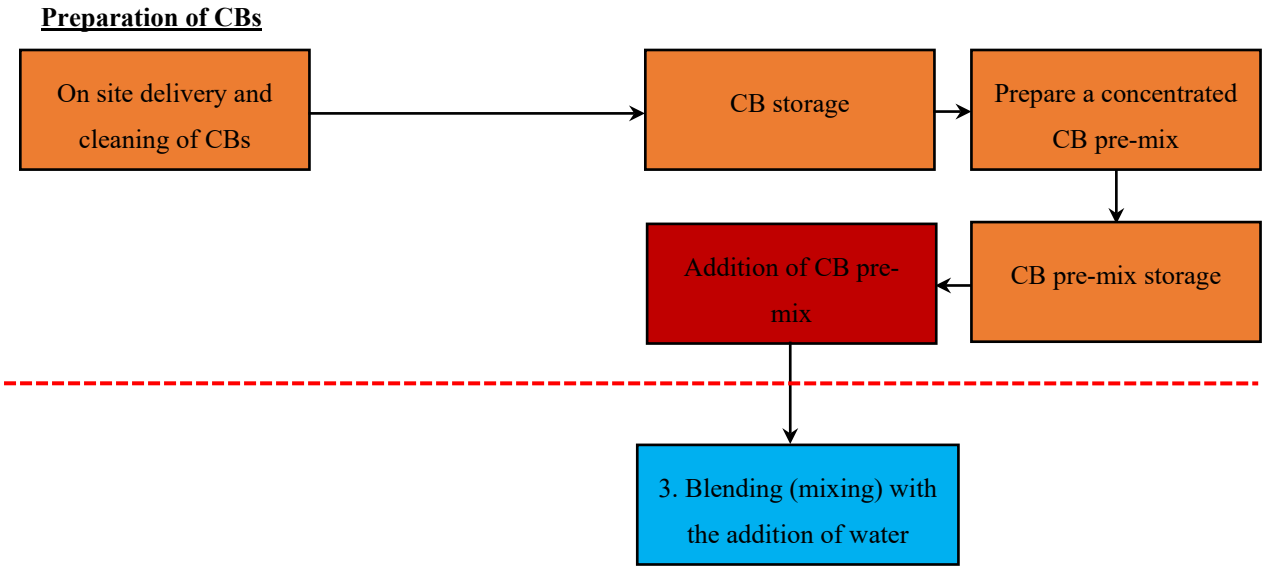


Fig. 7.4. Method 3—Addition of a concentrated CB pre-mix into the Automatic Manufacturing Process.

7.5.1.4. Appropriate Personal Protective Equipment for the Automatic Manufacturing Process

There is a possibility when dealing with CBs that the chemicals present in CBs may be distributed into the air with other dust particles. It is therefore essential that all workers exposed to excessive dust particles be equipped with dust masks. As mentioned in the above methods, if moisture is added to the mixture at the same time as the introduction of CBs, then the risk of inhalation of airborne particles is drastically reduced or possibly mitigated. However, there is the potential for leachates contaminating surfaces, such as conveyor belts, and mills, etc., due to the potential for the moisture to introduce metals to the environment [251]. As a result, workers encountering this mixture or associated surfaces should also wear protective gloves and long-sleeve clothing.

7.5.1.5. Odor Issues and Reduction Methods for the Automatic Manufacturing Process

Odor can be a significant issue introduced through the incorporation of CBs into the automatic brick making process. Odor is caused by the diffusion of chemicals entrapped within CBs being released, such as volatile organic compounds (VOCs) [252-254]. Odor can be displaced by air current or diffusion, both of which rely upon environmental conditions and time. The smell is a biochemical reaction where VOCs attach themselves to the olfactory receptors located in the nose. These receptors then send an electrical signal to the brain that is perceived as odor [255]. Not all VOCs have an odor

associated with it, for example, methane, which is an odorless chemical [159]. Due to the smell being the human perception of the physical world, it is hard to physically or chemically measure and quantify smell as even the smallest amount of VOC can produce an odor. This is a problem because CBs are introduced in the blending stage, but the odor can persist until the firing of the CB bricks (stages 3-7).

Odor contains four attributes that humans can detect and describe. These attributes include the detection threshold, intensity, character, and hedonic tone. The detection threshold is the minimum amount of chemicals needed to detect odor; intensity is how strong the odor is upon and after detection; character is defined as what the odor smells like, and hedonic tone is how pleasant the odor is to smell [256]. Out of the four attributes, only two of them are quantifiable – detection threshold and intensity. The detection threshold is defined as the point at which 50% of the population (normally a panel of 6 people) can detect an odor and is recorded numerically as 1 odor unit per meter cube ($1\ \mathrm{o.u./m^3}$) [257]. The intensity of an odor can be quantified using the Weber-Fechner law, in which an increase in concentration is perceived as a large increase in intensity when the concentration value is small. The equation intensity versus concentration is a logarithmic relation (Eq. (7.6)), which means that for CBs in the brick manufacturing process, a small increase in CB content can lead to a large increase in the intensity of the CB odor [257].

Intensity versus CB% curve,

$$I = K_w \, ln\left(\frac{CB}{CB_0}\right) \tag{7.2}$$

Where;

I = Intensity of CB Odor

K_w = Constant based on how sensitive the human nose is at detecting odor from CB

CB_o = Cigarette butt content (%) to which 50% of the population can detect an odor (AMOL)

CB = Cigarette butt content

7.5.1.6. Controlling Odor Level

The potency of the odor is dependent on the volume of chemicals present in the brick factory. This would be the rate of diffusion of the volume of CBs present, which will vary according to the ventilation rate. [253]. Every brick factory has its own unique ventilation rate, which is based on the environmental conditions and the design of the factory [258]. The ventilation rate will fluctuate a little but will remain constant, and, therefore, it is recommended to change the CB content as it is an independent variable subject to change by the manufacture. The process of changing the CB content within the mix should be an incremental one where manufactures start with a small number of CBs then assess the outcome. From there, if it is found that the Acceptable Max Odor Limit (AMOL) has not been reached, then the CB content should be increased by a small amount. The process should then be repeated until AMOL

is reached. The Acceptable Max Odor Limit (AMOL) is defined as the cigarette butt content at which 50% of the workers surveyed can detect an odor. Once AMOL is reached, it is recommended to use a previous CB increment. This solution can also be implemented in conjunction with the other solutions suggested below.

Increasing the ventilation rate by installing exhaust fans or fume extractors at specific points along the production chain is a viable solution. The odor can either be vented or captured using filters (such as activated carbon). Unlike controlling the CB content in the mix, increases in ventilation require some capital to be invested. The benefit of this solution is that it provides a direct and permanent increase in odor reduction within the factory.

7.5.1.7. Odor Elimination Using UVC Light

A possible elimination method for cigarette odor is to treat the odor with UVC light, which could oxidise the odors emitted from the cigarette butts. It does this by shattering the molecular bonds through the impact energy and then allowing the elements to oxidise to form new molecules that are odorless or have less odor [259]. UVC light has the capacity to break down Carbon hydrogen bonds, which have a bonding strength of 413 kJ mol^{-1}. UVC with a wavelength of 254 nm can be used to break down the bond due to the photon containing 470 kJ mol^{-1} of energy [259]. Oxidation can then be achieved by breaking down the dioxide molecules found within the air. In order to break down the oxygen bond, which has a bonding strength of 498 kJ mol^{-1}, UVV light is required. UVV with a wavelength of 185 nm can be used to break down the bond due to the photon containing 645 kJ mol^{-1}, which is enough to shatter the bond [259].

7.5.2. Incorporation into Manual Processes

CBs in the manual brickmaking process should only be incorporated if handled with care, and strict OH&S procedures are followed where workers always utilise the required PPE. The problem of odor should also be addressed as discussed above.

The addition of CBs to manual manufacturing processes is quite simple, as the mixture is easily accessible and is available at all stages leading up to the moulding and firing of the bricks. To incorporate the CBs into the manual clay brick making process, two methods are suggested: Method 1—Addition of whole CBs with no modification, and Method 2 – Addition of a concentrated CB pre-mix (with a high % content of CB).

Like the automatic manufacturing process, once CBs have been delivered on-site, they will be stored, cleaned, and handled strictly according to OH&S requirements.

7.5.2.1. Method 1—Addition of Whole CBs with no Modification

Fig. 7.5 demonstrates the stages of the Manual Process and the proposed stage to add the CBs. This paper suggests that whole CBs can be incorporated into the mixture between stages 2 and 3, after sufficient water has been added. This will reduce the risk of inhalation of dust particles containing any toxic chemicals from the CBs. Jrup (2003) states that environmental contamination by heavy metals predominately occurs via pathways leading to the air (such as processing of materials, combustion), soil (leaching into groundwater), and water (via surface runoff) [251]. As a result, it is expected that there will be no exposure to airborne contamination if the CBs are added when the mixture is wet.

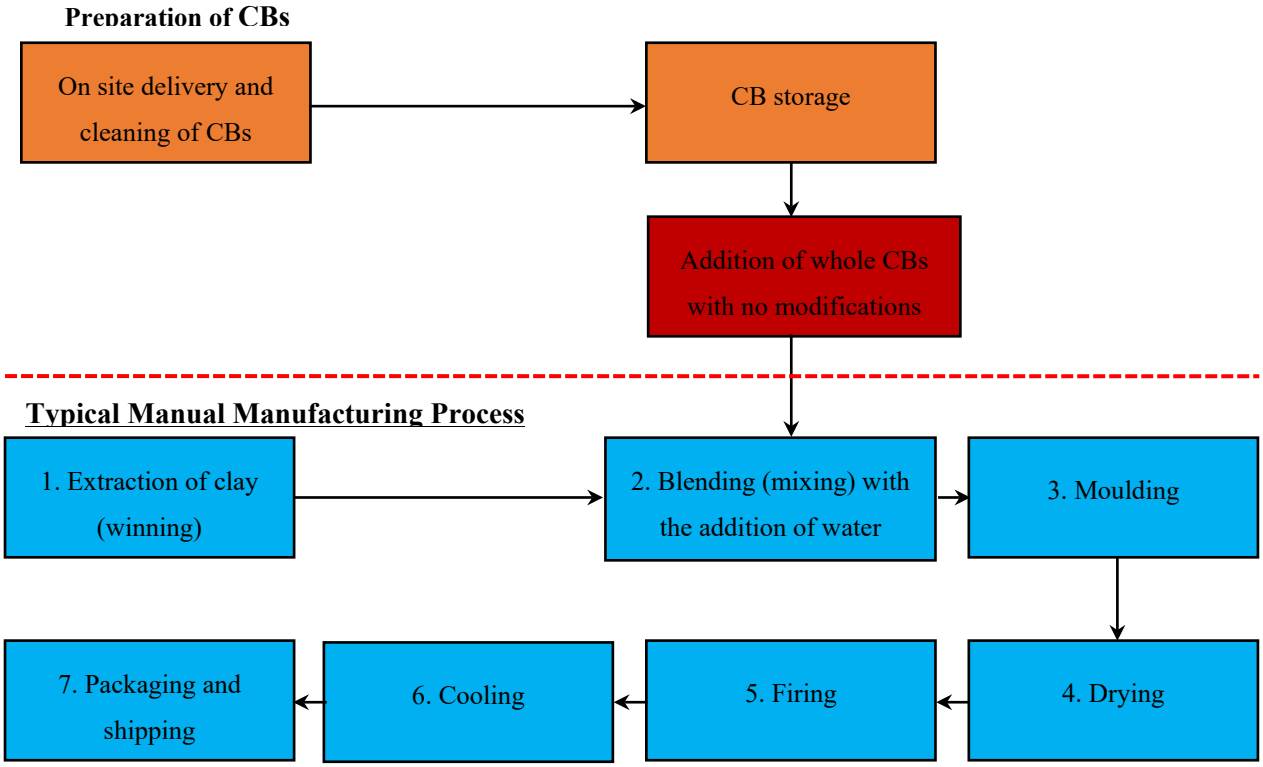


Fig. 7.5. Method 1—Addition of whole CBs into the Manual Process.

7.5.2.2. Method 2—Addition of a Concentrated CB pre-mix (With a High % Content of CB)

Fig. 7.6 demonstrates the proposed stage to make and add the concentrated CB pre-mix. As only the preparation stage is different, refer to Fig. 7.4 for the shared common stages. This paper suggests that the concentrated CB pre-mix should be added between stages 2 and 3. This method may be more appropriate for the manual process, where the pre-mix can be stored in appropriate containers on-site and ready for use.

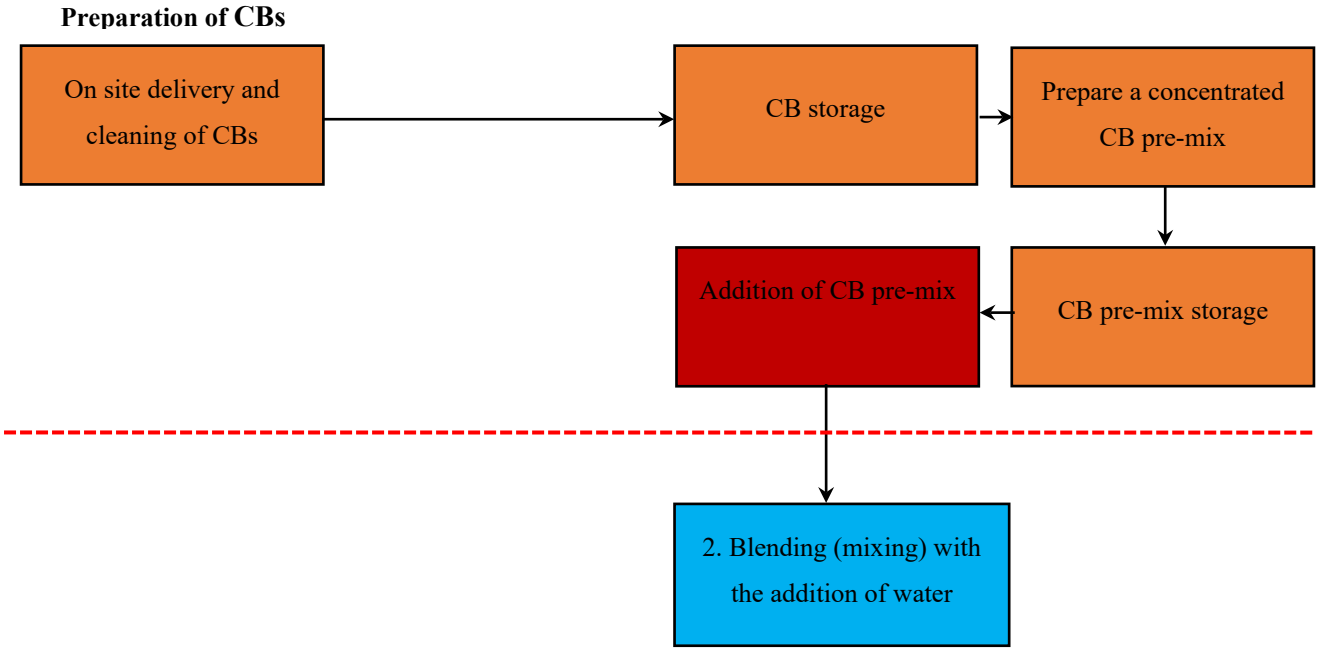


Fig. 7.6. Method 2—Addition of a concentrated CB pre-mix into the Manual Process.

7.5.2.3. Appropriate PPE for the Manual Process

The negative effects on human health from heavy metals are linked to lead, cadmium, mercury, and arsenic [251]. Studies have shown that three out of the four heavy metals stated above are present in CB smoke [238].

Therefore, it is essential that any worker who comes into direct contact with this moist mixture wears heavy-duty personal protective equipment (PPE). Examples of PPE include heavy-duty gloves and long-sleeve clothing for those mixing via hand and gumboots for those utilising their feet.

7.5.2.4. Odor Issues and Reduction Methods for the Manual Process

As stated before, odor is an issue in the production and use of CB bricks. In the manual process, odor is a problem in the moulding and drying stages (stages 3-4) [250]. The drying stage, which could last for up to two weeks, takes place in an open environment where the sun has access to the moulds. Gradually introducing CBs in the brick making process and assessing for AMOL would be an ideal low-cost solution. This is considering that the bricks would be placed in an open air environment where the odor can be carried away by the air currents. However, if the environment is densely populated the CBs should be treated using effective UV light or other methods before manufacturing. If UV treatment cannot be used it is not recommended that CBs be incorporated into the manual brick making process.

7.6. Recycling CBs on an Industrial Scale: Collection and Processing

7.6.1. CB Collection Systems

The huge environmental problem presented by CB litter has steadily gained focus across the world. To reduce the number of CBs being incorrectly disposed of, several companies in Australia have developed CB disposal systems. These companies provide receptacles (Fig. 7.7) for the disposal of CBs, in addition to servicing regimes and the removal and final disposal of CB waste. Currently, much of this waste is placed in landfills or incinerated at the expense of the collection company. In all cases, clients (local governments and other organisations) can pay a periodic rental fee per receptacle, which includes the cost of collection, servicing, and final disposal. Receptacles take many forms, including wall-mounted, post-mounted, and free-standing [260].



Fig. 7.7. Typical CB Receptacles: free standing (top), post mounted (bottom).

It is suggested that these companies deliver CBs (after appropriate preparations) directly to the brick manufacturing facilities instead of landfill or incineration sites. The advantages of delivering CBs directly to brick manufactures are as follows: firstly, collection companies can save large amounts of money by not having to pay for incineration or landfill of CB waste. Secondly, the negative environmental effects of incineration and landfill can be avoided. Finally, brick manufacturers can gain

access to existing collection systems without having to pay for services or having to implement their own collection schemes. It should be noted, however, that upon wide-scale introduction into bricks, CBs may be considered a valuable source of energy and that when demand is sufficient, collection companies may decide to charge brick manufacturers for delivery. The safe storage and handling of used CBs on-site are discussed in the following section of this paper.

7.6.2. Preliminary Bacteriological Investigations

A serious concern in the recycling of used CBs is that they may be contaminated with bacteria or viruses from contact with people or the environment. As a result, this study undertook two pilot investigations to determine whether the tested species of bacteria were present and if these bacteria posed a health risk to the waste collection and processing personnel.

The first pilot investigation focused on determining if any of several common bacteria were present on either the used or unused cigarette filters. The purpose of this investigation was to determine if it was plausible that bacteria could be present and survive on discarded CBs. The second pilot investigation provided further results where additional species of bacteria were tested. The second test was conducted on used, unused (control), and dried CBs, as well as CBs that were in contact with two naphthalene balls (mothballs).

7.6.2.1. Pilot Investigation 1

This investigation consisted of two samples and one control group: Control group (unused CBs); Sample 1 - Used CBs, 7-day old sample; and Sample 2 - Used CBs, collected on the day sample.

The control group for the study consisted of generic cigarettes, which were bought and cut down to a length consisting of the filter plus 1.5 cm (for a total length of 3.5 cm). This was to ensure that the tobacco content of the control group closely resembled that of a smoked cigarette. A previous study has shown that some *Pseudomonas* spp can thrive on the nicotine in solid tobacco waste [261]. As a result of this finding, it was essential to ensure that similar volumes of tobacco were present in all the samples (Fig. 7.8).

The used CBs for this investigation were collected from the tops of bins around Victoria's Melbourne Central Station. Collecting CBs from the bins increased the chances that the samples were fresh. Melbourne Central Station is a central hub for travel in Melbourne's CBD, and, therefore, it provides reasonably representative samples (i.e. people from many areas and socio-economic backgrounds travel through this area).

In this preliminary bacteriological investigation, the presence of *Salmonella* spp, *Escherichia coli, Pseudomonas aeruginosa, Enterococcus faecalis,* Coagulase +ve *Staphylococcus* spp, and *Streptococcus* spp were investigated.



Fig. 7.8. CB samples utilised for preliminary bacteriological study 1. The dates indicate when the used samples were collected in 2017.

Escherichia coli live naturally in the intestinal tract of humans and animals, and, in most cases, is harmless [262]. However, some *E. coli* are pathogenic and can cause diarrhea or other illnesses [262]. Although the most vulnerable include the elderly and young children, anyone can be infected. The bacteria can be spread through contact with unclean surfaces that are contaminated with faecal matter, such as human hands or door handles [262], and infections start when this bacteria is consumed. As a result, it is possible that *E. coli* can be spread to CBs from unwashed hands that have encountered contaminated surfaces, or they may even be present within the mouth when contaminated food is consumed. Thus, the presence of *E. coli* was investigated.

Salmonella spp commonly causes gastro, and is spread through the ingestion of improperly cooked food, water, or through hands that have encountered animal faeces [263]. As a result, it is plausible that *Salmonella* spp could be present on used CBs if smokers have not washed their hands after coming into contact with animals, another infected person, or a contaminated material such as improperly cooked food. As this bacterium enters the body through the consumption of contaminated food, it is also possible that it is present within the mouth.

Pseudomonas aeruginosa is primarily responsible for the infection of humans by *Pseudomonas* strains of bacteria. Healthy people who are infected generally present only mild symptoms, but those with a weak immune system (such as those who are already sick) can suffer severe infections that can be fatal. *Pseudomonas aeruginosa* can cause ear infections, eye infections, and skin rashes in healthy people, whilst people with a weak immune system may contract blood infections and pneumonia [264]. It is

primarily spread by hand or from contact with contaminated equipment. *Pseudomonas aeruginosa* thrives in moist environments.

Most *Enterococcus* species are not pathogenic and are found naturally in the intestinal tract of humans and animals. However, some strains, including *Enterococcus faecalis,* are known pathogens that can cause several illnesses, including urinary tract infections and infections of open wounds. *Enterococcus* species are difficult to kill because they have developed a resistance to antibiotics, including penicillin [265]. From the intestinal tract, *Enterococcus* species could spread to surfaces and hands, resulting in contamination, where it can be distributed through direct contact [266]. It is entirely possible that these bacteria could be spread to the surface of CBs through contaminated hands, hence it was investigated.

Staphylococcus bacteria are found naturally on the skin and in/around the nose [267]. It is estimated that around one third of humans carry *Staphylococcus* [268]. In most cases these bacteria cause no harm but can become pathogenic if they enter the body. Its population growth can result in an infection. Serious issues include skin impetigo, wound infection, cellulitis, pneumonia, septic arthritis, sepsis, and endocarditis [267]. Some strains are particularly dangerous as they are resistant to antibiotics. It is likely that contact with the skin around the mouth or the hands of people that have touched their nose could result in contamination of CBs by *Staphylococcus* bacteria. As contaminated food enters the body, *Staphylococcus* bacteria may also be present in the mouth, which may be transferred to CBs.

Streptococcus bacteria are found in the upper respiratory tract and skin of humans. There are two types, Group A and Group B. Group A can cause the common 'strep throat', in addition to other illnesses, such as scarlet fever, impetigo, and cellulitis. Group B can cause blood infections, pneumonia, and meningitis [269]. The bacteria can be spread from person to person through sneezing and contact with contaminated surfaces. As a result, it was determined that these bacteria could be present on used CBs.

Table 7.3. Preliminary bacteriological test results for used and unused CBs (Pilot investigation 1).

Sample	Salmonella spp /15g	Escherichia coli MPN/g	Pseudomonas aeruginosa MPN/g	Enterococcus spp MPN/g	Coagulase +ve Staphylococcus spp cfu/g	Streptococcus spp cfu/g
Control (unused CBs)	Not Detected	<2	<2	<2	<100	<100
Sample 1	Not Detected	<2	<2	2	<100	<100
Sample 2	Not Detected	<2	<2	<2	800	<100

Note: the number of CBs in each sample were: control (75), sample 1 (72), sample 2 (71).

The results in Table 7.3 show that Coagulase +ve *Staphylococcus* spp were sufficient in number in sample 2, whilst there were insufficient numbers to be detected in sample 1. The presence of Coagulase +ve *Staphylococcus* is shown in Fig. 7.9. This bacterium was not detected in sample 1, which could be an indication that *Staphylococcus* bacteria struggle to survive on CBs for an extended period. Alternatively, it could indicate that *Staphylococcus* was only present on the CBs of sample 2. Despite

the confirmation of the presence of these bacteria, it has been reported that *Staphylococcus* requires a colony count exceeding 100,000 to infect humans [270]. However, it is possible that these bacteria could enter via the mouth or through cuts in the hands of workers, where the colony could flourish to an infectious concentration. As a result, it is recommended that gloves should be used when handling CBs.

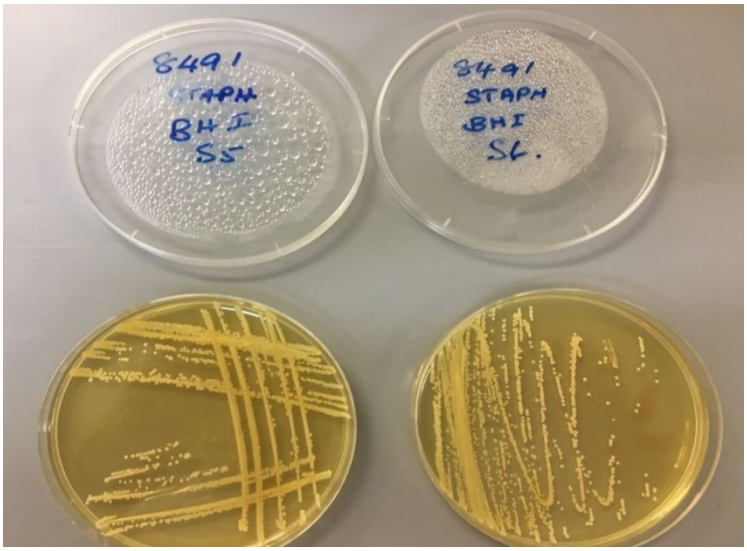


Fig. 7.9. Confirmation of presence of Staphylococcus spp in sample 2.

A very small concentration of *Enterococcus faecalis* was also detected in sample 1, which reinforces the prediction that unclean hands can transfer intestinal bacteria to CBs. It is possible that the original concentration was higher, and that the bacteria had slowly died over time. It is also plausible that the original concentration was just as low and that the bacteria can effectively survive on the surface of the CBs for a few hr. In either case, the concentration is a negligible one, but shows that the bacteria are capable of surviving on the surface of used CBs for an unknown period.

7.6.2.2. Pilot Investigation 2

This investigation consisted of one control group and four additional samples (Fig. 7.10): Control group (unused CBs); Sample 1- Used CBs, fresh CBs collected the day before the test; Sample 2 - Used CBs, older samples; Sample 3 - Used CBs, older samples that were dried at 105 °C for 6 h; and Sample 4 - Used CBs, fresh CBs in contact with mothballs for 24 h.

Similar to the pilot investigation 1, the control group for pilot investigation 2 was prepared using the same method and preparations. The sample size was approximately double when compared to investigation 1 as more tests were required to investigate the larger variety of bacteria. The control group was prepared a day before the test.

For sample 1, fresh CBs for this investigation were collected from the tops of bins and their tilt trays as well as ashtrays from restaurants/cafes around Melbourne's CBD. Fresh CBs were collected in this manner to reduce the chance of the butts being very old and the chance that the butts encountered the ground. The samples were collected on the day before the testing. For sample 2, old CBs that were a few months old were collected from RMIT storage. Sample 3 comprised old CBs that were a few months old and collected from RMIT storage; however, these CBs were placed in an oven at 105 degrees for 6 h. Sample 4 were fresh CBs collected in the same manner as sample 1 but had been treated with mothballs. The treatment consisted of placing two naphthalene (mothballs) balls inside a bag full of CBs and leaving for 24 h before removing them from the bag for testing. Samples 3 and 4 were two of the tested treatment methods for reducing the bacterium counts.

In this preliminary bacteriological investigation, the previous six bacteria were investigated: *Salmonella* spp, *Escherichia coli, Pseudomonas aeruginosa, Enterococcus faecalis,* Coagulase +ve *Staphylococcus* spp, and *Streptococcus* spp. Furthermore, an additional four types of bacteria were investigated: *Listeria* spp, *Legionella* spp, *Bacillus* spp, and *Clostridia* spp.



Fig. 7.10. CB samples utilised for preliminary bacteriological study 2.

Listeria spp are present in a wide variety of environments, including the soil, in animals, and in areas with water [271]. Most species of this bacteria group are not pathogens; however, there is one pathogen within this species that is harmful to humans, the '*Listeria monocytogenes' [271]*. This pathogen is harmful as it can cause listeriosis. Although most healthy people who are infected generally present only mild symptoms, it can prove fatal to people with a weak immune system, the elderly, and can cause pregnant women to undergo a miscarriage [271]. *L. monocytogenes* can exist and grow in food. Through

consumption, *L. monocytogenes* may exist within the mouth, where it is possible that it may spread to CBs. Therefore, the presence of this species has been investigated.

Legionella spp are naturally present in water sources of the environment. This pathogen is harmful when it grows within artificial water systems, such as pipes, that are common within households [272]. *Legionella* spp can cause two main illnesses – pneumonia and Pontiac fever [272]. Most healthy people who encounter *Legionella* spp are not affected. Those vulnerable include the elderly, smokers, and people with a weak immune system. *Legionella* spp can enter the body via respiration and aspiration. If a colony is large enough and is thriving within an artificial water system, it is possible that *Legionella* spp may be present in the evaporated water that people can breathe [272]. Though less frequent, people may also get sick when drinking water contaminated with *Legionella* spp as it could accidentally enter the lung [272]. It is possible that this pathogen is spread to CBs when exposed to a water environment, and, thus, the presence of this species has been investigated.

Bacillus spp are generally present in soil environments [273]. They are most commonly found in raw vegetables, water, organic matter, dust, and some flora species [273]; however, there have been scenarios where *Bacillus* spp was found in other types of food groups, such as meats and rice, and in hospital environments [273]. Most of these species are non-pathogenic. When *Bacillus* spp contaminates food, it typically survives the cooking procedure, and, as a result, food poisoning is likely to occur upon consumption [274]. This can be prevented through proper food handling. Gastrointestinal illness can develop if a high quantity of *Bacillus* spp is consumed [274]. It is possible that *Bacillus* spp could spread to CBs when exposed to soil, contaminated hands, contaminated surfaces, or the mouth, hence the presence of this species was investigated.

Clostridia spp are mainly present in soil environments where organic matter exists [275]. Other locations include the sediments of aquatic environments, and the intestinal tract and normal microbial flora of animals and humans [275, 276]. Pathogenic species that are harmful to humans include, but are not limited to, *C. botulinum* – which can cause food-borne botulism, *C. tetani* – which can cause tetanus, *C. perfringens* – which can cause wounds, surgical infections, gas gangrene, and food poisoning, and, lastly, *C. difficile* – which can cause antibiotic-associated diarrhoea, colitis, and pseudomembranous colitis [276, 277]. An external method for *Clostridia* spp to enter the body is from the consumption of improperly cooked and stored food, in particular, meat and improperly heated canned food [275]. It is possible that *Clostridia spp* could spread to CBs when encountering soil, contaminated hands, contaminated surfaces, or the mouth, hence the presence of this species was investigated.

Table 7.4. Preliminary bacteriological test results for used, unused, and dried used CBs and CBs with mothballs (Pilot investigation 2).

Sample	Salmonella spp /15g	Escherichia coli MPN/g	Pseudomonas aeruginosa MPN/g	Enterococcus spp MPN/g	Coagulase +ve Staphylococcus spp cfu/g	Streptococcus spp cfu/g	Listeria spp /25g	Bacillus spp cfu/g	Clostridia spp /g	Total Leginella spp cfu/g
Control (unused CBs)	Not Detected	<2	<2	4	<100	<100	Not Detected	70,000	Not Detected	<1000
Sample 1	Not Detected	<2	2	<2	<100	<100	Not Detected	70,000	Not Detected	<1000
Sample 2 Stored CBs	Not Detected	<2	<2	<2	<100	<100	Detected	39,000	Not Detected	<1000
Sample 3 Dried CBs	Not Detected	<2	<2	<2	<100	<100	Not Detected	440	Not Detected	<1000
Sample 4 with Mothballs	Not Detected	<2	<2	<2	<100	<100	Not Detected	4,100	Not Detected	<1000

The results in Table 7.4 show that *Salmonella* spp, *E.coli*, Coagulase +ve *Staphylococcus* spp, *Clostridia* spp, and *Legionella* spp were not found in any of the samples. The investigation of Coagulase +ve *Staphylococcus* spp for sample 1 (the freshest CBs of this investigation) was not present in comparison to pilot investigation 1, where it was present in sample 2 (the fresh CBs of that investigation). This could be because the sample was slightly fresher or that the bacteria were only present within the sample collected. In this investigation, a small count of *Pseudomonas aeruginosa* was found in sample 1. As this species of bacteria is primarily spread by hand, its presence may be due to unclean hands that transfer the bacteria to CBs. A small count of *Enterococcus faecalis* was found in the control sample, which was likely contaminated by the preparation environment, as this bacterium was not present in the control sample of pilot investigation 1.

In sample 2, *Listeria* spp was detected from the older CBs. *Listeria* spp is known to survive in waste water and sewerage sludge for a long period of time [278]. This sample may have been contaminated by this species of bacteria, where its characteristics of surviving for a long period of time can be observed.

Lastly, for *Bacillus* spp, counts of 70,000 cfu/g were found on control sample 1, 39,000 cfu/g were found on sample 2, and 440 cfu/g were found on sample 3 (Fig. 7.11). These higher quantities are not unusual for soil. As *Bacillus spp* have been recorded to grow on fresh and cured tobacco leaves [279], this could be a plausible reason for its presence. When comparing the dried CBs from sample 3 with the other samples, the count of *Bacillus spp* was significantly lower. This indicates that the majority of *Bacillus spp* present did not survive the drying process. It was noted that sample 4's preparation method was easier than sample 3's and was almost as effective at eliminating the *Bacillus* spp, counts.

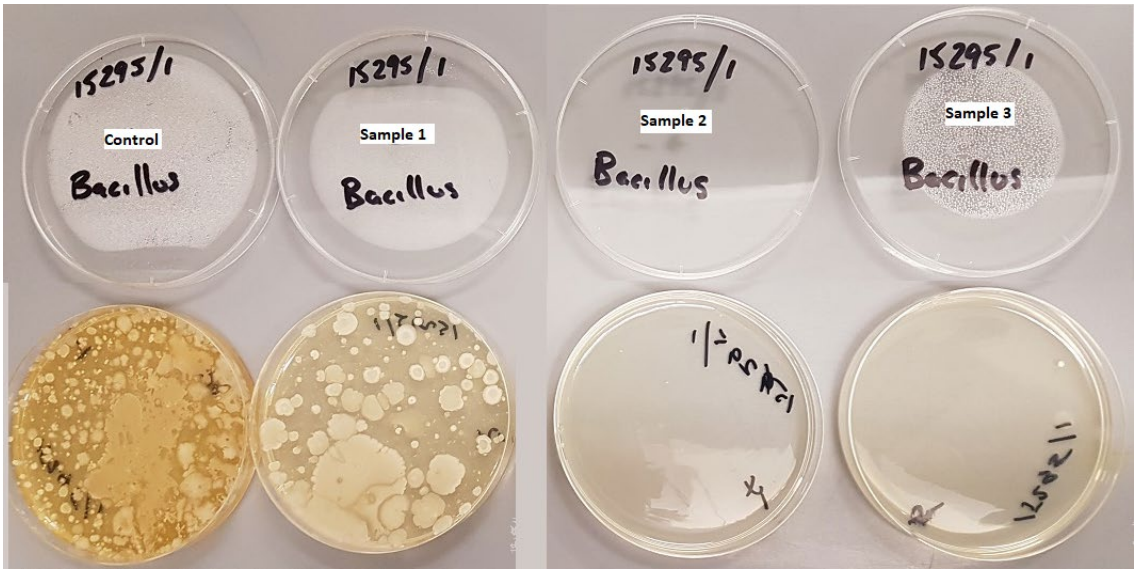


Fig. 7.11. Comparison of Bacillus spp present in samples.

The workers dealing with cigarette butts should wear the appropriate PPE and follow OH&S procedures to mitigate the potential risk that the pathogens and chemicals in CBs pose. Some of the PPE recommended include wearing long sleeves and gloves and the usage of an appropriate mask.

The results of both pilot investigations 1 and 2 should in no way be interpreted as a comprehensive study. Only six types of bacteria were investigated for pilot investigation 1, and 10 types were investigated for pilot investigation 2. In addition, only a small number of cigarettes were tested. About 75 CBs were collected for each sample for pilot investigation 1, and 150 CBs were collected for each sample for pilot investigation 2. This quantity was then divided to investigate each species of bacteria, thus limiting the scope of the findings.

This paper suggests that further investigation be conducted in this area, particularly given the positive confirmation that *Enterococcus faecalis,* Coagulase +ve *Staphylococcus* spp, *Pseudomonas aeruginosa, Listeria* spp, and *Bacillus* spp were present in such a small population of CBs.

Attempts at determining a way of investigating the presence of viruses failed, as it is impractical to develop a method to utilise for testing and almost impossible to verify if the produced results would be accurate or that the virus presented a viable infectious threat.

7.6.2.3. Presence of Viruses

The possible presence of viruses, such as hepatitis A, B, and C, as well as HIV on CBs was initially investigated. However, hepatitis A can only be spread through the ingestion of food or water contaminated by the faecal matter of an infected person and is rarely fatal [280]. As a result, it is incredibly unlikely that any CB would be contaminated by the virus. Similarly, hepatitis B has been

found in trace amounts in saliva (and therefore could contaminate CBs), however, this concentration is considered far too insignificant to transmit the virus [281]. Hepatitis C can only be spread through an infected person's blood directly entering the body of a non-infected person [282]. As CBs will not be ingested or come into contact with any worker's orifices (as they will be wearing PPE) Hepatitis C can be disregarded as a threat. HIV/AIDS is also a serious virus, which can only be spread through blood-to-blood contact or through contact with an infected person's various bodily fluids (not including saliva) [283]. As a result, HIV is highly unlikely to be present on CBs, and cannot be spread if handlers are protected with PPE.

7.6.3. Potential Cleaning Methods for CBs

7.6.3.1. Naphthalene

Naphthalene is a white crystalline powder that produces a well-recognised smell (mothballs). The chemical is isolated from coal tar and has a relatively short half-life in the atmosphere of 3 to 8 h [284]. Naphthalene is the major active ingredient in 'toilet cakes' and is used as a fumigant [285]. Naphthalene is a suspected carcinogen, and exposure to very high concentrations can lead to damage to blood cells and haemolytic anaemia. However, exposure to low concentrations in the open air should be safe for adults and children older than three [286]. Naphthalene and its derivatives have proven antimicrobial capacities [287]. In addition, a common dye, β-naphthol, possesses effective antimicrobial properties [288]. Mkpenie et al. (2008) confirmed this, testing azo-2 naphthol and 2-naphthol on five pathogenic microorganisms: *Escherichia coli*, *Staphylococcus aureus*, *streptococcus faecalis*, *pseudomonas aeruginosa,* and *bacillus subtilis [289]*. The results demonstrated that both chemicals were highly effective at killing all the microorganisms tested. Considering Naphthalene's short half-life in the atmosphere and antimicrobial abilities, in addition to the ease in which 'mothballs' and 'toilet cakes' can be located and purchased, these products are potentially a very quick, easy, and cheap method for cleaning CBs, as seen in Section 7.6.2.2 (Table 7.4). This paper suggests that toilet cakes or mothballs could be placed into the bags that dry CBs are collected in (i.e. by the collector) and tied. The fumigation effects would then likely kill any possible bacteria. Care should be taken not to inhale the fumes released when the bag is opened, as per the reasons stated above. The fumes produced should be minimal if only a few toilet cakes/mothballs are used. This method would be cheap, simple and effective.

The following methods are more suitable for large scale-cleaning plants.

7.6.3.2. Ozone

Ozone (O_3) is an unstable gas with powerful oxidising properties. It is predominately produced naturally as a result of ultraviolet rays irradiating oxygen molecules in the stratosphere, where up to

90% of all ozone is located [290]. Research has shown that ozone is an effective antibacterial agent and that it is useful against both gram positive and gram-negative bacteria, as well as spores and vegetative cells [291]. This antimicrobial power is predominately a result of the oxidising potential of ozone, where molecular ozone or its decomposition products, such as radical hydroxyl ions, react with bacteria/virus cells and disrupt normal functions, thereby resulting in inactivation or destruction [292]. Ozone generators are widely available, and therefore, could be considered easily accessible in most situations. Ozone has been used to sterilise equipment and containers in the food industry for many years, including in Japan, where the method has been utilised in food manufacturing facilities since 1982 [293]. Using ozone to sanitize surfaces, equipment, and even food is quite safe, as even though ozone is highly toxic, it quickly disassociates into oxygen [294].

Ozone is effective as an antimicrobial due to its potent oxidisation capabilities [295]. The third oxygen atom in ozone is only loosely bonded, and, as a result, is unstable and highly reactive. It will attach itself to organic matter, such as bacteria, and destroy them through the process of oxidisation [296]. Ozone has also been shown to have a capacity to destroy a wide variety of viruses, including hepatitis A and influenza A [297]. Treating fruits and vegetables with ozone has also been shown to increase the shelf life of the products [298, 299]. It is essential to remember that ozone is a toxic gas, and can be lethal in sufficiently high doses [300]. Therefore, it is essential that system designs ensure that humans have no exposure to ozone.

Ozone sterilisation of surfaces can occur in two main forms: gaseous ozone and ozonated water. Objects can be submerged in ozonated water for treatment or can be exposed to continuously generated gas-phase ozone. Rooms can be sealed whilst gaseous ozone is generated for sterilisation. The ozone will decompose into oxygen atoms at a rate that is influenced by both room temperature and the surface composition of any catalyst [301]. No person can be present inside a sealed room during the gaseous ozone sterilisation process for the reasons stated above. A gaseous ozone concentration of 25 parts-per-million (ppm) in the air at a Relative Humidity (RH) of 90% was found to be bactericidal after an exposure time of 20 min [294]. A large volume of research shows that ozone has substantial oxidising, antibacterial, and antiviral capabilities, which suggests that it could be applied for the treatment of any potential bacteria, viruses, and fungal spores on CBs. Considering the need to control the moisture content of mixes, and the relatively complex process of producing large volumes of ozonated water, this paper suggests that gas phase ozone is a more appropriate method.

7.6.3.3. Hydrogen Peroxide

Hydrogen peroxide (H_2O_2) is a weak acid with strong oxidising capabilities, and, therefore, can be utilised as a powerful bleaching agent [302]. Hydrogen peroxide is commonly utilised as a sterilisation technique in hospital settings, where it is used to disinfect rooms, in addition to re-usable medical and

biomedical equipment [303, 304]. It is also commonly used in food manufacturing and industrial settings and, more recently, in water treatment [305]. Hydrogen peroxide quickly decomposes into water and oxygen (its constituent atoms) and is, therefore, a non-toxic and environmentally friendly replacement for room sterilisation by formaldehyde, which is an expensive and carcinogenic chemical [306]. Hydrogen peroxide can be employed in either a liquid or gaseous form (vaporized hydrogen peroxide). As a liquid, it is commonly used as a surface disinfectant and, in some concentrations, as an antiseptic medicine [305].

Vaporised hydrogen peroxide (VHP) is used for the disinfection of surfaces, re-usable medical equipment, and room fumigation [303, 307]. Hydrogen Peroxide, as an oxidising agent, acts by forming HO. Free radicals react with bacteria/virus cells and thus prevent the correct functioning of these cells. This results in the inhibition of infection and replication processes [308]. An advantage of vaporised hydrogen peroxide when compared to gaseous ozone is that it has relatively low toxicity to humans and breaks down spontaneously into harmless components [305]. Vaporised hydrogen peroxide generation systems are widely available. As a result of its availability and sterilising properties, hydrogen peroxide may be an effective treatment method for any viruses, bacteria, or fungi that may be present on CBs.

7.6.3.4. Non-Ionising Ultraviolet Light Radiation

Short wave ultraviolet light (250 to 260 nm) has been used successfully to inactivate microorganisms in water and food in addition to those located on surfaces [309]. Additionally, UVC light has been successfully utilised to disinfect high-touch surfaces in hospital settings [310]. Research also indicates that the effective reduction of bacteria on stainless steel and polypropylene surfaces can be increased by utilising dry heat during the UV sterilisation procedure [311].

UV light comprises wavelengths between 100 and 400 nm. These wavelengths are non-ionising. Short wave ultraviolet light (UVC) refers to UV wavelengths of between 200 and 280nm [312]. UVC is commonly termed the ‘germicidal spectrum’ because it is the band width of UV radiation that demonstrates the greatest biocidal effects on bacteria [313]. The lethal effect of UVC light is due to direct alteration of microbial DNA, resulting from the formation of dimers. Once the DNA is damaged, the cells can no longer reproduce and are inactivated [314].

Short wave UV lamps are widely available and are identical to fluorescent lamps; however, they lack a phosphor coating, which allows the glass to transfer UVC. It is important to note that short wave UV lamps that produce radiation of 260 nm or lower will also produce ozone and, therefore, must be monitored for safety [309]. UV light, predominately UVB, but in some circumstances UVA, can penetrate human skin and cause cyclobutene pyrimidine dimers (mainly thymine dimers), which result in mutations found in skin cancers [315]. In contrast, UVC light is absorbed by the outer layer of human skin (epidermis) and is not known to form CPD (cyclobutane pyrimidine dimers) in human skin [316].

Despite this, exposure of the eye to UVC can cause incredibly painful corneal burns [316] or result in the development of CPD in the first 10% of the cornea [317]. Therefore, whilst UVC lamps are relatively safe to use, it is essential to ensure that workers are not exposed to UV irradiation systems whilst they are in operation.

7.6.3.5. Dry and Moist Heat Treatment

Heat treatment for the sterilisation of food in the food industry has been utilised for over 90 years [318], as well as in healthcare settings [319]. There are two heat treatment methods, dry heat sterilisation, and moist heat sterilisation. Dry heat sterilisation involves purely heated air with no or little moisture, which kills cells via oxidisation. Temperatures usually exceed 180 °C and exposure times generally last up to 60 min [320]. Temperature increases result in a reduced time of exposure required to achieve sterilisation. Alternatively, moist treatment sterilisation involves lower temperatures than dry heat sterilisation and is applied as steam under pressure, resulting in what is considered as being a generally more efficient sterilisation method in that highly resistant spores can be destroyed with temperatures of around 121 °C after approximately 30 min [320]. Dry heat sterilisation requires higher temperatures and longer exposure times because bacteria are far more resistant to dry heat sterilisation than moist heat sterilisation [318]. This is predominately a result of the thermal conductivity of water vs. air, where water allows more heat to be transferred into the bacteria cell.

However, steam will only destroy cells on contact with the surface of an object, whereas radiant heat will still result in cell death [319]. The implications of this are that large volumes of CBs may not be successfully treated with the moist heat sterilisation method, as it is unlikely steam would be able to penetrate/access all surfaces. However, moist heat sterilisation could be utilised if relatively small volumes of CBs are placed in an autoclave at any given time. This would introduce additional handling times (i.e. repetitive loading/unloading of small samples, etc.).

The ignition temperature of paper has been reported to be in the range of 218 to 246 °C [321], and, therefore, under dry heat temperatures of 180 °C CBs would not be expected to combust. However, great care must be ensured if dry heat sterilisation is utilised for the treatment of CBs, as the presence of various chemicals may alter the ignition temperature.

7.6.3.6. Notes on Nicotine

Nicotine is the most common chemical found in CBs. It is highly addictive and a known carcinogen. Furthermore, the toxicity of Nicotine does lend it some antimicrobial power, and high concentrations of nicotine, diluted in phosphate-buffered saline have been found to be effective at killing a number of common bacteria [322] including *Escherechia coli*. However, the same study found that nicotine had little or no effect on *Staphylococcus aureus.* Similarly, it has been reported that both aqueous and

ethanolic extracts of Nicotiana tabacum were effective at inhibiting the growth of M.tubercolosis [323]. Further study found that six extracts of Nicotiana tabacum (Ethanol, Acetone, Butanol, Water, n-Hexane, and Ethyl acetate) were effective in inhibiting the growth of a number of gram positive and gram-negative bacteria, many of which are known pathogens [324].

Historically, tobacco was used to treat several ailments. A review of 128 cases of tobacco treatment, published between 1785 and 1860, identified several allegedly successful applications of the plant to treat a number of ailments. These included the treatment of tetanus, rodent ulcer, general ulcers, and wounds [325]. Additionally, in 1924, a salve composed of burnt tobacco leaves and lanolin was pronounced to have antiseptic effects on superficial ulcers and wounds [326]. Before its introduction to Europe, tobacco was also utilized by the Native American populations. Fernando Ocaranza summarised the medicinal uses of tobacco by Native Americans in Mexico as being for the relief of pain and the treatment of wounds and burns [326].

Nicotine is a toxic chemical and lethal to humans. Although the literature generally accepts that a dose of 30-60mg is lethal, growing evidence supports 0.5 g as being a lethal dose[327].

Considering the findings discussed here, it would be expected that due to the toxicity of nicotine and other chemicals in CBs and CB smoke, that they would present an incredibly hostile environment for bacteria to survive in. However, our initial microbiological pilot study into the presence of bacteria on CBs has shown that bacteria can possibly survive on CBs, albeit for an unknown period. As a result, further investigations into this area should be undertaken.

7.6.4. Safe Handling of CBs

Prior to discussing the safe handling of CBs, it is known that high levels of dust exposure can lead to respiratory problems, most notably in a brick factory environment where workers are likely to encounter dust particles, which is seen in a large variety of studies conducted across the world. One such study found that exposure to high levels of dust in two brick manufacturing plants in Croatia had led to an increased prevalence of chronic respiratory symptoms in exposed workers when compared to the control group [328]. Similar research was conducted in Cape Town, South Africa, where 268 brick workers from five brickworks were studied. It was found that both smoking and exposure to dust had an impact on the prevalence of chronic respiratory disease. However, smoking generally had a weaker effect when compared to dust exposure, and only affected early onset symptoms [329]. Similarly, brick workers in Nepal and Pakistan were found to suffer significant volumes of respiratory problems that were predominately linked to particulate matter exposure [330, 331]. Additionally, studies have linked silica exposure in the brick manufacturing industry to silicosis, tuberculosis, and lung cancer. Fishwick et al. (2015) states that current workplace exposure contributes to the risk of developing silicosis [332]. As it is difficult to control the levels of dust generated within brick manufacturing plants and their

environments, it is suggested that employees should always wear protective masks when exposed to dust, particularly where no effort has been made to reduce the volume of dust generated.

When handling dry CBs, it is essential to wear appropriate masks. This is to protect against any possible inhalation of VOC or exposure to any pathogens that may be present in the sample, and is especially important in a brick factory due to the dust exposure alone. Hand protection is required to ensure that pathogens do not enter cuts by chemicals getting absorbed through the skin. In addition to this, it is very important to wear gloves when handling CBs that are moist, as heavy metals, such as lead, cadmium, and arsenic, can leach off the butts.

It is recommended that CBs be sterilised immediately upon or prior to delivery to the manufacturing site. Several appropriate methods have been discussed which have been deemed to be suitable for industrial scale sterilisation of used CBs. For small volumes of CBs, the use of mothballs or toilet cakes that contain Naphthalene can be cost efficient, as, when placed in the bag containing CBs, the Naphthalene will fumigate all or some of the bacteria that may be present (further investigation is recommended in this area). This solution however is considered viable for small volumes and should be used in the interim until a larger scale sterilisation method can be implemented

Previous investigations into the leachates from manufactured CB bricks show that, after firing, only trace amounts of metal leachates were found in bricks with 2.5, 5.0, 7.5, and 10.0% content of CBs [1]. These concentrations were lower than the regulatory limits set by USEPA (1996) and EPAV (2005) [243, 244].

Odor from the incorporation of CBs into brick production should also be addressed. The issue stems from VOCs being emitted from CBs and can either be addressed prior to entering the manufacturing site or during manufacturing of CB bricks. It should be noted that odor disappears after firing of the bricks occurs.

As always, it is essential that local regulations and standards regarding the handling of municipal waste be adhered to.

7.7. Implementation Guide

The following steps may be followed for the implementation of recycling CBs in fired-clay bricks.

i. For CB collection, aim to develop a close relationship with CB collection companies to facilitate the delivery of CBs to manufacturing sites. These CBs are normally collected from modern bins or receptacles.
ii. Once the CBs are collected, a sterilisation method (Section 7.6) should be used to clean the CBs from bacteria. The odor may be purged from CBs in this stage (Section 7.5). If mothballs (containing naphthalene) are used, they should be put into the bags containing CBs to inactivate

any bacteria that may be present. This can be done by the collector or by workers on-site. Care should be taken to not breathe in fumes when the bags are opened. The CBs will then be stored on-site.

iii. When ready, CBs can be incorporated into the brick clay mix through a method suggested in Section 7.5 for incorporation into Automatic Processes, and 7.5.2 for incorporation into Manual Processes. Once the CBs have been incorporated, the remaining steps that are common within the brick manufacturing process can be followed.

iv. Always ensure that relevant OH&S standards are followed, the correct PPE is worn, and that the fumes are not breathed. Refer to Sections 7.5.1.4 and 7.5.2.3 for more detail on the safe handling of CBs.

7.8. Conclusion

This study has presented and discussed some of the results of an ongoing study on recycling CBs in fired-clay bricks. The results of a CB energy contribution study, bacteriological investigation of CBs, and the properties of bricks incorporating CBs have been discussed. Also, the implementation procedure for recycling CBs in bricks and the pre-processing issues of CBs are analysed and discussed. The main findings are summarised and discussed below.

Energy savings: The calorific value of CBs was investigated. It was found that the average gross calorific value of CBs with remnant tobacco was 16.5 MJ kg^{-1}. This is a high value compared to the calorific value of black coal, which is 20 MJ kg^{-1}. If 2.5% of all the bricks produced annually around the world include 1% CB content, the energy consumption of the process can be reduced by over 20 billion MJ. This approximately equates to the power used by one million homes every year in the State of Victoria, Australia. Therefore adding 1% CBs (about 20 kg/m^3) to the brick clay results in significant energy savings, solves the CB pollution problem globally, and tangibly reduces the environmental impact of the brick manufacturing industry, while, concurrently providing financial savings to manufacturers through the reduction in energy for firing bricks. Fired-clay bricks are excellent for recycling waste materials. There is a need for a specially dedicated brick factory in every state/country for recycling cigarette butts and other similar waste materials. Due to the high calorific value of CBs, it is discouraged large amounts of CBs to be incorporated in clay bricks. During the firing phase, lime lumps may occur at elevated temperatures within the brick structure due to the increased organic matter found in CBs.

Bacteriological investigations: A serious concern in the recycling of used CBs is that they may be contaminated with bacteria or viruses from contact with people or the environment. As a result, this study undertook two pilot investigations. The presence of ten common bacteria was investigated on unused, fresh, and old CBs. In the first investigation, used CBs that were one week old, and used CBs

that were collected on the day of the testing day were compared with an unused control sample. A small colony of Coagulase +ve *Staphylococcus* spp was detected on the same-day sample. For the second investigation, used CBs that were collected a day before the testing, used old CBs, and used CBs dried at 105 °C for 6 hr were tested and compared with the control sample. The most significant finding from this investigation was that *Listeria* spp was detected in the used old CB sample and that all used and unused CB samples except the dried sample had significant counts of *Bacillus* spp. It was also found that the effect of the contact of naphthalene balls with fresh used CBs on the reduction of *Bacillus* spp, bacteria, was very significant. This study suggests that naphthalene-containing mothballs be used to fumigate bags used to collect and store CBs. Further study is recommended in this area. To address the potential for the presence of pathogens in CBs, several suitable sterilisation methods have been reviewed and discussed.

Sterilisation methods: Following the two bacteriological investigations, several industrial sterilisation measures have been studied, including ozone, vaporized hydrogen peroxide, moist heat sterilisation, and dry heat sterilisation, Ultraviolet Light Radiation (UV), and Naphthalene. These methods could be employed to sterilise large volumes of waste materials. Hydrogen peroxide may be an effective and practical sterilising method for any viruses, bacteria, or fungi that may be present on CBs for large scale-cleaning plants.

Properties of bricks incorporating CBs: The compressive strength of 1% CB content bricks with 15.5% and 17.5% moisture content averaged 27.5 MPa and 25.8 MPa, respectively, down from the 43.2 MPa exhibited by the 15% moisture content control bricks. The standard compressive strength for bricks for low-medium rise residential structures in Australia is between 10 and 20 MPa. The decrease in compressive strength as a result of the increase in CB content, establishes a limitation regarding the content of CB that can be incorporated in fired clay bricks. The dry density of the bricks decreased from 2134 kg m^{-3} to 1964 kg m^{-3} (8% reduction). This results in bricks with lower thermal conductivity, which are lighter and therefore reduce the 'dead-weight' of potential structures. Lower thermal conductivity can have a significant influence on the energy consumption of a home, as it reduces the need to use air-conditioning systems to adjust the temperature inside the house. The water absorption and initial rate of absorption of the brick increased as expected due to the reduction in density of the samples. However, the changes were minor and well within acceptable limits. These results confirm the previous results published by Mohajerani et al. (2016).

Implementation procedures: To incorporate the CBs into the automatic brick manufacturing process, three processing methods have been suggested. Shredding CBs is expected to improve uniformity in the particle size and distribution of CBs within the mix, thereby leading to clay bricks with more uniform properties. However, shredding of the CBs must be performed in a fully enclosed environment,

such as an enclosed industrial shredder. This is necessary as it keeps the dust and chemicals from spilling out into the environment.

CB collection systems: In Australia, several companies have developed CB disposal systems. These companies provide receptacles for the disposal of CBs. Currently, the collected CBs are placed in landfill or incinerated by the collection company. There are different types of receptacles, including wall-mounted, post-mounted, and free standing.

CB odor: The odor from VOC emissions in CBs can be an issue for recycling CBs in bricks or any other application. This has been discussed and analysed. UV light has the capacity to oxidise and shatter the molecules in VOC elements from CBs to form new molecules that are non-odorous or less odorous. It is possible to deodorize and sterilize CBs at the same time using a UV light with an appropriate wavelength. This was studied, and UVC or UVV is suggested for this dual treatment. A comprehensive experimental study is recommended in this area.

7.8.1. Proposal

About 6 trillion cigarette butts are littered worldwide every year, resulting in over 1.2 million tonnes of toxic to highly toxic waste contaminating our cities and the environment. Considering the combined risks from hundreds of highly toxic chemicals and possible pathogens in cigarette butts, it is proposed that the littering of this waste anywhere in cities and the environment be strictly prohibited, and that offenders be heavily fined. This strategy should be supported by appropriate education, guidelines, and advertising. Our cities, parks, waterways, beaches, and oceans have been contaminated for many years, with millions of tonnes of unsightly and toxic cigarette butts. Effective and strong laws and guidelines by governments for solving this global pollution problem are overdue and need to be established urgently. This action should be supported by an adequate number of receptacles installed at critical locations in cities and public places by local governments for the collection of CBs for recycling. The sincere, effective, and urgent support and cooperation of smokers, governments, educators, and waste management industries are essential for ending the littering of cigarette butts in the environment.

Chapter 8: Application of UVC light to minimize the odor generated from volatile organic compounds during the production of fired clay bricks incorporating cigarette butts

Abstract

The increasing trend in global cigarette consumption in many parts of the world has led to a tremendous volume of waste being generated in the form of used cigarette butts (CBs). Mohajerani et al. (2016) proposed a novel method involving the incorporation of CBs in fired clay bricks (FCBs). However, an implementation issue has surfaced, where strong odors are being released through the emission of volatile organic compounds (VOCs) during the storage of CBs. To address this problem, the objective of this study was to determine techniques to detect and analyze the VOCs generated from CBs, along with investigating methods to minimize or eradicate the odors generated during the production processes of FCBs incorporating CBs. Charcoal sorbent tubes were used to collect the gas components, and gas chromatography-mass spectrometry (GC-MS) was used to identify and quantify the VOCs generated. UVC light was implemented to determine its effect on odor reduction and mitigation. The main hazardous constituents in freshly discarded CBs were found to be benzaldehyde, benzyl alcohol, p-xylene, styrene, and 2,5-dimethylanisole. The results revealed that UVC radiation could diminish the release of the toxic components and minimize the odor generated from CBs during the storage and manufacturing process of FCBs incorporating CBs. Furthermore, it was found the emission of VOCs during the mixing, compaction, and firing processes were much higher for untreated CBs as opposed to the UVC light treated CBs.

The research presented in this chapter has been submitted to the following:

Kurmus, H., Mohajerani, A., and Suntovski, J., 2021. Application of UVC light to minimize the odor generated from volatile organic compounds during the production of fired clay bricks incorporating cigarette butts. Clean Technologies and Environmental Policy. (under review)

8.1 Introduction

One of the most common types of littered waste globally is cigarette butts (CBs). It was estimated by The Tobacco Atlas, that in 2016 alone, about 5.7 trillion cigarettes were smoked worldwide [333]. This amounts to over 1.2 million tonnes of CB waste produced annually. This is indeed a significant amount of waste, and it is expected to rise as the global population increases [334].

CBs are manufactured with filters to reduce the tar intake by smokers. The filter is made from cellulose acetate, which is a synthetic compound derived from acetylation of the plant substance cellulose [335], which can take 18 months or longer to degrade naturally [336]. CBs are a nuisance, as they are predominantly littered and tend to create a plethora of waste with the potential of leaching harmful chemicals and heavy metals into the environment, which can be detrimental. A number of heavy metals found in CBs which are potentially leached into the environment include chromium, cadmium, nickel, arsenic, and lead [336, 337]. Therefore, it is paramount that waste materials be managed efficiently [338]. One such way of managing and reducing CBs in the environment is through recycling the waste in the production of construction materials.

In recent years, a novel method has been proposed to incorporate CBs into the production of fired clay bricks (FCBs) [1, 2, 22, 23, 115, 126, 176, 339]. Based on several years of research, Mohajerani et al. [1] proposed the CB problem can potentially be abolished by incorporating 1% CBs into 2.5% of the world's total FCB production. Numerous mechanical and physical tests were conducted on the FCB samples incorporating various percentages of CBs, including compressive strength, density, water absorption, initial rate of absorption, efflorescence, shrinkage, energy savings, thermal conductivity, and gas emissions and the results were promising. Moreover, a comprehensive leachate analysis of heavy metals and polycyclic aromatic hydrocarbons (PAHs) in FCBs incorporating CBs was conducted to investigate the potential leachate of heavy metals and toxic chemicals during the use and disposal of the material. The modified FCBs were found to be below the regulatory threshold limits and categorized as non-hazardous waste when compared to the EPA Victoria industrial waste guidelines.

An implementation issue with the addition of CBs in FCBs is the high volume of odors generated from the mass storage of CBs in industrial settings and during the manufacturing process of FCBs incorporating CBs. The odor released from waste materials has received increasing attention, as the consensus has shifted towards an environmental-centric approach. Odors are perceived as atmospheric pollutants, hence why it is a source of complaint in many industrial processes.

Concerns associated with the strong odor of CBs and the potential release of volatile organic compounds (VOCs) during the storage and manufacturing process of CBs in FCBs need to be addressed and investigated. Practical techniques need to be adopted to measure the VOCs released from CBs. Therefore, the aim of this research was to determine techniques to detect and analyze the emission of

VOCs generated from CBs, along with investigating methods to minimize or eradicate the odors generated from CBs. Charcoal sorbent tubes were used to collect the gas components, and gas chromatography-mass spectrometry (GC-MS) was used to identify and quantify the VOCs generated. Industrial methods and techniques adopted to eliminate CB odor on an industrial scale were reviewed and analysed, including ultraviolet-C (UVC) light, ozone, and vaporized hydrogen peroxide. Several factors were considered when selecting the most effective and efficient equipment. High safety and affordability of the proposed equipment were other significant characteristics taken into consideration when selecting the preferred device. Based on an extensive review, the UVC was implemented to determine its effect on odor reduction and mitigation. The purpose of this study includes:

1. Determination of VOCs in freshly smoked CBs and UVC-treated smoked CBs.
2. Determination of VOCs during the mixing process of raw clay soil, water, oven-dried CBs, and UVC-treated oven-dried CBs.
3. Determination of VOCs during the compaction process of clay bricks incorporating oven-dried CBs and UVC-treated oven-dried CBs.
4. Determination of VOCs during the firing process of clay bricks incorporating oven-dried CBs and UVC-treated oven-dried CBs.

8.2 Odor

8.2.1 Quantifying Odor

Smell is a biochemical reaction where odorous molecules attach themselves to olfactory receptors located in the nose. The chemicals are required to be in gaseous form in order to both enter and attach to the olfactory receptors. The nose also filters out the large particulate matter with nostril cavities lined with mucus. Once the molecules attach to the olfactory receptors, the receptors then send an electrical signal to the brain, which we perceive as odor [340]. Not all molecules can attach to olfactory receptors, which leads to some molecules being odorless; for example, methane is considered an odorless chemical [341]. However, it is difficult to chemically or physically measure and quantify odor, as smell is a human perception that varies based on the individual [342].

8.2.2 Composition of Cigarette Odor

The composition of cigarette odor is complex as cigarettes can emit over 4000 different chemicals [336]. A combination of different chemicals can enter the nose and create smells, with some chemicals having more pungent odor whilst others are present in higher concentrations [342]. In order to simplify the research, it is best to describe odor using functional groups rather than listing each individual molecule. Functional groups are particular combinations of atoms within molecules that have their own distinctive properties, irrespective of the other atoms present in a molecule [343].

Cigarette smoke contains the following odor causing functional groups, aromatic hydrocarbons, saturated hydrocarbons, terpenes, aromatic hydrocarbons, cyclic hydrocarbons, alcohols, organic acids, esters, unsaturated hydrocarbons, N-containing compounds, ketones, and O-containing compounds [344]. The majority of the tobacco odor is made up of aldehydes which contributed to 50-90% of the odor, depending on environmental conditions [344]. The individual aldehyde compounds such as acetaldehyde (1.2~26%), butylaldehyde (2.3~18%), furfural (7.6~58), valeraldehyde (1.1~6.7%), and hexaldehyde (1.4~41%) contributed most to the odor. The next highest group, which is organic acids, only contributed 0.92-6% of the odor [344].

Therefore, if aldehydes were to be reduced, the cigarette odor intensity would also reduce. It is to be noted that even if aldehydes were to reduce, the other pungent odor groups such as alcohols, organic acids, and ketones would also have to be monitored so that the chemical reactions don't result in the increased pungent odor groups.

However, the above aforementioned functional groups may not be present in CBs as they are too volatile or can be retained and are unable to be remitted. Aldehydes may become trapped but then remitted quickly out of CBs due to the volatile nature of aldehydes. Lower molecular weight chemicals tend to break through the filters faster. Some of the medium volatile chemicals that are present and can be emitted from CBs over a long period of time include nicotine, polycyclic aromatic hydrocarbons (PAHs), hydrogen cyanide, and some organic gasses [345].

8.2.3 Diffusion and Flow

Diffusion is the main cause of odor emission from the CBs, where odor molecules travel from an area of high concertation to low concentration [346]. CBs, although not an effective adsorptive material, can still capture some particulate matter, and the filter may reemit some of the particles back into the environment as third-hand smoke, similar to how some smoke room walls reemit smoke after a given period of time [347, 348]. Nicotine and tar particulate matter are considered heavy and sticky, with the median diameter size of the particulate matter being 300 nm [346]. This could explain why nicotine smoke odor lingers on CBs well after combustion has ended and stays static within the environment. When compared to e-cigarette vapor which can vaporize in seconds, juxtaposed to that regular cigarette odor stays in the environment for a long time depending on the ventilation of the room. The reemitting rate is dependent on environmental conditions such as temperature, relative humidity, and volatility of the chemicals being emitted [345]. Other possible movement methods of the particles include gravitational settlement, electrostatic effects, thermophoresis, and ventilation [346].

8.2.4 Measurement of Odor

Measurement of odor can be achieved in two unique ways. Method one involves counting the individual molecules by weight using a specialised detection method such as the gas chromatography-mass-spectrometry (GC-MS) to give specific results. The second method is evaluating the odor as a whole and giving it a rating based on the odor threshold and intensity of the odor. A device such as an olfactometer does this by diluting fresh air with odorous air until it detects the odor [349]. Method one considers all the various VOCs and counts each VOC compound, whilst method two is an overall evaluation of odor based on the odor threshold and intensity. In general, in the last decade, the use of olfactometers has gained attention in the field of research due to lower operational costs, ease of operation, and accuracy in results, however, the initial setup cost is substantially higher compared to other methods [350].

8.2.5 Techniques for Odor Reduction

CBs are considered lethal wastes as they consist of heavy metals, toxic chemicals, and bacteria and therefore require treatment prior to being handled by humans when being used for recycling purposes [351-353]. Moreover, the odor emitted from CBs in the form of VOCs is a health issue when potentially stored in mass amounts in an industrial setting. Therefore, it is vital to find a method capable of treating CBs on a large scale which can potentially be adopted in the brick-making industry. A few of the popular and widely employed techniques utilised to eliminate odor on an industrial scale include UVC light, ozone, and vaporized hydrogen peroxide.

8.2.5.1 Ultraviolet Light-C (UVC)

The application of UVC light is a very popular technique utilised worldwide in the food industry and for disinfection and sanitisation purposes due to its low environmental impacts and high safety standards as it emits radiation in the range of 200 to 280 nm (Fig. 8.1) [354]. Initially, mercury was employed in the production of UV lamps as the principal source of light emission, however, the import or export of products containing mercury was globally prohibited due to the environmental health risks the heavy metal posed [354]. An alternative to the mercury-based UV lamps was the use of light emitting diode (LED) UV lamps which became adopted extensively due to it being affordable and eco-friendly [354].

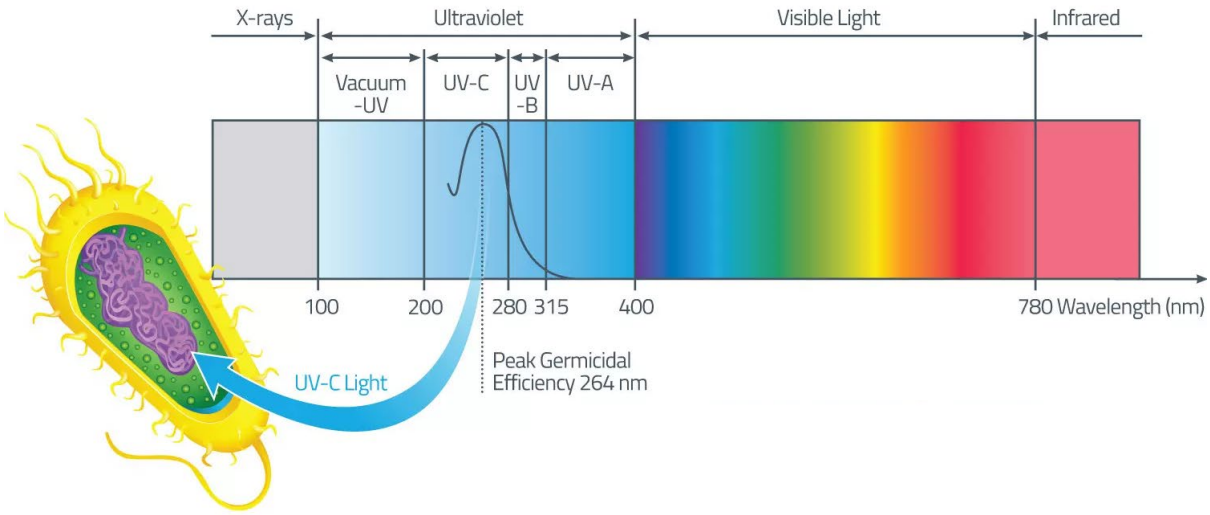


Fig. 8.1. Spectrum of light and UVC light radiation range.

The technique of sanitising or disinfecting a surface or product through the use of UVC light requires an exposure period ranging from 10 – 30 min, however, the exposure time depends on the characteristics of the material [355]. It is recommended plastic and fiber-based materials, similarly to CBs require an exposure period of roughly 5 min to UVC light to treat the material [356].

8.2.5.2 Ozone

Ozone is well-known for its oxidation characteristics and is produced through ozone generator devices which break down oxygen molecules into two single oxygen atoms [357]. The two single atoms then bind with molecules existing in the air to produce ozone molecules [358]. Car dealer services and the hospitality industry prominently utilise ozone gas to eradicate the unpleasant smells of smoke and paint. Moreover, ozone is applied for disinfection purposes and for taste and odor removal in wastewater [359]. Therefore employing ozone on waste materials to remove odor is potentially a valid method, especially to eradicate or mitigate the harmful smells emitted from CBs.

8.2.5.3 Vaporised Hydrogen Peroxide

Hydrogen peroxide (H_2O_2) has been implemented in sanitation since its discovery by the French chemist Louis-Jacques Thénard in the early 1800s [360]. There has been a plethora of reiteration techniques and methodologies over the years to refine the use of hydrogen peroxide as a disinfectant agent. H_2O_2 has become an integral component in surface disinfectants in the food and pharmacology industries. A technique to improve the delivery of H_2O_2 as a sterilising agent was first developed by Francis Moore and Leon Perkinson in 1977, which was based on a cold sterilization process using the vaporized hydrogen peroxide sterilisation method [361].

In its vapor form, the chemical is transparent, water-soluble, and transparent [360]. It is an excellent disinfectant due to its acidic nature, and when found in high concentrations, the substance acts as an oxidizing agent and is therefore effective at nullifying bacteria, viruses, VOCs, and odor. Based on research, it was found the vapor form of hydrogen peroxide is much more effective and adequate as opposed to the aqueous form. This is due to the gaseous form consisting of greater kinetic energy and therefore proceeding to a faster penetration rate [360].

Implementation of vaporized hydrogen peroxide can be further divided into atmospheric pressure or vacuum conditions. An atmospheric pressure condition is generally implemented for the disinfection of large-area surfaces such as rooms and isolators [362]. The equipment to conduct such work ranges from handheld devices to industrial instruments, which can be integrated into ventilation systems. These instruments tend to create a vapor or a fog of a diluted solution of hydrogen peroxide with varying concentrations (eg. 25% and 35%), which can be used to treat and disinfect objects or an open area.

In this section, three industrial methods and techniques were reviewed and analysed to determine the most effective and efficient process to apply to eliminate CB odor/smell on an industrial scale, including UVC light, ozone, and vaporized hydrogen peroxide. Several factors were considered when selecting the most practical equipment. High safety and affordability of the proposed equipment were other significant characteristics taken into consideration when selecting the preferred device. Based on an extensive review, UVC light was selected to be implemented in this study as UVC apparatuses are simple, efficient, and effective equipment, with minimal training required to operate. The use of UVC light to treat CBs and the smell emitted from littered CBs will require a sanitization device/chamber that is commercially available. Therefore, a Lumin Multipurpose Sanitizer was purchased for the purpose of this study.

8.3 Materials and methods

8.3.1 Materials and reagents

This study utilised CBs of various brand sizes supplied by Buttout Australia Pty Ltd. The raw clay soil was obtained by PGH Bricks and Pavers (VIC, Australia). For the preparation of standard solutions, benzene-d_6, toluene-d_8, and the 48-component indoor air standard were purchased from Sigma-Aldrich Pty. Ltd. (NSW, Australia). For the sampling of VOCs, an AirCheck5000 pump and charcoal sorbent tubes were purchased (Active Environmental Solutions, VIC, Australia). A Lumin Multipurpose Sanitizer (VIC, Australia) was used to treat the samples with UVC light. The fundamental properties and functions of the materials and equipment utilised for this study can be seen in Table 8.1.

Table 8.1. Fundamental properties and functions of raw materials, standards and equipment.

Materials	Fundamental properties		Functions or purposes
Soil	Chemical composition (%)	SiO2 (62.84), Al2O3 (19.95), Fe2O3 (6.15), K2O (4.87), Na2O (2.46), MgO (1.62), TiO2 (1.19), CaO (0.27)	Main raw material for fired clay bricks
	Mineral composition	Quartz, Muscovite, Clinochlore, Albite, Mikasaite	
	Soil classification	Sandy silty clay (MC)	
	Sample (%) < 75 µm	77.5	
	Liquid limit (%)	34	
	Plastic limit (%)	25	
CBs			Acts as a pore forming agent
Benzene-d_6	Purity (analytical pure, %)	99	Internal standard
	Chemical formula	C6H6	
	Molecular weight	84.15	
	Concentration (µg/ml)	2000	
Toluene-d_8	Purity (analytical pure, %)	99.6	Internal standard
	Chemical formula	C7H8	
	Molecular weight	100.19	
	Concentration (µg/ml)	2000	
48-component indoor air standard	Concentration (µg/ml)	1000	Calibration standard
AirCheck5000 pump	Flow (ml/min)	5-500	Pump gas into sorbent tube
Coconut charcoal sorbent tubes	Size (mm)	110	Gas collection
	Sorbent (mg)	200/400	
	Mesh	20/40	
Lumin Sanitiser	Wavelength (nm)	254	Expose samples to UVC light
	5-minute cycle		

8.3.2 Production process

The production process can be divided into the exposure of CBs to UVC rays, oven drying of raw materials at 105 °C for 24 h, mixing of raw materials with water for 25 min, compaction of samples at 240 kPa, and firing of brick samples for 3 h at 1050 °C. Detailed information regarding the production process of FCBs incorporating CBs can be found in Kurmus and Mohajerani [176]. The exposure of CB samples to UVC-rays in the initial step of the process was carried out using a 5 min exposition time. The UVC lamp emitted rays at lengths of 254 nm. Changes in the concentration of VOCs were determined in the CB samples before and after the exposition to UVC light rays. The processing technology of unexposed CB FCB samples is the same as that of exposed CB FCB samples, except for the use of the Lumin sanitiser in the initial step. The necessary details of the preparation process, such as the control parameters, are shown in Fig. 8.2. The samples collected at each stage comprised of both

UVC treated/untreated CB samples to compare and analyse the effect of UVC light radiation on reducing the emission of VOCs.

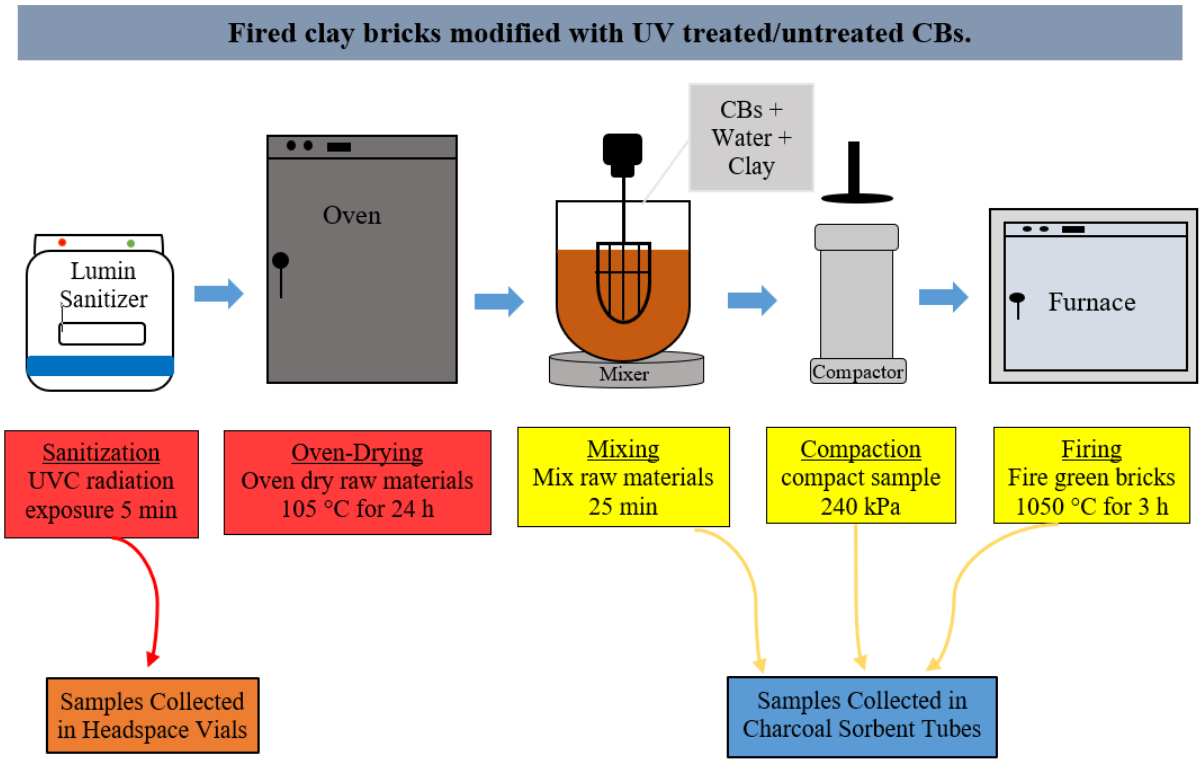


Fig. 8.2. Laboratory process flow chart of FCBs incorporating treated/untreated CBs and sample collection methods.

8.3.3 VOC collection process

8.3.3.1 Headspace Vials

Freshly smoked CB and UVC-treated smoked CB samples were placed in 20 mL vials and analysed with headspace analysis to determine the mass emitted from the butts into the headspace air of the vial. The method followed is a modified version of Method 5021A [363].

8.3.3.2 Charcoal Sorbent Tubes

Charcoal sorbent tubes were used for the solvent extraction method [364]. The sorbent tube cartridges are glass tubes with both ends glass sealed and contain two sections of 20/40 mesh of foam wool to separate the adsorbing section from the back-up section containing 400 mg and 200 mg of charcoal. Samples were collected through active sampling at the large end of the tube. A personal sample pump was used to pump air samples at a flow rate of 500 mL min^{-1}. During the mixing process of the raw materials and water, the sampler was positioned above the Hobart mechanical mixer for effective

sampling. Throughout the compaction process, the sampler was located at the breathing zone of the person (1.5 m above the ground), and during the firing process, the sampler was located right above the opening valve of the electric furnace.

After sampling, the charcoal tubes were detached from the pump, and the two open sides were firmly sealed with PTFE caps to prevent unwanted desorption. The samples were stored in a freezer at -5 °C until the analysis, to avoid a possibility of deterioration and contamination. Weekly field blanks were tested by placing a charcoal tube with broken ends to the sampler but without passing any air sample through, to identify any possible contamination. The blank tube was then collected and analysed.

In preparation for the analysis, each charcoal tube was scored with a quartz blade in the adsorbing section of charcoal and broken open. The glass wool was removed and discarded. The charcoal in the adsorbing section was transferred to a headspace screw top 20 mL clear vial. The separating foam was removed and discarded, and the backup section was transferred to another capped vial. These two sections were analysed separately.

To desorb the samples, 1 mL of desorbing solution was pipetted into each sample vial. The desorbing solution consisted of 25 µL of internal standard solution in carbon disulphide. The sample vials were capped with PTFE/Sil magnetic headspace caps as soon as the solution was added. Immediately after the solvent process, 250 µL of sample extract was injected into a split/splitless inlet port. Chromatographic conditions are detailed in the following section.

8.3.3.3 Chromatographic conditions

In both methods separation and detection were performed in an Agilent 7000 Series GC-MS triple quad system using two Agilent DB-5MS columns 5 m long and 0.25 mm in diameter. Analyses was performed by injecting 250 µL of extract in a split/splitless port at 250 °C with a split of 7.5 mL min^{-1}. The oven temperature of the GC was initially held at 40 °C for 5 min then raised to 260 °C at a rate of 20 °C min^{-1} and held at that temperature for 8 min.

8.3.3.4 Calibration Standards

Quantitative analysis was only conducted on the samples collected from the sorbent charcoal tubes. Calibration for the charcoal sorbent tubes was done by enriching the sorbent tubes with the internal and calibration standards. Concentrations between 3.125 to 50 µg ml^{-1} of standard solutions of the forty-eight target chemicals and internal standards benzene-d_6 and toluene-d_8 were injected into the sorbent tubes. The syringe was placed in the sorbent for 20 s and immediately placed in the headspace vials and analysed as described above.

8.4 Results and discussion

8.4.1 Components in headspace vials

The process of removing VOCs through UVC radiation is called photodegradation which involves two pathways [365]. The first pathway being photolysis whereby the VOC molecules are broken down by UVC light photons and one or more covalent bond is cleaved resulting from the absorption of light. The second pathway is photooxidation, whereby the UVC irradiation excites the VOC molecules resulting in an oxidation reaction. Fig. 8.3 shows the composition of the VOCs generated from UVC light treated and untreated CBs. The peak heights of different substances are different at various retention times. The higher peak value represents the high content of the component. Fig. 8.3 shows that the peak heights of freshly smoked CBs are generally higher than those of UVC light treated CB samples. Especially in benzaldehyde, benzyl alcohol, p-xylene, styrene and 2,5-Dimethylanisole, the peak value of untreated CBs was significantly higher than that of treated CBs. This finding indicates that UVC light can effectively reduce the release of volatile gases in CBs. Similarly, Włodarczyk-Makuła [366] found the exposure of UVC-rays on the removal of polycyclic aromatic hydrocarbons (PAHs) in wastewater to be effective. A decrease of 65% in PAHs was achieved when the wastewater was exposed to radiation, the exposure furthermore resulted in the disinfection of the wastewater samples.

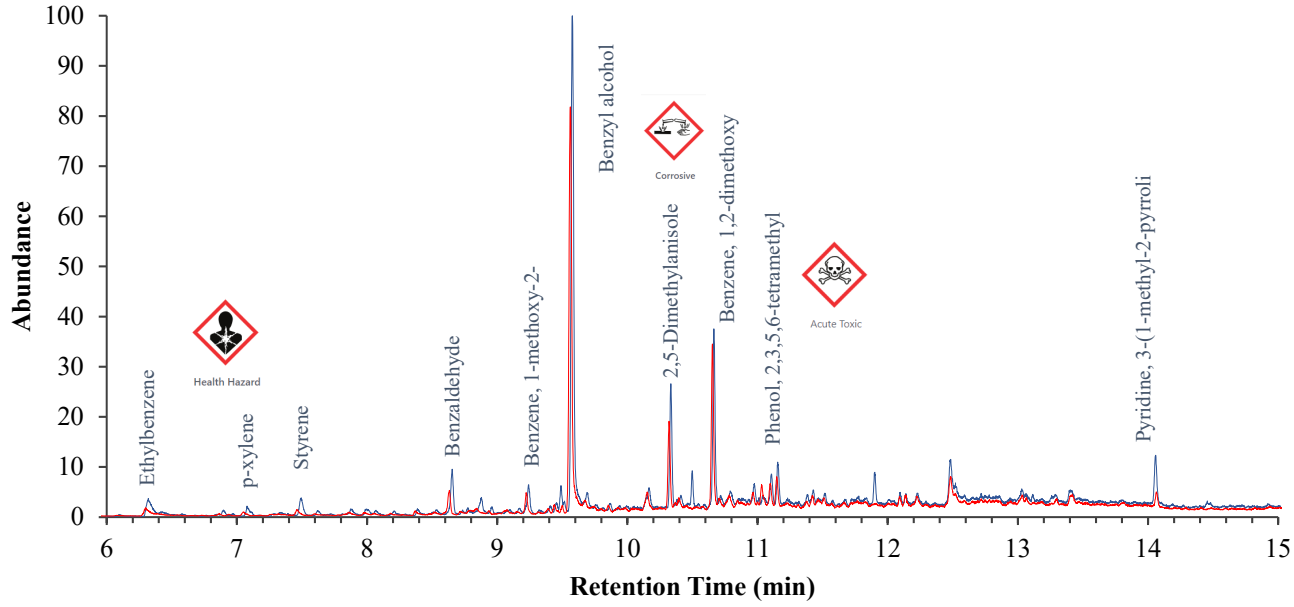


Fig. 8.3. Mass spectrogram of VOCs in UVC light treated and untreated CB samples.

8.4.2 Particles filtered by charcoal sorbent tubes

The selected GC-MS quantification ion chromatograms of the 48 indoor compounds can be seen in Fig. 8.4 and Table 8.2. The chromatogram displays excellent separation, sensitivity and peak shape. The

internal standards Benzene d_6 and Toluene d_8 were used to quantify acetone, benzene, Bromodichloromethane, 1-Butanol, Ethyl alcohol, Chloroform, Dibromochloromethane, Toluene, Decanal, 1,4-Dichlorobenzene, *m*-Xylene, Decane, 4-Ethyltoluene, 1,2-Dichloroethane, 1,3,5-Trimethylbenzene, Dichloromethane, 1,2-Dichloropropane, 2-Butanone, 2,4-Dimethylpentane, Dodecane, 3-Ethyltoluene, Pure, Ethylbenzene, 2-Ethyltoluene, Heptane, 2,2,4-Trimethylpentane, Hexane, (*R*)-(+)-Limonene, 4-Methyl-2-pentanone, Nonanal, 1,2,3-Trimethylbenzene, Nonane, Trichloroethylene, Octane, Pentadecane, (1S)-(-)-alpha-Pinene, 1-Propanol, 1,2,4-Trimethylbenzene, 2-Propanol, Styrene, Tetrachloroethylene, Hexadecane, Tetradecane, 1,2,4,5-Tetramethylbenzene, (-)-beta-Pinene, Tridecane, Undecane, *o*-Xylene and *p*-Xylene. The results behaved linearly over a concentration range from 3.125 µg ml^{-1} to 50 µg ml^{-1} for all VOCs.

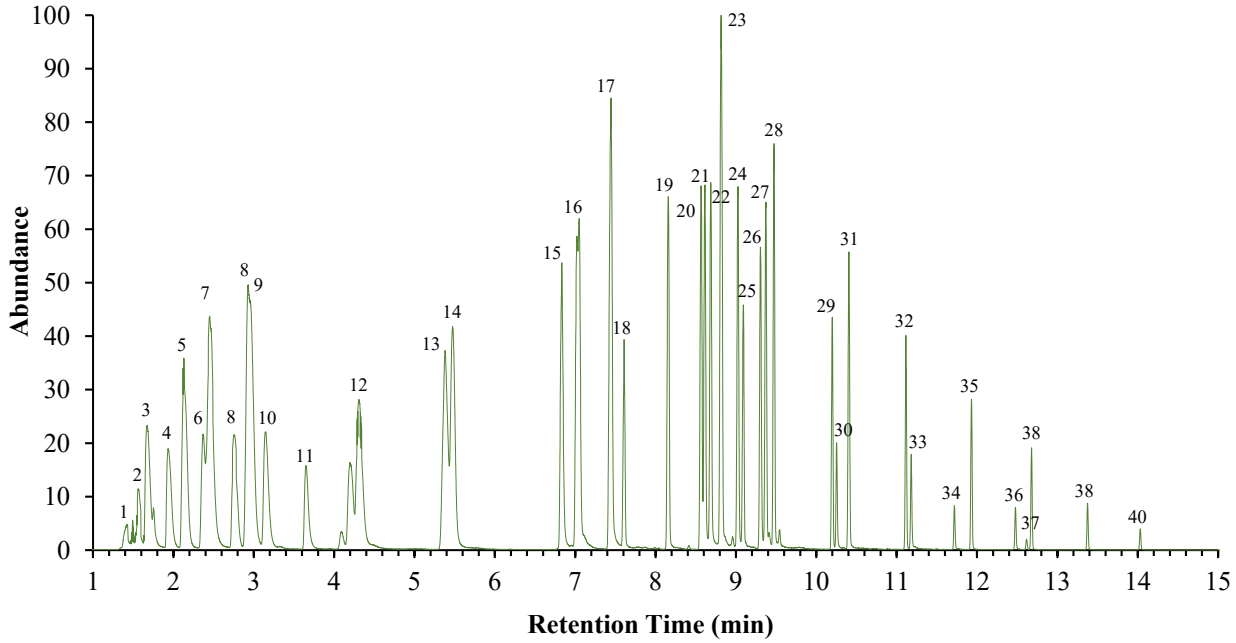


Fig. 8.4. A typical chromatogram of the 48 air indoor VOC compounds.

Table 8.2. VOCs evolved from the 48 indoor compounds.

1	1-Propanol	15	Tetrachloroethylene	29	Undecane
2	2-Propanol	16	m-Xylene	30	Nonanal
3	Dichloromethane	17	Styrene	31	1,2,4,5-Tetramethylbenzene
4	1-Butanol	18	Decane	32	Dodecane
5	Chloroform	19	Nonane	33	Decanal
6	1,2-Dichloroethane	20	(1S)-(-)-alpha-Pinene	34	Heptane
7	Benzene	21	3-Ethyltoluene	35	Hexane
8	2,2,4-Trimethylpentane	22	2-Ethyltoluene	36	Hexadecane
9	1,2-Dichloropropane	23	4-Ethyltoluene	37	Octane

10	Trichloroethylene	24	1,2,3-Trimethylbenzene	38	2,4-Dimethylpentane
11	Bromodichloromethane	25	1,2,4-Trimethylbenzene	39	Pentadecane
12	Ethylbenzene	26	1,4-Dichlorobenzene	40	Tetradecane
13	Toluene	27	1,3,5-Trimethylbenzene		
14	Dibromochloromethane	28	(R)-(+)-Limonene		

In the following part only thirty of the forty-eight compounds were examined and quantified. Table 8.3 shows the total emissions of VOCs during the mixing, compaction and firing process of FCBs incorporating 2% CBs by weight. The emissions of trichloroethylene, 2,2,4-trimethylpentane and nonanal were not detectable. The various volatile compounds are grouped into aromatics, alkanes, halides, aldehydes, ketones, terpenes, and chlorinated hydrocarbons. The alkane compounds including hexane, nonane, heptane, octane, undecane, dodecane, tridecane, tetradecane and pentadecane were found in concentrations ranging from 2.78 to 61.12 μg ml^{-1} for all process's during the production of FCBs incorporating CBs. While the aromatics were found in concentrations ranging from 2.21 to 14.65 μg ml^{-1}. Compounds including benzene, toluene, xylenes, ethylbenzene and styrene were similarly found in a study involving the analysis of VOCs generated from the smoke of cigarettes [367]. The halide chloroform remained constant in concentration throughout each process for both UVC light treated and untreated CBs, although it is a known constituent of tobacco smoke.

The emission of VOCs during the various processes were much higher for untreated CBs as opposed to the UVC treated CBs. For ethylbenzene a 50% reduction was observed during the firing process when UVC treated CBs were used, similarly a 57% reduction in the emission of styrene was observed with the use of UVC treated CBs. This displays the effectiveness of the UVC light radiation on breaking down the covalent bonds of the organic compounds in the CB samples and therefore diminishing the odor generated during the storage and production process of FCBs incorporating CBs.

In general, the emission of VOCs during the firing process was greater than the mixing and compaction processes. This is because of the high temperatures (1050 °C) the FCB samples incorporating CBs are subjected during the firing process to which results in the evaporation of the organic compounds and, therefore its release through the valve located on the top of the furnace to the external environment. The mixing process of the raw materials (clay, CBs, and water) also resulted in a high quantity VOCs being generated due to the release of particles from the fast-mixing process in the mechanical mixer. For the aromatics like 1,2,4-trimethylbenzene, a concentration of 3.64 μg ml^{-1} was released during the firing process for UVC light treated CBs as opposed to 9.76 μg ml^{-1} during the mixing process.

Table 8.3. VOC emission concentrations (μg ml^{-1}) during the mixing, compaction and firing process of FCBs incorporating CBs.

Compounds	Untreated CBs	UVC light treated CBs

	Mixing process	Compaction process	Firing process	Mixing process	Compaction process	Firing process
Aromatics						
Toluene	2.30 ± 0.51	2.82 ± 0.18	2.70 ± 0.27	3.74 ± 0.30	4.21 ± 0.36	2.21 ± 0.18
Ethylbenzene	3.82 ± 0.25	5.69 ± 1.29	4.93 ± 1.37	2.84 ± 0.93	2.41 ± 0.67	2.39 ± 0.18
o-Xylene	3.79 ± 0.46	10.05 ± 8.04	8.01 ± 5.50	3.91 ± 0.33	4.14 ± 0.67	4.70 ± 0.13
Styrene	12.37 ± 0.67	12.72 ± 4.86	14.65 ± 6.17	7.80 ± 0.28	6.57 ± 0.81	6.23 ± 1.17
1,2,4-trimethylbenzene	5.66 ± 3.02	6.41 ± 3.72	10.79 ± 9.03	9.76 ± 0.58	6.95 ± 5.13	3.64 ± 0.08
Alkanes						
Nonane	7.50 ± 1.54	8.56 ± 0.98	11.06 ± 4.42	3.98 ± 0.73	3.90 ± 0.35	2.78 ± 0.27
Undecane	5.32 ± 0.59	8.01 ± 4.28	7.76 ± 1.94	7.45 ± 1.26	6.31 ± 1.21	6.51 ± 1.29
Dodecane	11.14 ± 2.05	11.50 ± 1.94	20.94 ± 13.15	5.25 ± 0.03	6.44 ± 0.81	5.22 ± 0.71
Tridecane	7.16 ± 1.87	51.33 ± 59.17	9.22 ± 0.60	6.83 ± 0.63	8.01 ± 0.48	5.73 ± 0.15
Tetradecane	10.26 ± 4.91	41.75 ± 26.30	28.39 ± 27.41	8.63 ± 0.38	19.86 ± 18.09	8.08 ± 2.92
Heptane	7.98 ± 3.84	47.53 ± 7.11	56.27 ± 17.10	19.29 ± 1.06	61.12 ± 44.56	15.92 ± 4.33
Hexane	12.97 ± 5.51	7.29 ± 0.12	9.17 ± 3.39	8.29 ± 1.16	9.94 ± 1.61	9.03 ± 0.51
2,2,4-Trimethylpentane	n.d	n.d	n.d	n.d	n.d	n.d
Octane	9.21 ± 2.38	38.77 ± 23.60	52.09 ± 41.91	30.81 ± 4.28	25.00 ± 3.58	9.47 ± 3.42
Halides						
Dibromochloromethane	3.95 ± 2.03	22.06 ± 20.97	4.54 ± 0.17	3.05 ± 2.56	24.39 ± 21.68	3.33 ± 3.30
Tetrachloroethylene	8.03 ± 0.09	14.73 ± 6.04	23.16 ± 15.38	6.70 ± 0.16	10.86 ± 1.27	9.23 ± 0.22
Trichloroethylene	n.d	n.d	n.d	n.d	n.d	n.d
Dichloromethane	37.28 ± 26.61	39.95 ± 17.85	13.55 ± 9.28	10.85 ± 5.99	8.73 ± 3.13	17.98 ± 10.74
Chloroform	6.32 ± 0.09	7.04 ± 1.24	6.44 ± 0.18	6.24 ± 0.07	6.42 ± 0.29	6.62 ± 0.46
Bromodichloromethane	8.91 ± 4.30	27.33 ± 4.16	18.45 ± 3.46	6.40 ± 0.02	10.73 ± 2.99	6.03 ± 0.68
Aldehydes						
Nonanal	n.d	n.d	n.d	n.d	n.d	n.d
Decanal	22.73 ± 13.52	60.98 ± 24.09	72.26 ± 53.69	37.66 ± 8.94	68.07 ± 74.37	30.66 ± 25.58
Ketones						
2-Butanone	7.71 ± 0.70	7.49 ± 0.32	6.74 ± 0.53	6.43 ± 0.10	7.28 ± 0.48	6.62 ± 0.08
4-Methyl-2-pentanone	13.72 ± 10.11	17.28 ± 13.93	14.65 ± 5.80	7.89 ± 1.67	10.54 ± 4.24	7.25 ± 0.15
Terpenes						
Limonene	8.98 ± 0.88	44.31 ± 18.89	46.69 ± 39.93	23.96 ± 1.65	19.93 ± 10.09	7.50 ± 0.66
Chlorinated hydrocarbons						
1,2-dichloroethane	7.54 ± 0.92	6.30 ± 0.02	6.54 ± 0.28	7.07 ± 0.05	6.83 ± 0.38	6.66 ± 0.10

8.5 Conclusion

This research mainly aimed to determine the chemical composition of cigarette butt (CB) emissions and incorporate Ultraviolet-C (UVC) light radiation to retard the release of hazardous gases and odor to solve a reoccurring issue involving the release of toxic compounds during the storage and production of fired clay bricks (FCBs) incorporating CBs for the successful implementation on an industrial scale. It was found by using charcoal sorbent tubes, all kinds of volatile organic compounds (VOCs) can be

effectively collected and easily evaluated with chemical test analysis. The gas chromatography-mass spectrometry (GC-MS) test reveals that the hazardous organic constituents in CB emissions mainly include benzaldehyde, benzyl alcohol, p-xylene, styrene, and 2,5-Dimethylanisole. The VOC emissions in UVC treated CB samples are significantly less than that of untreated CB samples, therefore showing UVC radiation can effectively reduce the generation of hazardous gases. The emission of VOCs during the various processes was much higher for untreated CBs as opposed to the UVC treated CBs, displaying the effectiveness of the UVC light radiation in breaking down the covalent bonds within the complex VOCs. For ethylbenzene a 50% reduction in emissions was observed during the firing process when UVC treated CBs were used, similarly, a 57% reduction in the emission of styrene was observed with the use of UVC treated CBs. In general, the emission of VOCs during the firing process was greater than the mixing and compaction processes. This is due to the high temperatures the samples are subjected to, resulting in the evaporation and release of VOCs. However, it is recommended further research is conducted utilizing distinct methods to capture the VOC emissions during the various processes of manufacturing FCBs incorporating various percentages of CBs.

Chapter 9: Conclusion and Recommendations

9.1 Conclusions

This study investigated recycling 0.5%, 1%, 1.5%, and 2% cigarette butts (CBs) by weight (wt) in fired clay bricks (FCBs) and compared them with those of controlled FCBs. In alignment with the research objectives, a series of systematic investigations and detailed studies were conducted to develop environmentally and economically sustainable and practical techniques to recycle CB waste in the production of FCBs on an industrial scale. All objectives relative to the study have been achieved, and a summary of each chapter has been concluded as follows.

In Chapter 2, a comprehensive review was conducted on various studies involving the toxicity and valorization options of CB waste and the feasibility and sustainability of the recycling methods adopted. Eleven CB valorization proposals and practices were studied, including the incorporation of CBs in FCBs and asphalt concrete. Based on the extensive research, it was determined a potential solution to the CB waste conundrum could be through the recycling of CBs in the production of FCBs.

In Chapter 3, numerous physical and mechanical tests were conducted, including water absorption, compressive strength, efflorescence, initial rate of absorption (IRA), density, and shrinkage. It was found compressive strength decreased from 48.6 MPa to 30.8 MPa with the addition of 1% CBs to FCBs, similarly, dry density decreased from 2114 kg m^{-3} to 1983 kg m^{-3} with the addition of 1% CBs. Furthermore, the increase in CB content resulted in an increase in porosity, consequently resulting in an increase in water absorption. The advantages of lightweight construction materials include cheaper production, lower overall life cycle energy usage, and design flexibility. The results found in this study complete and confirm the previous findings on recycling CBs in fired clay bricks at RMIT University and the published proposal that all CBs can be sustainably recycled in FCBs by incorporating 1% CBs in 2.5% of the world's brick production.

Chapter 4 presented the energy savings, thermal conductivity, porosity, micro- and macro-structural analysis results of FCBs incorporating CBs. Based on the novel method developed to monitor energy consumption, it was found for bricks incorporating 1% CBs by wt., a minimum energy saving of 8% is achievable. This means that an estimated $53.15 million in terms of electricity usage could be saved in Australia every year, thereby leading to economic construction. The porosity values ranged from 3.96% to 20.57% for bricks incorporating 0% to 2% CBs by wt. The loss of organic content in the CBs during

the firing process resulted in the formation of macropores within the brick structures. The 3D models of the bricks incorporating CBs demonstrated a homogenous distribution of pores within the brick structures. A reduction of up to 15.12% in thermal conductivity was achieved with the addition of 1% CBs by wt. in clay bricks compared to traditional bricks. That is a decrease from 0.463 W m^{-1} K^{-1} to 0.393 W m^{-1} K^{-1} in thermal conductivity due to the formation of pores from the burnt CBs, thereby improving the thermal performance. Consequently, this will decrease the energy required for cooling and heating in households.

Chapter 5 clarified issues regarding the possible leachate of heavy metals from (1) the use and disposal of bricks incorporating CBs (2) used, unused, and shredded used CBs and (3) paraffin wax and bitumen encapsulated CBs. It was found CBs are more likely to leach heavy metals in areas with highly acidic rain compared to the natural range of precipitation. In addition, more than 50% of all the tested heavy metals were immobilized after the firing process of FCBs incorporating CBs. Hence the addition of CBs in FCBs may be a feasible option to hinder the heavy metals present in CBs. The findings suggest that the variance in pH influences the behavior of the metal concentration leached from bitumen and wax encapsulated CBs for both broken and unbroken samples. The findings show that the leaching behavior of CBs is considerably lower after being encapsulated with wax and bitumen. This is the result of sealing the CBs with viscous and insoluble materials, which prevents the release of metals or the intrusion of any liquid. The leachate results for FCBs incorporating 0%, 0.5%, 1%, 1.5%, and 2% CBs and wax and bitumen encapsulated CBs were found to be insignificant compared to the limits set by the EPA Victoria solid industrial waste guidelines and, therefore, can be categorized as non-hazardous waste.

Chapter 6 presented the concentration levels of 16 EPA polycyclic aromatic hydrocarbons (PAHs) extracted from FCB samples, incorporating various percentages of CBs. The results showed that lower-molecular-weight PAHs (light PAHs), including naphthalene, fluorene, anthracene, pyrene, fluoranthene, and chrysene, with higher volatility and lower lipophilicity, were released at very low concentrations compared to the higher-molecular-weight PAHs (heavy PAHs) from FCB–CB samples. The concentrations of the mean ΣPAH for all samples were well below the EPA solid waste hazard categorization threshold limit for industrial waste. The results from the carbon, hydrogen, nitrogen, and sulfur (CHNS) and thermogravimetric (TG) analysis confirm an almost 100% combustion process during the firing process of FCBs. A content of less than 0.3% suggests that all carbon within the FCB–CB mixture relatively disappeared during the firing process up to 1050 °C. It can be concluded that the results presented in this study clarify the issues raised regarding the possible release of toxic PAHs into the environment during the use and disposal of FCBs incorporating CBs or the potential storage of carbon in the FCB–CB mixture.

Chapter 7 presented the results of a bacteriological investigation of CBs and the implementation procedure for recycling CBs in FCBs, and the pre-processing issues of CBs. The presence of ten common bacteria was investigated on unused, fresh, and old CBs. The most significant finding from the investigation was that *Listeria* spp was detected in the used old CB sample and that all used and unused CB samples except the dried sample had significant counts of *Bacillus* spp. It was also found that the effect of the contact of naphthalene balls with fresh used CBs on the reduction of *Bacillus* spp, bacteria, was very significant. It was suggested shredding of the CBs must be performed in a fully enclosed environment, such as an enclosed industrial shredder. This is necessary as it keeps the dust and chemicals from spilling out into the environment. The odor from volatile organic compound (VOC) emissions in CBs can be an issue for recycling CBs in bricks or any other application. Therefore it is suggested UV light with an appropriate wavelength is implemented to deodorize and sterilize the CBs.

Chapter 8 aimed to determine the chemical composition of CB emissions and incorporate UVC radiation to retard the release of hazardous gases and odor to solve a reoccurring issue involving the release of toxic compounds during the storage and production of FCBs incorporating CBs for the successful implementation on an industrial scale. It was found UVC radiation can effectively reduce the generation of hazardous gases. The pollutant gas production in UVC light treated CB samples is significantly less than that of untreated CB samples. The emission of VOCs during the various processes were much higher for untreated CBs as opposed to the UVC treated CBs. For ethylbenzene a 50% reduction in emissions was observed during the mixing process when UVC treated CBs were used, similarly a 63% reduction in the emission of styrene was observed with the use of UVC treated CBs. In general, the emission of VOCs during the firing process was greater than the mixing and compaction processes.

9.2 Recommendations

The future works recommended are listed as below:

i. A life-cycle assessment (LCA) is recommended to analyse the environmental impacts of incorporating CBs in the production of FCBs.
ii. It is recommended to investigate the leaching performance of toxic metals at the various liquid to solid ratios and for compacted granular material, as described in EPA Methods 1314 and 1315 to consider the leaching of bricks subjected to various disposal or use scenarios.
iii. Further research regarding the emission of volatile organic compounds (VOCs) during the storage of CBs and during the mixing, oven-drying, and firing process of producing FCBs incorporating various percentages of CBs should be conducted.
iv. Other deodorization and sterilization methods for CBs should be examined and investigated and compared with the current UVC light results to determine the ultimate treatment technique.

Chapter 10: References

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Appendix A: Publications

JOURNAL PUBLICATIONS

Thesis Related Articles

1. **Kurmus, H.** and Mohajerani, A., 2020. The toxicity and valorization options of cigarette butts. Waste Management, 104, pp. 104–118.
   (Author Contribution: Kurmus, H. (80%), Mohajerani, A. (20%)).
2. **Kurmus, H.** and Mohajerani, A., 2020. Recycling of cigarette butts in fired clay bricks: A new laboratory investigation. Materials, 13, p.790.
   (Author Contribution: Kurmus, H. (80%), Mohajerani, A. (20%)).
3. **Kurmus, H.** and Mohajerani, A., 2020. Leachate Analysis of Heavy Metals in Cigarette Butts and Bricks Incorporated with Cigarette Butts. Materials, 13(12), p. 2843.
   (Author Contribution: Kurmus, H. (80%), Mohajerani, A. (20%)).
4. **Kurmus, H.** and Mohajerani, A., 2021. Energy Savings, Thermal Conductivity, Micro and Macro Structural Analysis of Fired Clay Bricks Incorporating Cigarette Butts. Construction and Building Materials, p. 283.
   (Author Contribution: Kurmus, H. (80%), Mohajerani, A. (20%)).
5. Mohajerani, A., Qun Hui, S., Shen, C., Suntovski, J., Rodwell, G., **Kurmus, H**., Hana, M. and Rahman, M.T., 2020. Implementation of Recycling Cigarette Butts in Lightweight Bricks and a Proposal for Ending the Littering of Cigarette Butts in Our Cities. Materials, 13, p. 4023.
   (Author Contribution: Mohajerani, A. (20%), Qun Hui, S. (10%), Shen, C. (10%), Suntovski, J. (10%), Rodwell, G. (10%), **Kurmus, H**. (20%), Hana, M. (10%) and Rahman, M.T. (10%)).
6. **Kurmus, H.**, Mohajerani, A. and Grist, S., 2021. Polycyclic aromatic hydrocarbons in fired clay bricks incorporating cigarette butts. Materials, 14(8), p. 2032.
   (Author Contribution: Kurmus, H. (80%), Mohajerani, A. (10%), Grist, S. (10%))
7. Mohajerani, A., **Kurmus, H.**, Rahman, M.T., Smith, J., Woo, S.S., Jones, D., Dastjerdi, M.G., Pulitano, J., Nguyen, B.T., and Calderón, C. (2021). Bitumen and Paraffin Wax Encapsulated Cigarette Butt; Physical Properties and Leachate Analysis. International Journal of Pavement Research and Technology.
   (Author Contribution: Mohajerani, A. (23%), Kurmus, H. (21%), Rahman, M.T. (21%), Smith, J. (5%), Woo, S.S. (5%), Jones, D. (5%), Dastjerdi, M.G. (5%), Pulitano, J. (5%), Nguyen, B.T. (5%), and Calderón, C. (5%)).

8. **Kurmus, H.,** Mohajerani, A., and Suntovski, J., 2021. Application of UVC light to minimize the odor generated from volatile organic compounds during the production of fired clay bricks incorporating cigarette butts. Clean Technologies and Environmental Policy. (under review) (Author Contribution: Kurmus, H. (70%), Mohajerani, A. (20%) and Suntovski, J. (10%)).

Other Collaborative Articles

1. Mohajerani, A., Burnett, L., Smith, J.V., Markovski, S., Rodwell, G., Rahman, M.T., **Kurmus, H**., Mirzababaei, M., Arulrajah, A., Horpibulsuk, S. and Maghool, F., 2020. Recycling waste rubber tyres in construction materials and associated environmental considerations: A review. Resources, Conservation and Recycling, 155, 104679.
2. Mohajerani, A., Burnett, L., Smith, J.V., **Kurmus, H**., Milas, J., Arulrajah, A., Horpibulsuk, S. and Abdul Kadir, A., 2019. Nanoparticles in construction materials and other applications, and implications of nanoparticle use. Materials, 12(19), 3052. https://doi.org/10.3390/ma12193052
3. Mohajerani, A., **Kurmus, H**., Tran, L., Maha Arachchillegedar, C., Mirzababaei, M. and Arulrajah, A., 2018. Clegg impact hammer: an equipment for evaluation of the strength characteristics of pavement materials, turf, and natural and artificial playing surfaces: a review. Road Materials and Pavement Design, 21(2), 1-19. https://doi.org/10.1080/14680629.2018.1502092
4. Mohajerani, A. Lound, S. Liassos, G. **Kurmus, H**. Ukwatta, A. & Nazari, M. 2017. Physical, mechanical and chemical properties of biosolids and raw brown coal fly ash, and their combination for road structural fill applications. Journal of Cleaner Production, 166, 1-11. https://doi.org/10.1016/j.jclepro.2017.07.250
5. Mohajerani, A. Vajna, J. Cheung, T.H.H. **Kurmus, H**. Arulrajah, A. & Horpibulsuk, S. 2017. Practical recycling applications of crushed waste glass in construction materials: A review. Construction and Building Materials, 156, 443-467. https://doi.org/10.1016/j.conbuildmat.2017.09.005

PEER-REVIEWED CONFERENCE PUBLICATIONS

1. **Kurmus, H.** and Mohajerani, A., 2019. A Continuation Study on Recycling Cigarette butts in Fired Clay Bricks. In WEC2019: World Engineers Convention 2019 (p. 687). Engineers Australia, Melbourne, Australia.
2. **Kurmus, H.** and Mohajerani, A., 2020. A Comprehensive Leachate Study of Fired-Clay Bricks Incorporating Cigarette Butts, Paper Presented to GEOMATE; 10th Int. Conf. on Geotechnique, Construction Materials and Environment, Melbourne, Australia.